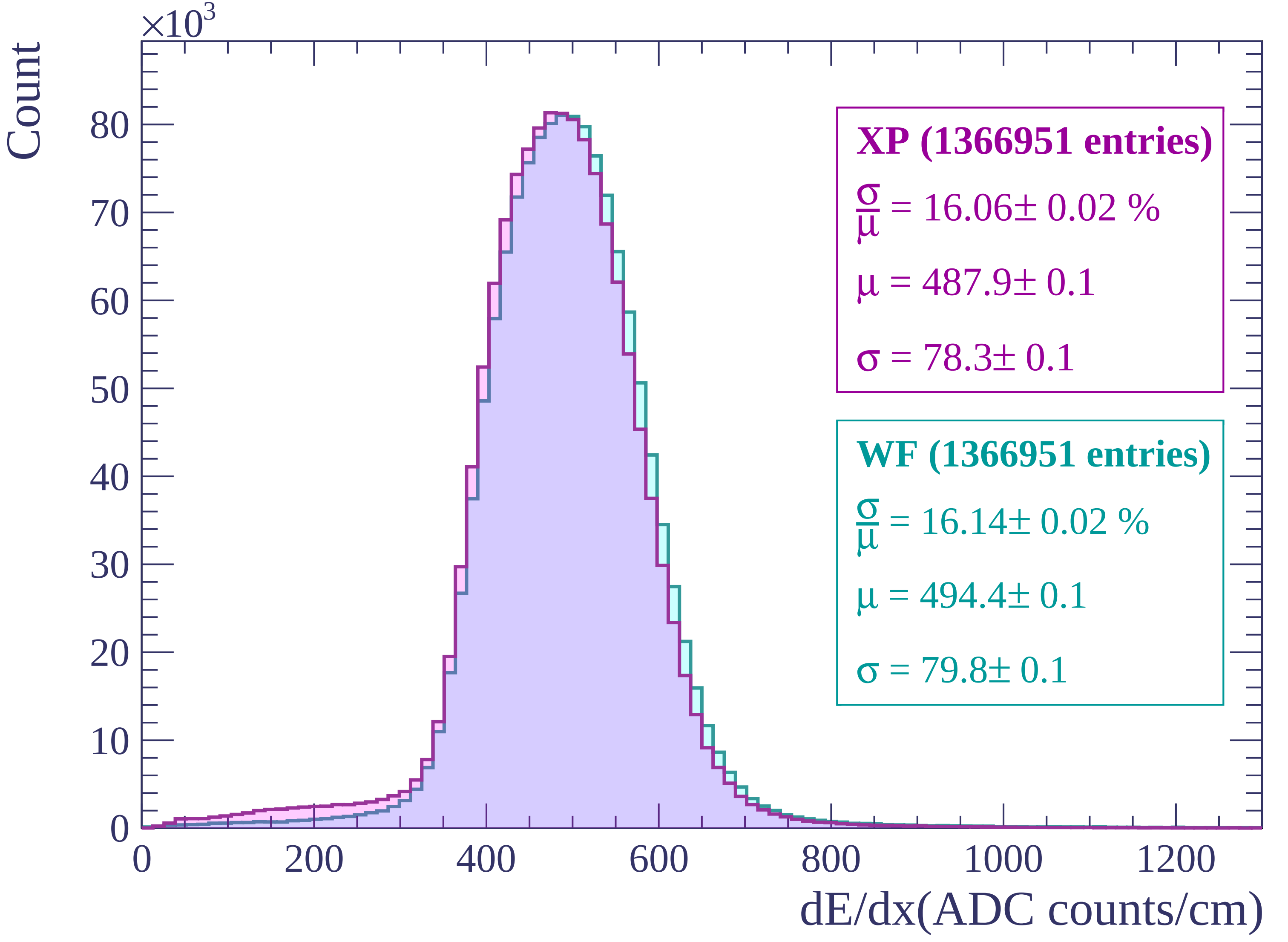
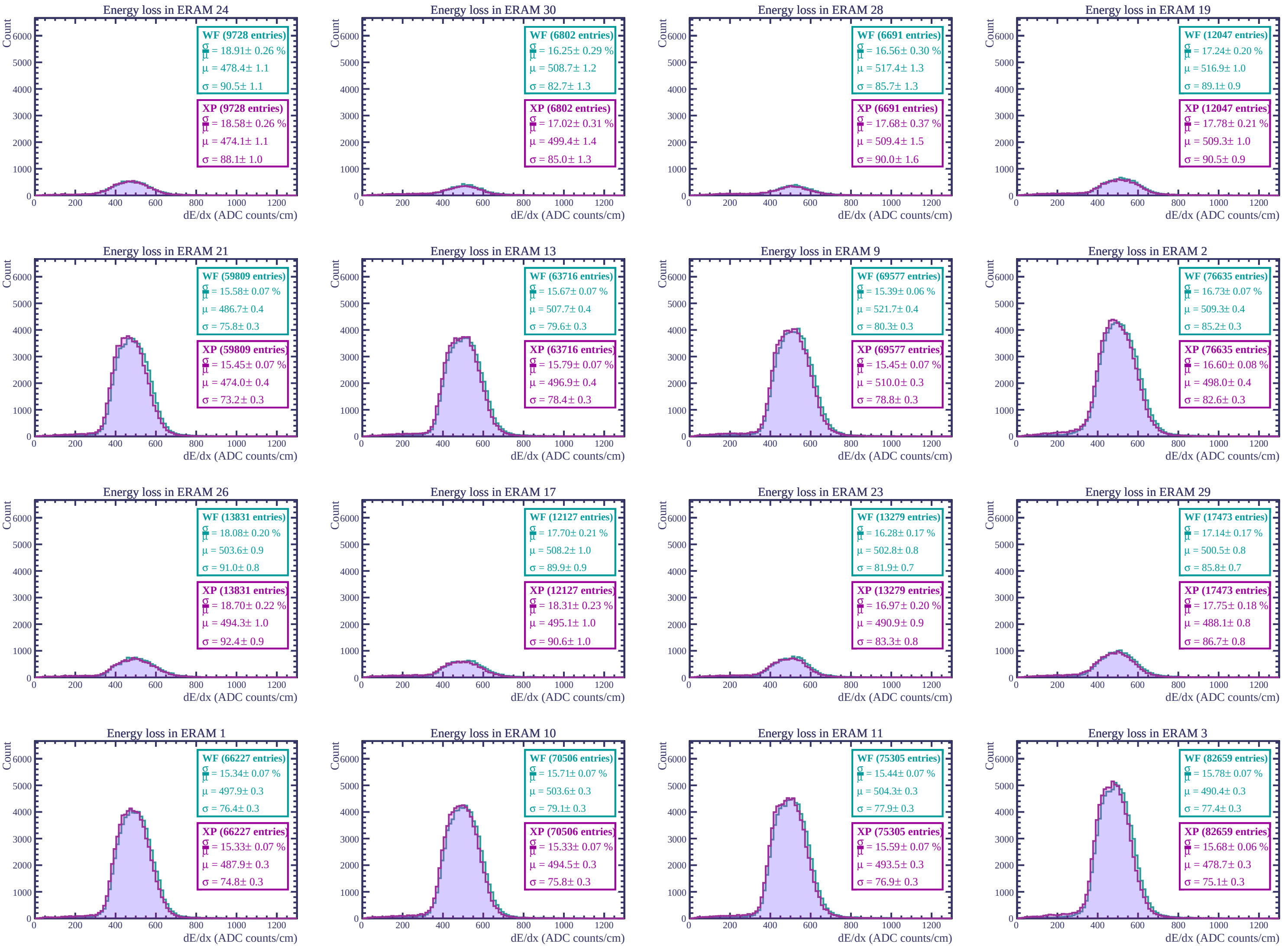
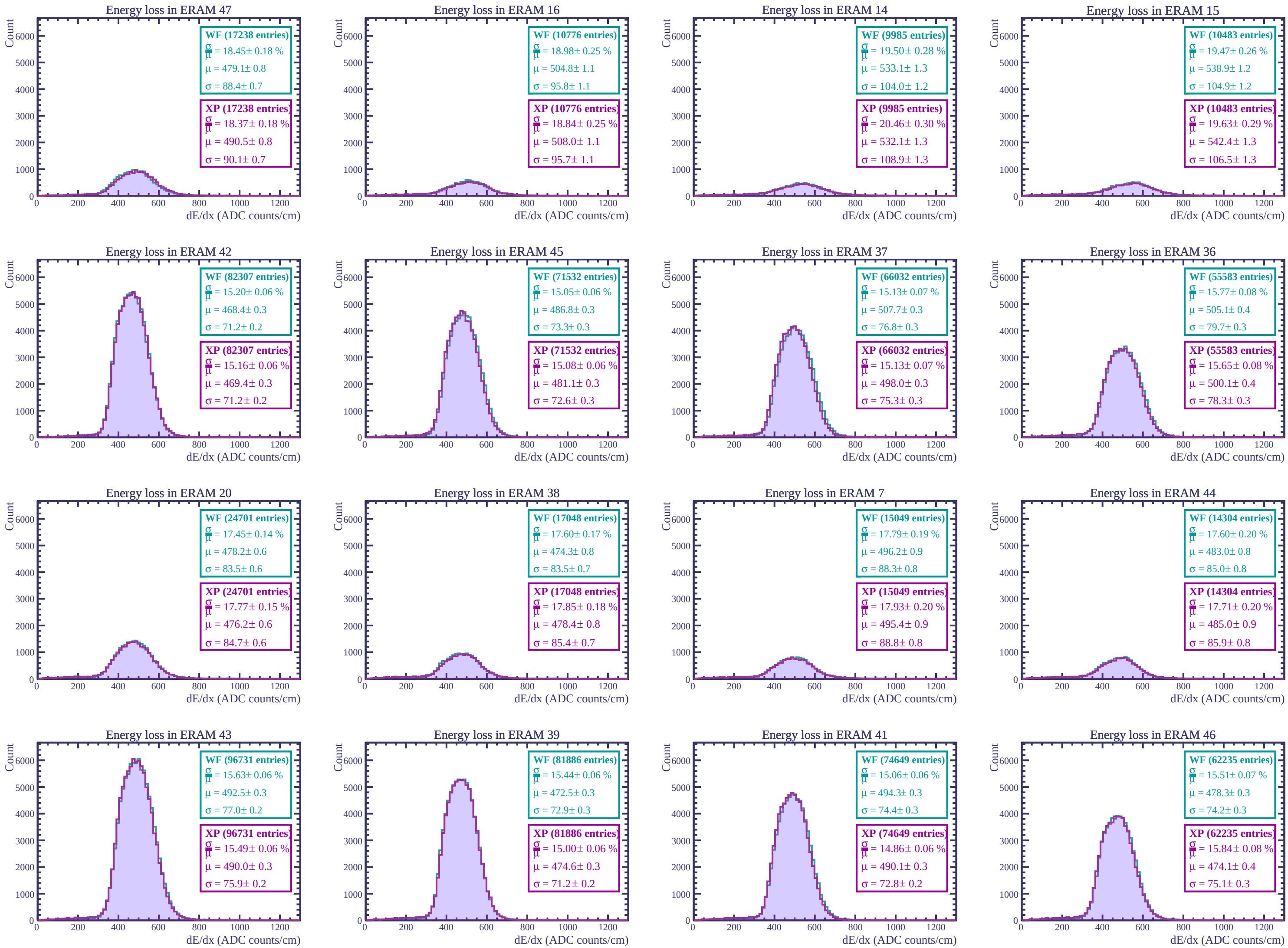


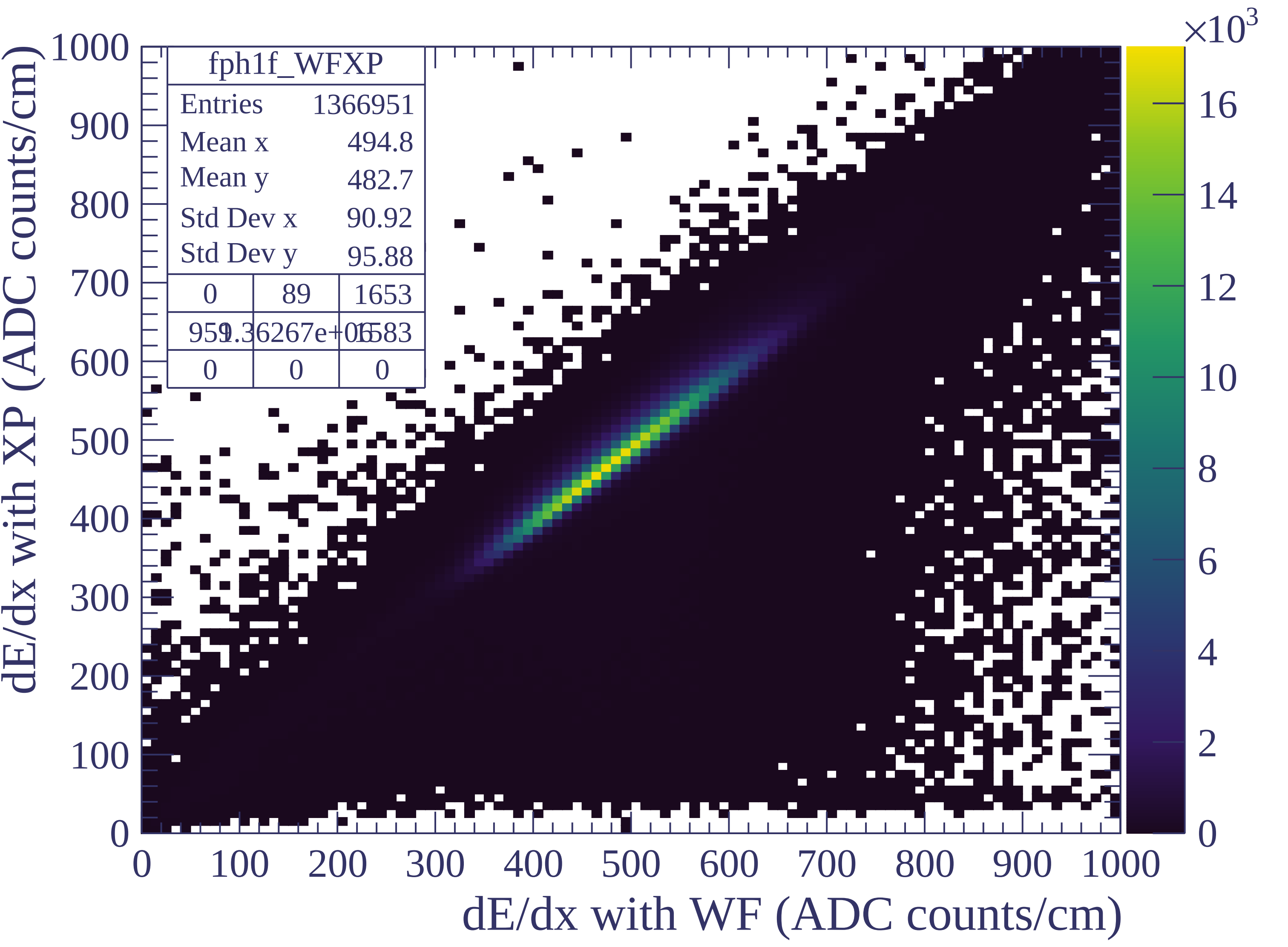
×10^3
Count
80
70
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (1366951 entries)
σ/μ = 16.06± 0.02 %
μ = 487.9± 0.1
σ = 78.3± 0.1
WF (1366951 entries)
σ/μ = 16.14± 0.02 %
μ = 494.4± 0.1
σ = 79.8± 0.1



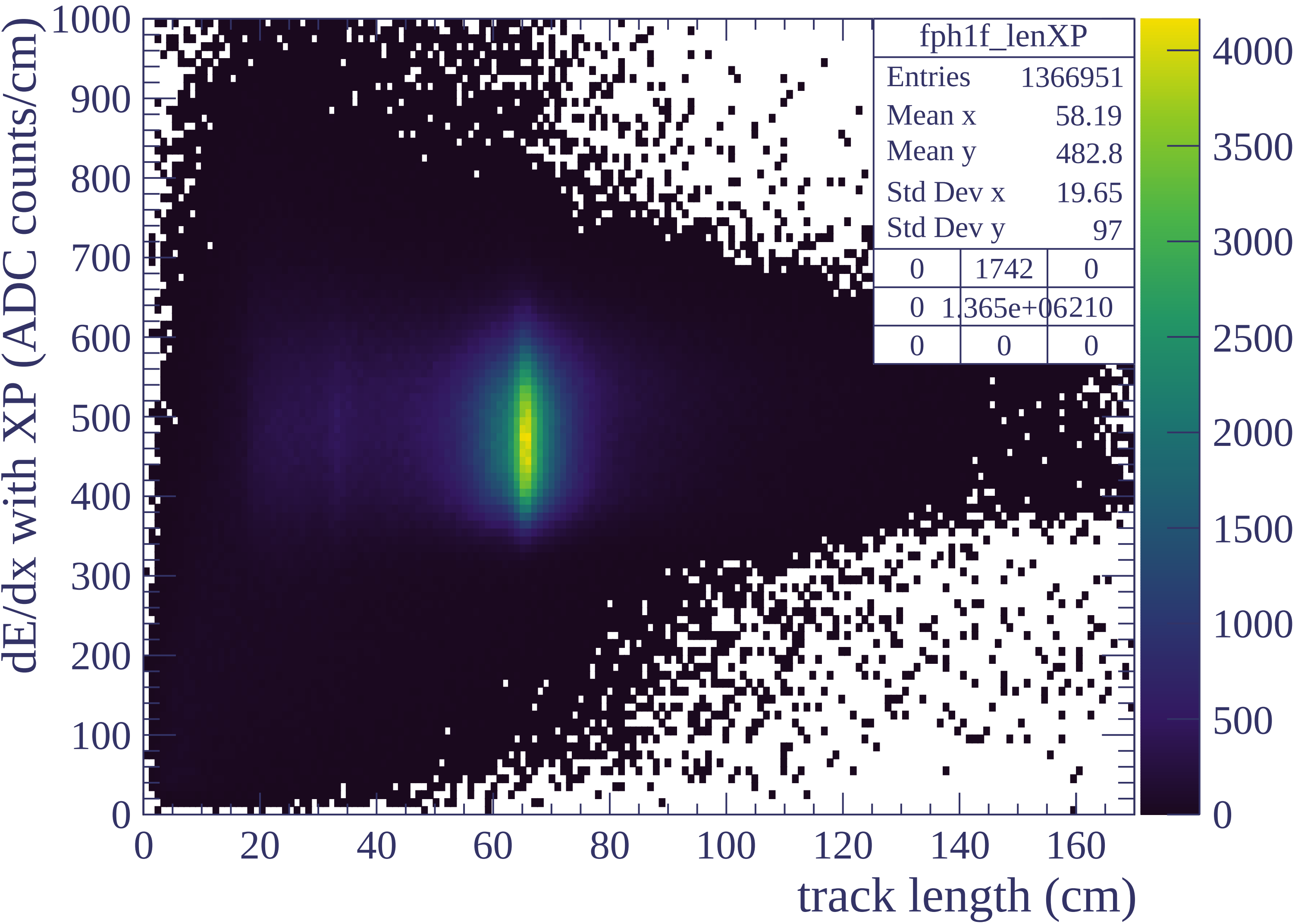
Energy loss in ERAM 24
WF (9728 entries)
σ/μ = 18.91± 0.26 %
μ = 478.4± 1.1
σ = 90.5± 1.1
XP (9728 entries)
σ/μ = 18.58± 0.26 %
μ = 474.1± 1.1
σ = 88.1± 1.0
Energy loss in ERAM 30
WF (6802 entries)
σ/μ = 16.25± 0.29 %
μ = 508.7± 1.2
σ = 82.7± 1.3
XP (6802 entries)
σ/μ = 17.02± 0.31 %
μ = 499.4± 1.4
σ = 85.0± 1.3
Energy loss in ERAM 28
WF (6691 entries)
σ/μ = 16.56± 0.30 %
μ = 517.4± 1.3
σ = 85.7± 1.3
XP (6691 entries)
σ/μ = 17.68± 0.37 %
μ = 509.4± 1.5
σ = 90.0± 1.6
Energy loss in ERAM 19
WF (12047 entries)
σ/μ = 17.24± 0.20 %
μ = 516.9± 1.0
σ = 89.1± 0.9
XP (12047 entries)
σ/μ = 17.78± 0.21 %
μ = 509.3± 1.0
σ = 90.5± 0.9
Energy loss in ERAM 21
WF (59809 entries)
σ/μ = 15.58± 0.07 %
μ = 486.7± 0.4
σ = 75.8± 0.3
XP (59809 entries)
σ/μ = 15.45± 0.07 %
μ = 474.0± 0.4
σ = 73.2± 0.3
Energy loss in ERAM 13
WF (63716 entries)
σ/μ = 15.67± 0.07 %
μ = 507.7± 0.4
σ = 79.6± 0.3
XP (63716 entries)
σ/μ = 15.79± 0.07 %
μ = 496.9± 0.4
σ = 78.4± 0.3
Energy loss in ERAM 9
WF (69577 entries)
σ/μ = 15.39± 0.06 %
μ = 521.7± 0.4
σ = 80.3± 0.3
XP (69577 entries)
σ/μ = 15.45± 0.07 %
μ = 510.0± 0.3
σ = 78.8± 0.3
Energy loss in ERAM 2
WF (76635 entries)
σ/μ = 16.73± 0.07 %
μ = 509.3± 0.4
σ = 85.2± 0.3
XP (76635 entries)
σ/μ = 16.60± 0.08 %
μ = 498.0± 0.4
σ = 82.6± 0.3
Energy loss in ERAM 26
WF (13831 entries)
σ/μ = 18.08± 0.20 %
μ = 503.6± 0.9
σ = 91.0± 0.8
XP (13831 entries)
σ/μ = 18.70± 0.22 %
μ = 494.3± 1.0
σ = 92.4± 0.9
Energy loss in ERAM 17
WF (12127 entries)
σ/μ = 17.70± 0.21 %
μ = 508.2± 1.0
σ = 89.9± 0.9
XP (12127 entries)
σ/μ = 18.31± 0.23 %
μ = 495.1± 1.0
σ = 90.6± 1.0
Energy loss in ERAM 23
WF (13279 entries)
σ/μ = 16.28± 0.17 %
μ = 502.8± 0.8
σ = 81.9± 0.7
XP (13279 entries)
σ/μ = 16.97± 0.20 %
μ = 490.9± 0.9
σ = 83.3± 0.8
Energy loss in ERAM 29
WF (17473 entries)
σ/μ = 17.14± 0.17 %
μ = 500.5± 0.8
σ = 85.8± 0.7
XP (17473 entries)
σ/μ = 17.75± 0.18 %
μ = 488.1± 0.8
σ = 86.7± 0.8
Energy loss in ERAM 1
WF (66227 entries)
σ/μ = 15.34± 0.07 %
μ = 497.9± 0.3
σ = 76.4± 0.3
XP (66227 entries)
σ/μ = 15.33± 0.07 %
μ = 487.9± 0.3
σ = 74.8± 0.3
Energy loss in ERAM 10
WF (70506 entries)
σ/μ = 15.71± 0.07 %
μ = 503.6± 0.3
σ = 79.1± 0.3
XP (70506 entries)
σ/μ = 15.33± 0.07 %
μ = 494.5± 0.3
σ = 75.8± 0.3
Energy loss in ERAM 11
WF (75305 entries)
σ/μ = 15.44± 0.07 %
μ = 504.3± 0.3
σ = 77.9± 0.3
XP (75305 entries)
σ/μ = 15.59± 0.07 %
μ = 493.5± 0.3
σ = 76.9± 0.3
Energy loss in ERAM 3
WF (82659 entries)
σ/μ = 15.78± 0.07 %
μ = 490.4± 0.3
σ = 77.4± 0.3
XP (82659 entries)
σ/μ = 15.68± 0.06 %
μ = 478.7± 0.3
σ = 75.1± 0.3
Count
dE/dx (ADC counts/cm)

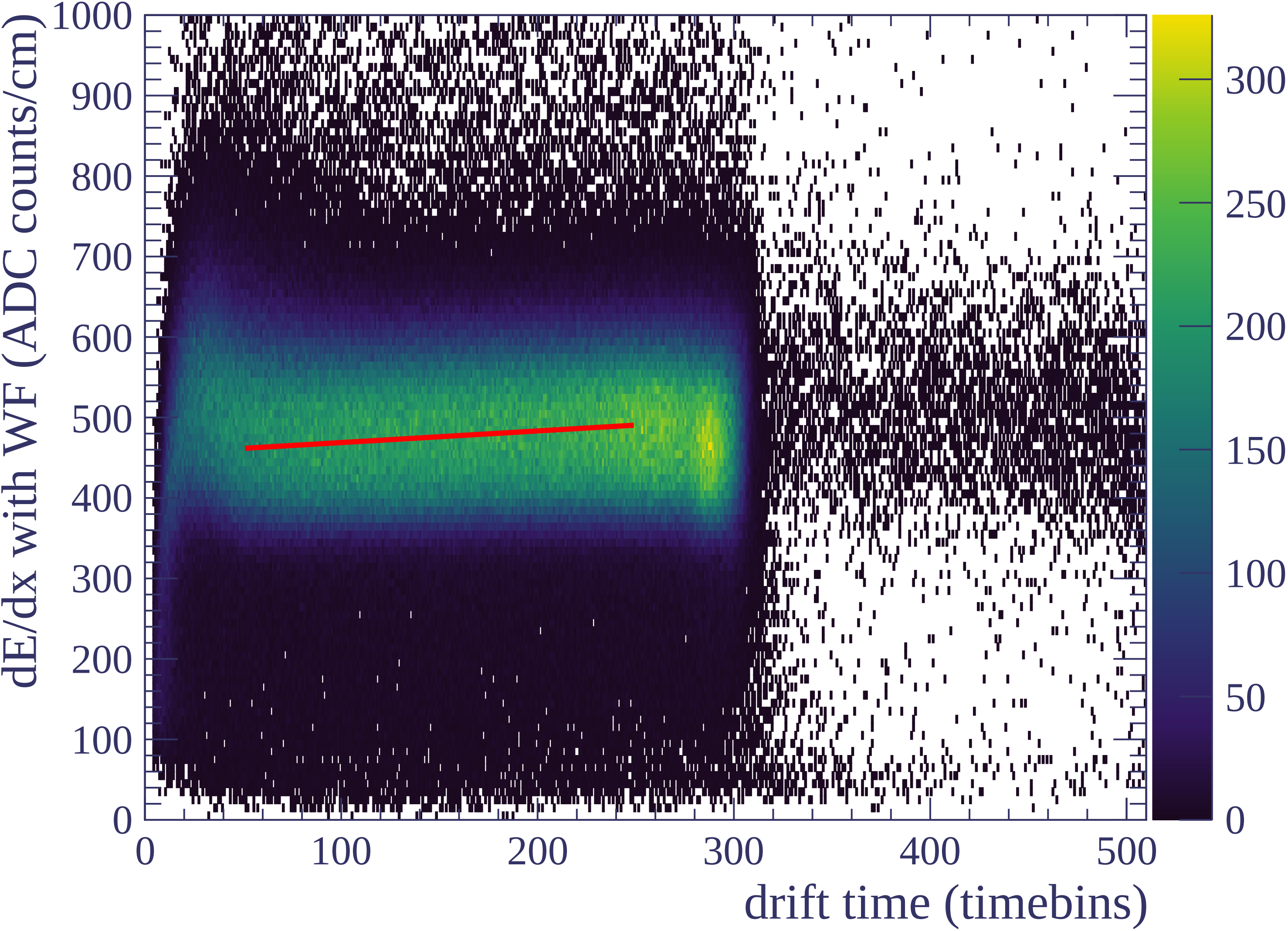


Energy loss in ERAM 47
WF (17238 entries)
σ/μ = 18.45± 0.18 %
μ = 479.1± 0.8
σ = 88.4± 0.7
XP (17238 entries)
σ/μ = 18.37± 0.18 %
μ = 490.5± 0.8
σ = 90.1± 0.7
Energy loss in ERAM 16
WF (10776 entries)
σ/μ = 18.98± 0.25 %
μ = 504.8± 1.1
σ = 95.8± 1.1
XP (10776 entries)
σ/μ = 18.84± 0.25 %
μ = 508.0± 1.1
σ = 95.7± 1.1
Energy loss in ERAM 14
WF (9985 entries)
σ/μ = 19.50± 0.28 %
μ = 533.1± 1.3
σ = 104.0± 1.2
XP (9985 entries)
σ/μ = 20.46± 0.30 %
μ = 532.1± 1.3
σ = 108.9± 1.3
Energy loss in ERAM 15
WF (10483 entries)
σ/μ = 19.47± 0.26 %
μ = 538.9± 1.2
σ = 104.9± 1.2
XP (10483 entries)
σ/μ = 19.63± 0.29 %
μ = 542.4± 1.3
σ = 106.5± 1.3
Energy loss in ERAM 42
WF (82307 entries)
σ/μ = 15.20± 0.06 %
μ = 468.4± 0.3
σ = 71.2± 0.2
XP (82307 entries)
σ/μ = 15.16± 0.06 %
μ = 469.4± 0.3
σ = 71.2± 0.2
Energy loss in ERAM 45
WF (71532 entries)
σ/μ = 15.05± 0.06 %
μ = 486.8± 0.3
σ = 73.3± 0.3
XP (71532 entries)
σ/μ = 15.08± 0.06 %
μ = 481.1± 0.3
σ = 72.6± 0.3
Energy loss in ERAM 37
WF (66032 entries)
σ/μ = 15.13± 0.07 %
μ = 507.7± 0.3
σ = 76.8± 0.3
XP (66032 entries)
σ/μ = 15.13± 0.07 %
μ = 498.0± 0.3
σ = 75.3± 0.3
Energy loss in ERAM 36
WF (55583 entries)
σ/μ = 15.77± 0.08 %
μ = 505.1± 0.4
σ = 79.7± 0.3
XP (55583 entries)
σ/μ = 15.65± 0.08 %
μ = 500.1± 0.4
σ = 78.3± 0.3
Energy loss in ERAM 20
WF (24701 entries)
σ/μ = 17.45± 0.14 %
μ = 478.2± 0.6
σ = 83.5± 0.6
XP (24701 entries)
σ/μ = 17.77± 0.15 %
μ = 476.2± 0.6
σ = 84.7± 0.6
Energy loss in ERAM 38
WF (17048 entries)
σ/μ = 17.60± 0.17 %
μ = 474.3± 0.8
σ = 83.5± 0.7
XP (17048 entries)
σ/μ = 17.85± 0.18 %
μ = 478.4± 0.8
σ = 85.4± 0.7
Energy loss in ERAM 7
WF (15049 entries)
σ/μ = 17.79± 0.19 %
μ = 496.2± 0.9
σ = 88.3± 0.8
XP (15049 entries)
σ/μ = 17.93± 0.20 %
μ = 495.4± 0.9
σ = 88.8± 0.8
Energy loss in ERAM 44
WF (14304 entries)
σ/μ = 17.60± 0.20 %
μ = 483.0± 0.8
σ = 85.0± 0.8
XP (14304 entries)
σ/μ = 17.71± 0.20 %
μ = 485.0± 0.9
σ = 85.9± 0.8
Energy loss in ERAM 43
WF (96731 entries)
σ/μ = 15.63± 0.06 %
μ = 492.5± 0.3
σ = 77.0± 0.2
XP (96731 entries)
σ/μ = 15.49± 0.06 %
μ = 490.0± 0.3
σ = 75.9± 0.2
Energy loss in ERAM 39
WF (81886 entries)
σ/μ = 15.44± 0.06 %
μ = 472.5± 0.3
σ = 72.9± 0.3
XP (81886 entries)
σ/μ = 15.00± 0.06 %
μ = 474.6± 0.3
σ = 71.2± 0.2
Energy loss in ERAM 41
WF (74649 entries)
σ/μ = 15.06± 0.06 %
μ = 494.3± 0.3
σ = 74.4± 0.3
XP (74649 entries)
σ/μ = 14.86± 0.06 %
μ = 490.1± 0.3
σ = 72.8± 0.2
Energy loss in ERAM 46
WF (62235 entries)
σ/μ = 15.51± 0.07 %
μ = 478.3± 0.3
σ = 74.2± 0.3
XP (62235 entries)
σ/μ = 15.84± 0.08 %
μ = 474.1± 0.4
σ = 75.1± 0.3
Count
dE/dx (ADC counts/cm)

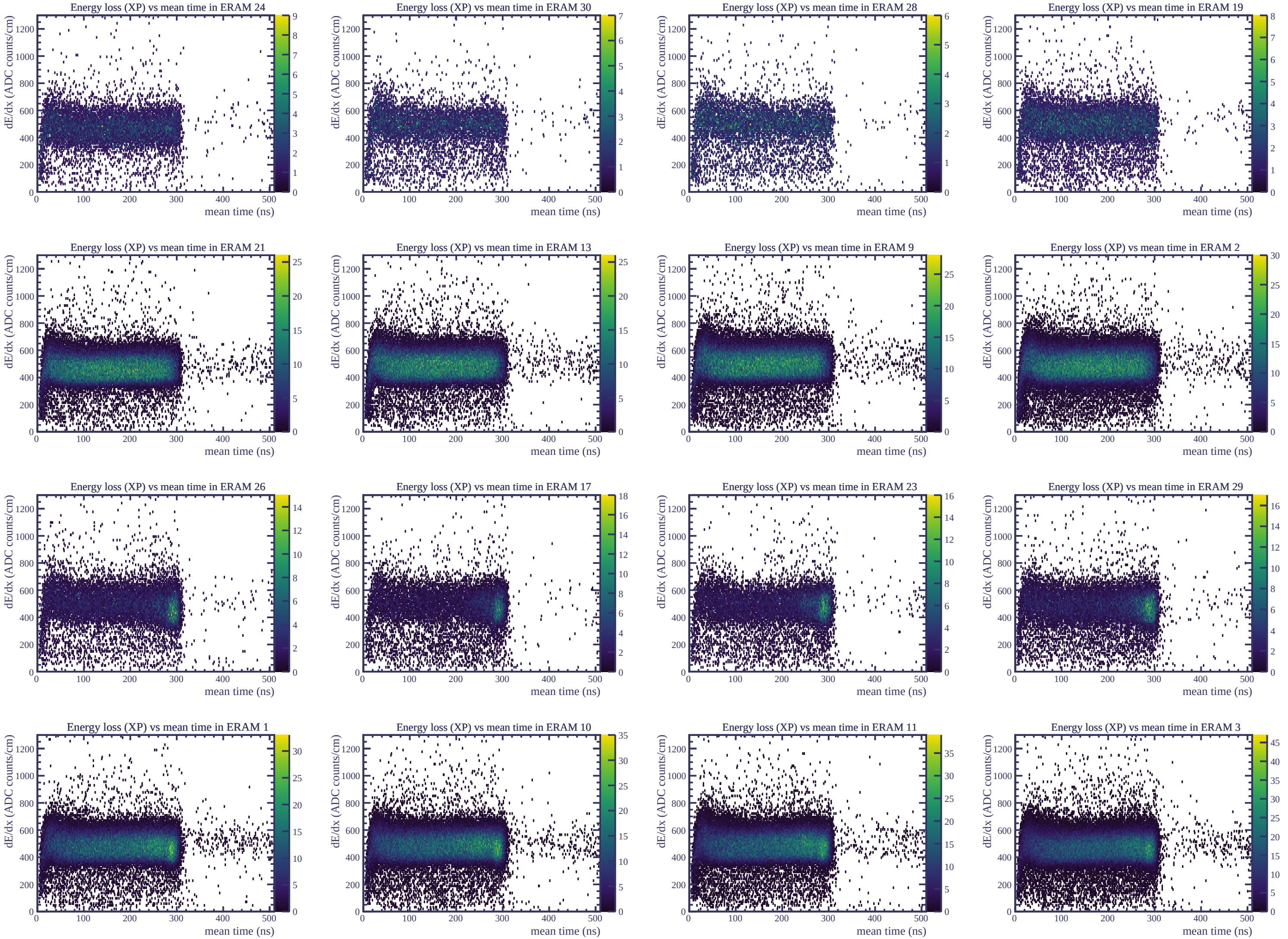


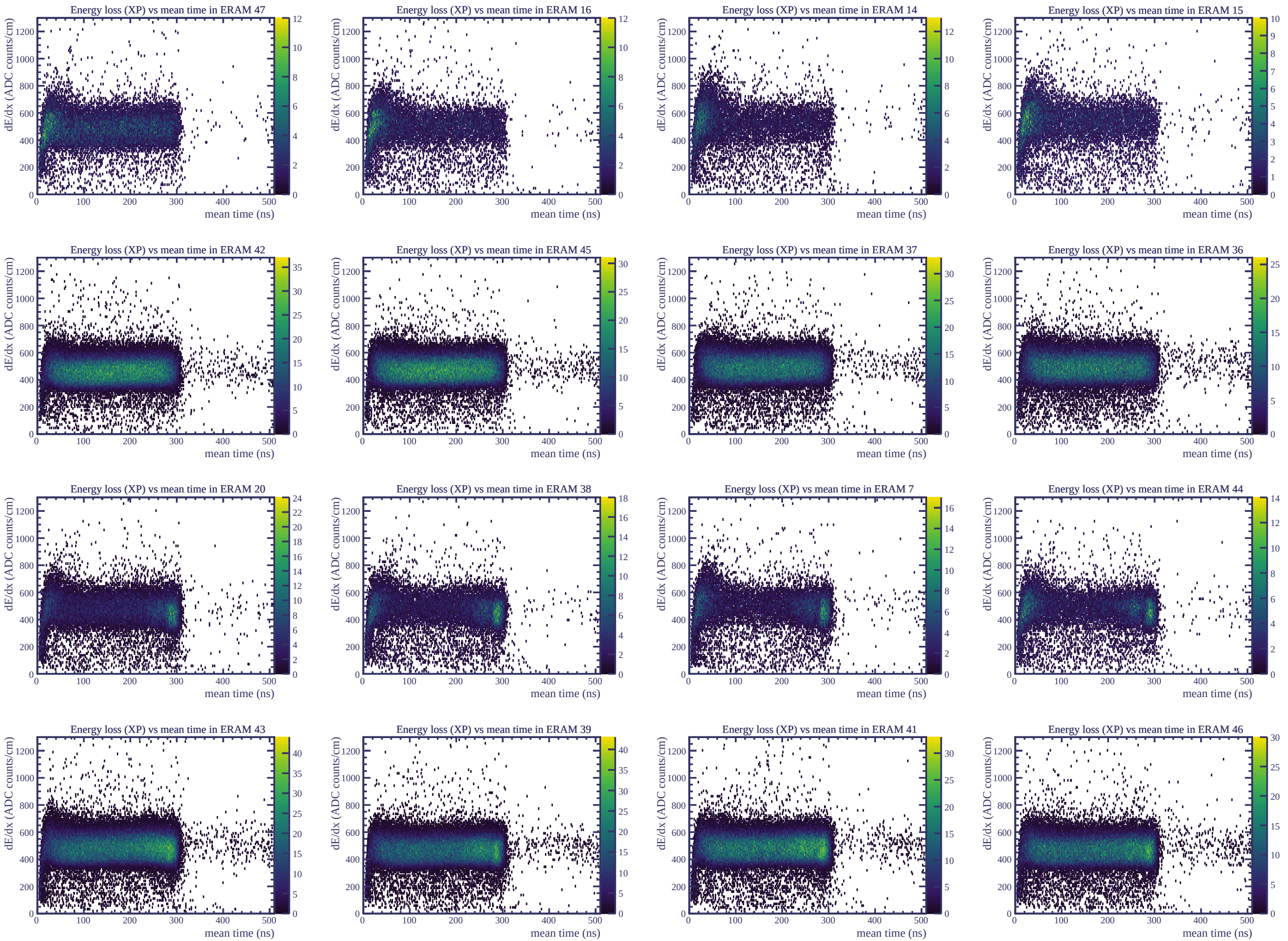
fph1f_WFXP
Entries 1366951
Mean x 494.8
Mean y 482.7
Std Dev x 90.92
Std Dev y 95.88
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
×10^3

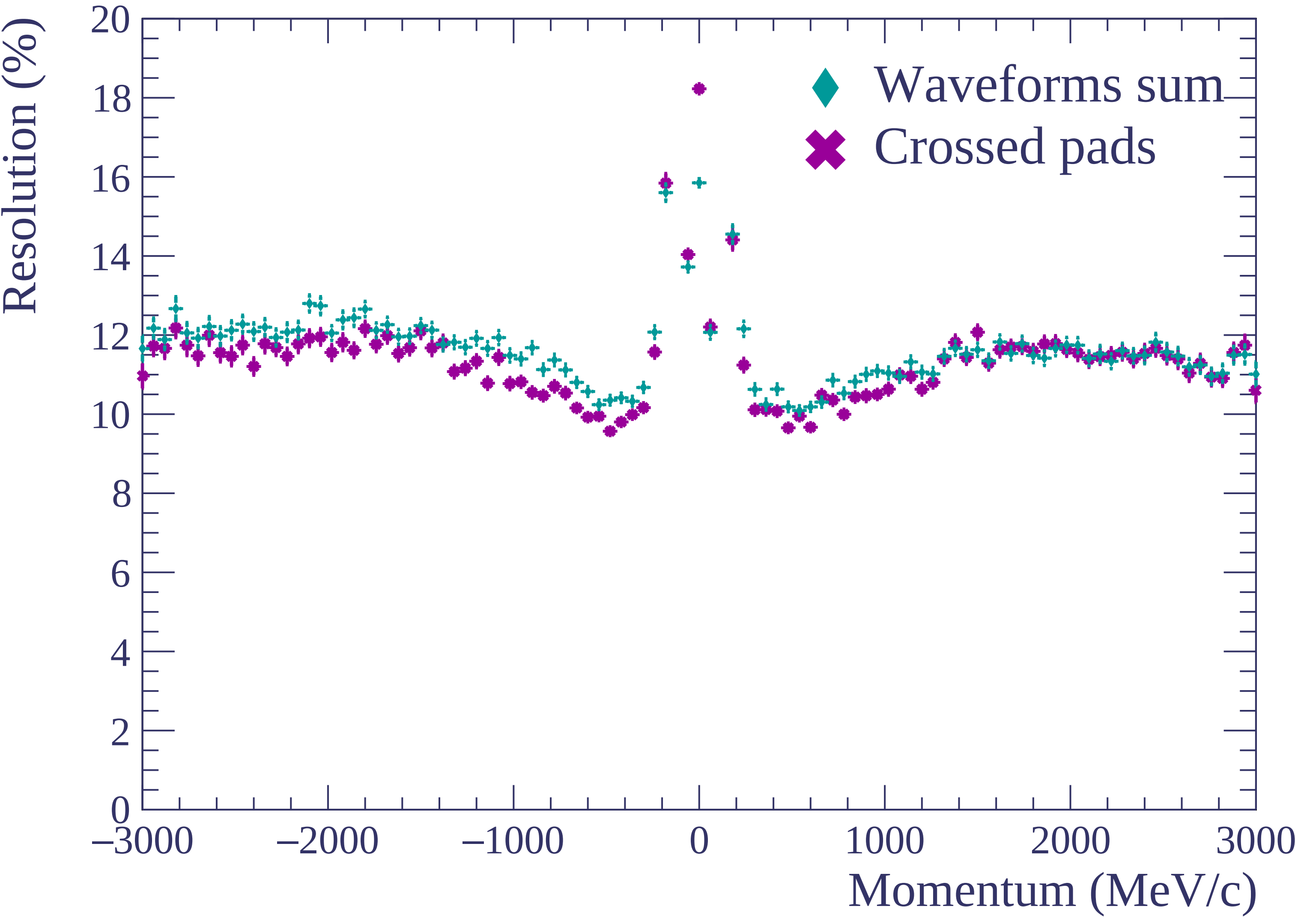

fph1f_lenXP
Entries 1366951
Mean x 58.19
Mean y 482.8
Std Dev x 19.65
Std Dev y 97
0 1742 0
0 1.365e+06210
0 0 0
dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600
700
800
900
1000
0
500
1000
1500
2000
2500
3000
3500
4000



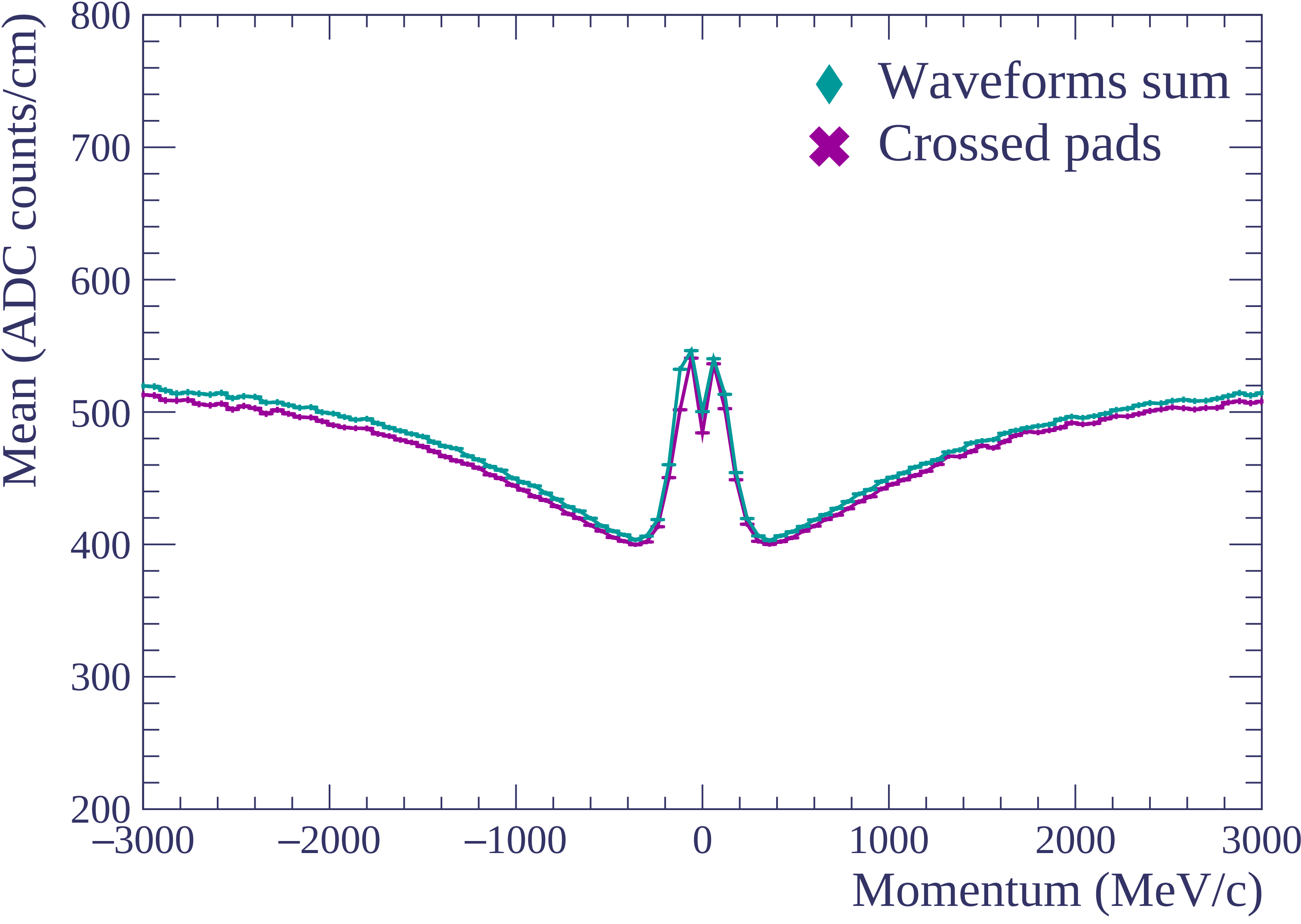
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
50
100
150
200
250
300


Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)

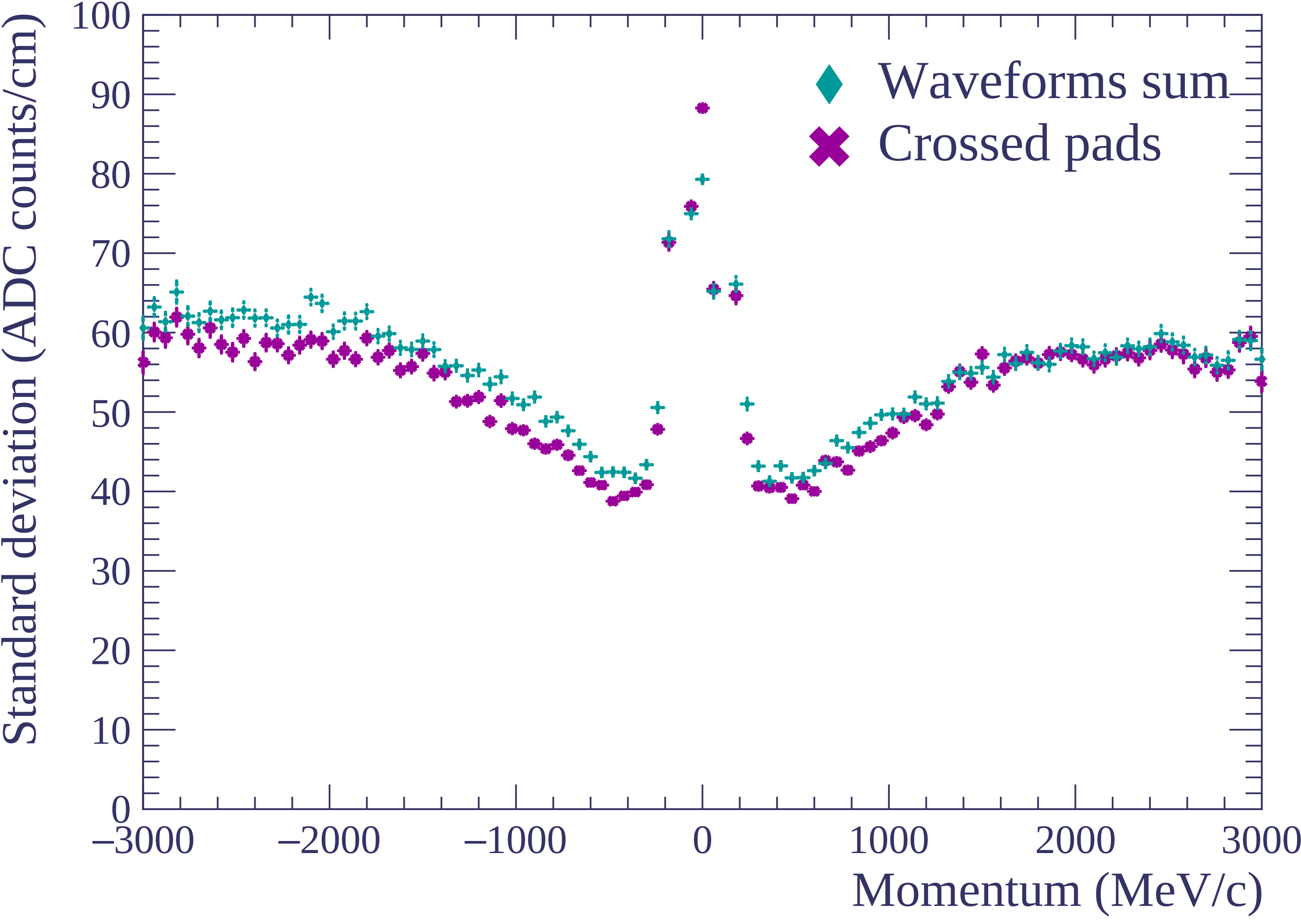

Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)



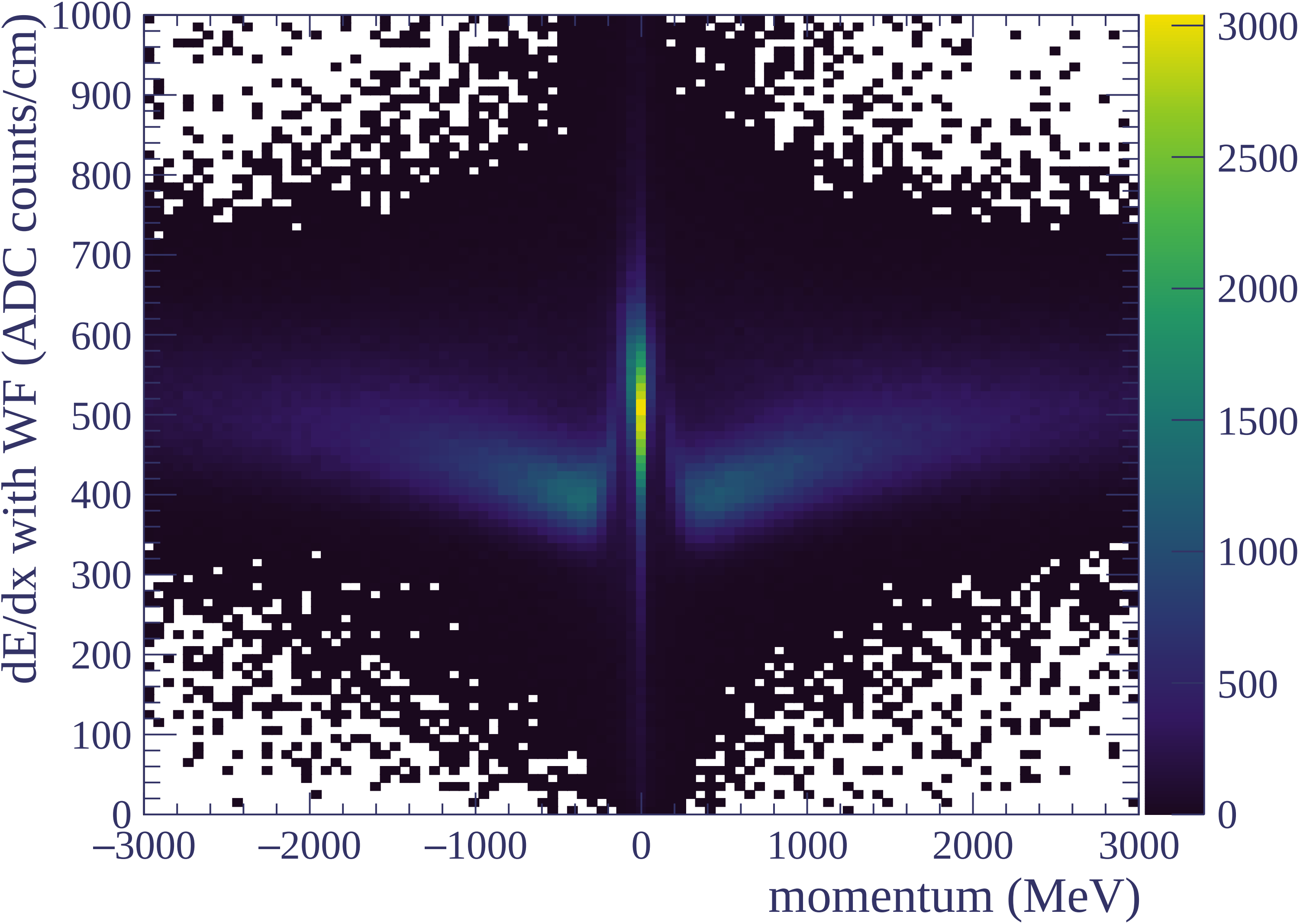
Waveforms sum
Crossed pads
Resolution (%)
Momentum (MeV/c)
0
2
4
6
8
10
12
14
16
18
20
−3000
−2000
−1000
0
1000
2000
3000

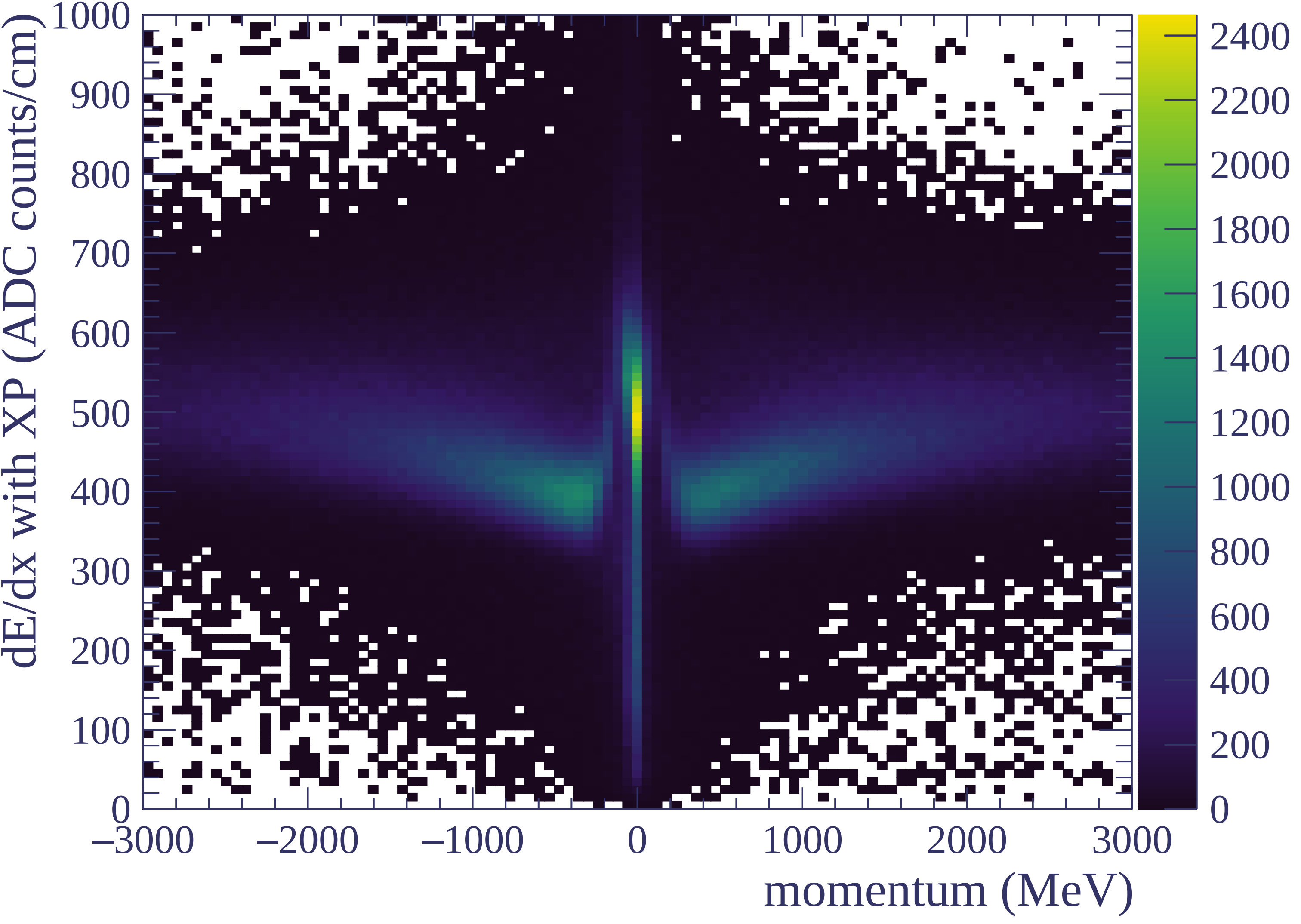


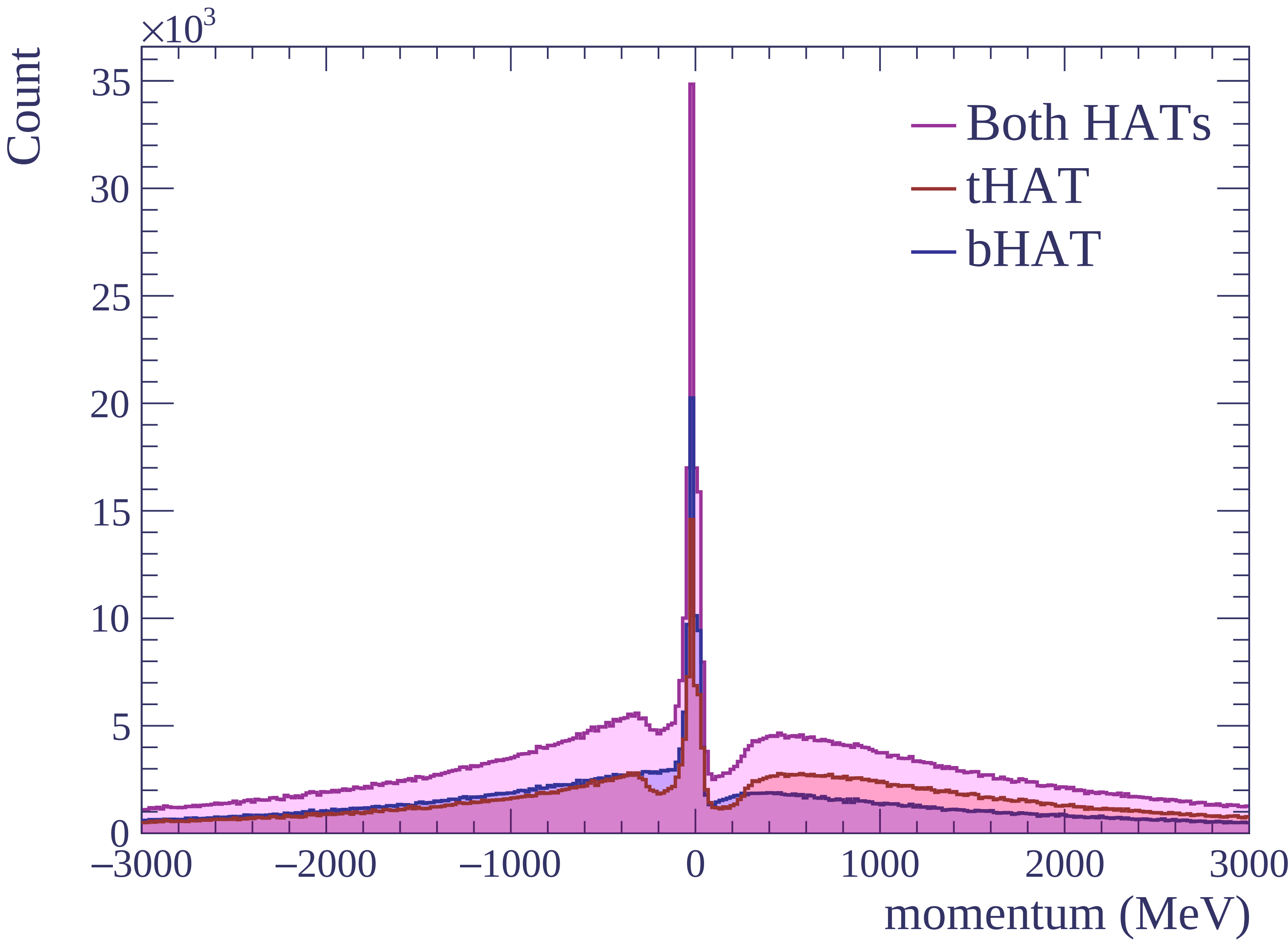
Waveforms sum
Crossed pads
Mean (ADC counts/cm)
Momentum (MeV/c)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000



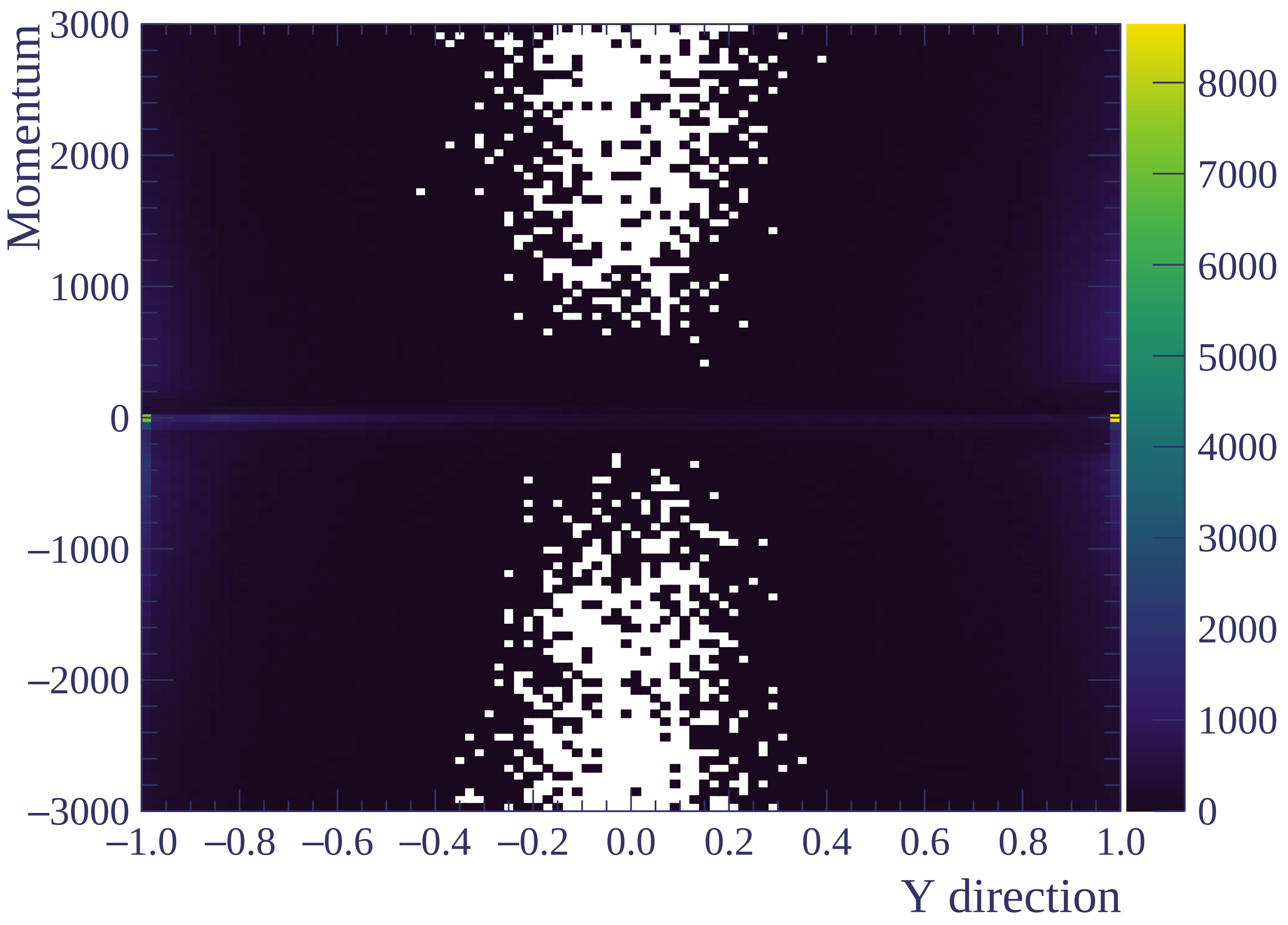
Standard deviation (ADC counts/cm)
Momentum (MeV/c)
Waveforms sum
Crossed pads
0
10
20
30
40
50
60
70
80
90
100
−3000
−2000
−1000
0
1000
2000
3000


dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
3000
2500
2000
1500
1000
500
0

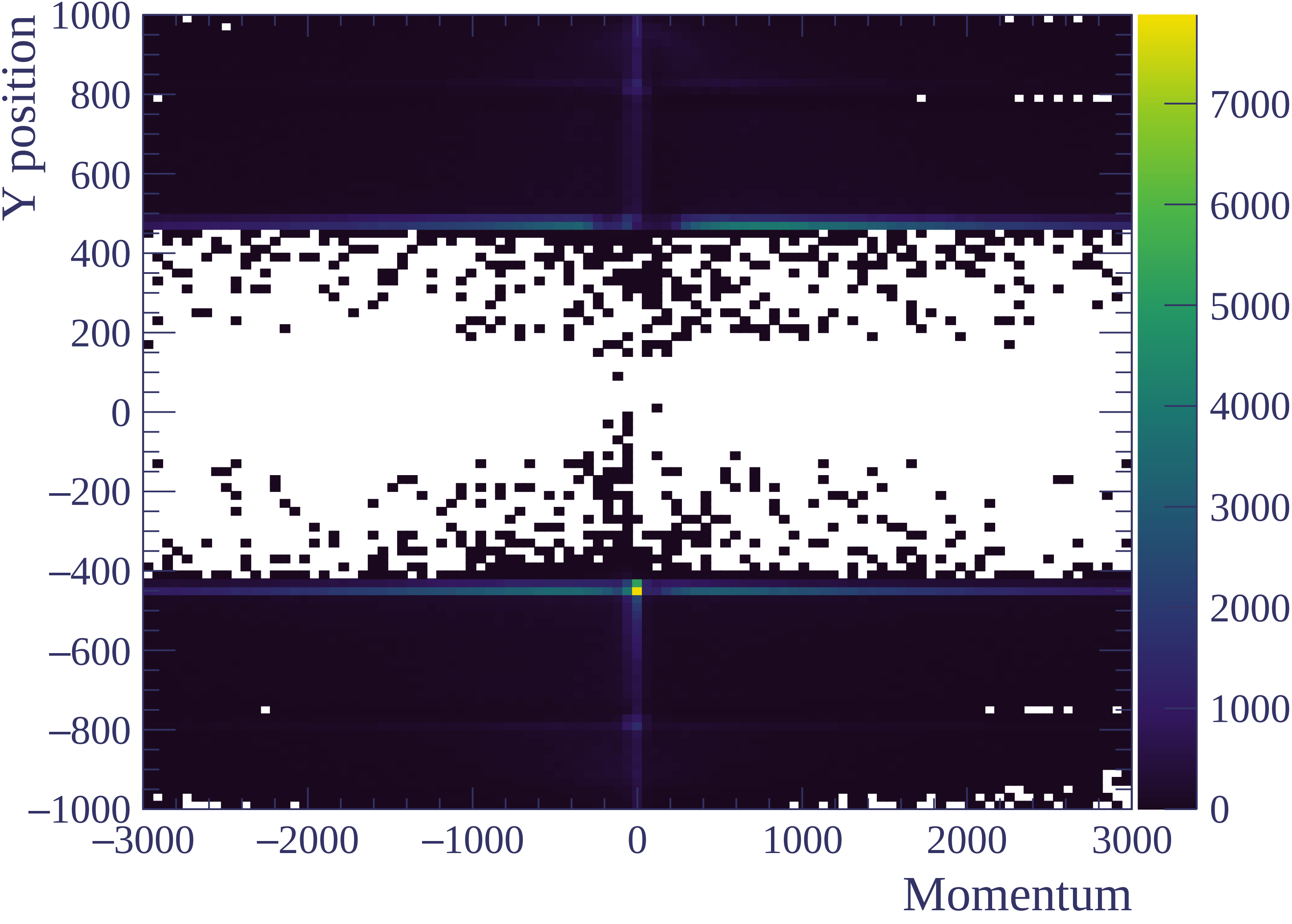

dE/dx with XP (ADC counts/cm)
momentum (MeV)
−3000
−2000
−1000
0
1000
2000
3000
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
2400



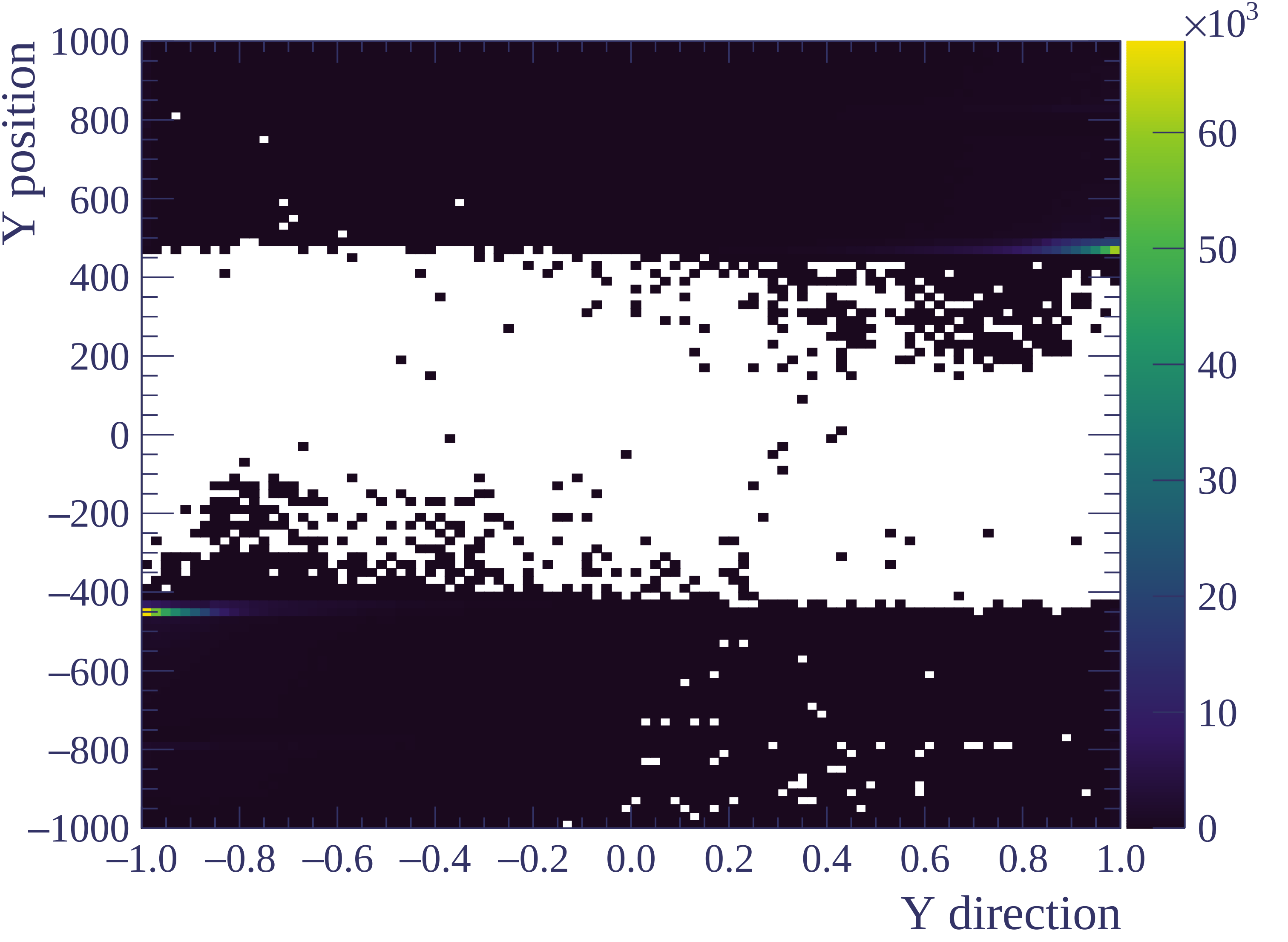
Count
×10^3
35
30
25
20
15
10
5
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT



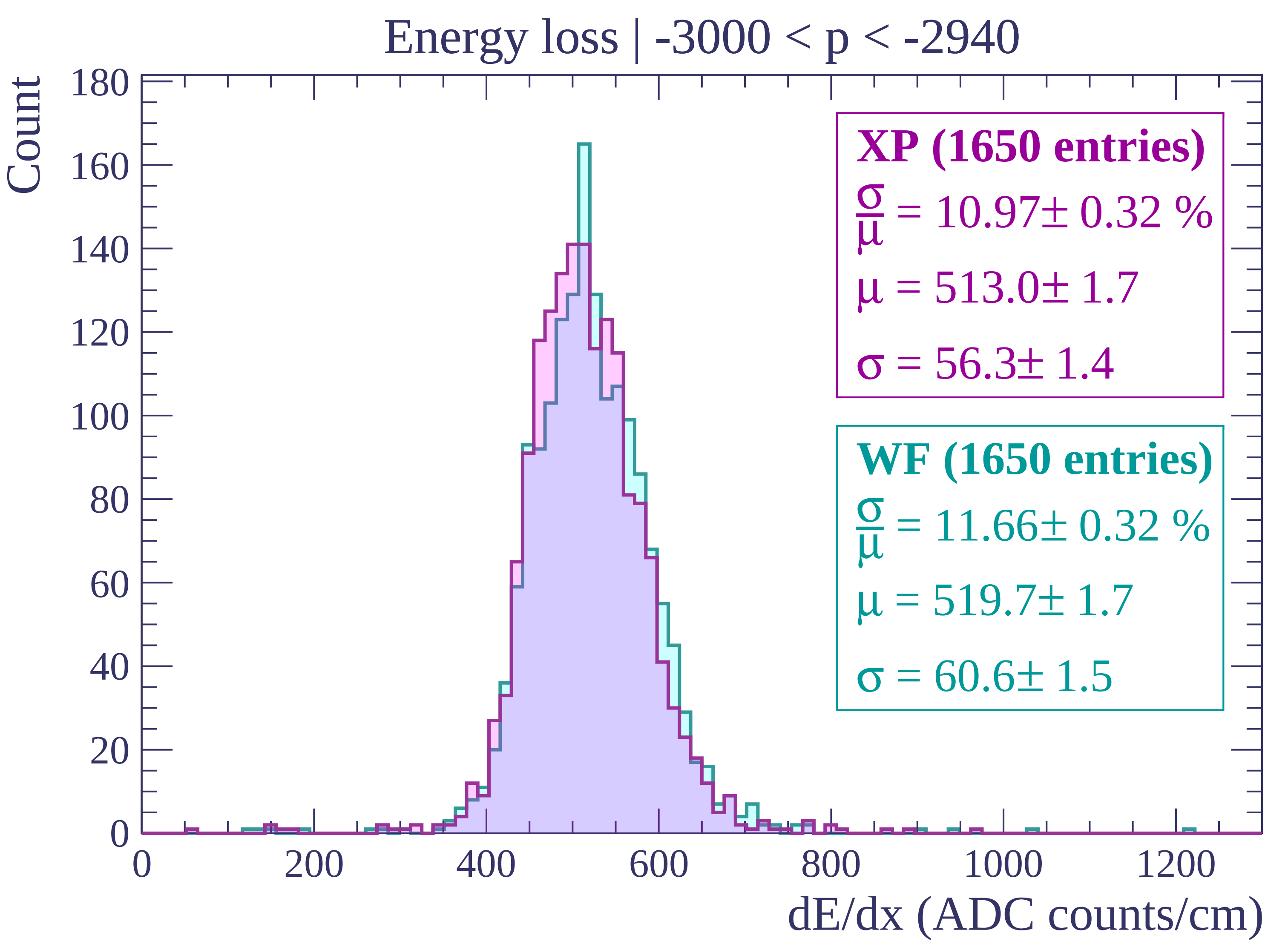
Momentum
3000
2000
1000
0
−1000
−2000
−3000
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction
8000
7000
6000
5000
4000
3000
2000
1000
0

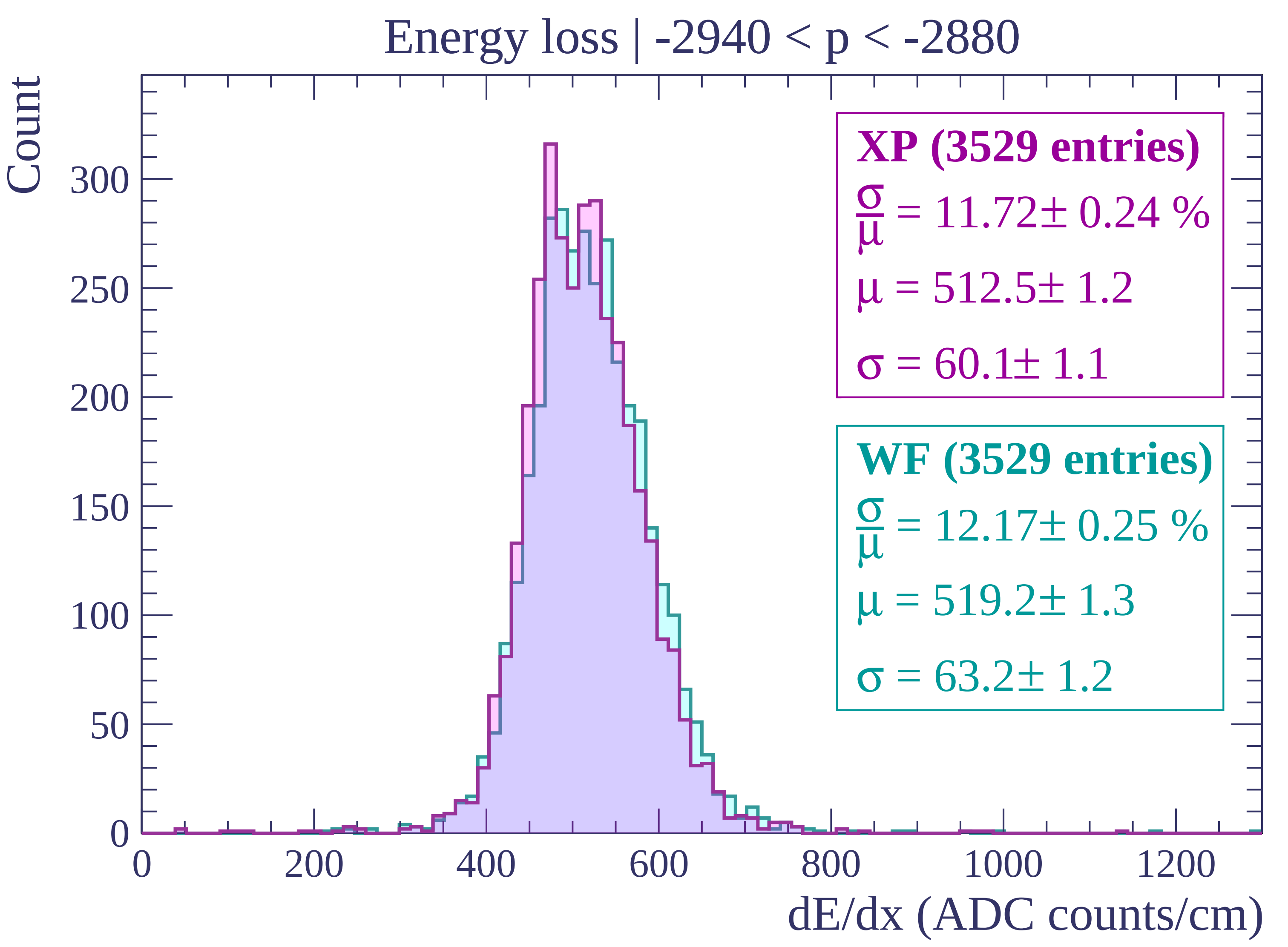


Y position
1000
800
600
400
200
0
−200
−400
−600
−800
−1000
−3000
−2000
−1000
0
1000
2000
3000
Momentum
7000
6000
5000
4000
3000
2000
1000
0

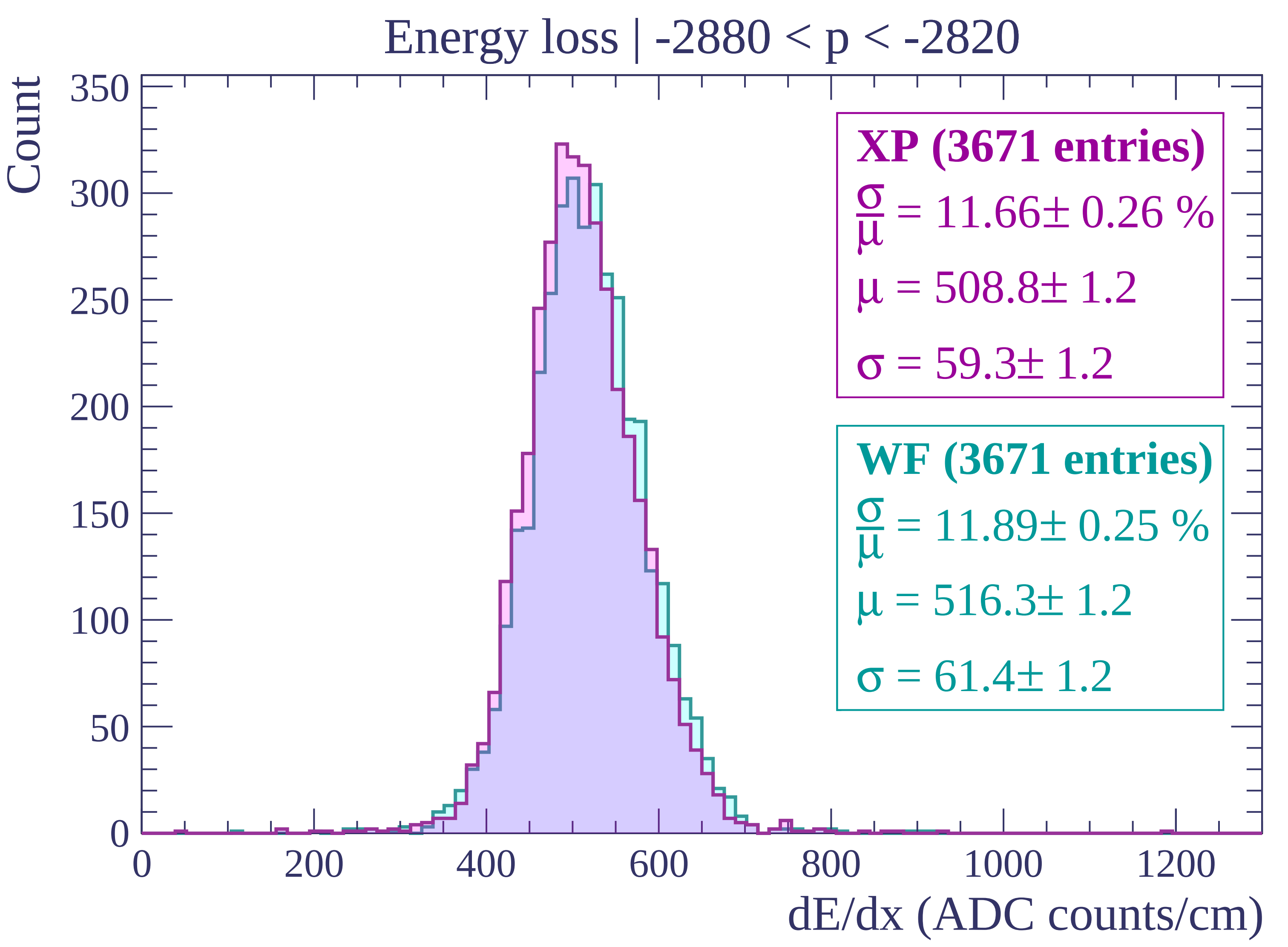


Y position
Y direction
×10^3
1000
800
600
400
200
0
−200
−400
−600
−800
−1000
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
60
50
40
30
20
10
0

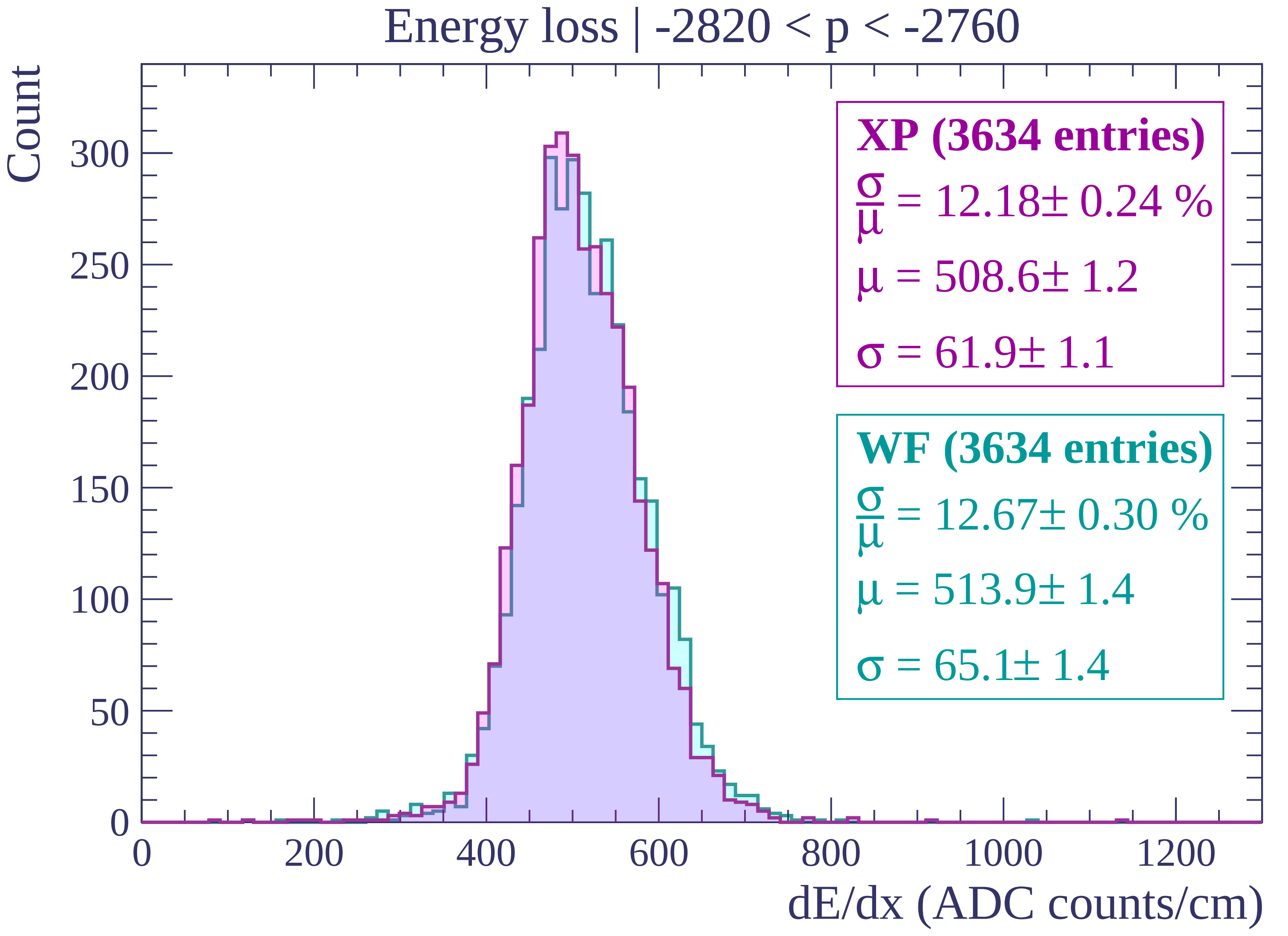

Energy loss | -3000 < p < -2940
Count
dE/dx (ADC counts/cm)
XP (1650 entries)
σ/μ = 10.97± 0.32 %
μ = 513.0± 1.7
σ = 56.3± 1.4
WF (1650 entries)
σ/μ = 11.66± 0.32 %
μ = 519.7± 1.7
σ = 60.6± 1.5



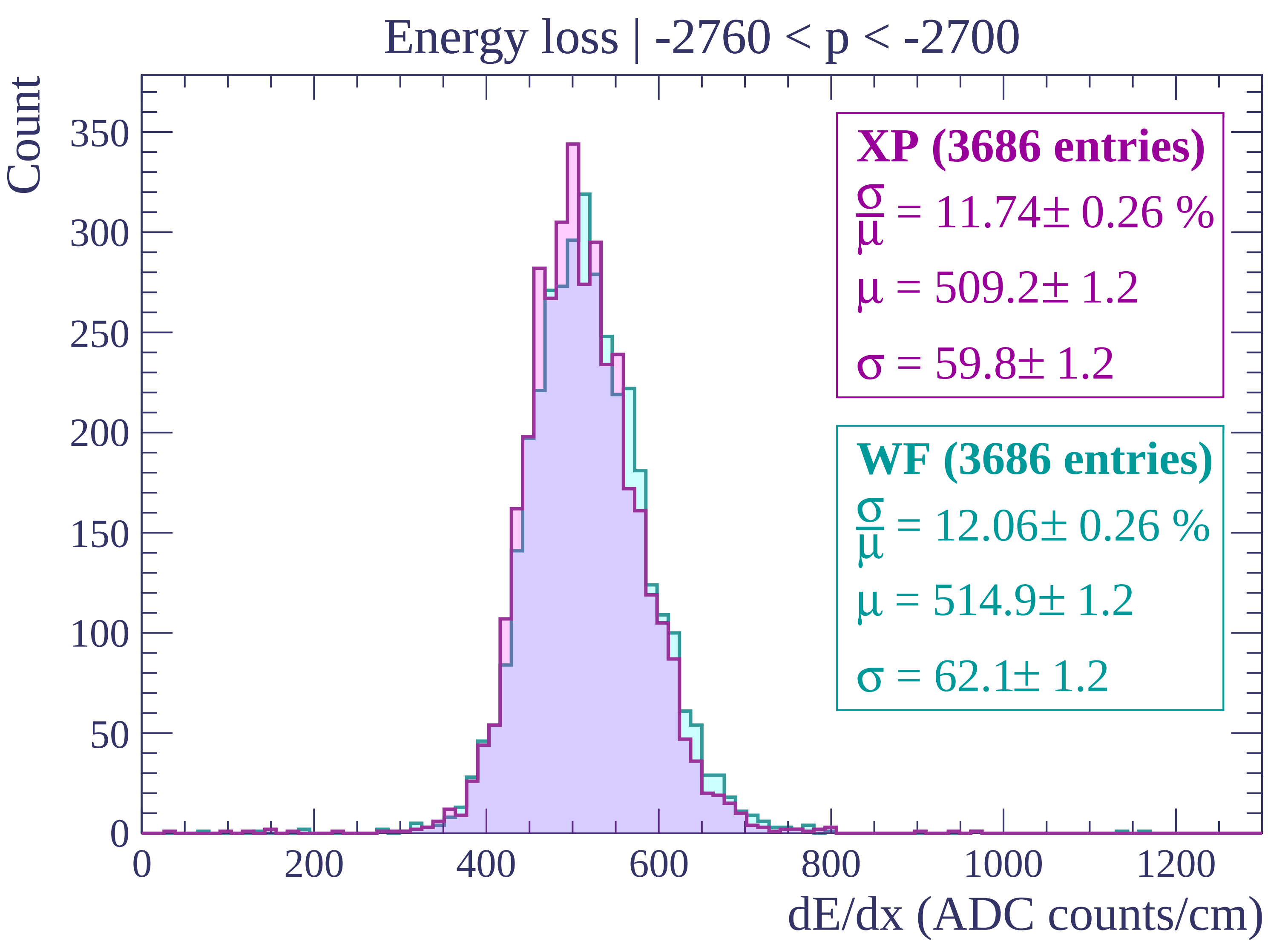
Energy loss | -2940 < p < -2880
Count
dE/dx (ADC counts/cm)
XP (3529 entries)
σ/μ = 11.72± 0.24 %
μ = 512.5± 1.2
σ = 60.1± 1.1
WF (3529 entries)
σ/μ = 12.17± 0.25 %
μ = 519.2± 1.3
σ = 63.2± 1.2
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200



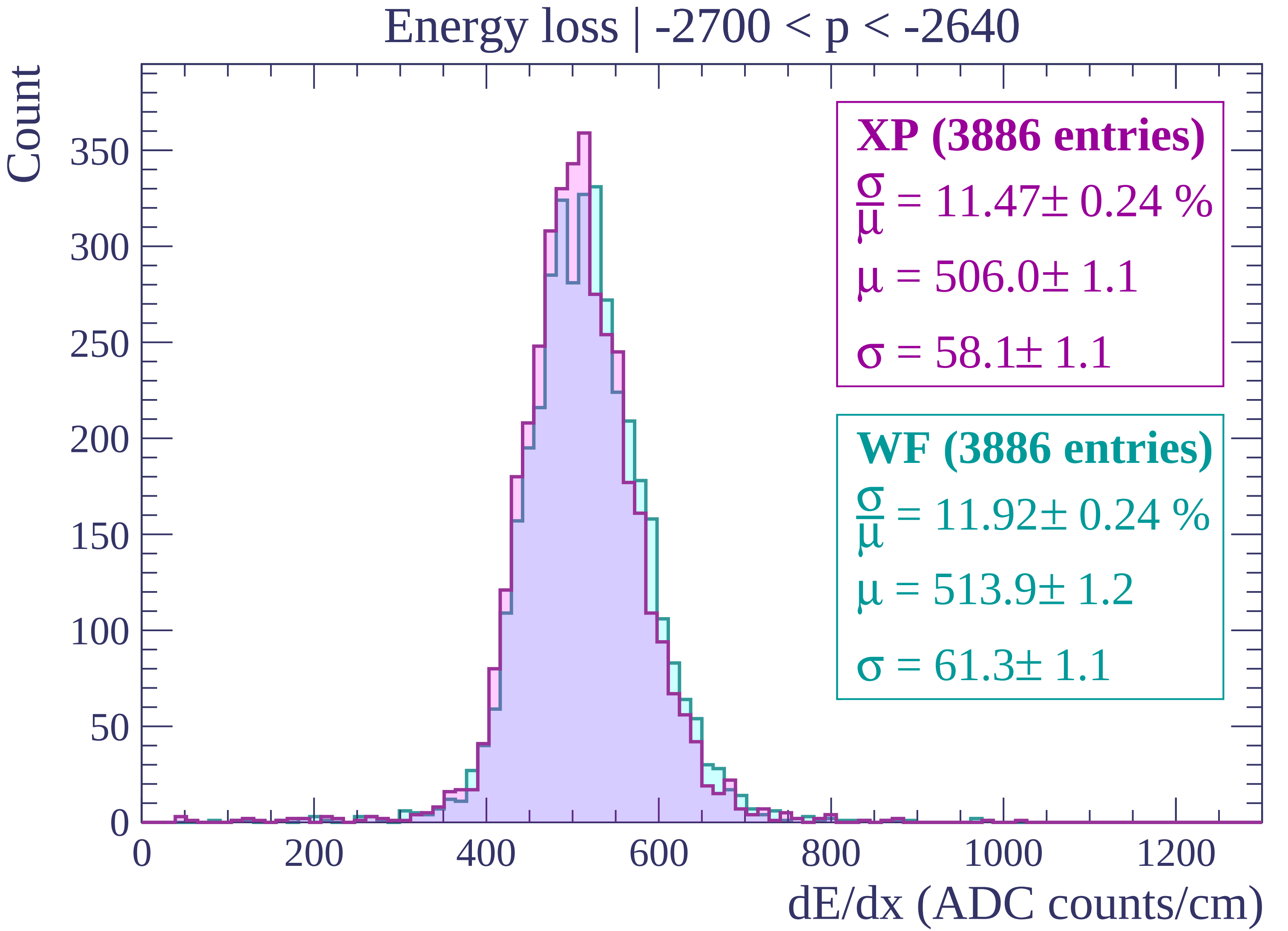
Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3671 entries)
$\frac{\sigma}{\mu}$ = 11.66± 0.26 %
μ = 508.8± 1.2
σ = 59.3± 1.2
WF (3671 entries)
$\frac{\sigma}{\mu}$ = 11.89± 0.25 %
μ = 516.3± 1.2
σ = 61.4± 1.2



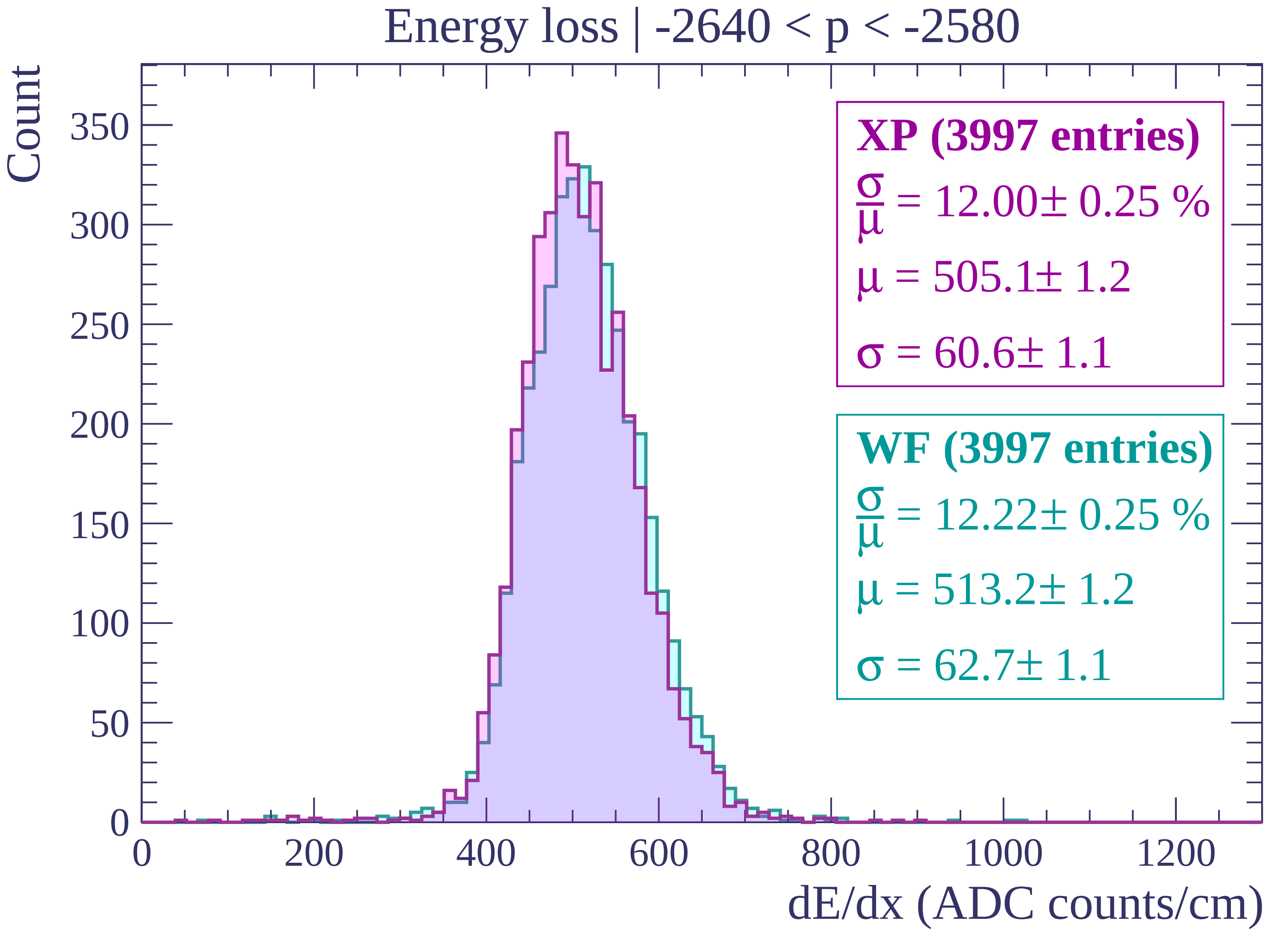
Energy loss | -2820 < p < -2760
Count
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3634 entries)
σ/μ = 12.18± 0.24 %
μ = 508.6± 1.2
σ = 61.9± 1.1
WF (3634 entries)
σ/μ = 12.67± 0.30 %
μ = 513.9± 1.4
σ = 65.1± 1.4



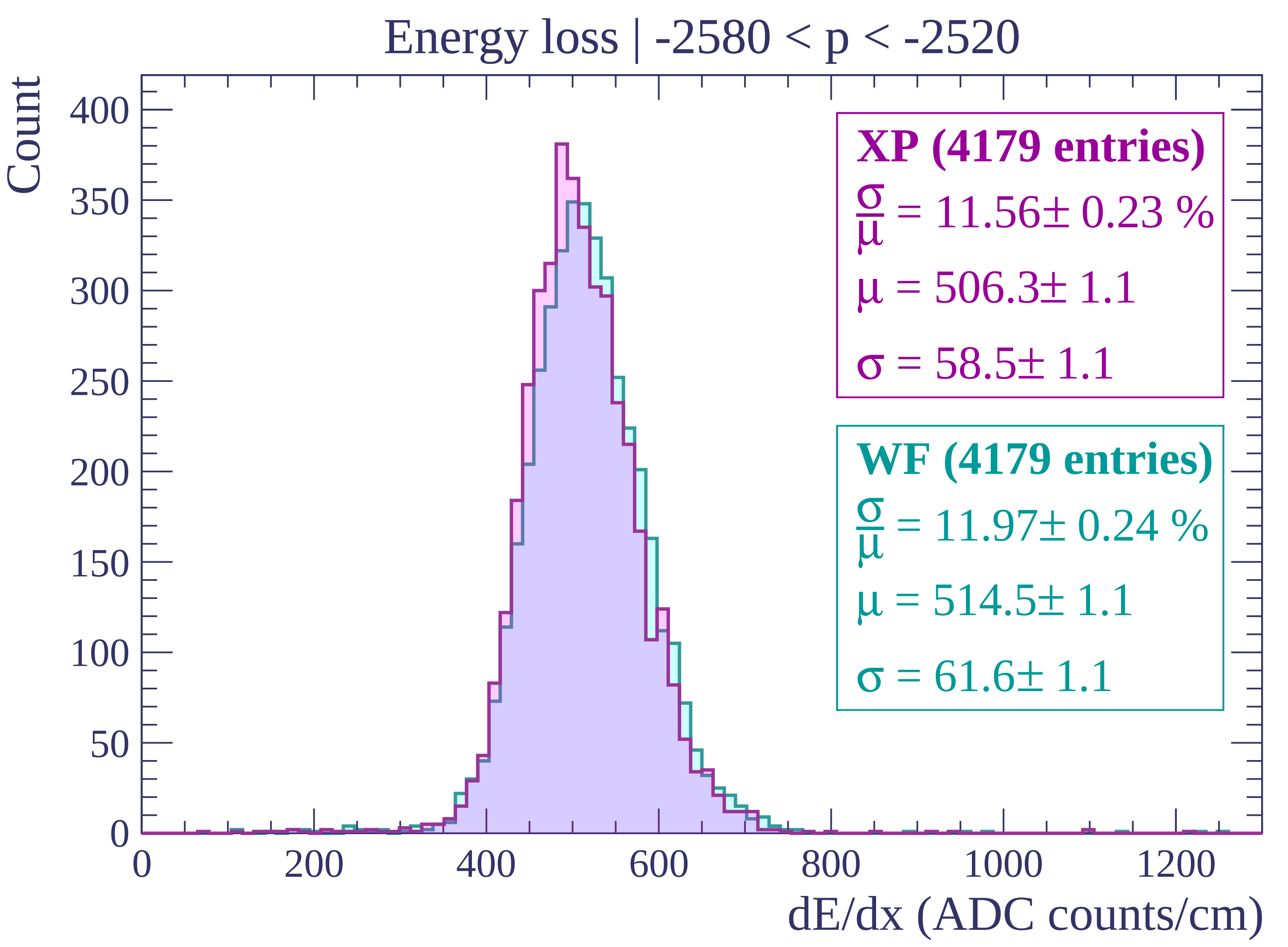
Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
XP (3686 entries)
$\frac{\sigma}{\mu}$ = 11.74± 0.26 %
μ = 509.2± 1.2
σ = 59.8± 1.2
WF (3686 entries)
$\frac{\sigma}{\mu}$ = 12.06± 0.26 %
μ = 514.9± 1.2
σ = 62.1± 1.2



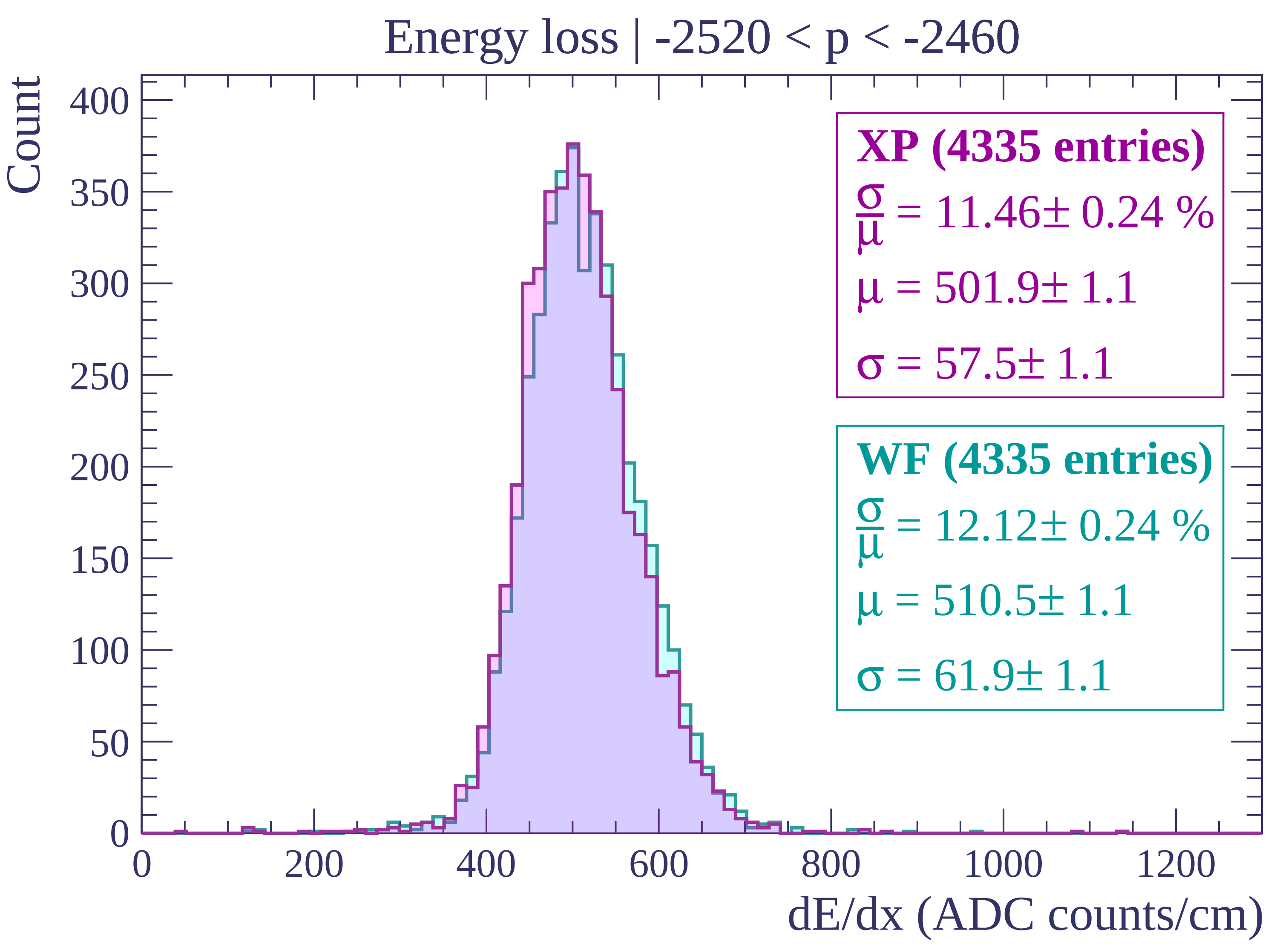
Energy loss | -2700 < p < -2640
Count
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3886 entries)
σ/μ = 11.47± 0.24 %
μ = 506.0± 1.1
σ = 58.1± 1.1
WF (3886 entries)
σ/μ = 11.92± 0.24 %
μ = 513.9± 1.2
σ = 61.3± 1.1

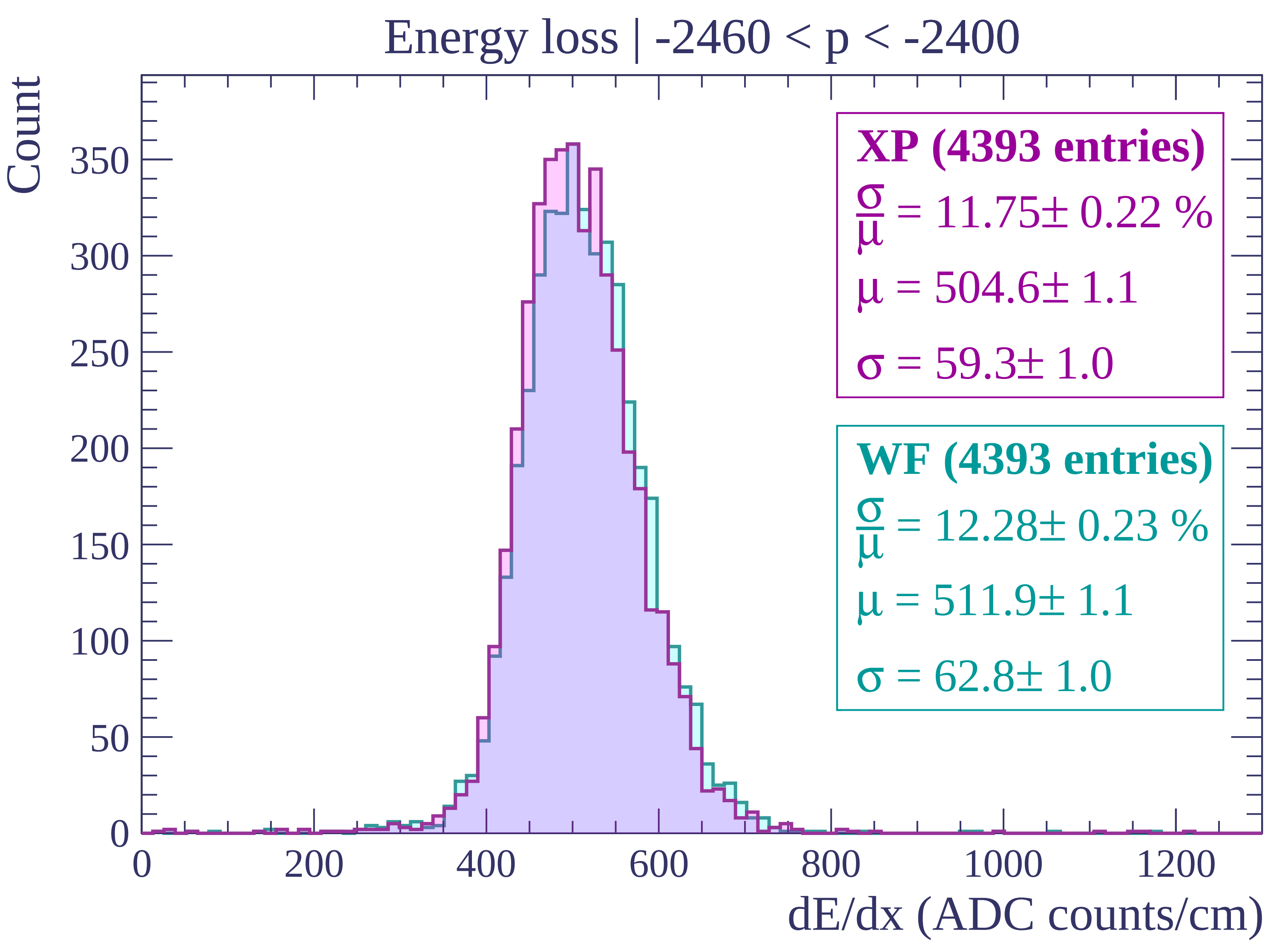


Energy loss | -2640 < p < -2580
Count
dE/dx (ADC counts/cm)
XP (3997 entries)
σ/μ = 12.00± 0.25 %
μ = 505.1± 1.2
σ = 60.6± 1.1
WF (3997 entries)
σ/μ = 12.22± 0.25 %
μ = 513.2± 1.2
σ = 62.7± 1.1

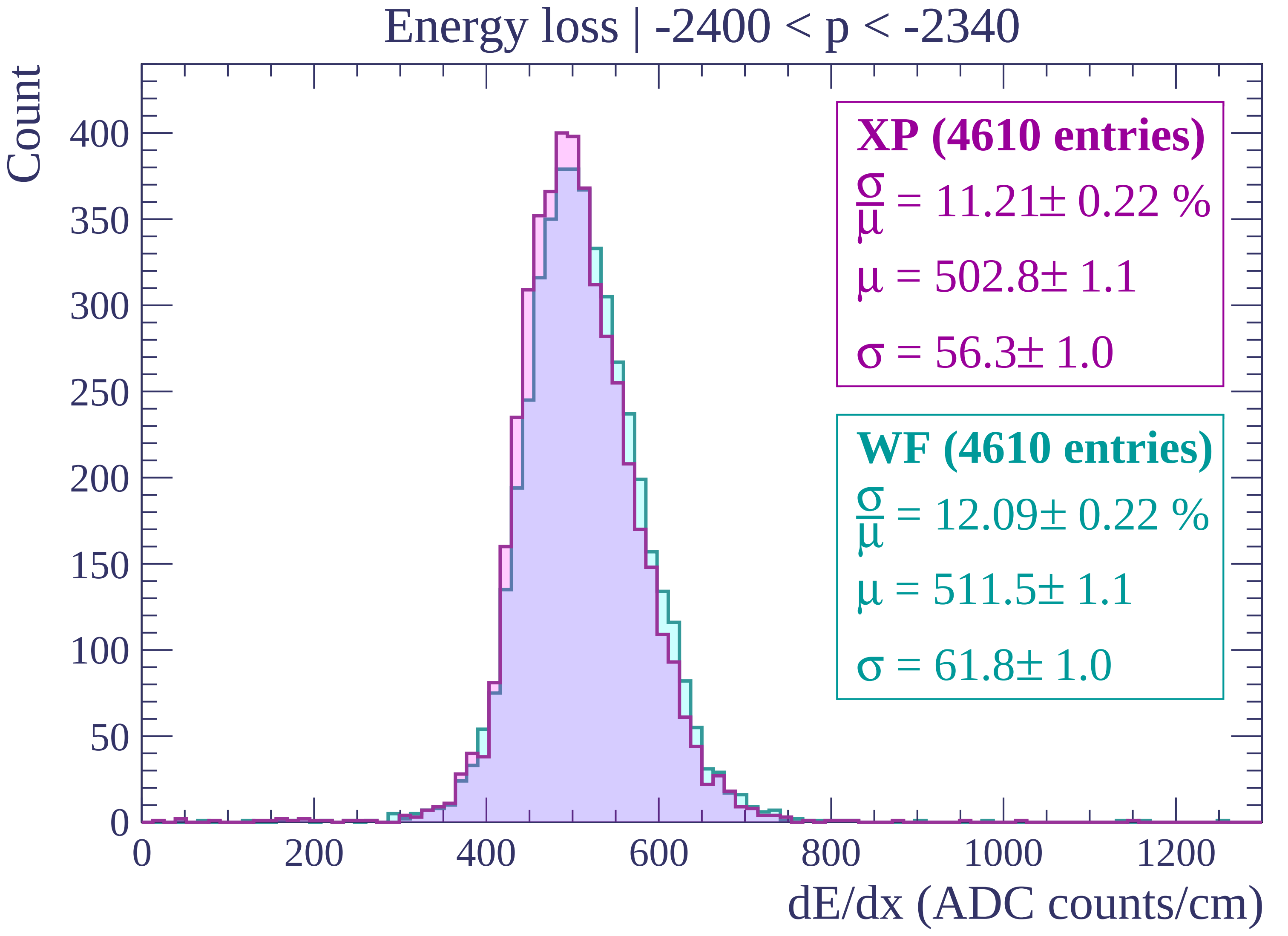


Energy loss | -2580 < p < -2520
Count
dE/dx (ADC counts/cm)
XP (4179 entries)
$\frac{\sigma}{\mu}$ = 11.56± 0.23 %
μ = 506.3± 1.1
σ = 58.5± 1.1
WF (4179 entries)
$\frac{\sigma}{\mu}$ = 11.97± 0.24 %
μ = 514.5± 1.1
σ = 61.6± 1.1

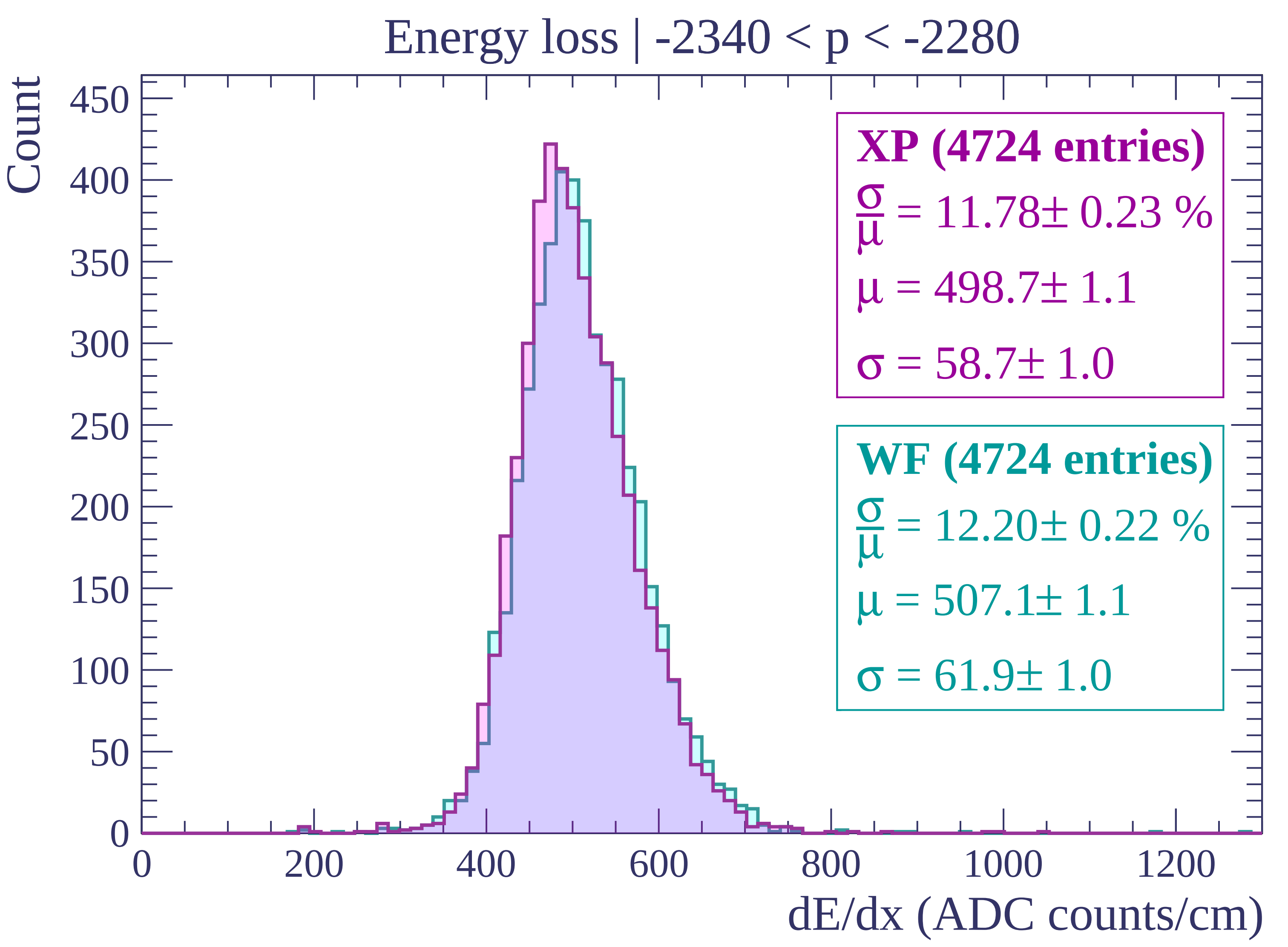

Energy loss | -2520 < p < -2460
Count
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4335 entries)
σ/μ = 11.46± 0.24 %
μ = 501.9± 1.1
σ = 57.5± 1.1
WF (4335 entries)
σ/μ = 12.12± 0.24 %
μ = 510.5± 1.1
σ = 61.9± 1.1

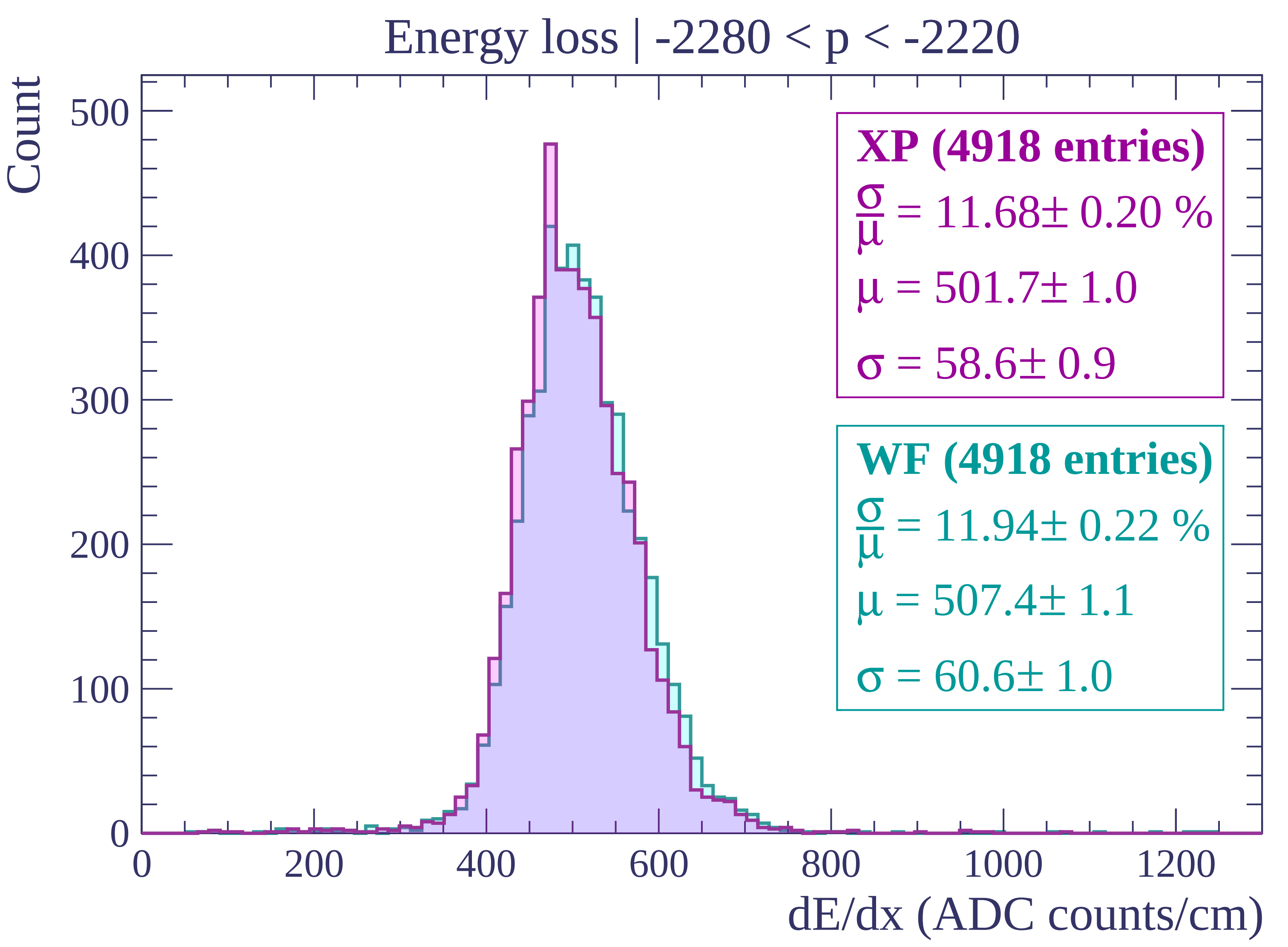


Energy loss | -2460 < p < -2400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
XP (4393 entries)
σ/μ = 11.75± 0.22 %
μ = 504.6± 1.1
σ = 59.3± 1.0
WF (4393 entries)
σ/μ = 12.28± 0.23 %
μ = 511.9± 1.1
σ = 62.8± 1.0

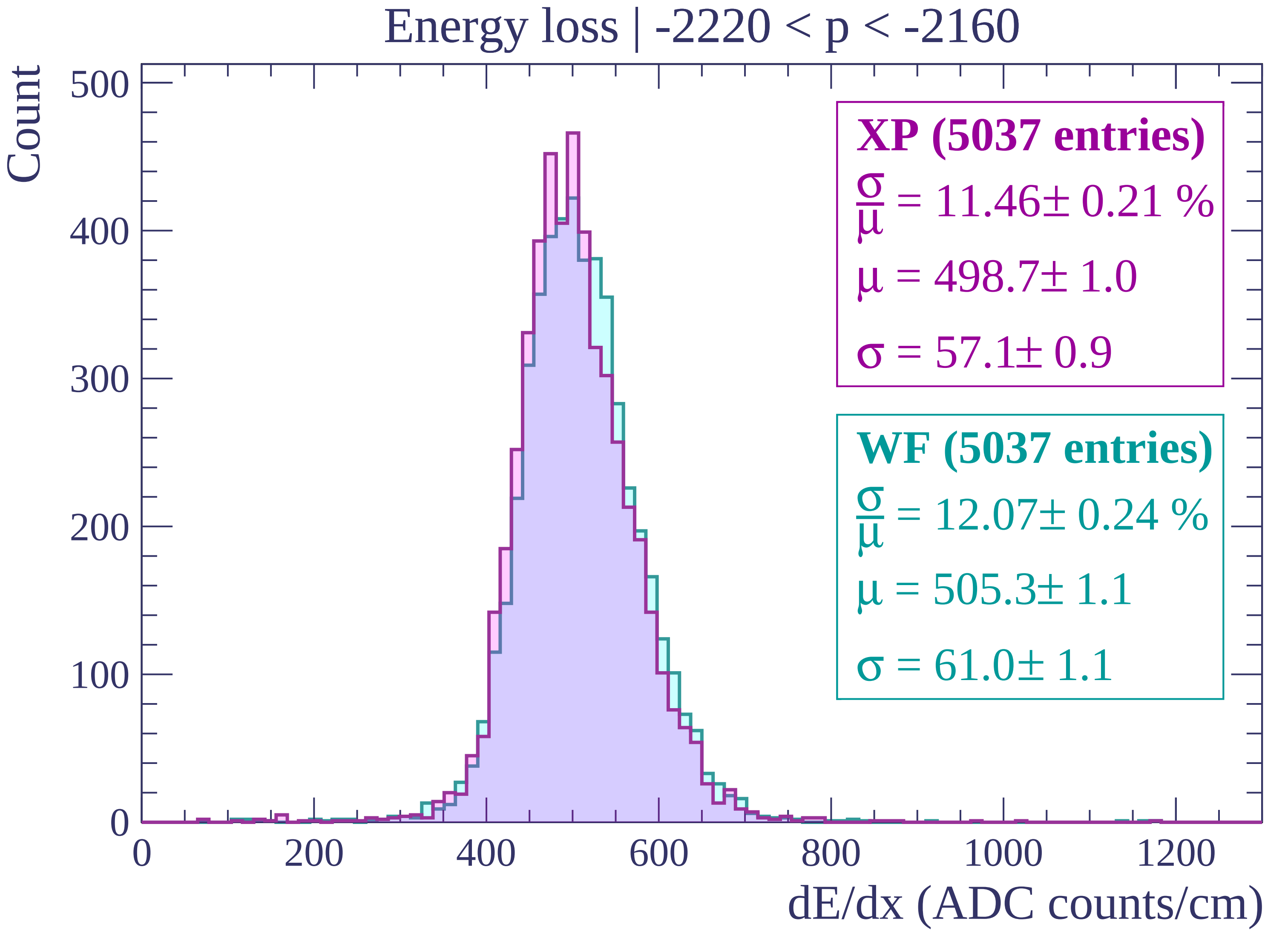


Energy loss | -2400 < p < -2340
Count
XP (4610 entries)
σ/μ = 11.21± 0.22 %
μ = 502.8± 1.1
σ = 56.3± 1.0
WF (4610 entries)
σ/μ = 12.09± 0.22 %
μ = 511.5± 1.1
σ = 61.8± 1.0
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200

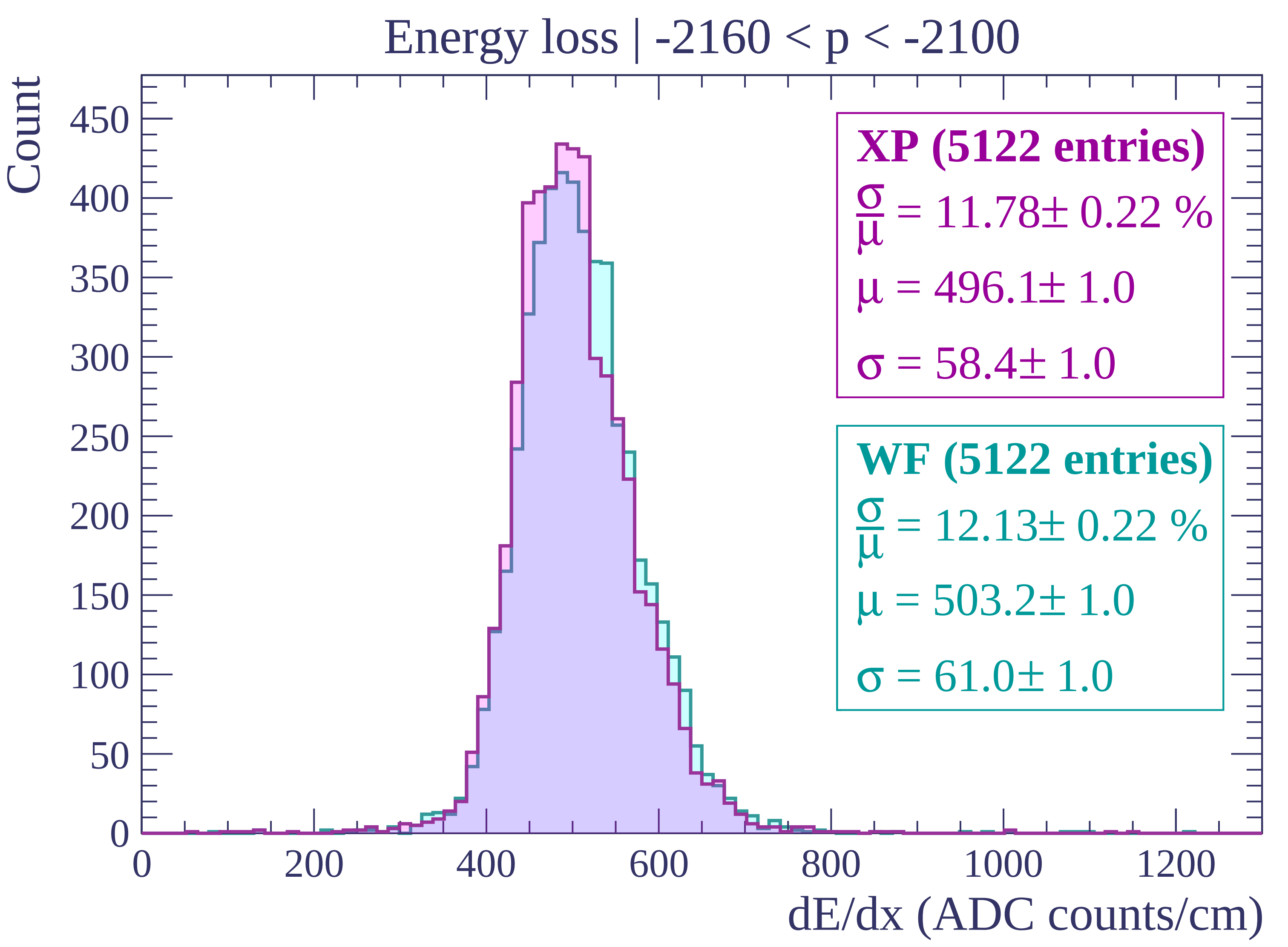

Energy loss | -2340 < p < -2280
Count
dE/dx (ADC counts/cm)
XP (4724 entries)
$\frac{\sigma}{\mu}$ = 11.78± 0.23 %
μ = 498.7± 1.1
σ = 58.7± 1.0
WF (4724 entries)
$\frac{\sigma}{\mu}$ = 12.20± 0.22 %
μ = 507.1± 1.1
σ = 61.9± 1.0
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200



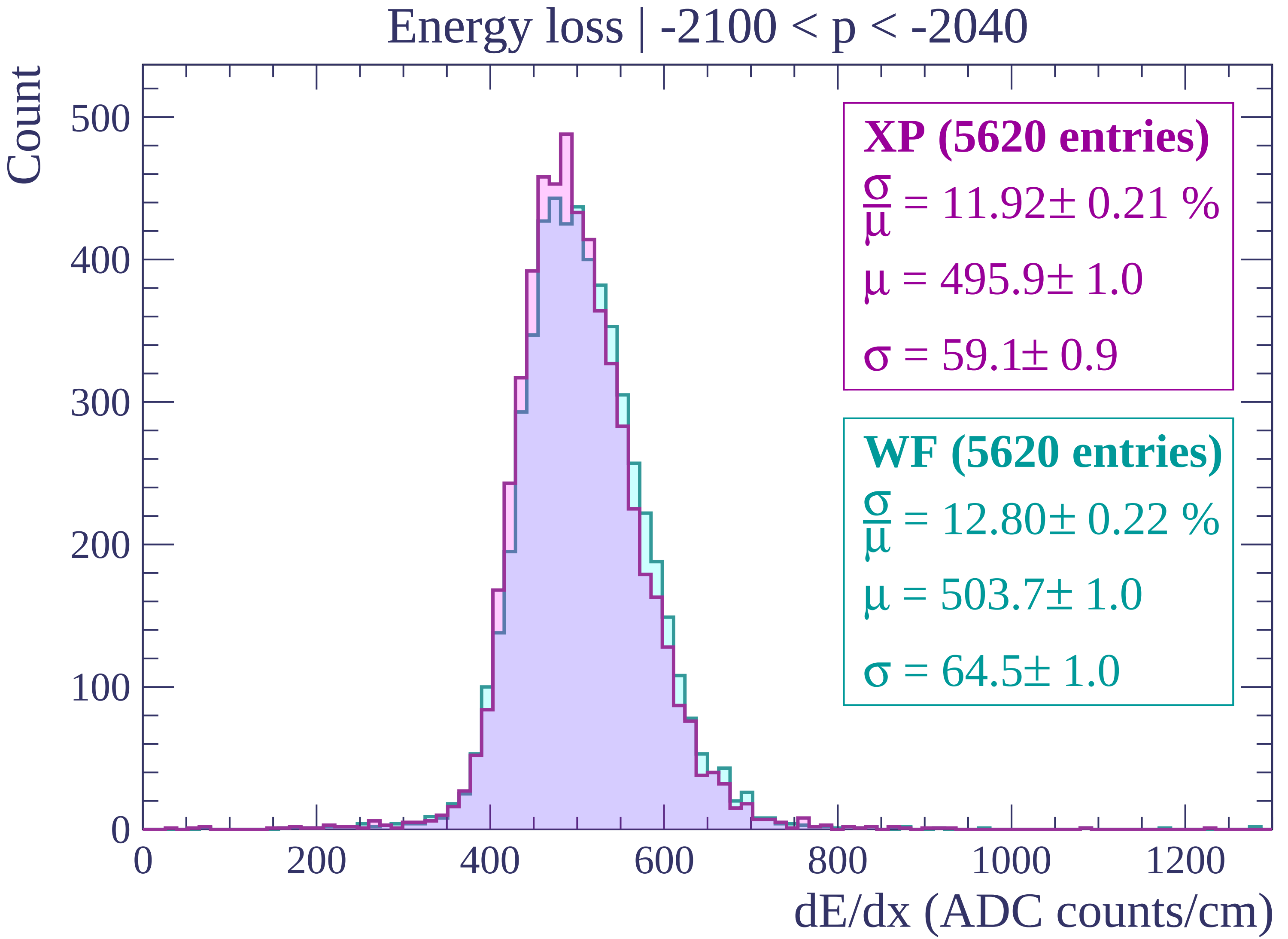
Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4918 entries)
σ/μ = 11.68± 0.20 %
μ = 501.7± 1.0
σ = 58.6± 0.9
WF (4918 entries)
σ/μ = 11.94± 0.22 %
μ = 507.4± 1.1
σ = 60.6± 1.0



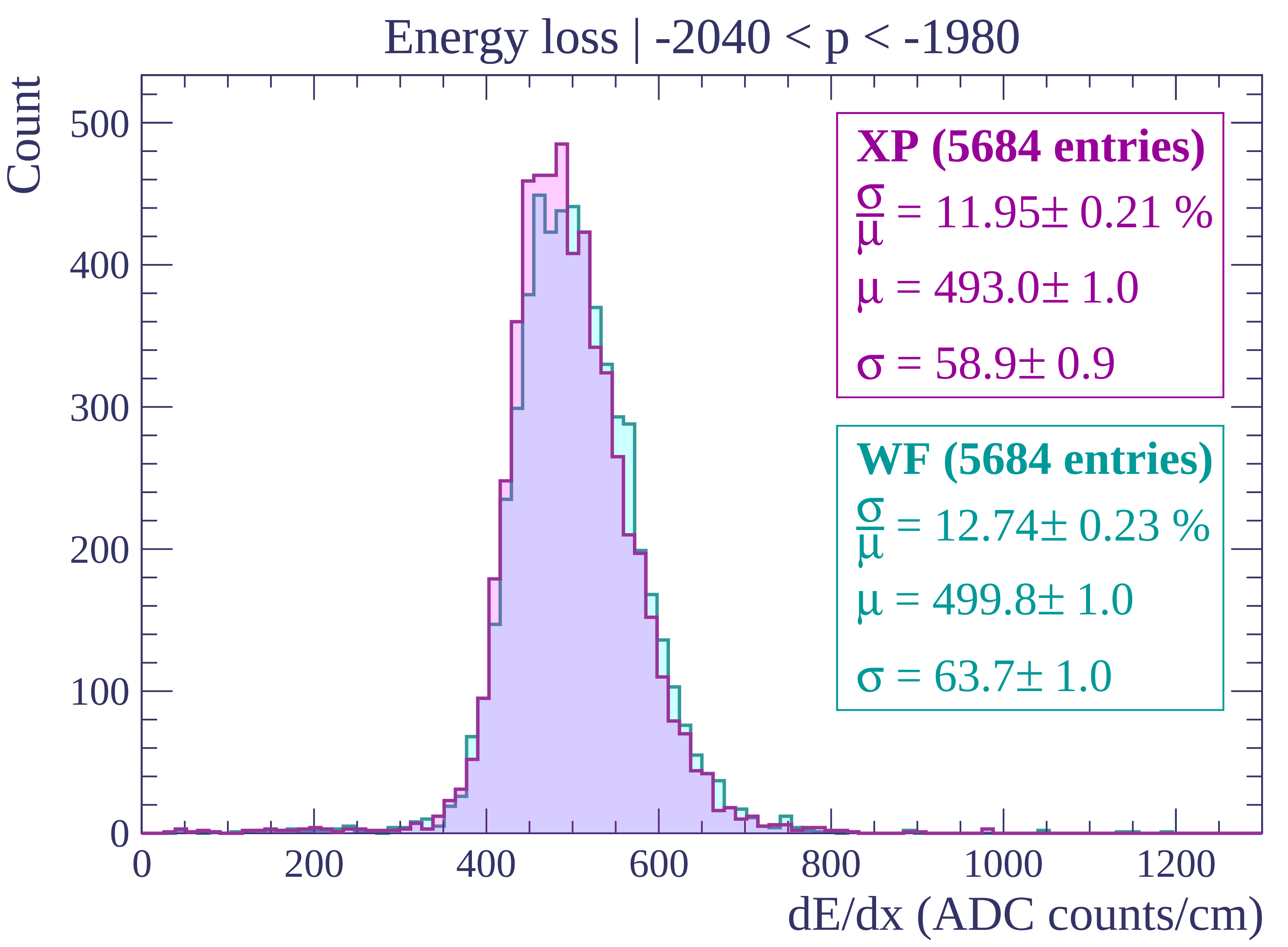
Energy loss | -2220 < p < -2160
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5037 entries)
σ/μ = 11.46± 0.21 %
μ = 498.7± 1.0
σ = 57.1± 0.9
WF (5037 entries)
σ/μ = 12.07± 0.24 %
μ = 505.3± 1.1
σ = 61.0± 1.1



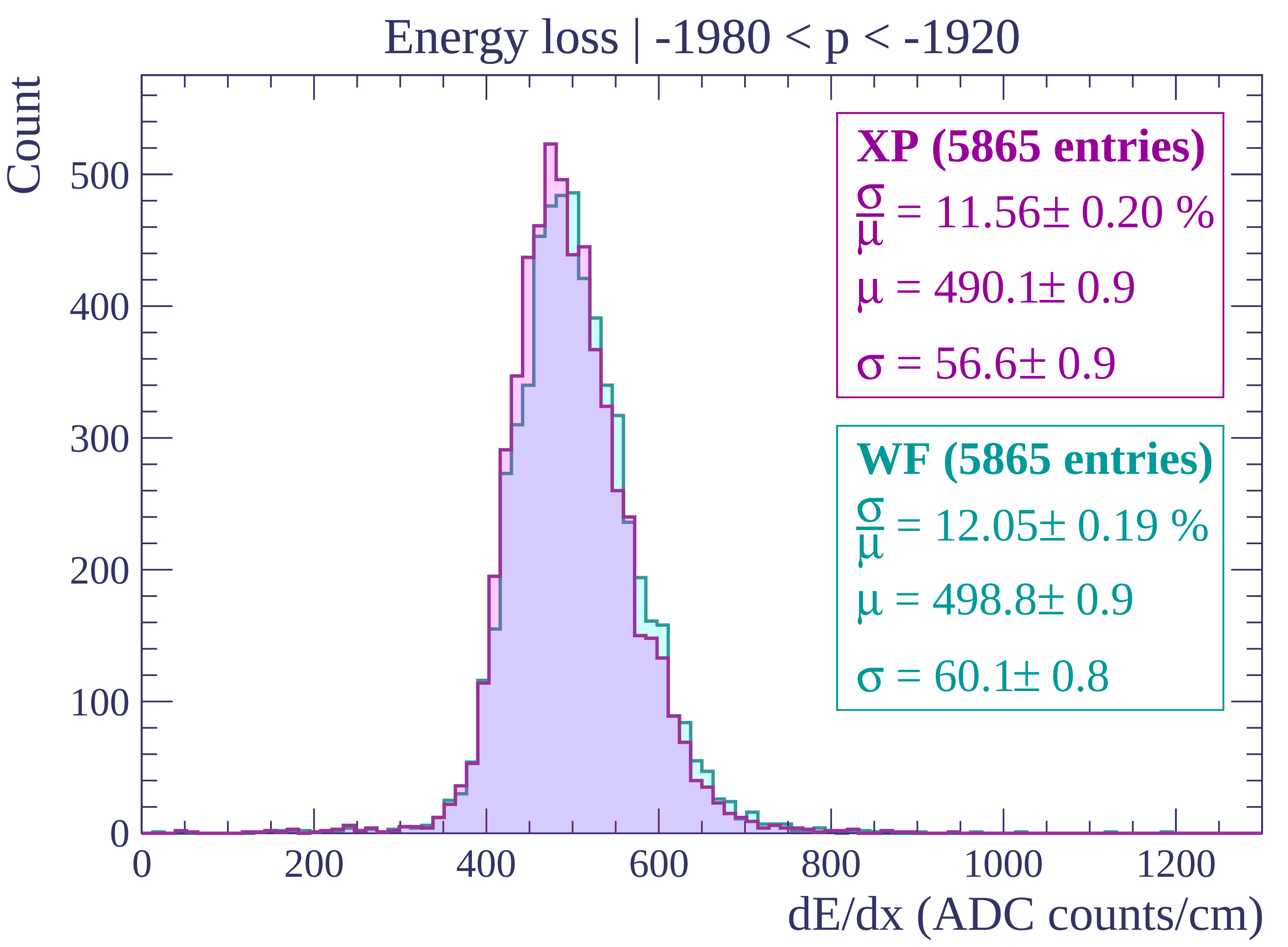
Energy loss | -2160 < p < -2100
Count
dE/dx (ADC counts/cm)
XP (5122 entries)
$\frac{\sigma}{\mu}$ = 11.78± 0.22 %
μ = 496.1± 1.0
σ = 58.4± 1.0
WF (5122 entries)
$\frac{\sigma}{\mu}$ = 12.13± 0.22 %
μ = 503.2± 1.0
σ = 61.0± 1.0



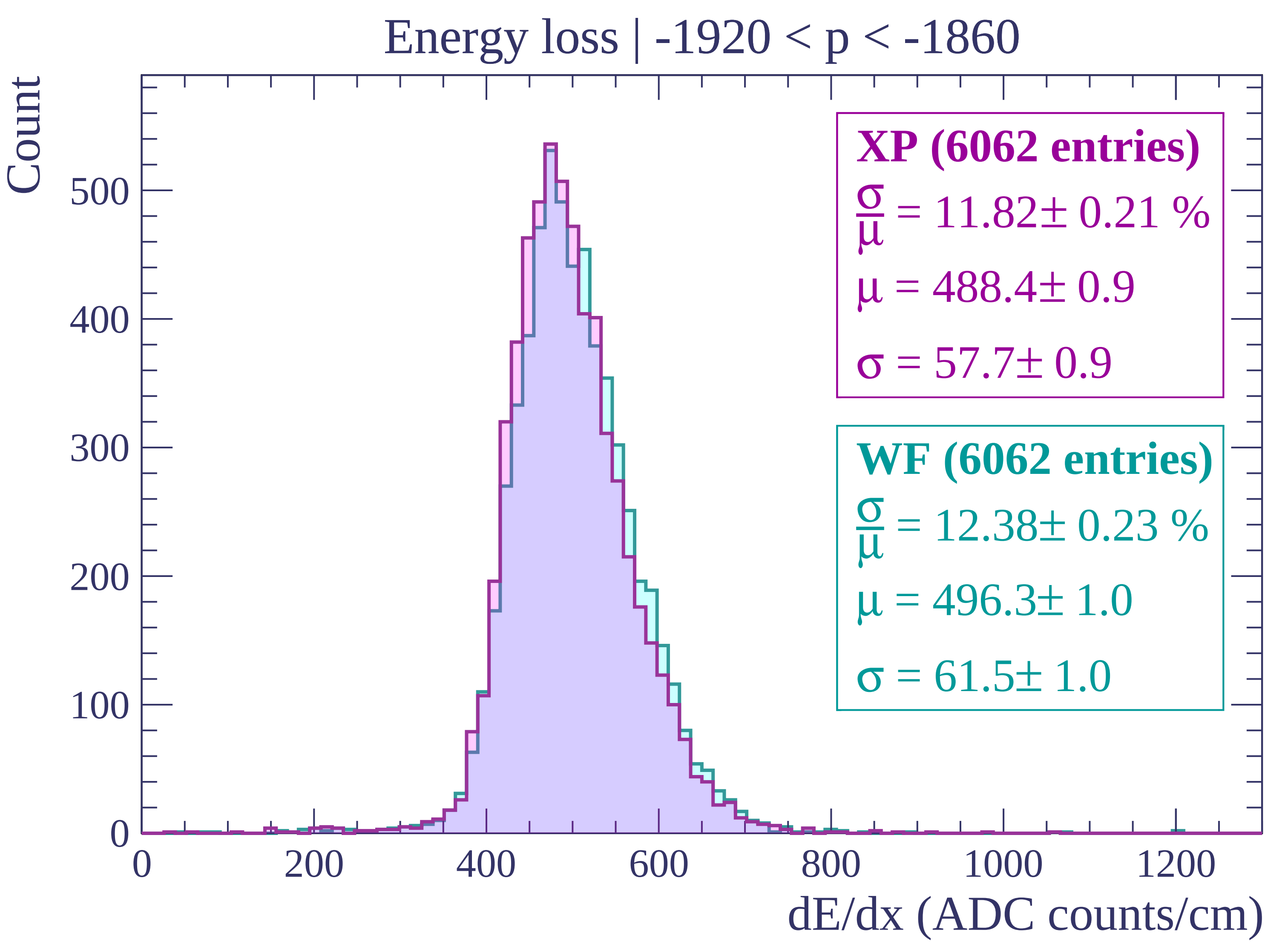
Energy loss | -2100 < p < -2040
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5620 entries)
$\frac{\sigma}{\mu}$ = 11.92± 0.21 %
μ = 495.9± 1.0
σ = 59.1± 0.9
WF (5620 entries)
$\frac{\sigma}{\mu}$ = 12.80± 0.22 %
μ = 503.7± 1.0
σ = 64.5± 1.0



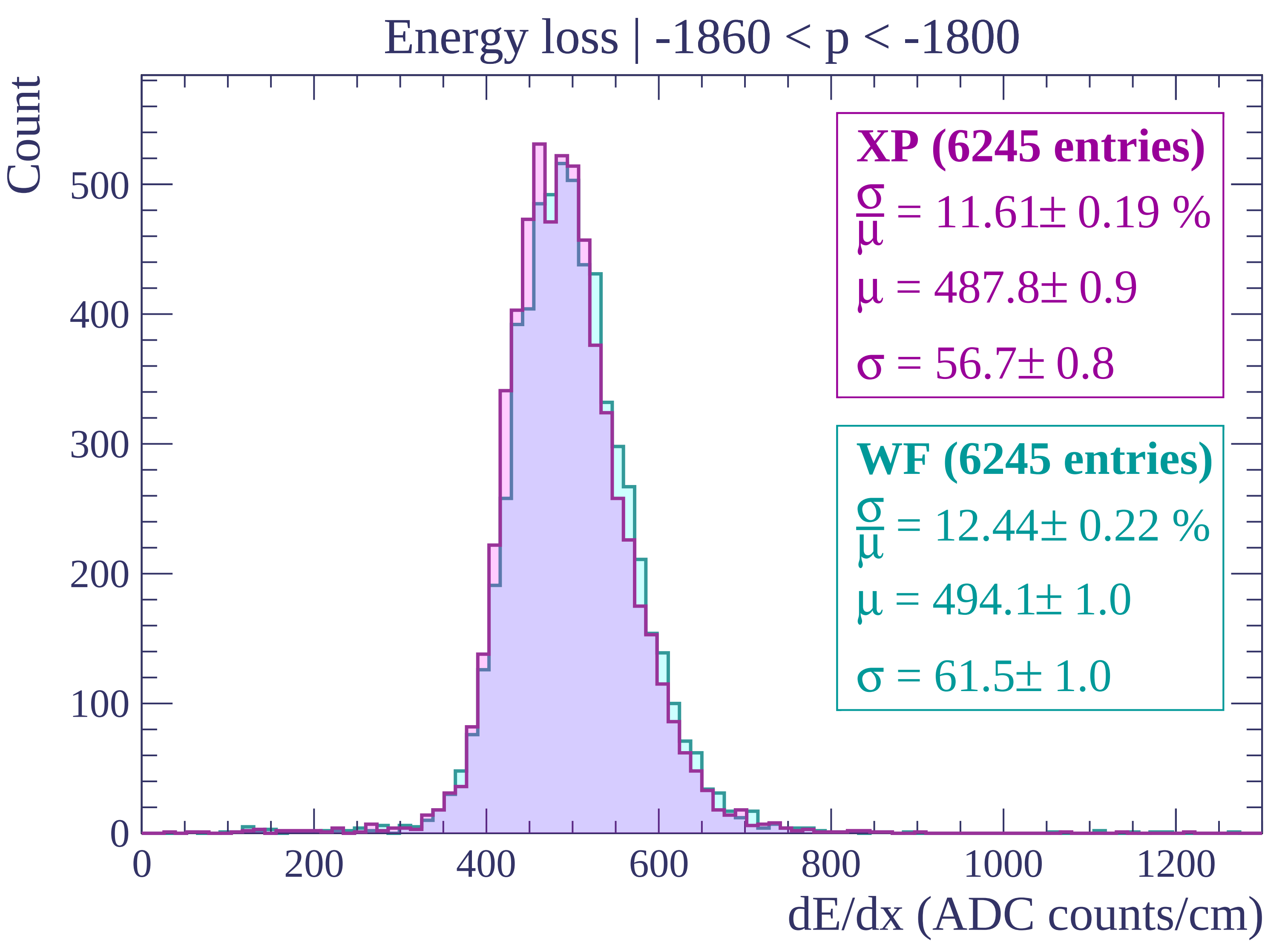
Energy loss | -2040 < p < -1980
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5684 entries)
$\frac{\sigma}{\mu}$ = 11.95± 0.21 %
μ = 493.0± 1.0
σ = 58.9± 0.9
WF (5684 entries)
$\frac{\sigma}{\mu}$ = 12.74± 0.23 %
μ = 499.8± 1.0
σ = 63.7± 1.0



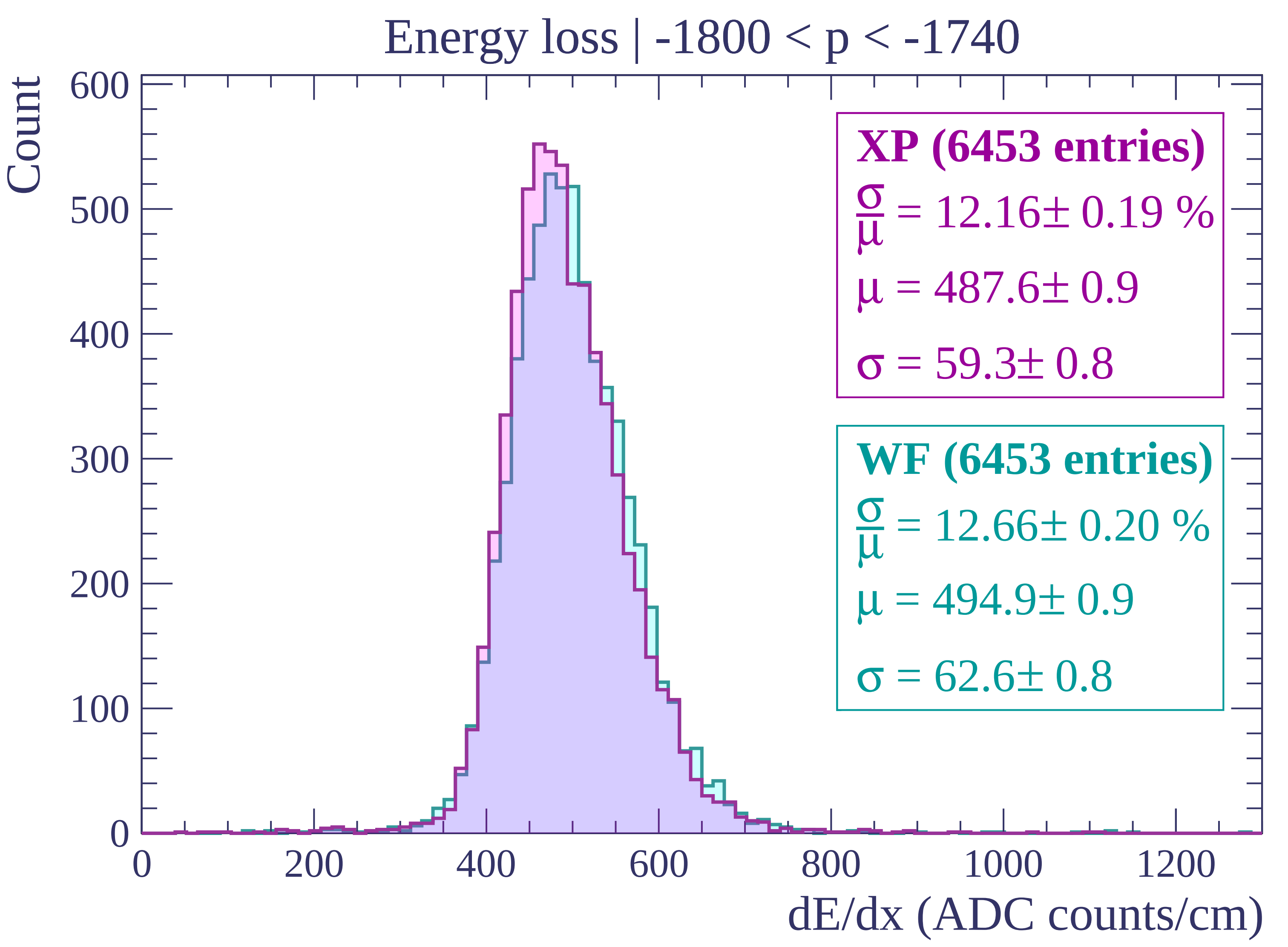
Energy loss | -1980 < p < -1920
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5865 entries)
σ/μ = 11.56± 0.20 %
μ = 490.1± 0.9
σ = 56.6± 0.9
WF (5865 entries)
σ/μ = 12.05± 0.19 %
μ = 498.8± 0.9
σ = 60.1± 0.8

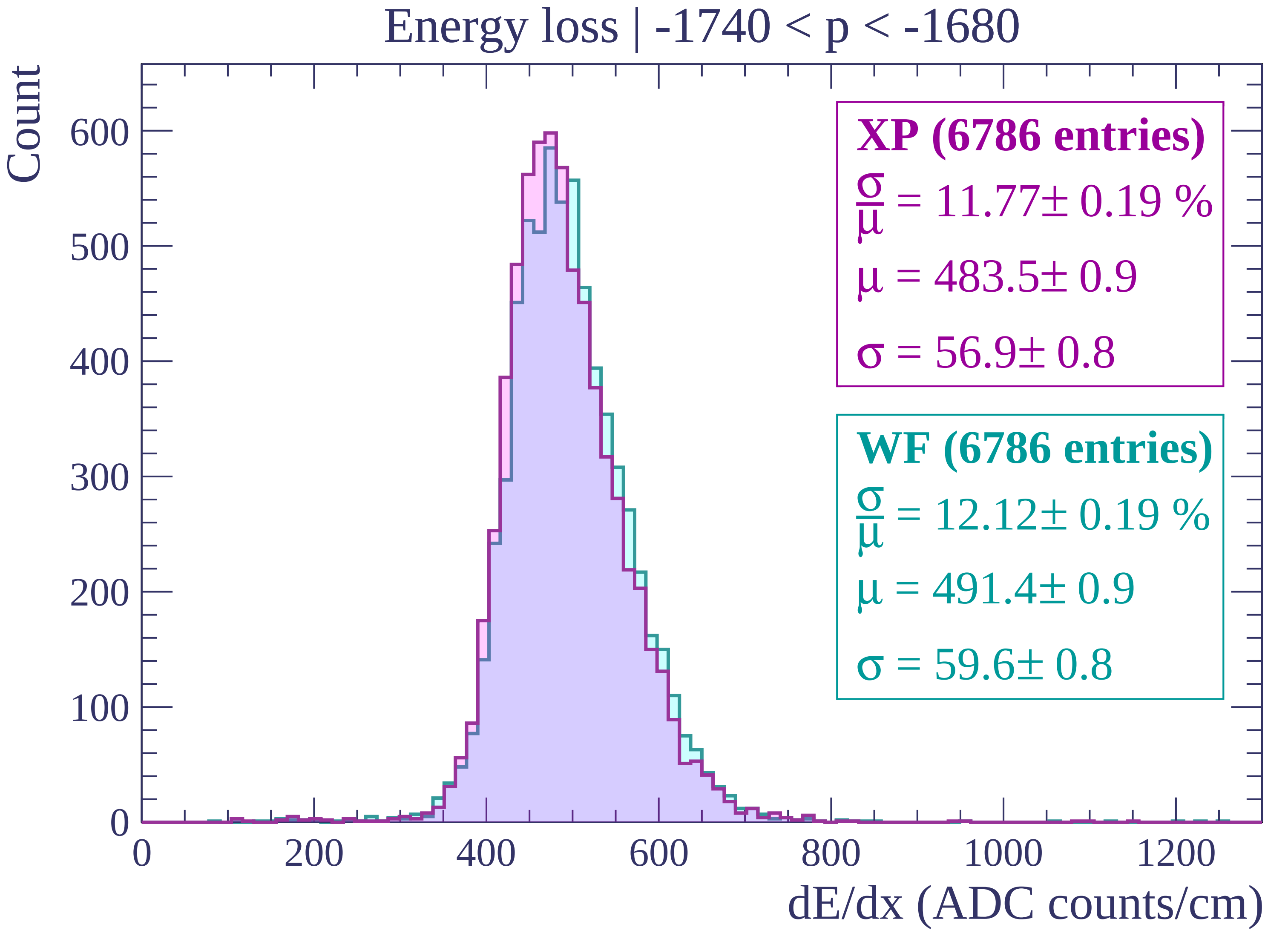


Energy loss | -1920 < p < -1860
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6062 entries)
σ/μ = 11.82± 0.21 %
μ = 488.4± 0.9
σ = 57.7± 0.9
WF (6062 entries)
σ/μ = 12.38± 0.23 %
μ = 496.3± 1.0
σ = 61.5± 1.0

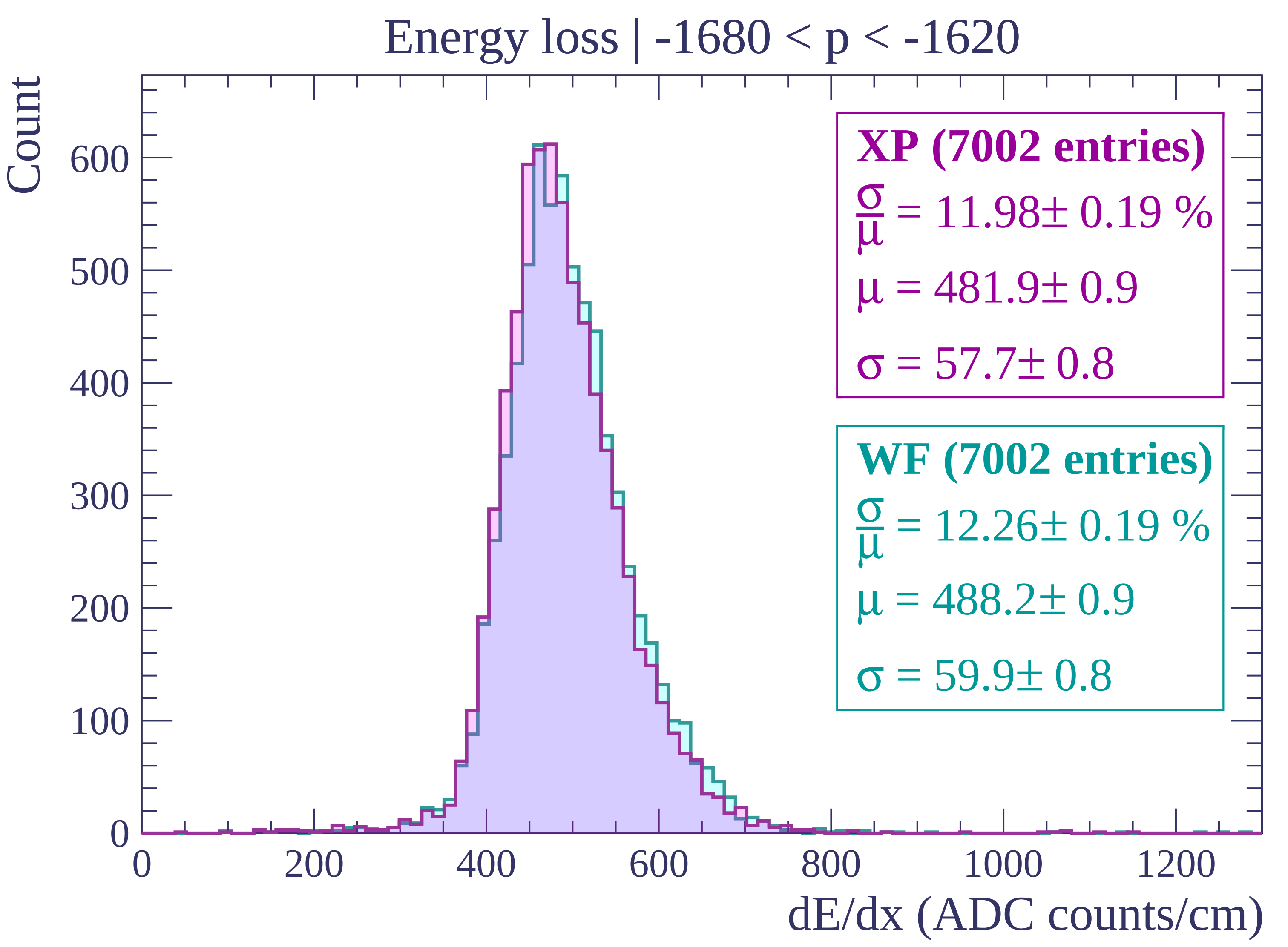


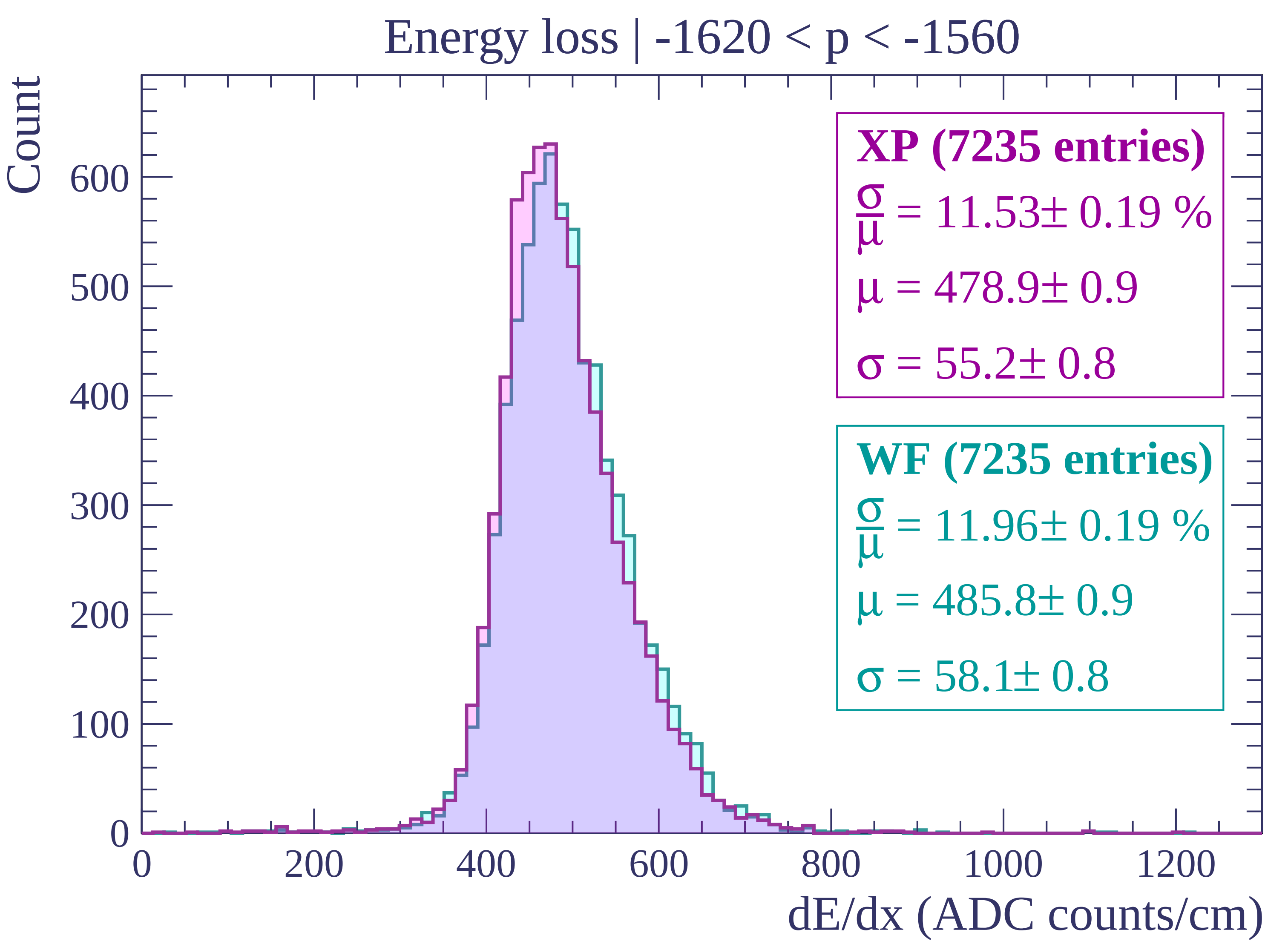
Energy loss | -1860 < p < -1800
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6245 entries)
$\frac{\sigma}{\mu}$ = 11.61± 0.19 %
μ = 487.8± 0.9
σ = 56.7± 0.8
WF (6245 entries)
$\frac{\sigma}{\mu}$ = 12.44± 0.22 %
μ = 494.1± 1.0
σ = 61.5± 1.0


Energy loss | -1800 < p < -1740
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6453 entries)
σ/μ = 12.16± 0.19 %
μ = 487.6± 0.9
σ = 59.3± 0.8
WF (6453 entries)
σ/μ = 12.66± 0.20 %
μ = 494.9± 0.9
σ = 62.6± 0.8

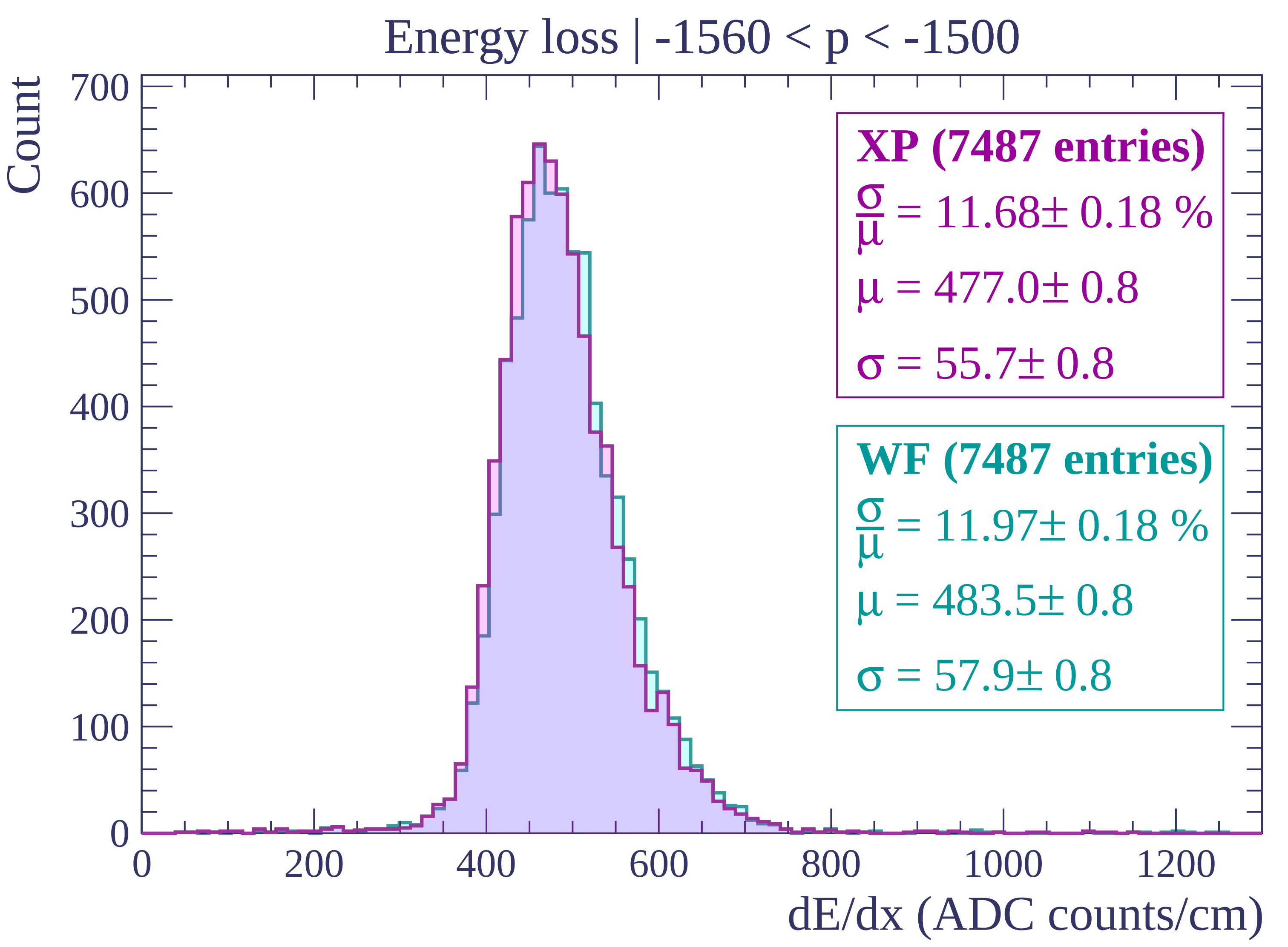


Energy loss | -1740 < p < -1680
Count
dE/dx (ADC counts/cm)
XP (6786 entries)
$\frac{\sigma}{\mu}$ = 11.77± 0.19 %
μ = 483.5± 0.9
σ = 56.9± 0.8
WF (6786 entries)
$\frac{\sigma}{\mu}$ = 12.12± 0.19 %
μ = 491.4± 0.9
σ = 59.6± 0.8

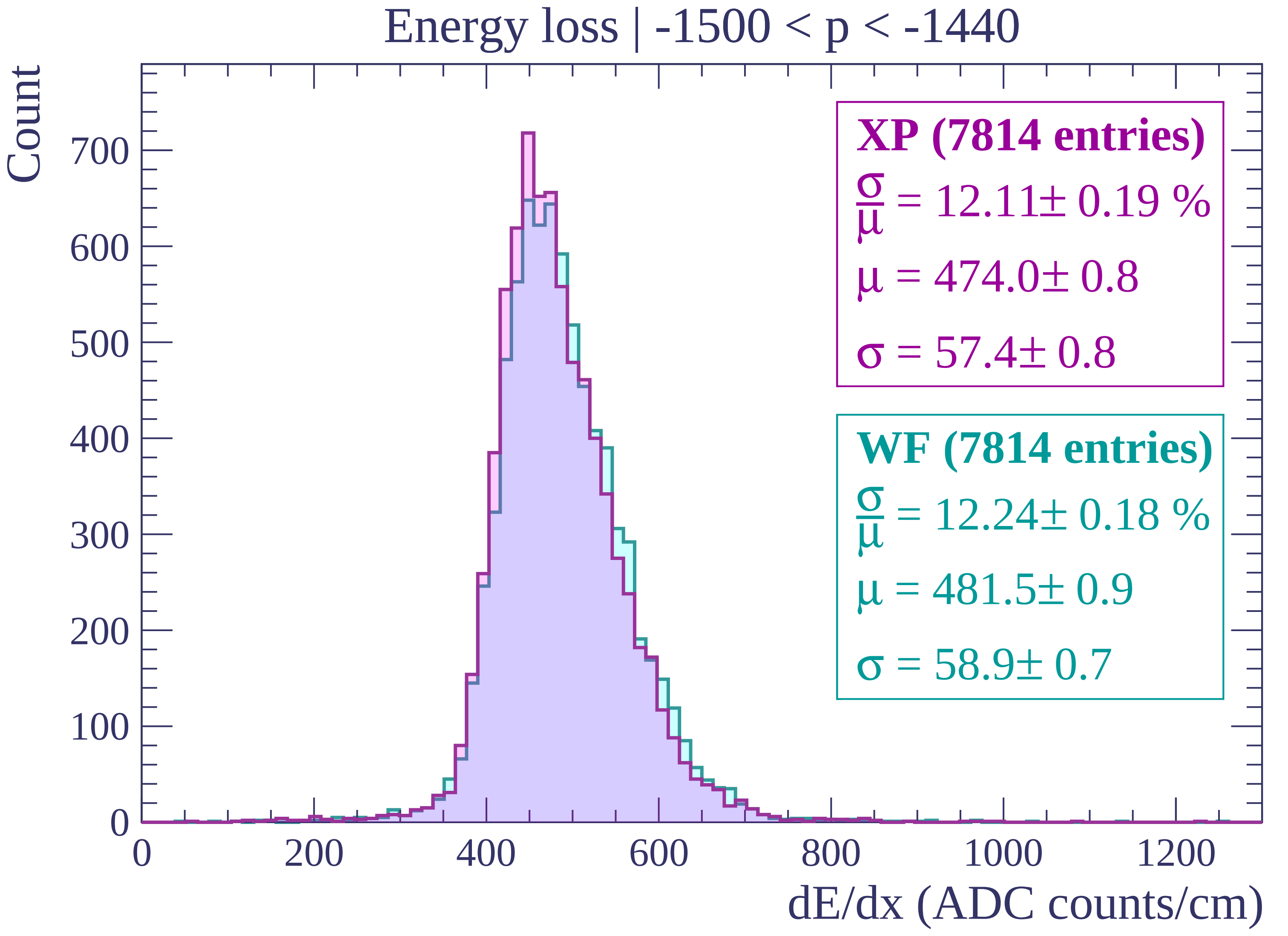

Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (7002 entries)
σ/μ = 11.98± 0.19 %
μ = 481.9± 0.9
σ = 57.7± 0.8
WF (7002 entries)
σ/μ = 12.26± 0.19 %
μ = 488.2± 0.9
σ = 59.9± 0.8



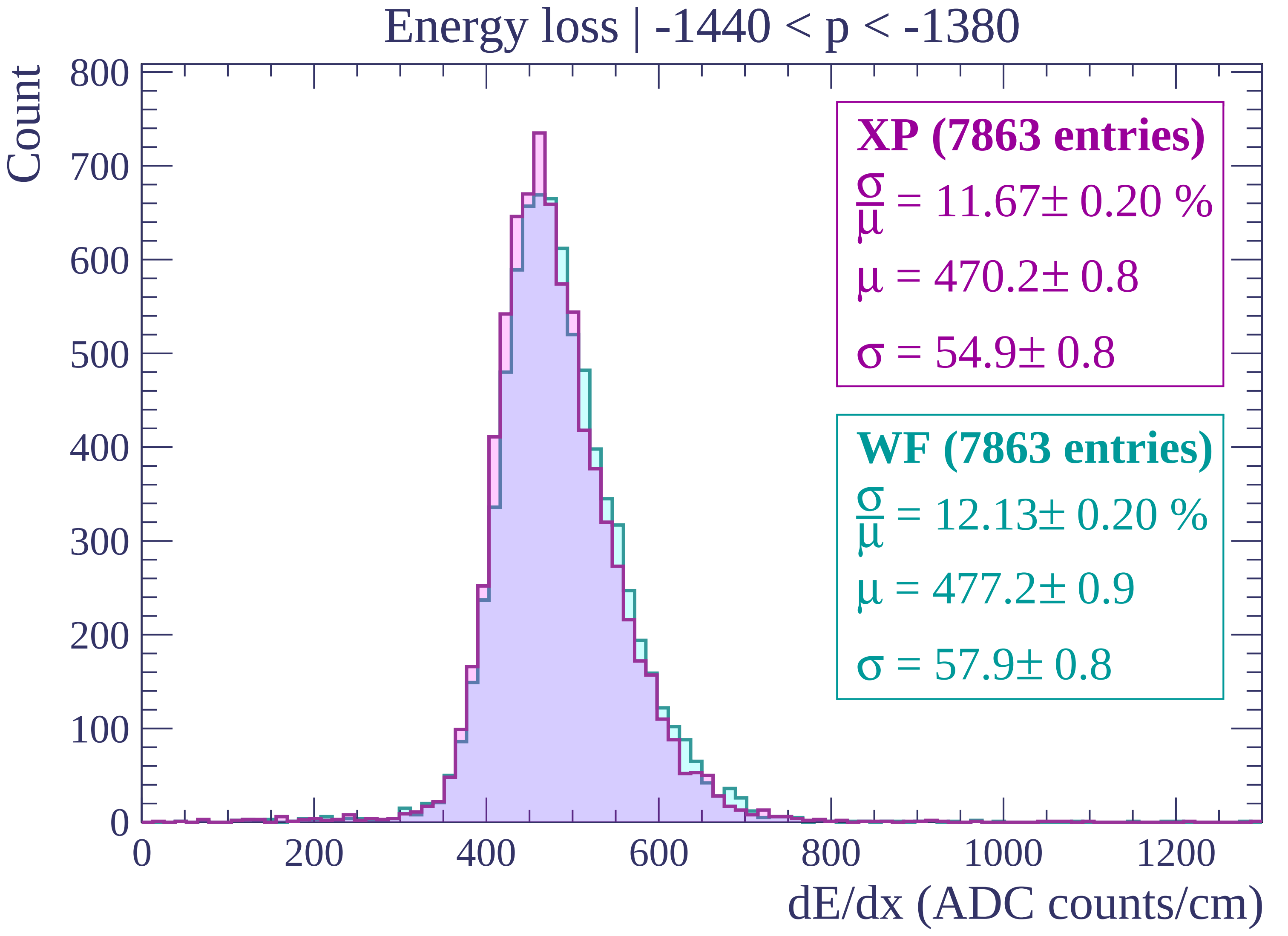
Energy loss | -1620 < p < -1560
Count
dE/dx (ADC counts/cm)
XP (7235 entries)
σ/μ = 11.53± 0.19 %
μ = 478.9± 0.9
σ = 55.2± 0.8
WF (7235 entries)
σ/μ = 11.96± 0.19 %
μ = 485.8± 0.9
σ = 58.1± 0.8
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200

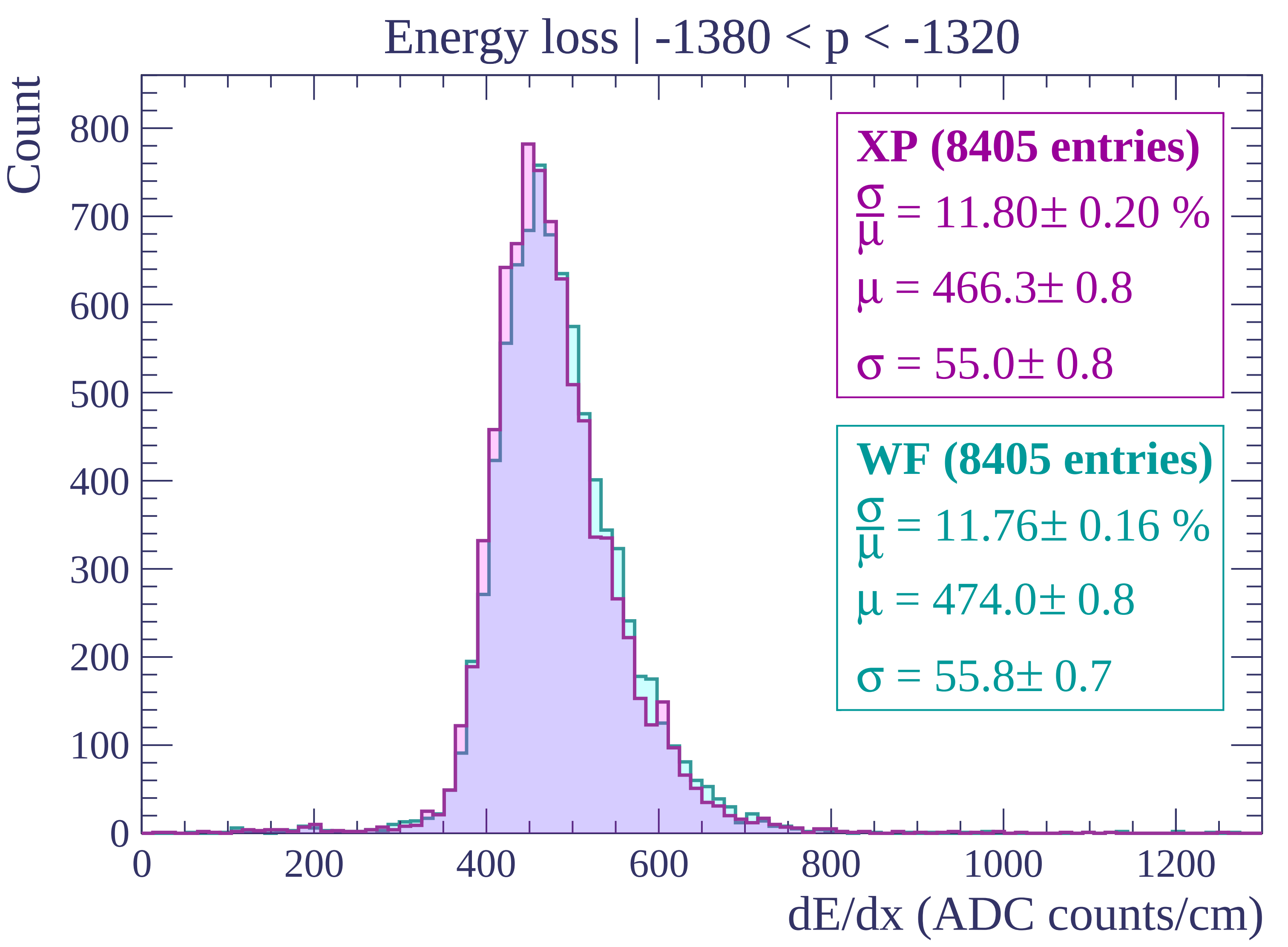


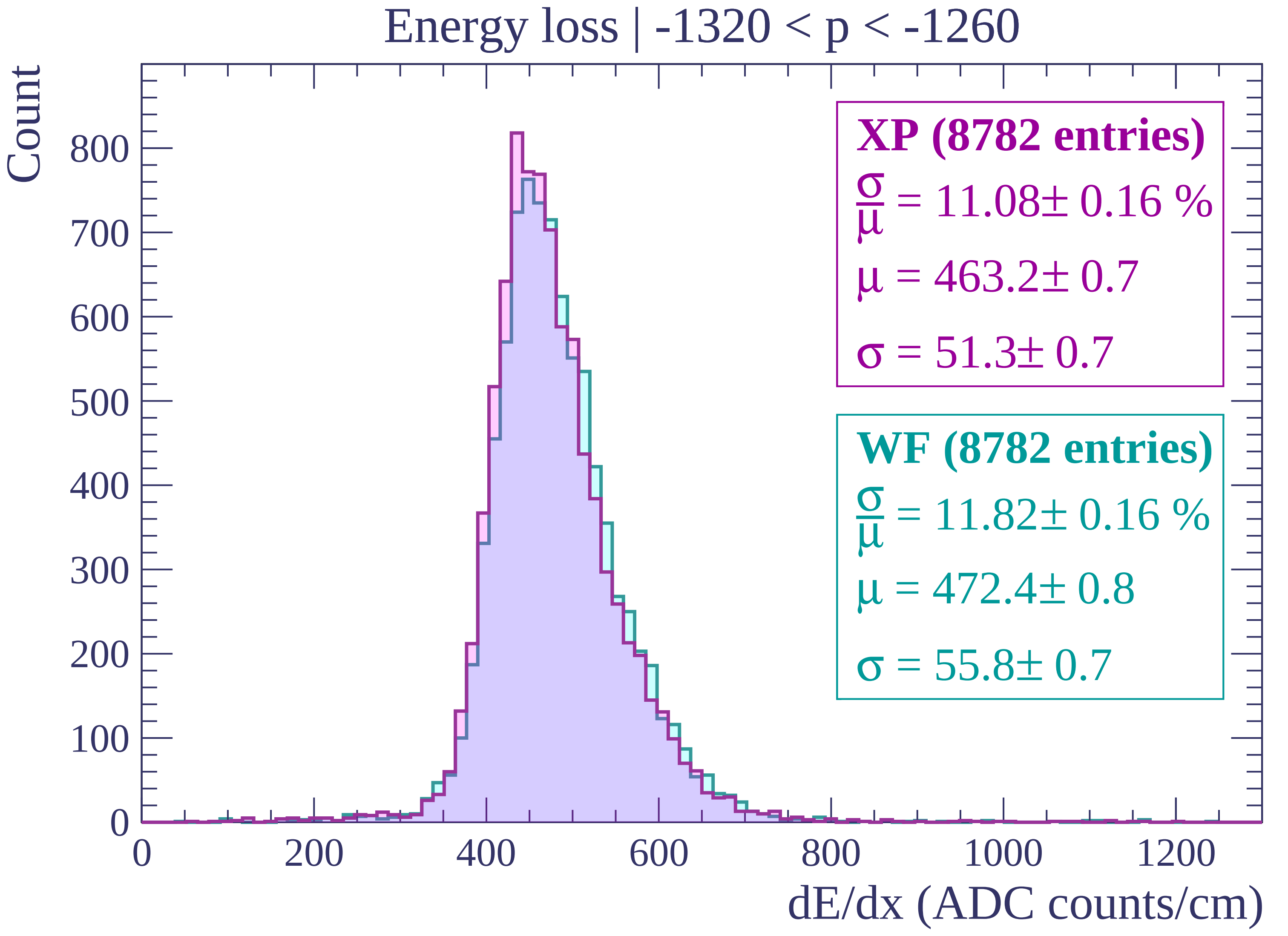
Energy loss | -1560 < p < -1500
Count
XP (7487 entries)
σ/μ = 11.68± 0.18 %
μ = 477.0± 0.8
σ = 55.7± 0.8
WF (7487 entries)
σ/μ = 11.97± 0.18 %
μ = 483.5± 0.8
σ = 57.9± 0.8
dE/dx (ADC counts/cm)



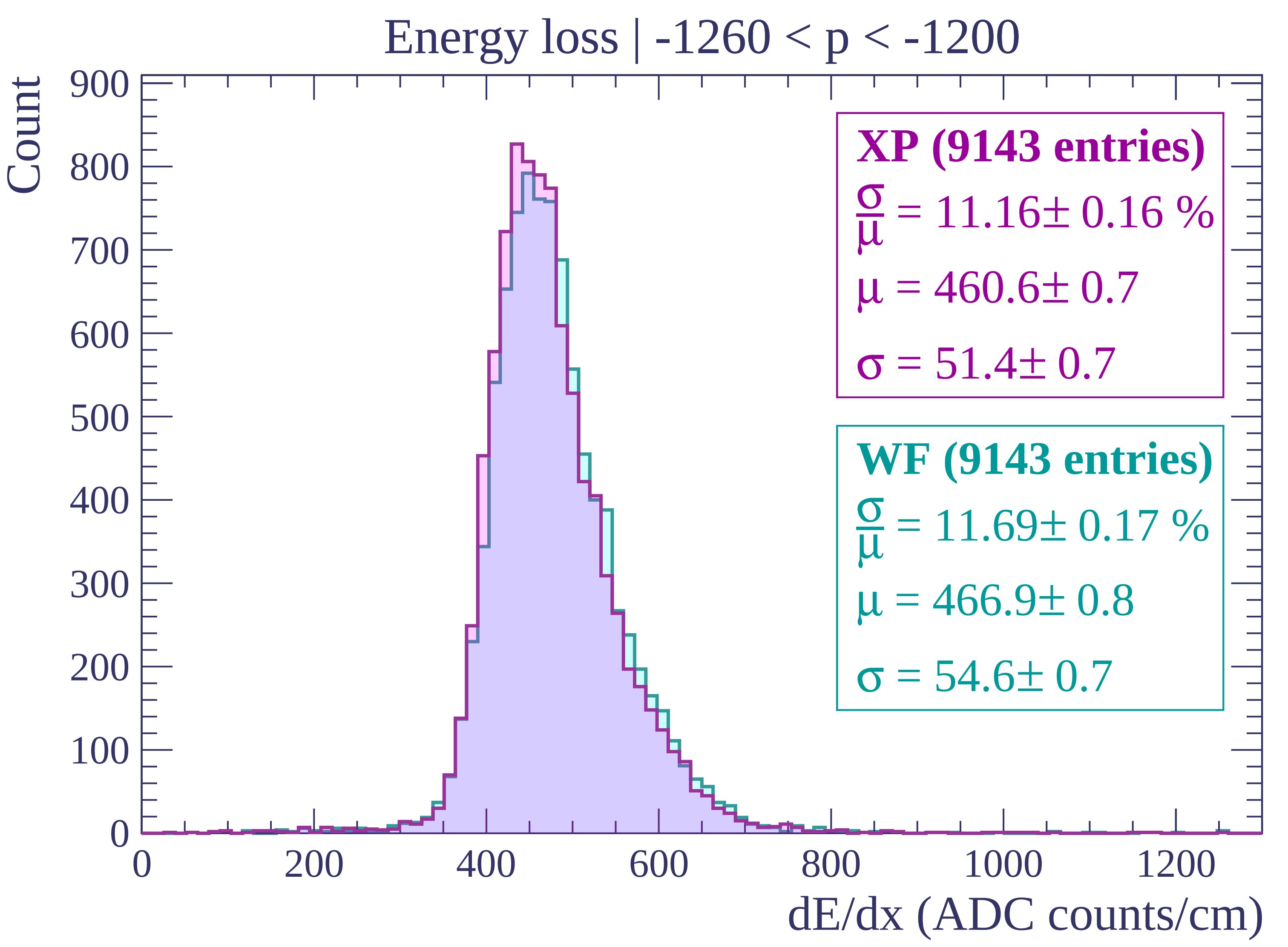
Energy loss | -1500 < p < -1440
Count
XP (7814 entries)
σ/μ = 12.11± 0.19 %
μ = 474.0± 0.8
σ = 57.4± 0.8
WF (7814 entries)
σ/μ = 12.24± 0.18 %
μ = 481.5± 0.9
σ = 58.9± 0.7
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

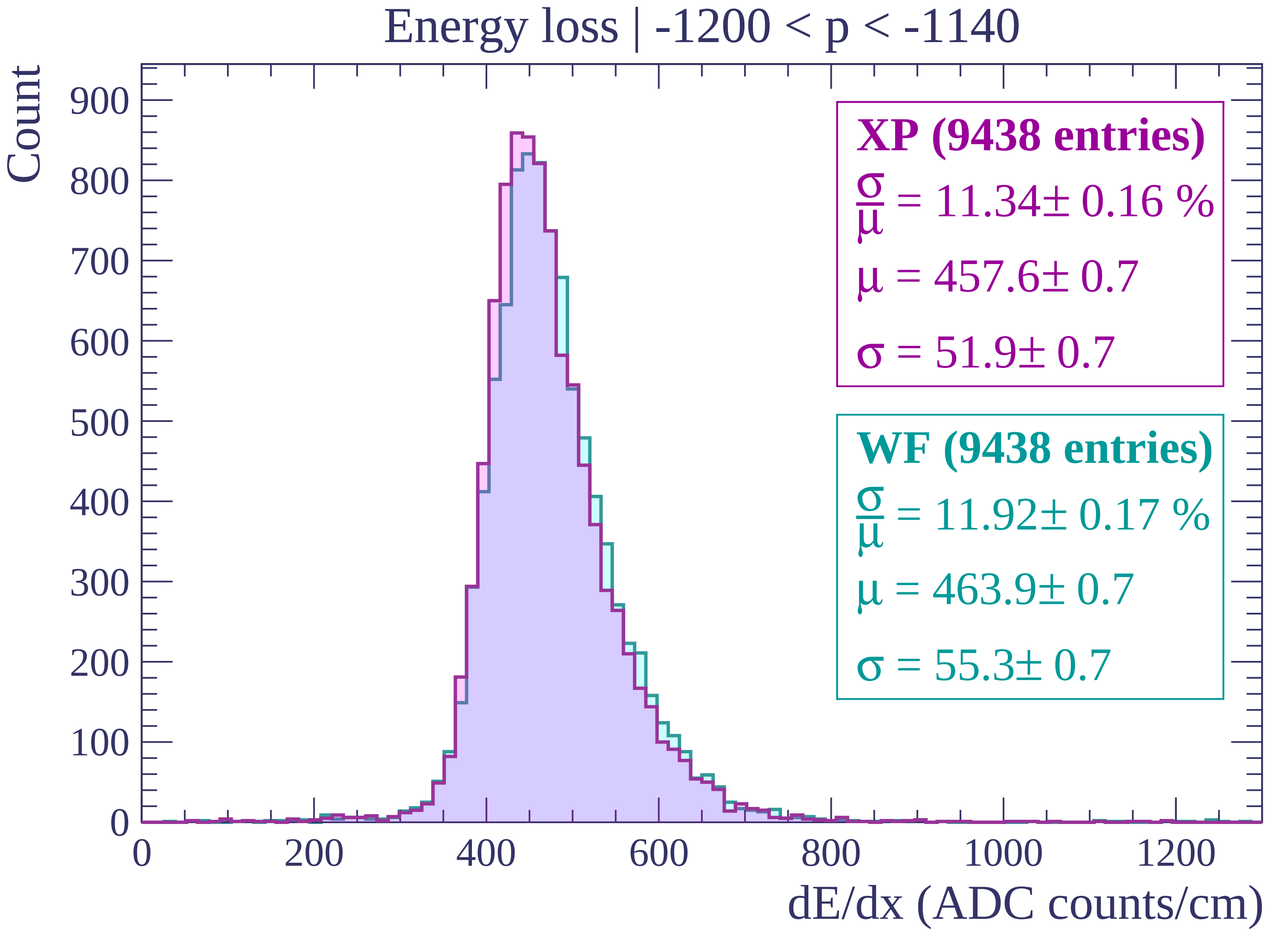

Energy loss | -1440 < p < -1380
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (7863 entries)
σ/μ = 11.67± 0.20 %
μ = 470.2± 0.8
σ = 54.9± 0.8
WF (7863 entries)
σ/μ = 12.13± 0.20 %
μ = 477.2± 0.9
σ = 57.9± 0.8


Energy loss | -1380 < p < -1320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP (8405 entries)
$\frac{\sigma}{\mu}$ = 11.80± 0.20 %
μ = 466.3± 0.8
σ = 55.0± 0.8
WF (8405 entries)
$\frac{\sigma}{\mu}$ = 11.76± 0.16 %
μ = 474.0± 0.8
σ = 55.8± 0.7

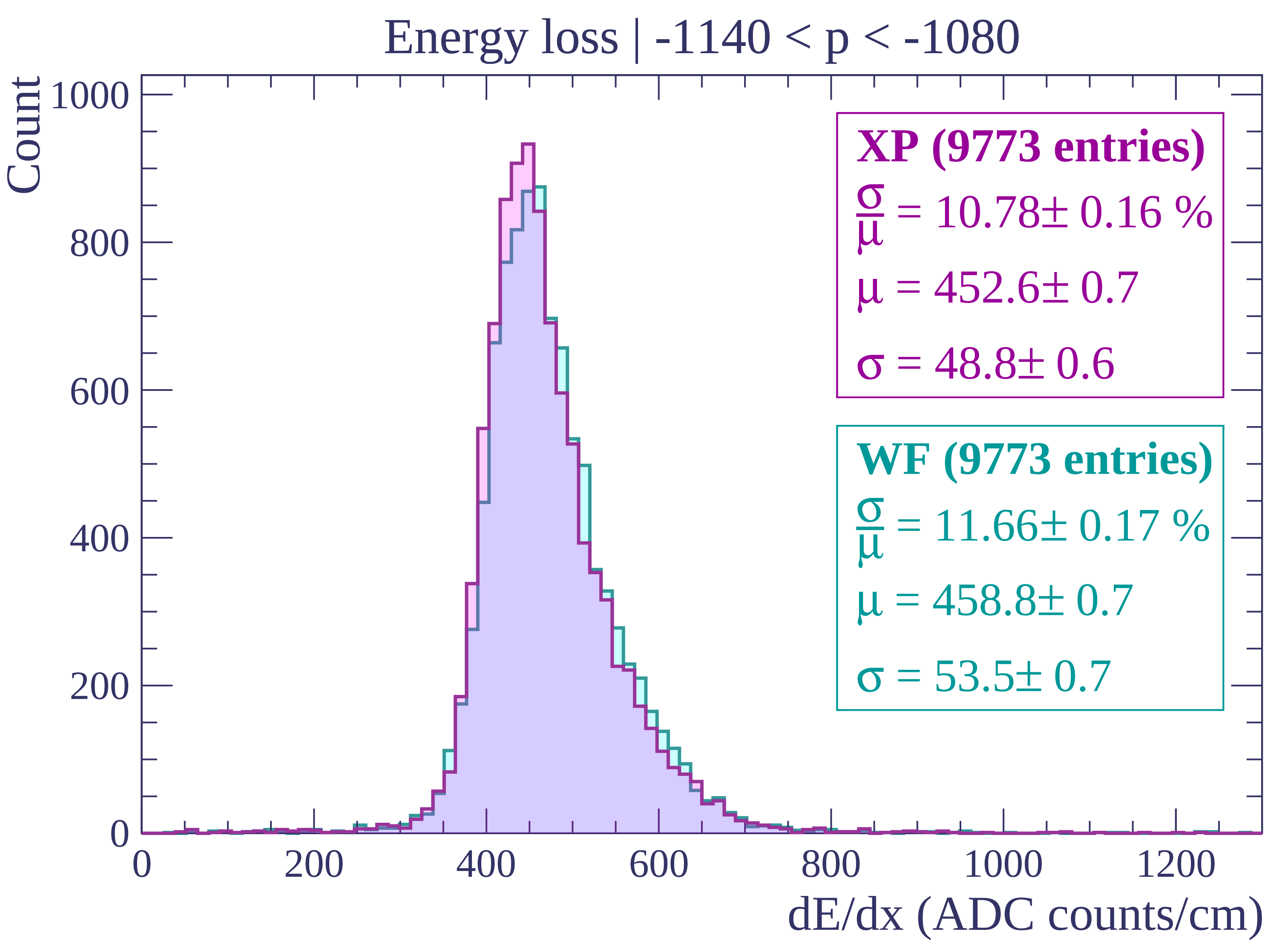


Energy loss | -1320 < p < -1260
Count
dE/dx (ADC counts/cm)
XP (8782 entries)
σ/μ = 11.08± 0.16 %
μ = 463.2± 0.7
σ = 51.3± 0.7
WF (8782 entries)
σ/μ = 11.82± 0.16 %
μ = 472.4± 0.8
σ = 55.8± 0.7

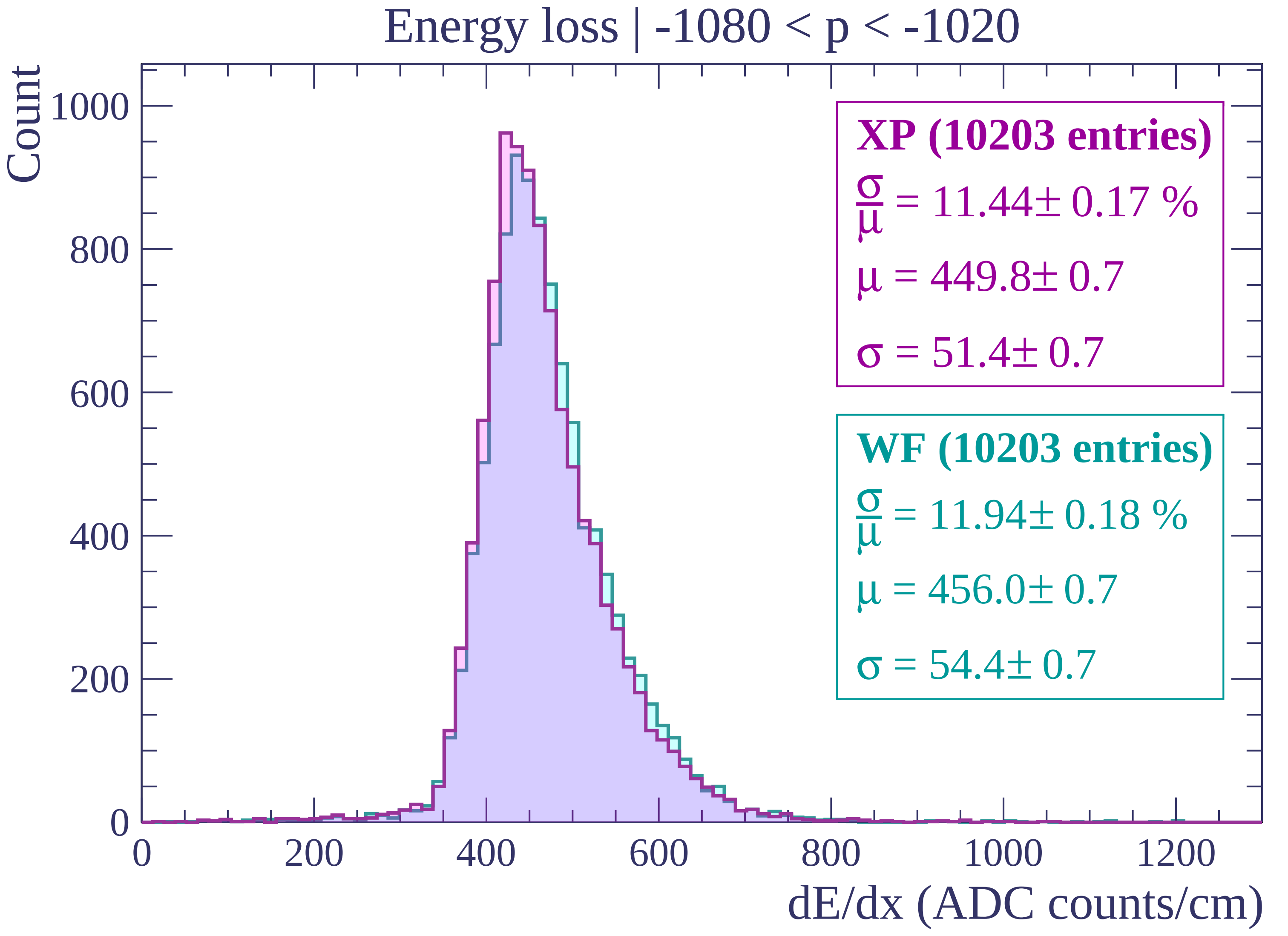

Energy loss | -1260 < p < -1200
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (9143 entries)
$\frac{\sigma}{\mu}$ = 11.16± 0.16 %
μ = 460.6± 0.7
σ = 51.4± 0.7
WF (9143 entries)
$\frac{\sigma}{\mu}$ = 11.69± 0.17 %
μ = 466.9± 0.8
σ = 54.6± 0.7



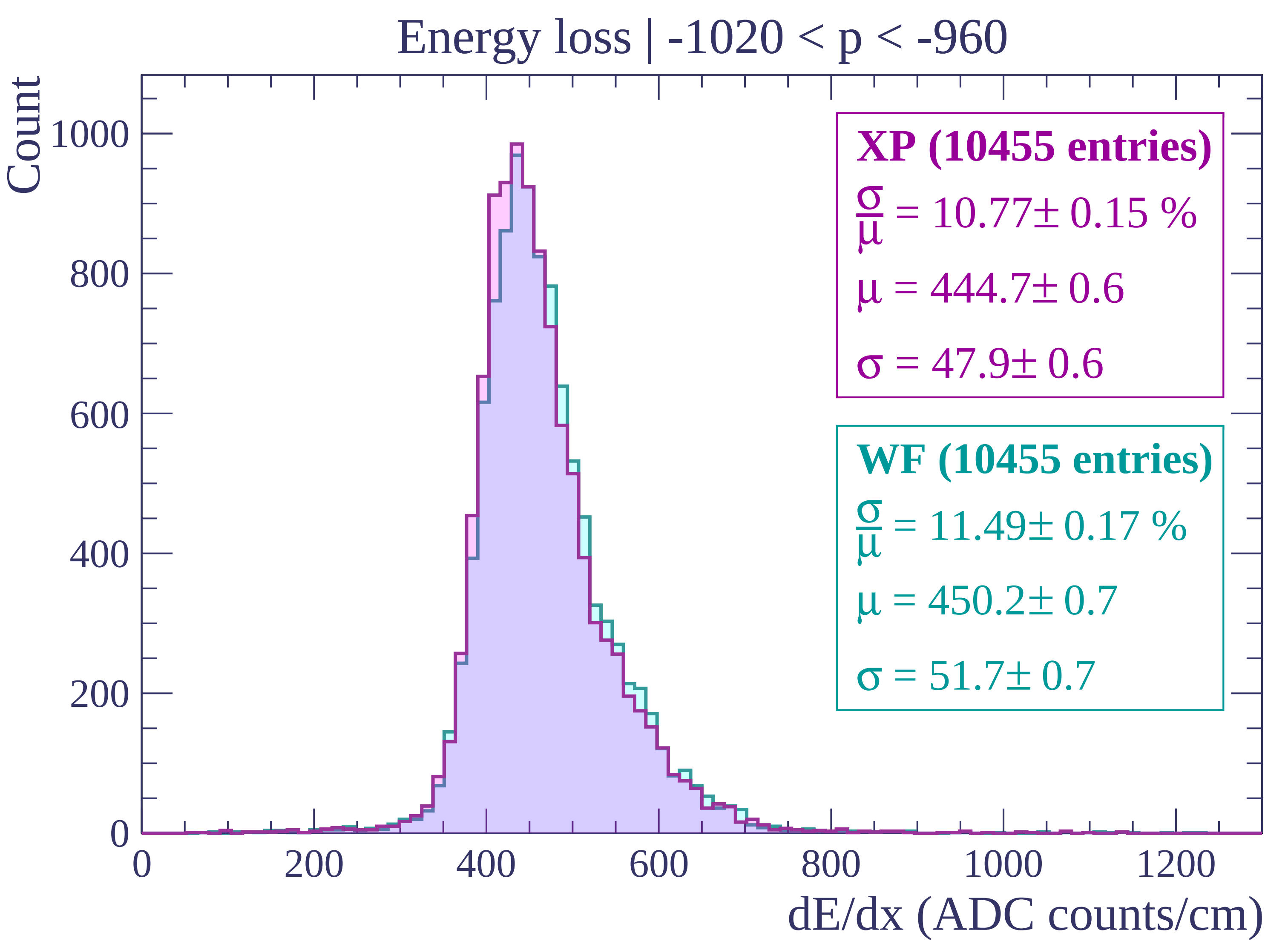
Energy loss | -1200 < p < -1140
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (9438 entries)
σ/μ = 11.34± 0.16 %
μ = 457.6± 0.7
σ = 51.9± 0.7
WF (9438 entries)
σ/μ = 11.92± 0.17 %
μ = 463.9± 0.7
σ = 55.3± 0.7



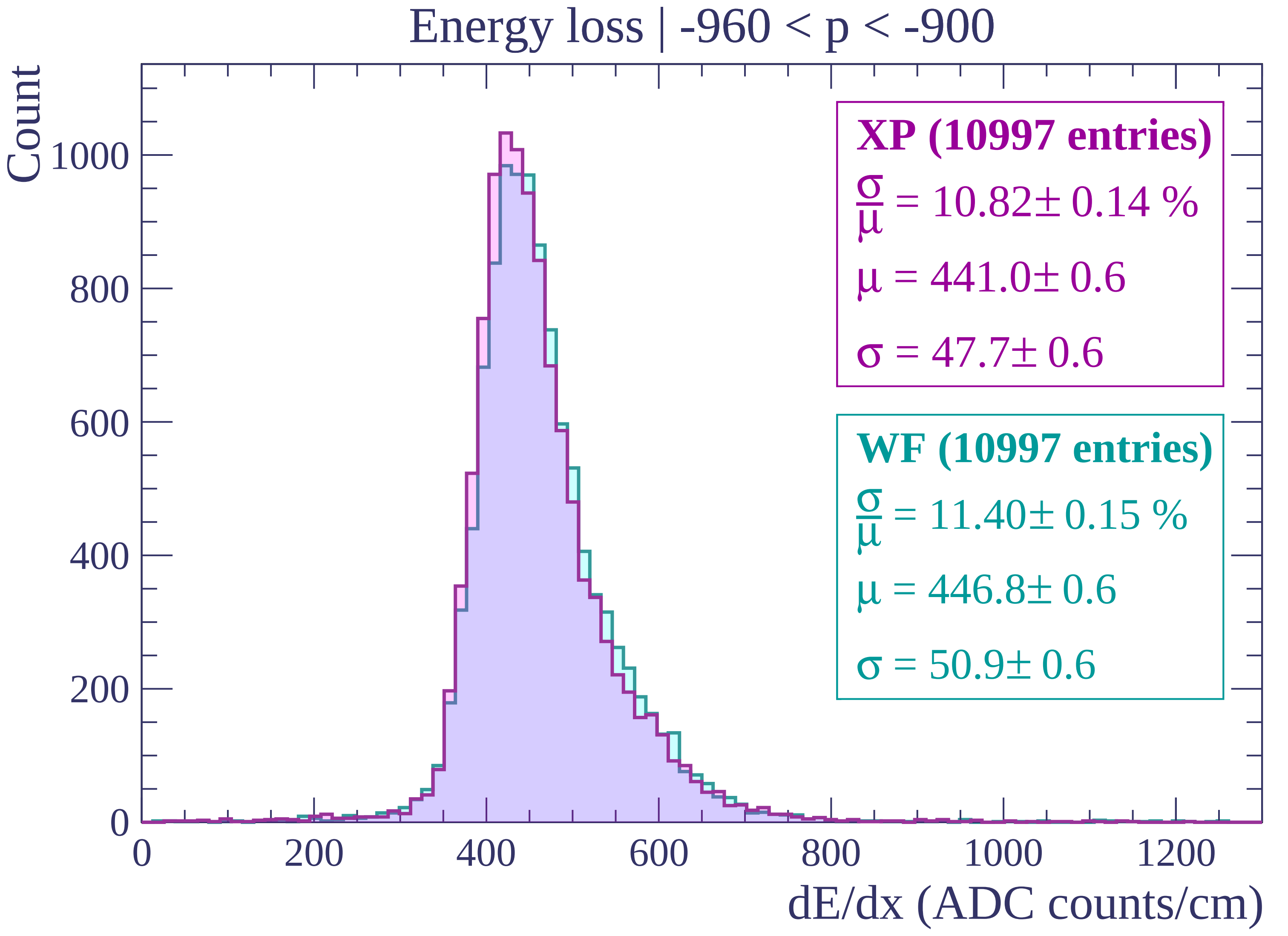
Energy loss | -1140 < p < -1080
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (9773 entries)
$\frac{\sigma}{\mu}$ = 10.78± 0.16 %
μ = 452.6± 0.7
σ = 48.8± 0.6
WF (9773 entries)
$\frac{\sigma}{\mu}$ = 11.66± 0.17 %
μ = 458.8± 0.7
σ = 53.5± 0.7

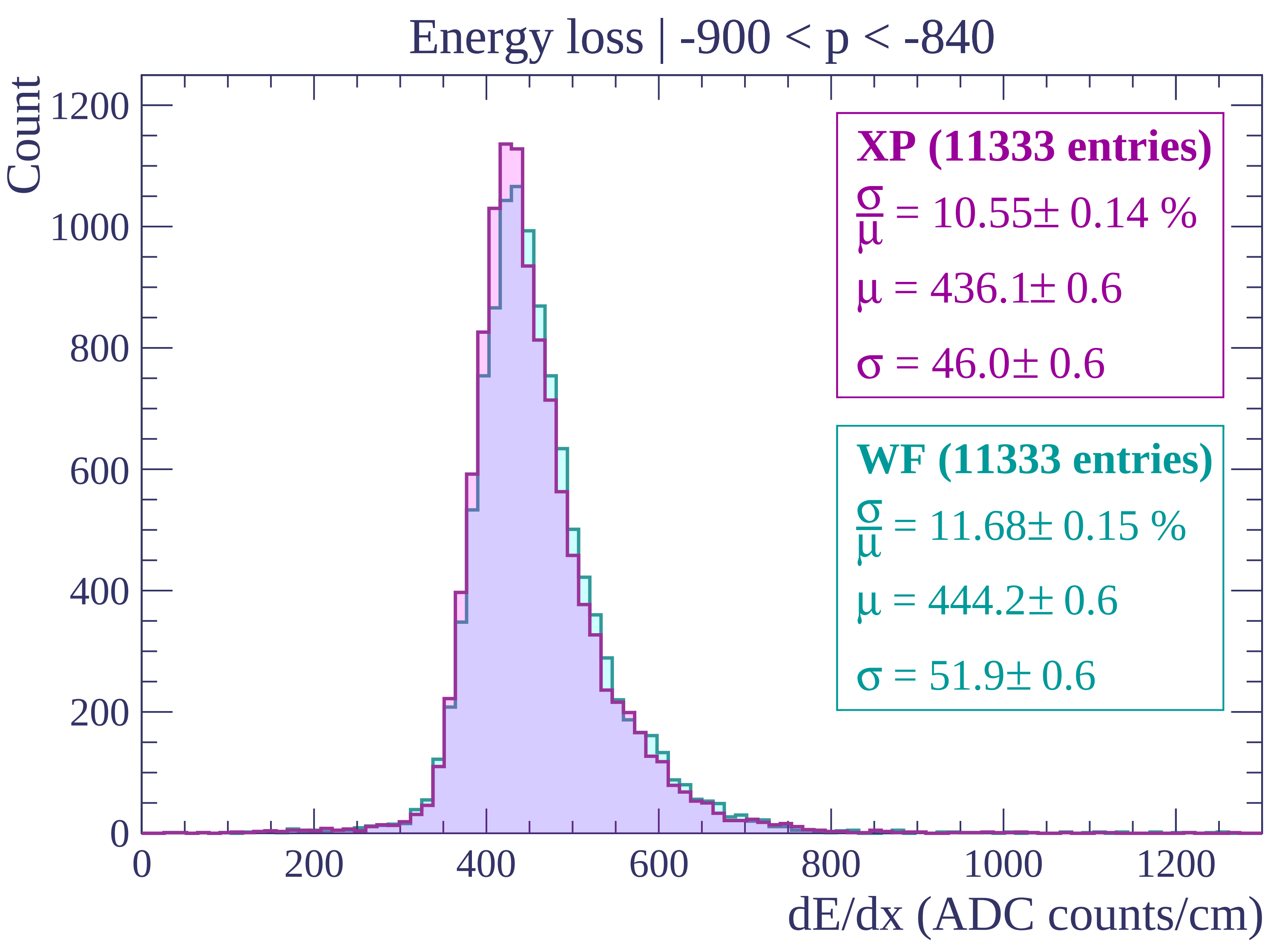


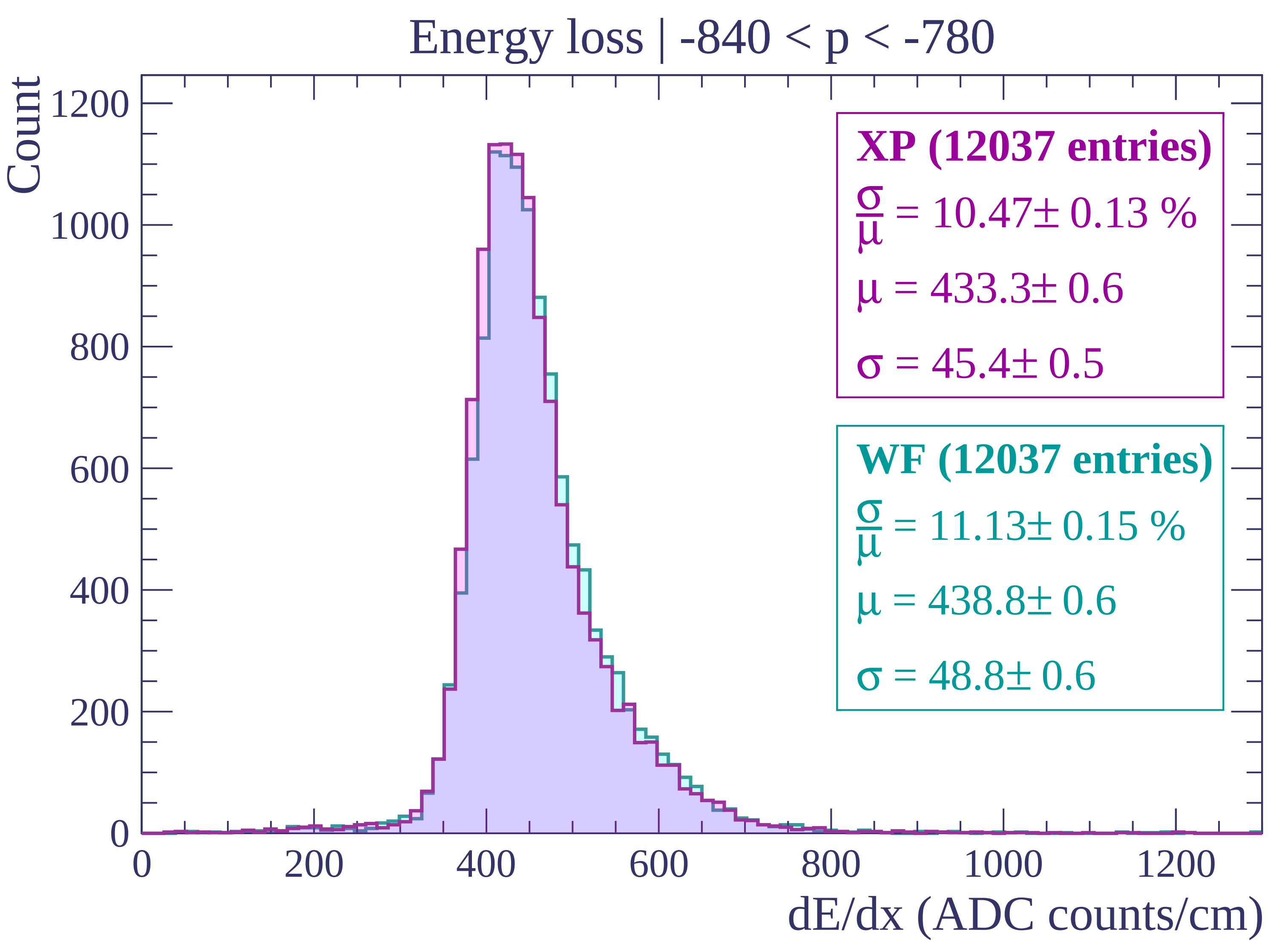
Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
XP (10203 entries)
σ/μ = 11.44± 0.17 %
μ = 449.8± 0.7
σ = 51.4± 0.7
WF (10203 entries)
σ/μ = 11.94± 0.18 %
μ = 456.0± 0.7
σ = 54.4± 0.7



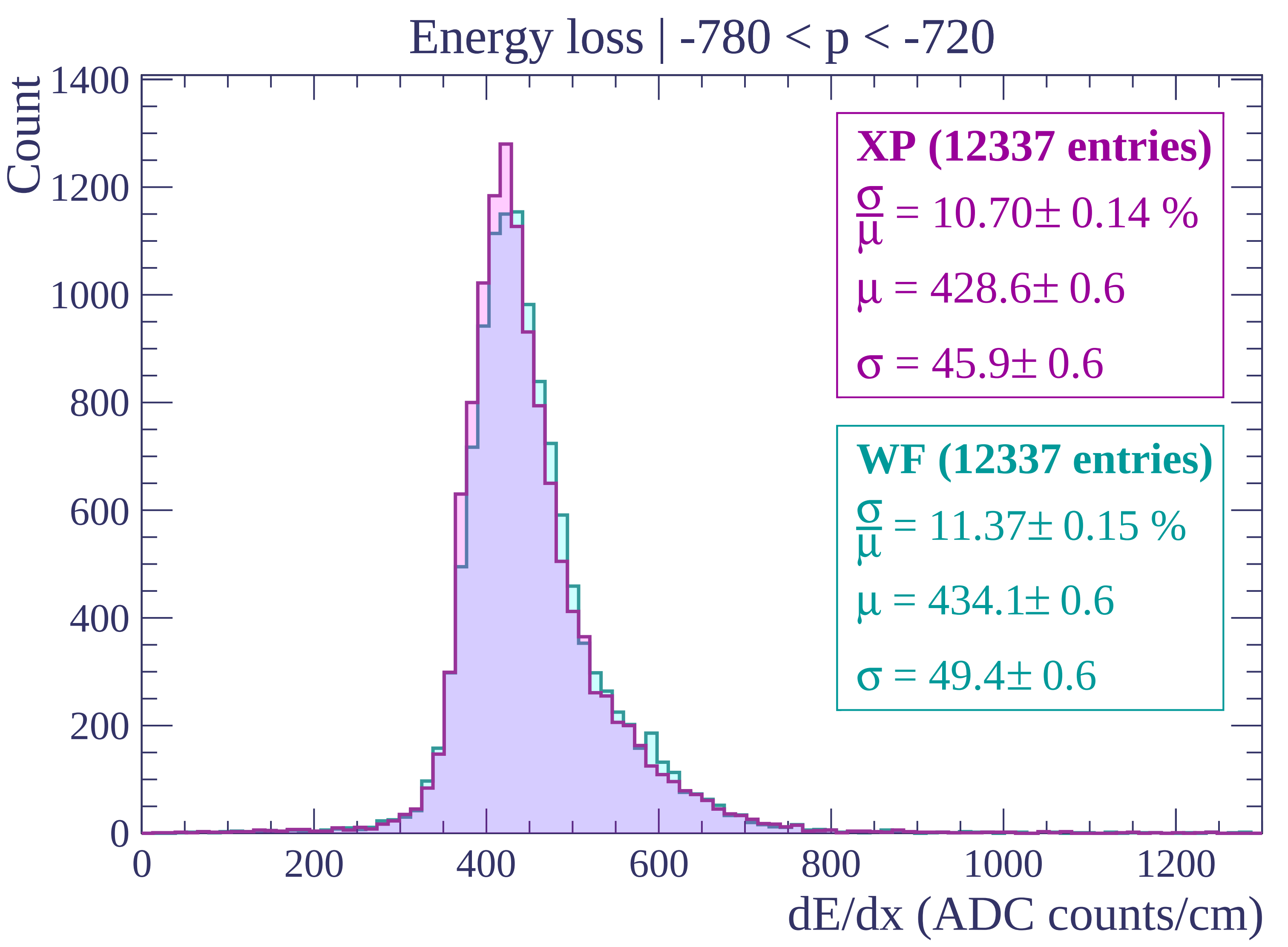
Energy loss | -1020 < p < -960
Count
dE/dx (ADC counts/cm)
XP (10455 entries)
σ/μ = 10.77± 0.15 %
μ = 444.7± 0.6
σ = 47.9± 0.6
WF (10455 entries)
σ/μ = 11.49± 0.17 %
μ = 450.2± 0.7
σ = 51.7± 0.7


Energy loss | -960 < p < -900
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (10997 entries)
σ/μ = 10.82± 0.14 %
μ = 441.0± 0.6
σ = 47.7± 0.6
WF (10997 entries)
σ/μ = 11.40± 0.15 %
μ = 446.8± 0.6
σ = 50.9± 0.6

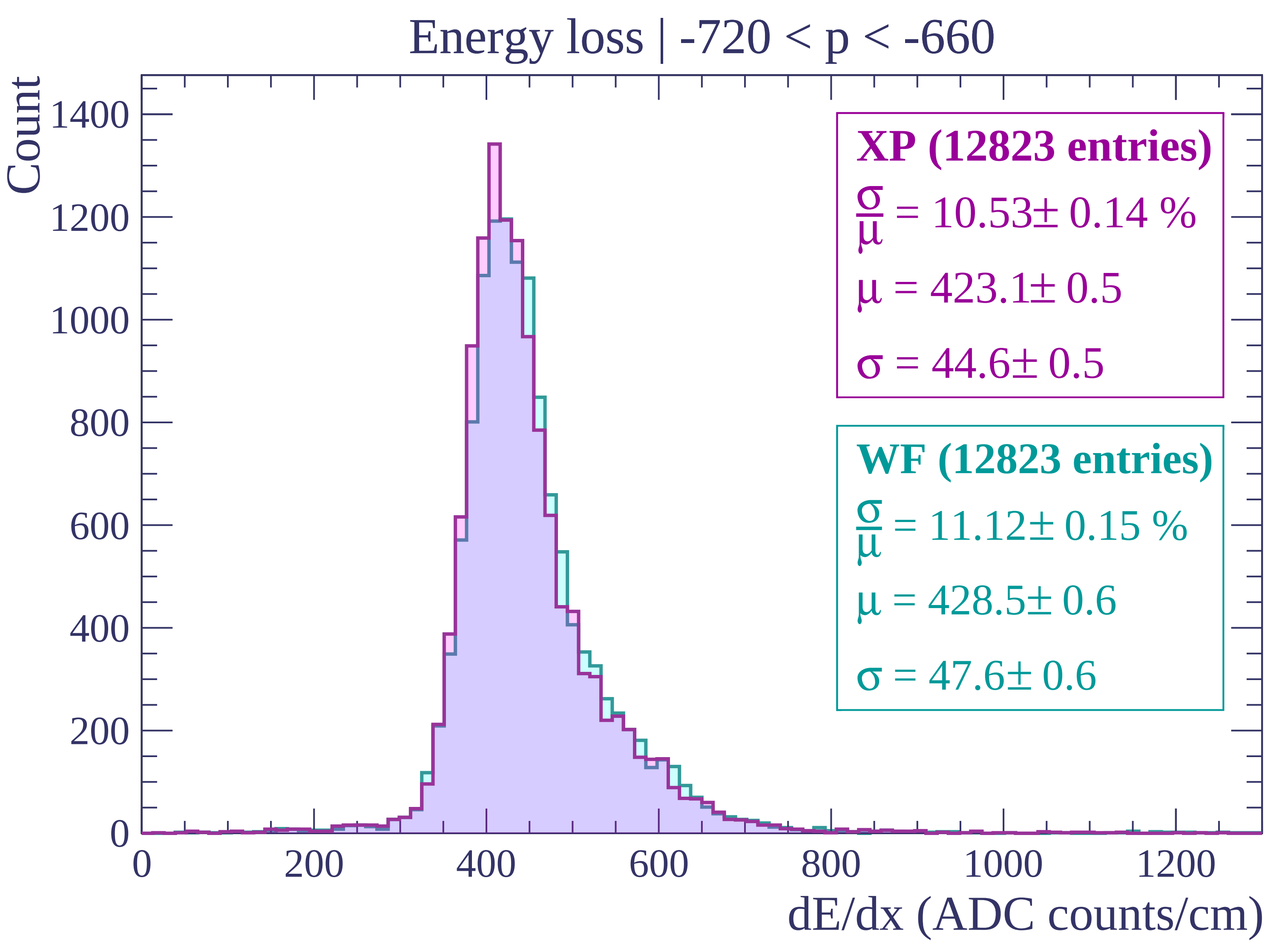

Energy loss | -900 < p < -840
Count
dE/dx (ADC counts/cm)
XP (11333 entries)
σ/μ = 10.55± 0.14 %
μ = 436.1± 0.6
σ = 46.0± 0.6
WF (11333 entries)
σ/μ = 11.68± 0.15 %
μ = 444.2± 0.6
σ = 51.9± 0.6



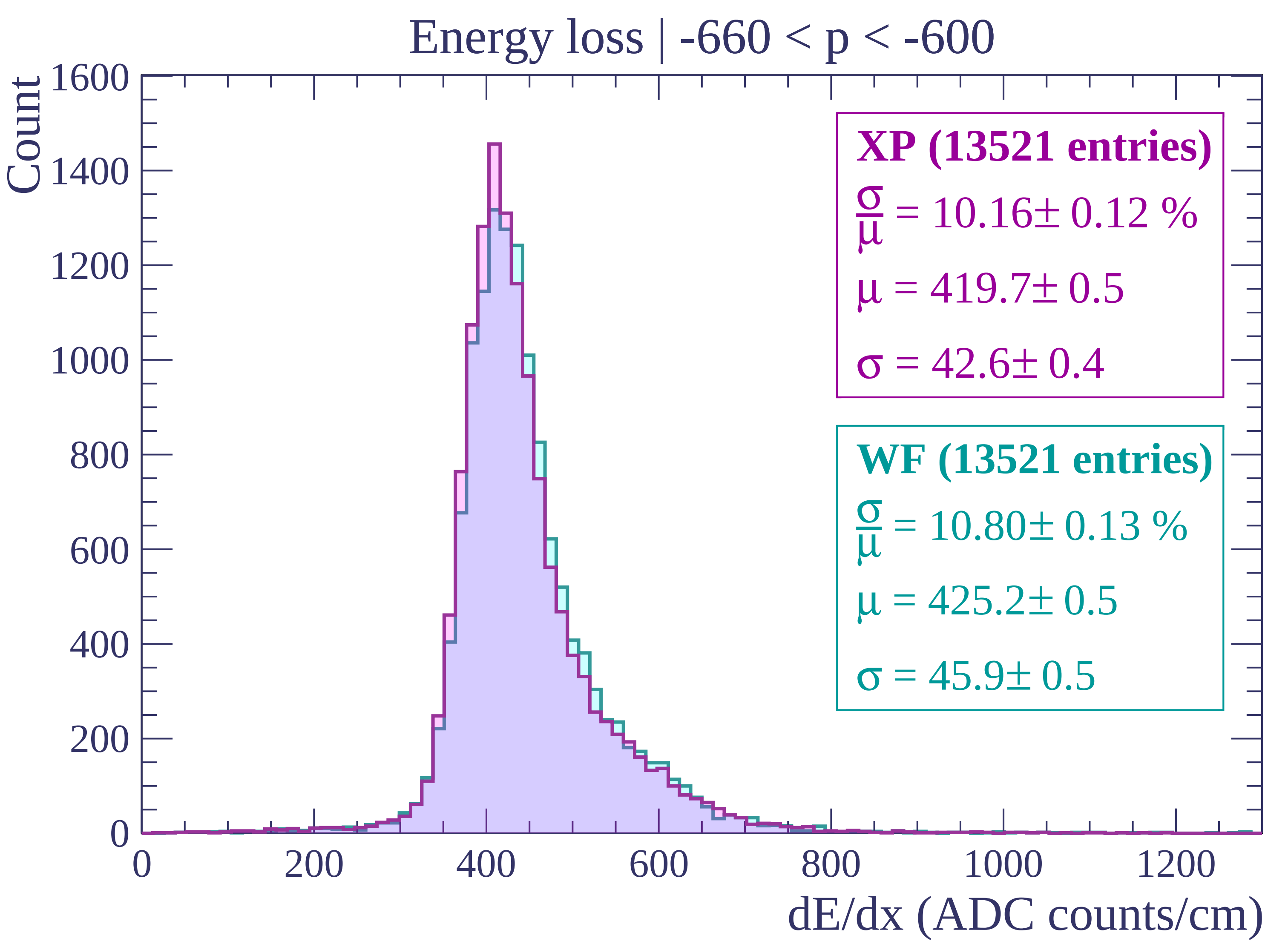
Energy loss | -840 < p < -780
Count
1200
1000
800
600
400
200
0
XP (12037 entries)
σ/μ = 10.47± 0.13 %
μ = 433.3± 0.6
σ = 45.4± 0.5
WF (12037 entries)
σ/μ = 11.13± 0.15 %
μ = 438.8± 0.6
σ = 48.8± 0.6
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



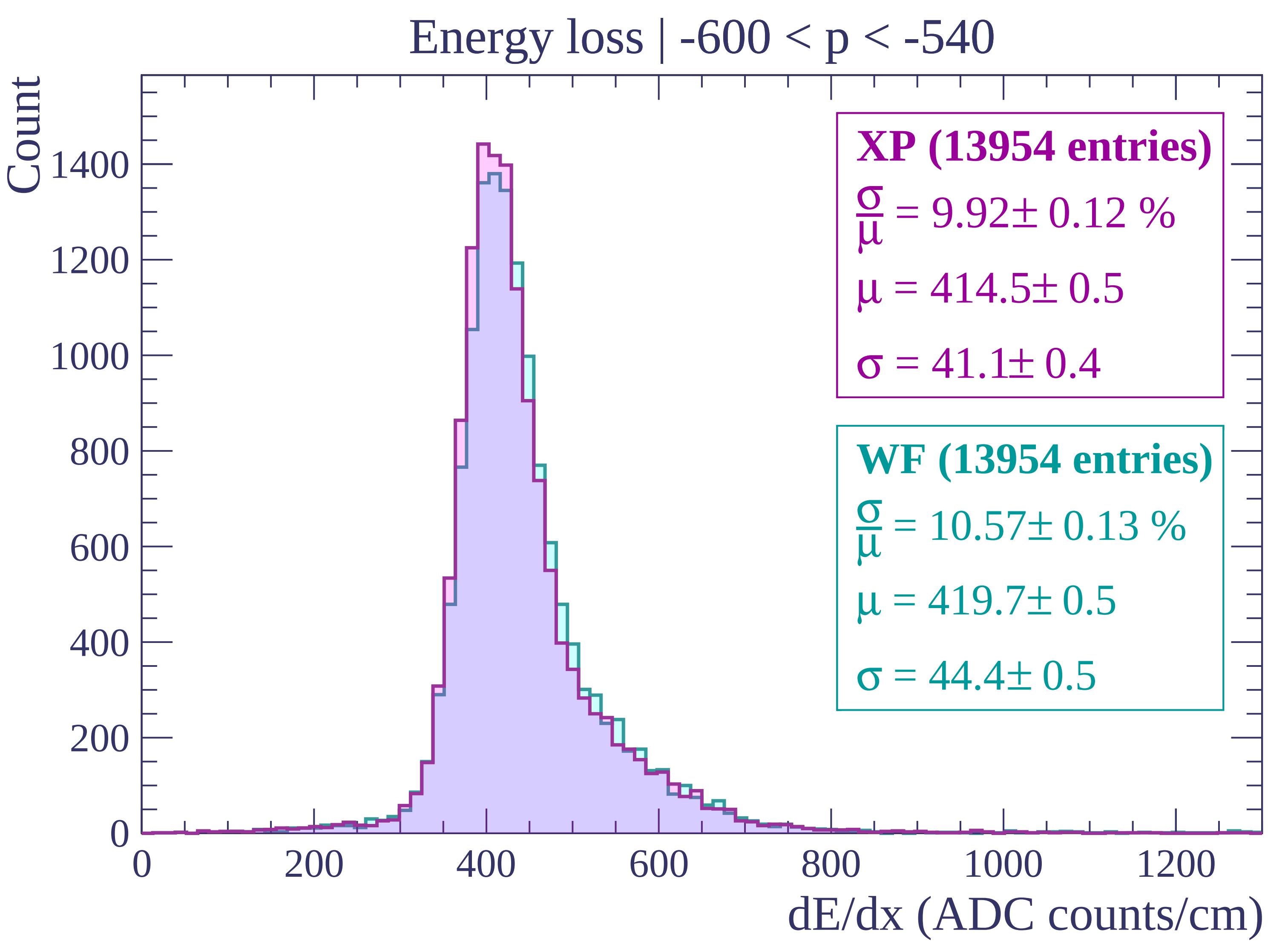
Energy loss | -780 < p < -720
Count
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (12337 entries)
σ/μ = 10.70± 0.14 %
μ = 428.6± 0.6
σ = 45.9± 0.6
WF (12337 entries)
σ/μ = 11.37± 0.15 %
μ = 434.1± 0.6
σ = 49.4± 0.6

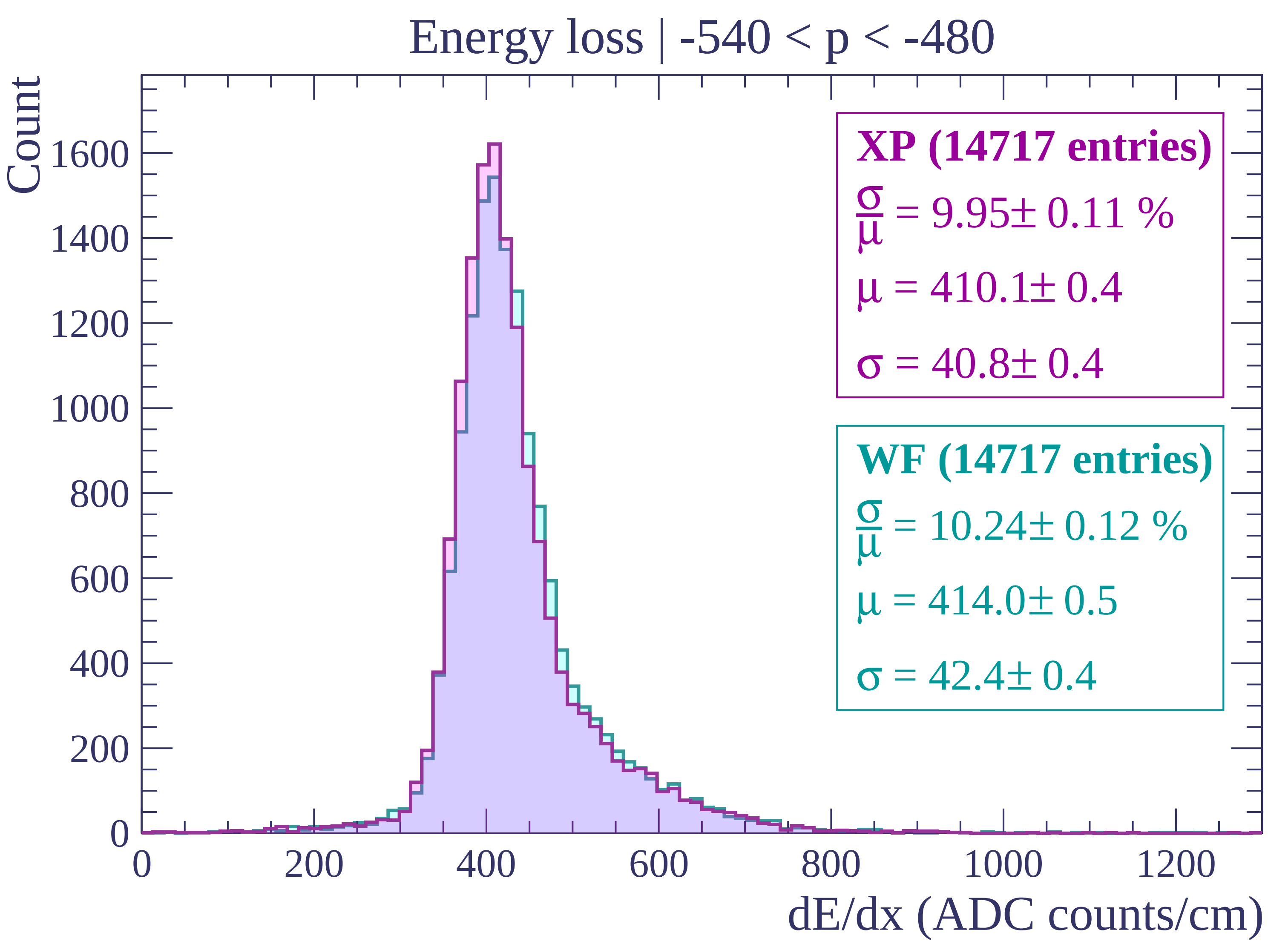


Energy loss | -720 < p < -660
Count
dE/dx (ADC counts/cm)
XP (12823 entries)
σ/μ = 10.53± 0.14 %
μ = 423.1± 0.5
σ = 44.6± 0.5
WF (12823 entries)
σ/μ = 11.12± 0.15 %
μ = 428.5± 0.6
σ = 47.6± 0.6

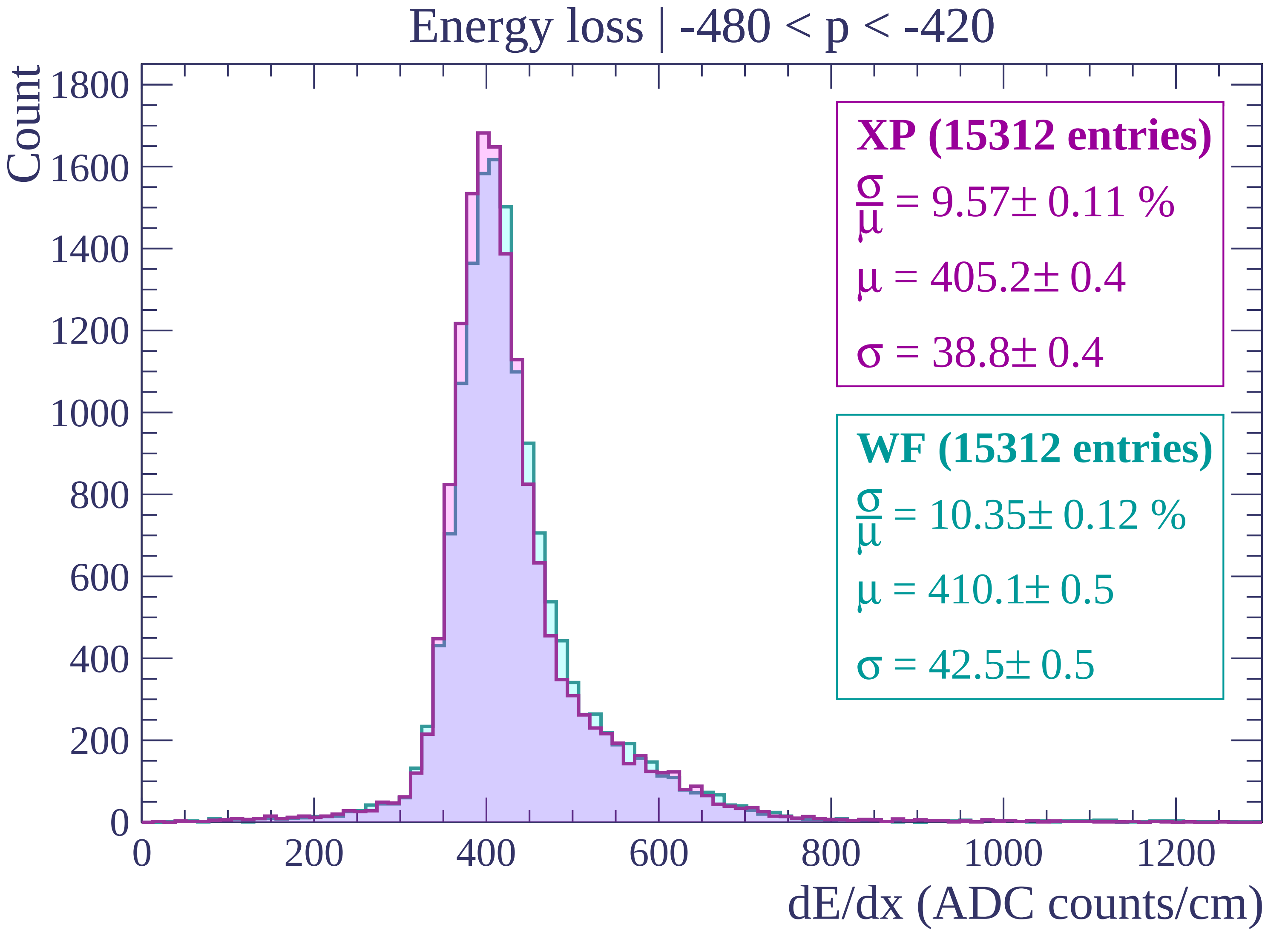


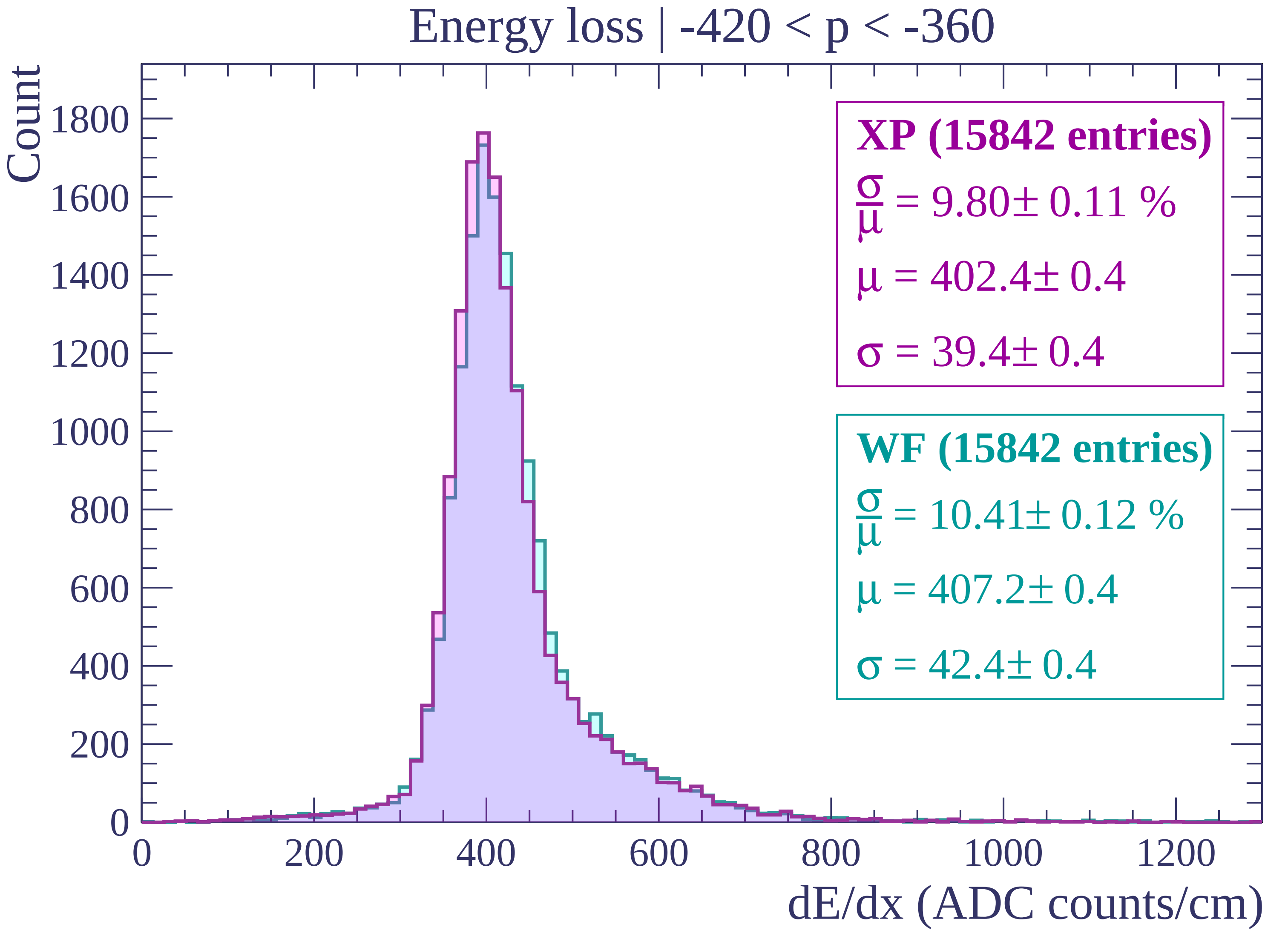
Energy loss | -660 < p < -600
Count
dE/dx (ADC counts/cm)
XP (13521 entries)
σ/μ = 10.16± 0.12 %
μ = 419.7± 0.5
σ = 42.6± 0.4
WF (13521 entries)
σ/μ = 10.80± 0.13 %
μ = 425.2± 0.5
σ = 45.9± 0.5
0
200
400
600
800
1000
1200
1400
1600
0
200
400
600
800
1000
1200

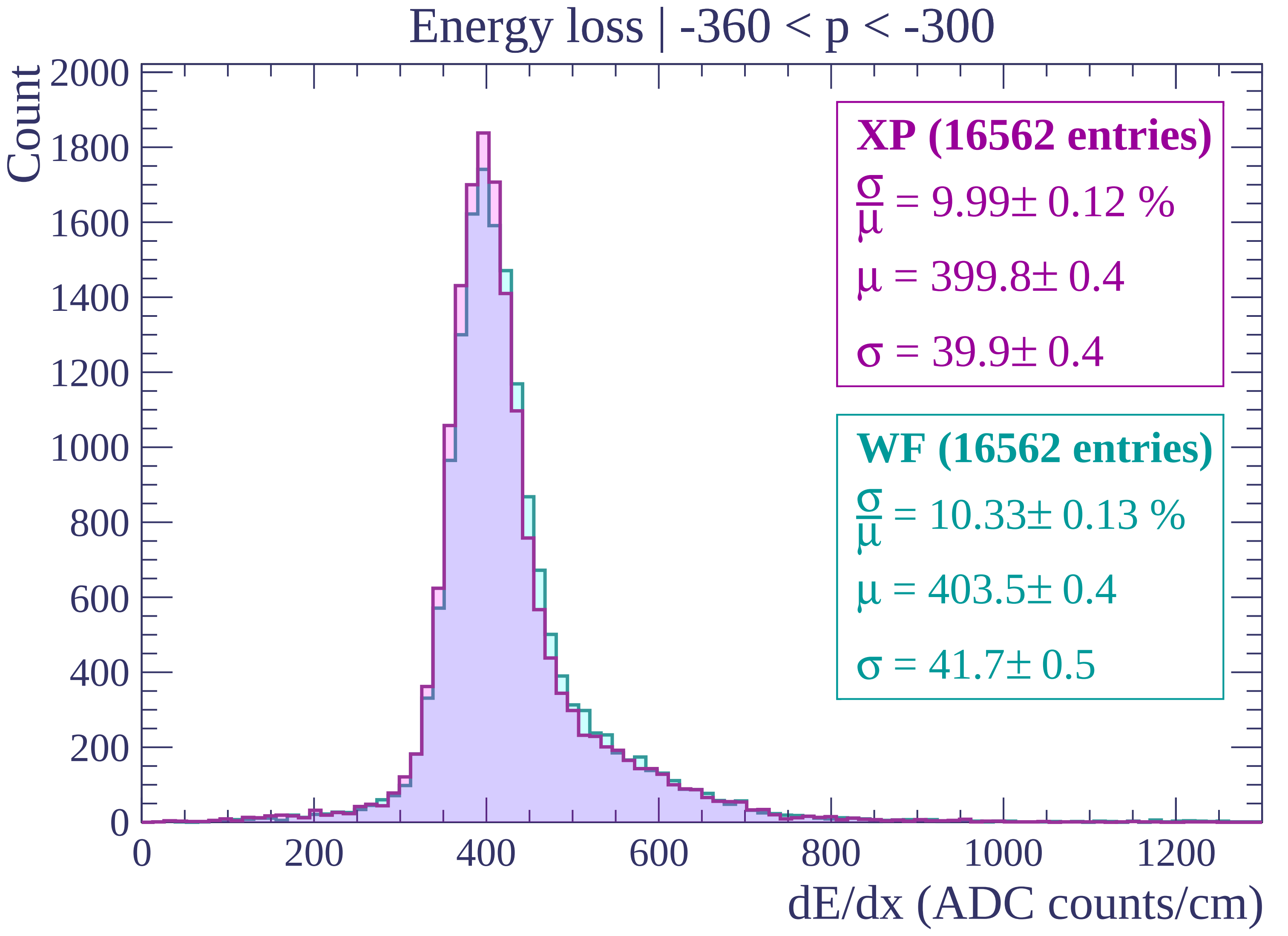

Energy loss | -600 < p < -540
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP (13954 entries)
σ/μ = 9.92± 0.12 %
μ = 414.5± 0.5
σ = 41.1± 0.4
WF (13954 entries)
σ/μ = 10.57± 0.13 %
μ = 419.7± 0.5
σ = 44.4± 0.5



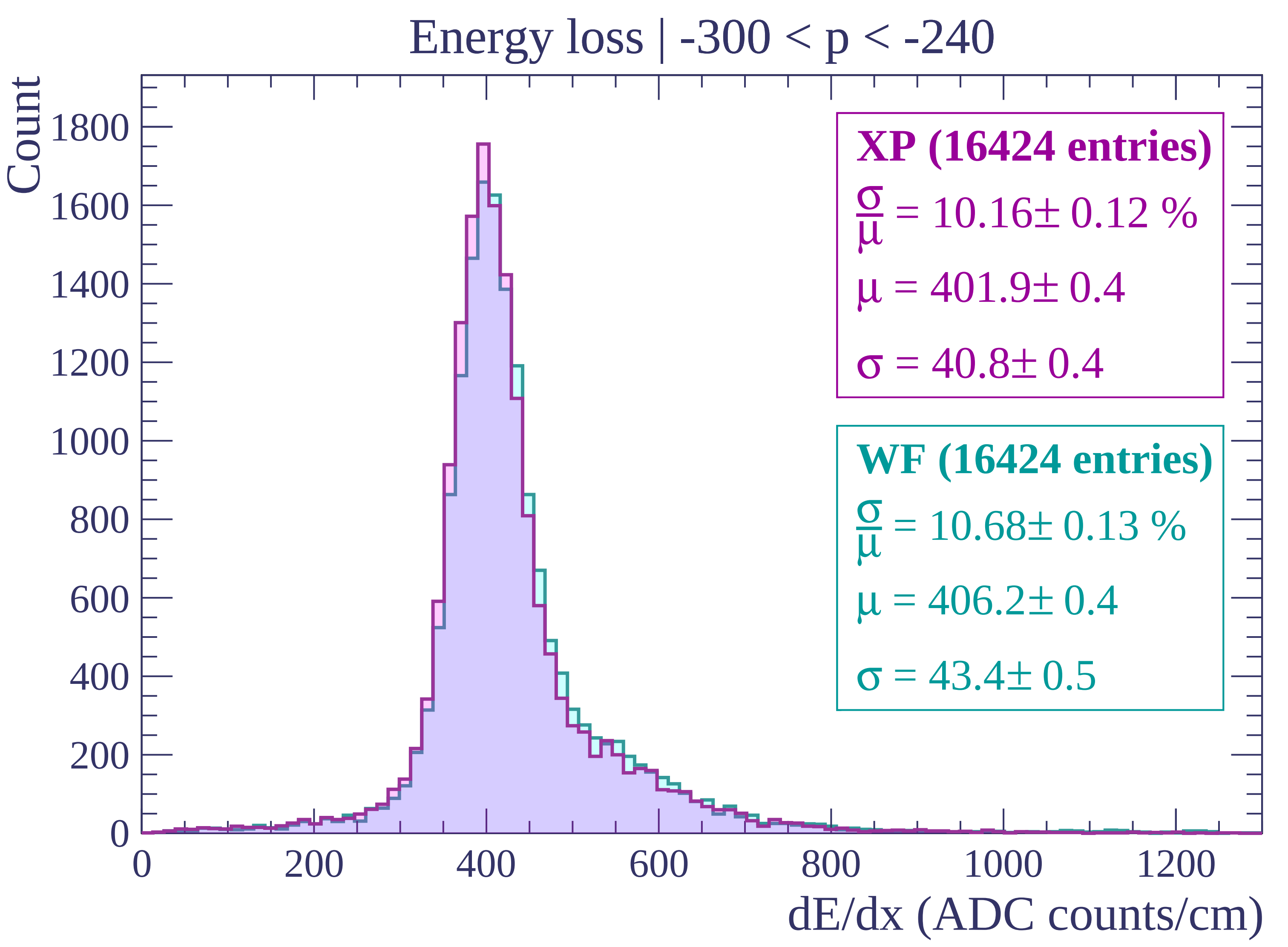
Energy loss | -540 < p < -480
Count
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (14717 entries)
σ/μ = 9.95± 0.11 %
μ = 410.1± 0.4
σ = 40.8± 0.4
WF (14717 entries)
σ/μ = 10.24± 0.12 %
μ = 414.0± 0.5
σ = 42.4± 0.4


Energy loss | -480 < p < -420
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
0
200
400
600
800
1000
1200
XP (15312 entries)
σ/μ = 9.57± 0.11 %
μ = 405.2± 0.4
σ = 38.8± 0.4
WF (15312 entries)
σ/μ = 10.35± 0.12 %
μ = 410.1± 0.5
σ = 42.5± 0.5

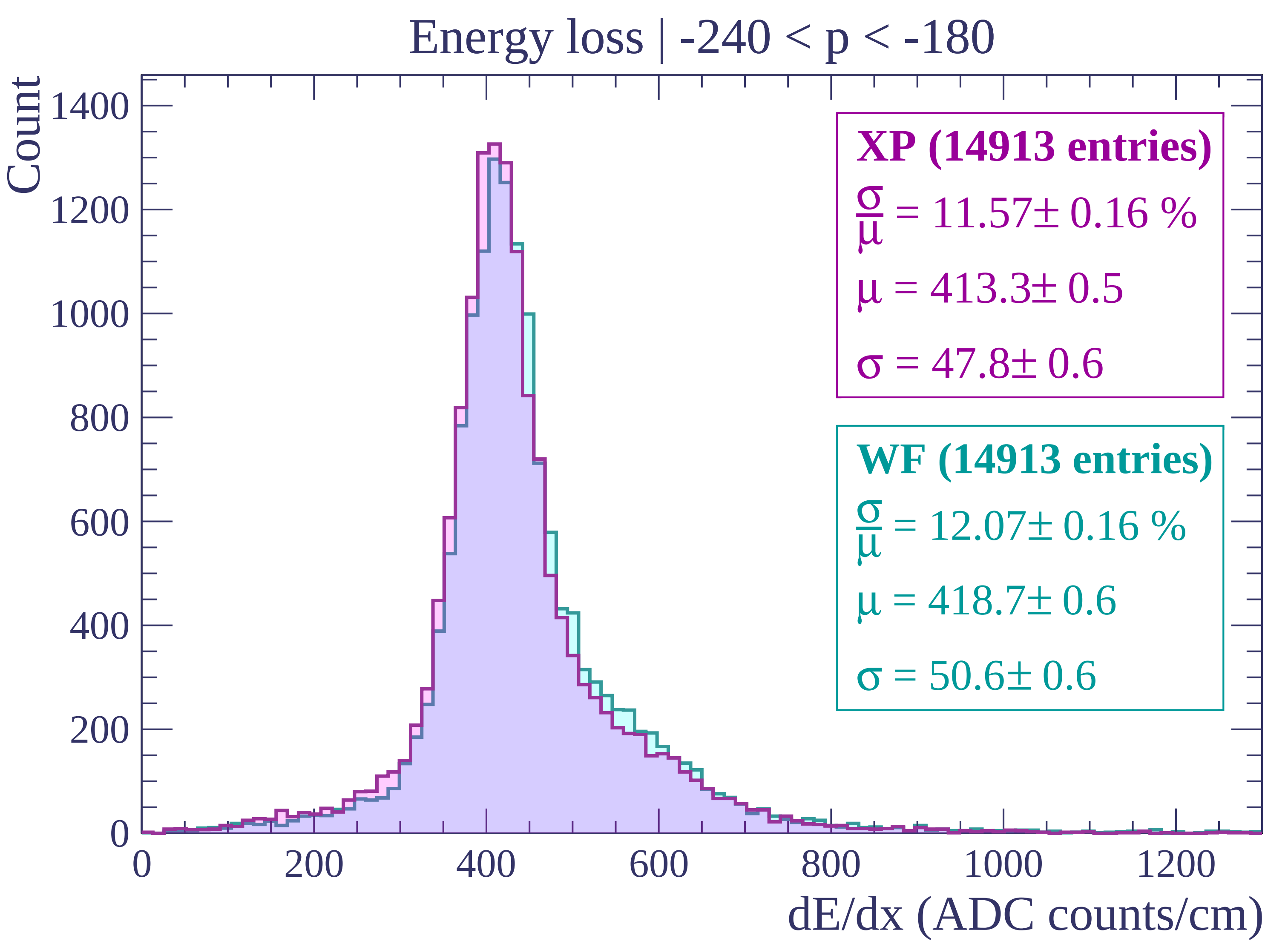

Energy loss | -420 < p < -360
Count
1800
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (15842 entries)
σ/μ = 9.80± 0.11 %
μ = 402.4± 0.4
σ = 39.4± 0.4
WF (15842 entries)
σ/μ = 10.41± 0.12 %
μ = 407.2± 0.4
σ = 42.4± 0.4



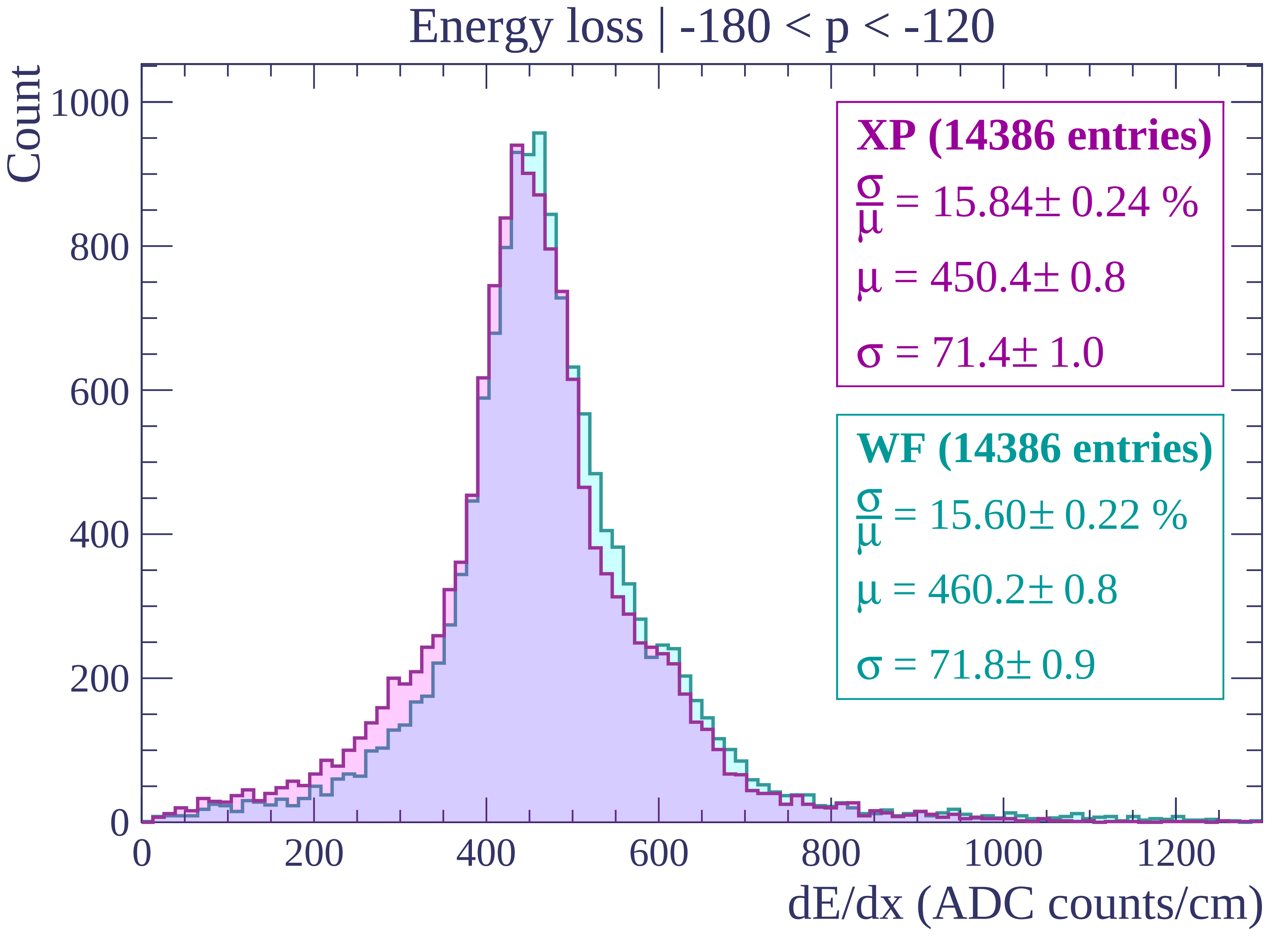
Energy loss | -360 < p < -300
Count
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (16562 entries)
$\frac{\sigma}{\mu}$ = 9.99± 0.12 %
μ = 399.8± 0.4
σ = 39.9± 0.4
WF (16562 entries)
$\frac{\sigma}{\mu}$ = 10.33± 0.13 %
μ = 403.5± 0.4
σ = 41.7± 0.5

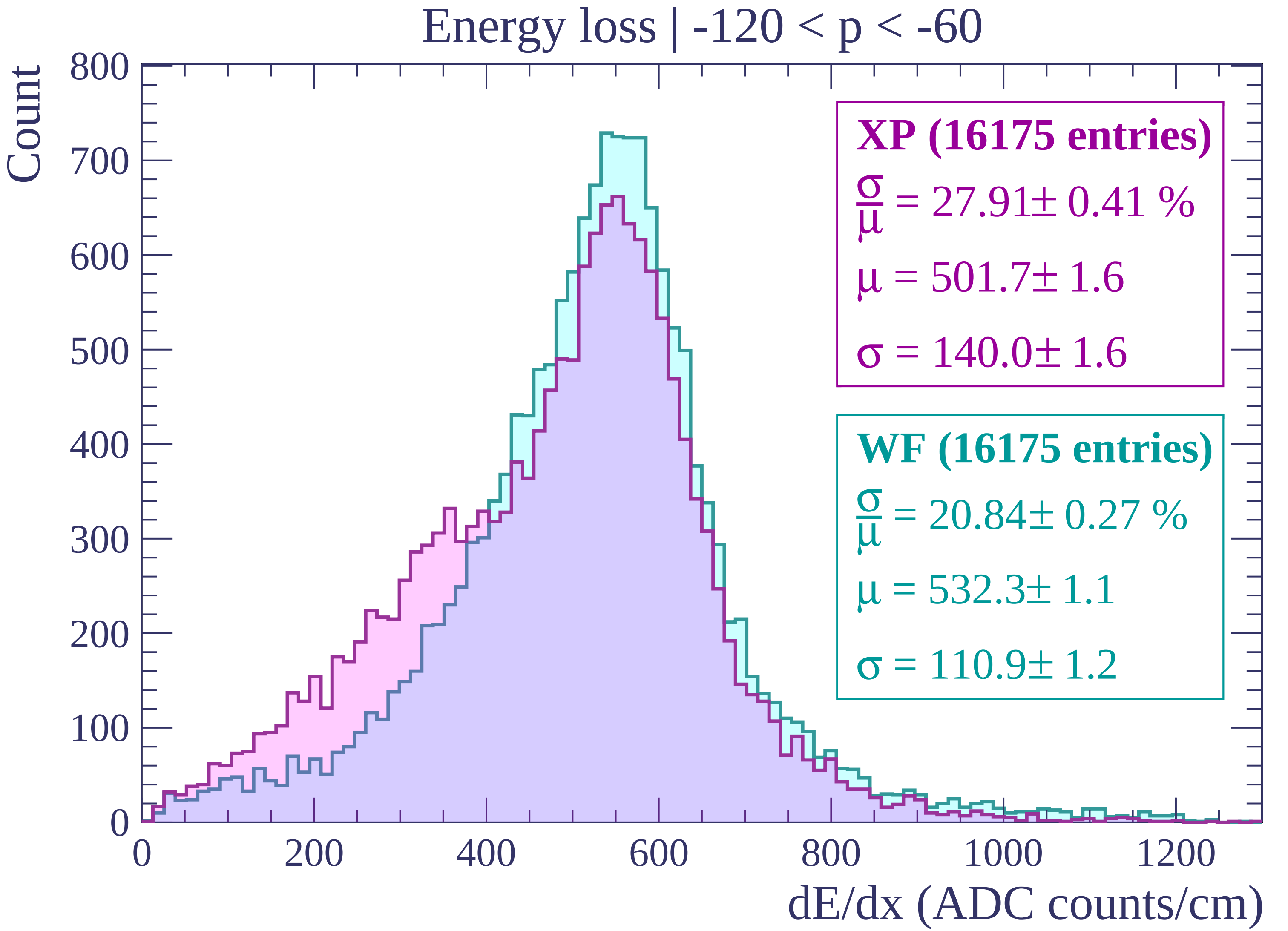


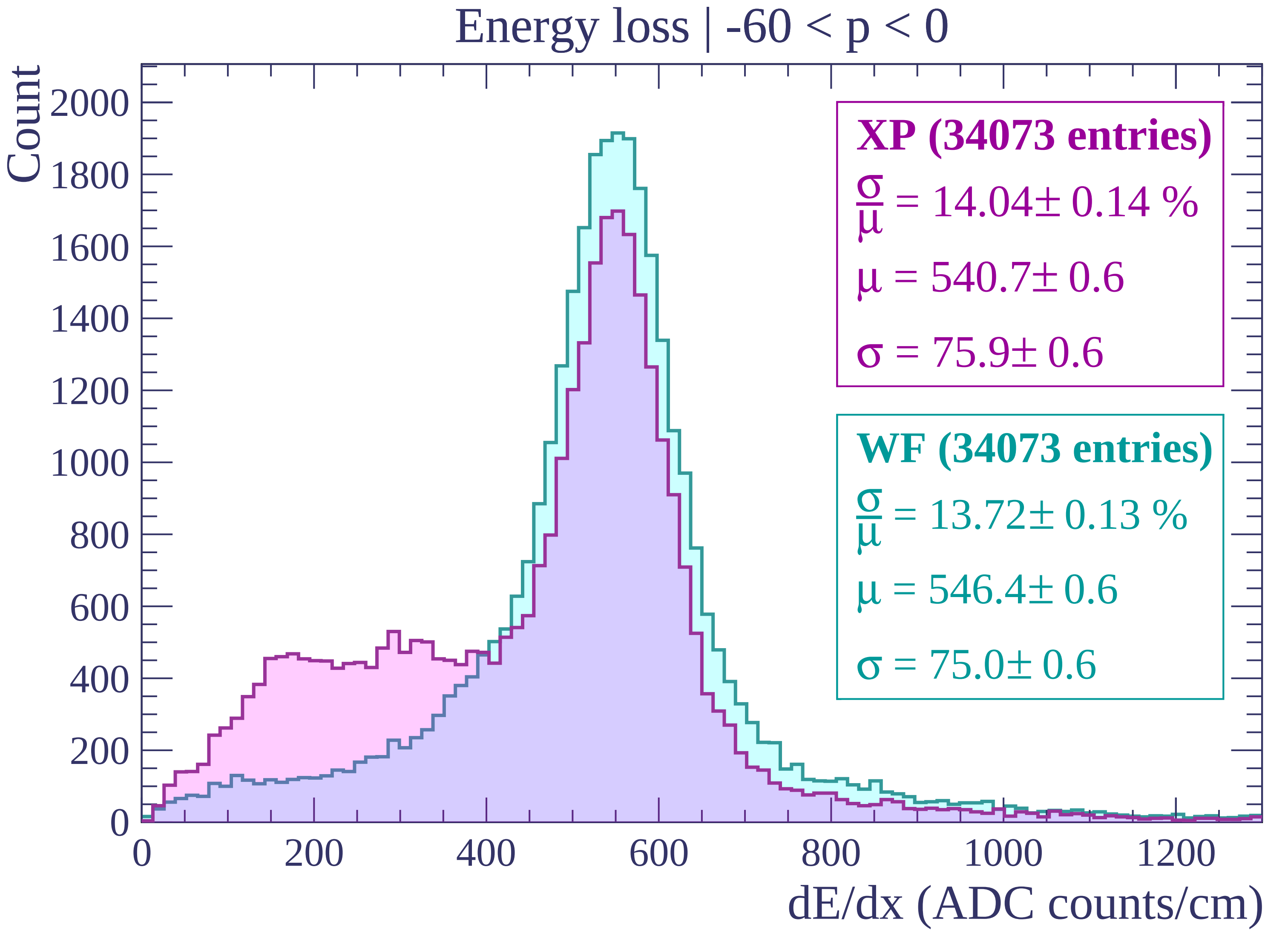
Energy loss | -300 < p < -240
Count
dE/dx (ADC counts/cm)
XP (16424 entries)
σ/μ = 10.16± 0.12 %
μ = 401.9± 0.4
σ = 40.8± 0.4
WF (16424 entries)
σ/μ = 10.68± 0.13 %
μ = 406.2± 0.4
σ = 43.4± 0.5



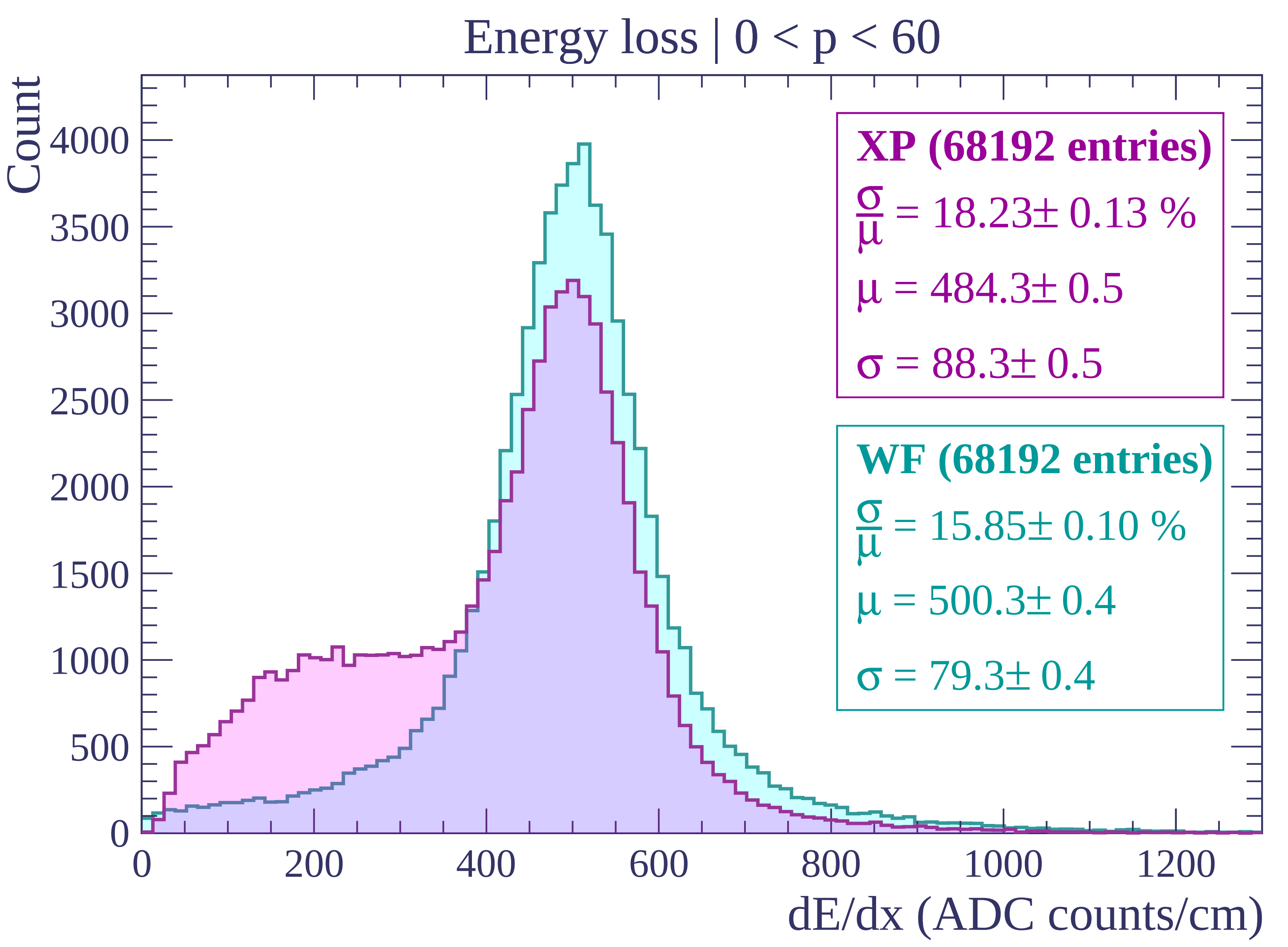
Energy loss | -240 < p < -180
Count
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (14913 entries)
σ/μ = 11.57± 0.16 %
μ = 413.3± 0.5
σ = 47.8± 0.6
WF (14913 entries)
σ/μ = 12.07± 0.16 %
μ = 418.7± 0.6
σ = 50.6± 0.6


Energy loss | -180 < p < -120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (14386 entries)
$\frac{\sigma}{\mu}$ = 15.84± 0.24 %
μ = 450.4± 0.8
σ = 71.4± 1.0
WF (14386 entries)
$\frac{\sigma}{\mu}$ = 15.60± 0.22 %
μ = 460.2± 0.8
σ = 71.8± 0.9

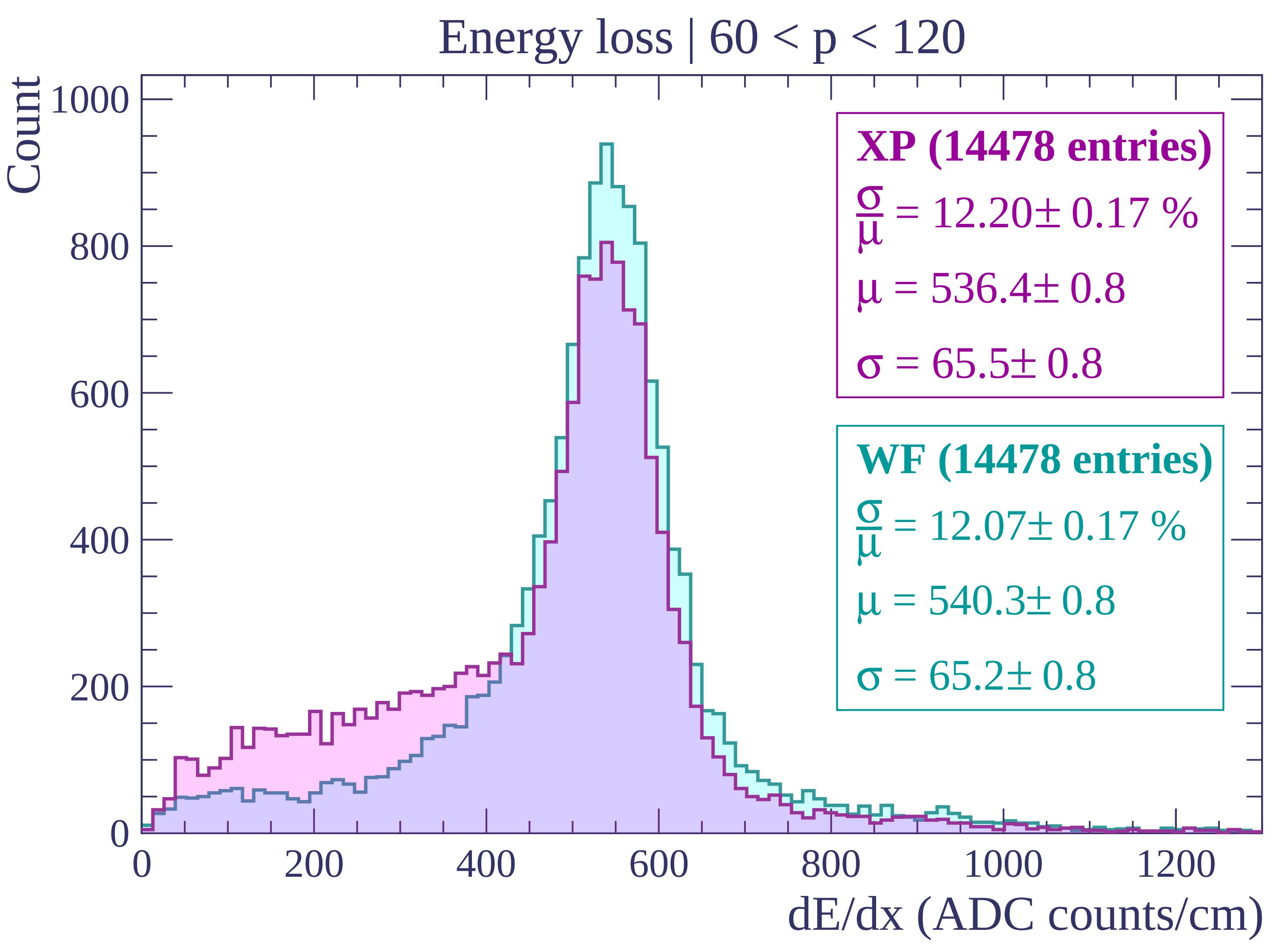

Energy loss | -120 < p < -60
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (16175 entries)
σ/μ = 27.91± 0.41 %
μ = 501.7± 1.6
σ = 140.0± 1.6
WF (16175 entries)
σ/μ = 20.84± 0.27 %
μ = 532.3± 1.1
σ = 110.9± 1.2



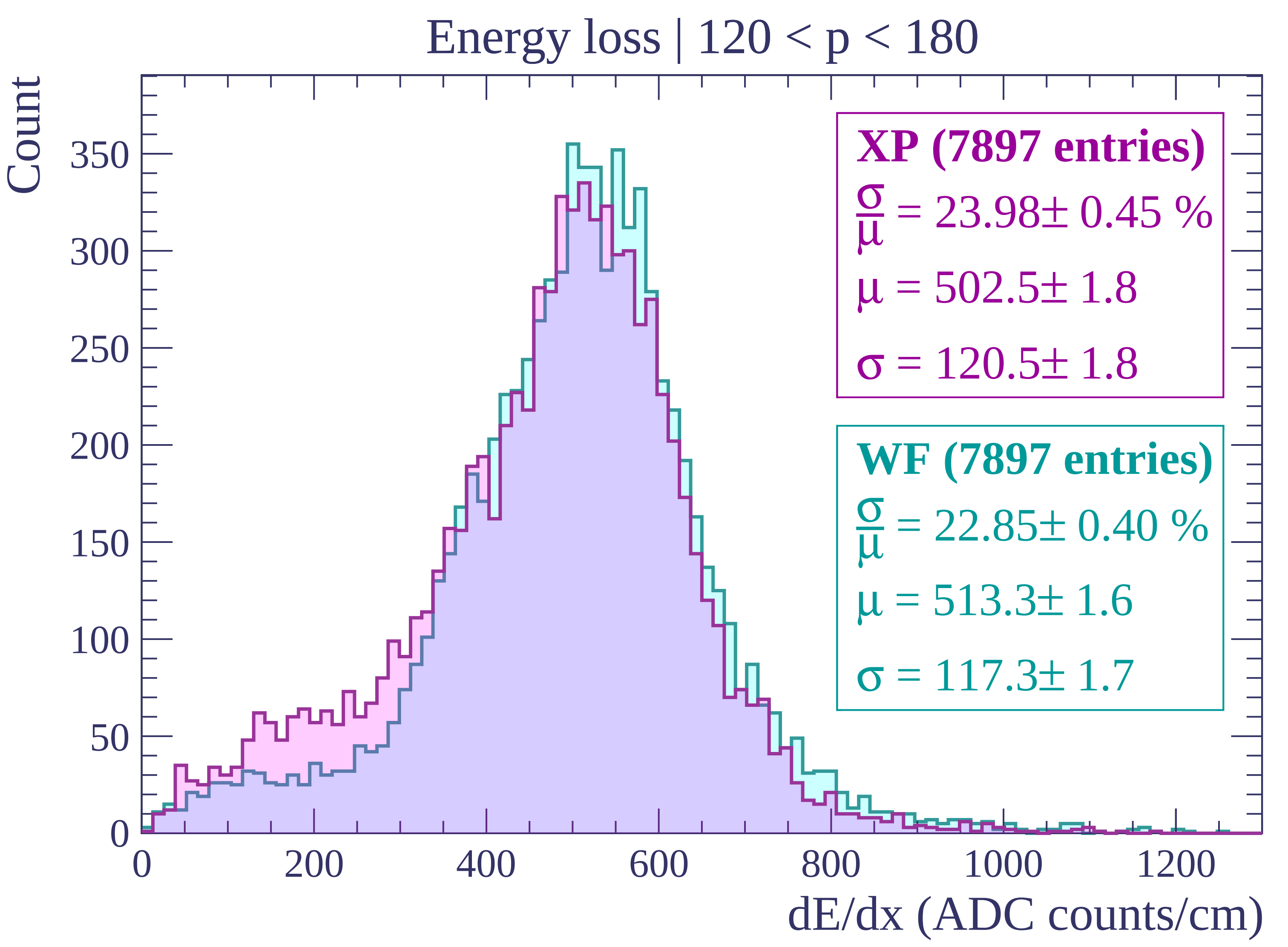
Energy loss | -60 < p < 0
Count
dE/dx (ADC counts/cm)
XP (34073 entries)
σ/μ = 14.04± 0.14 %
μ = 540.7± 0.6
σ = 75.9± 0.6
WF (34073 entries)
σ/μ = 13.72± 0.13 %
μ = 546.4± 0.6
σ = 75.0± 0.6
0
200
400
600
800
1000
1200
1400
1600
1800
2000



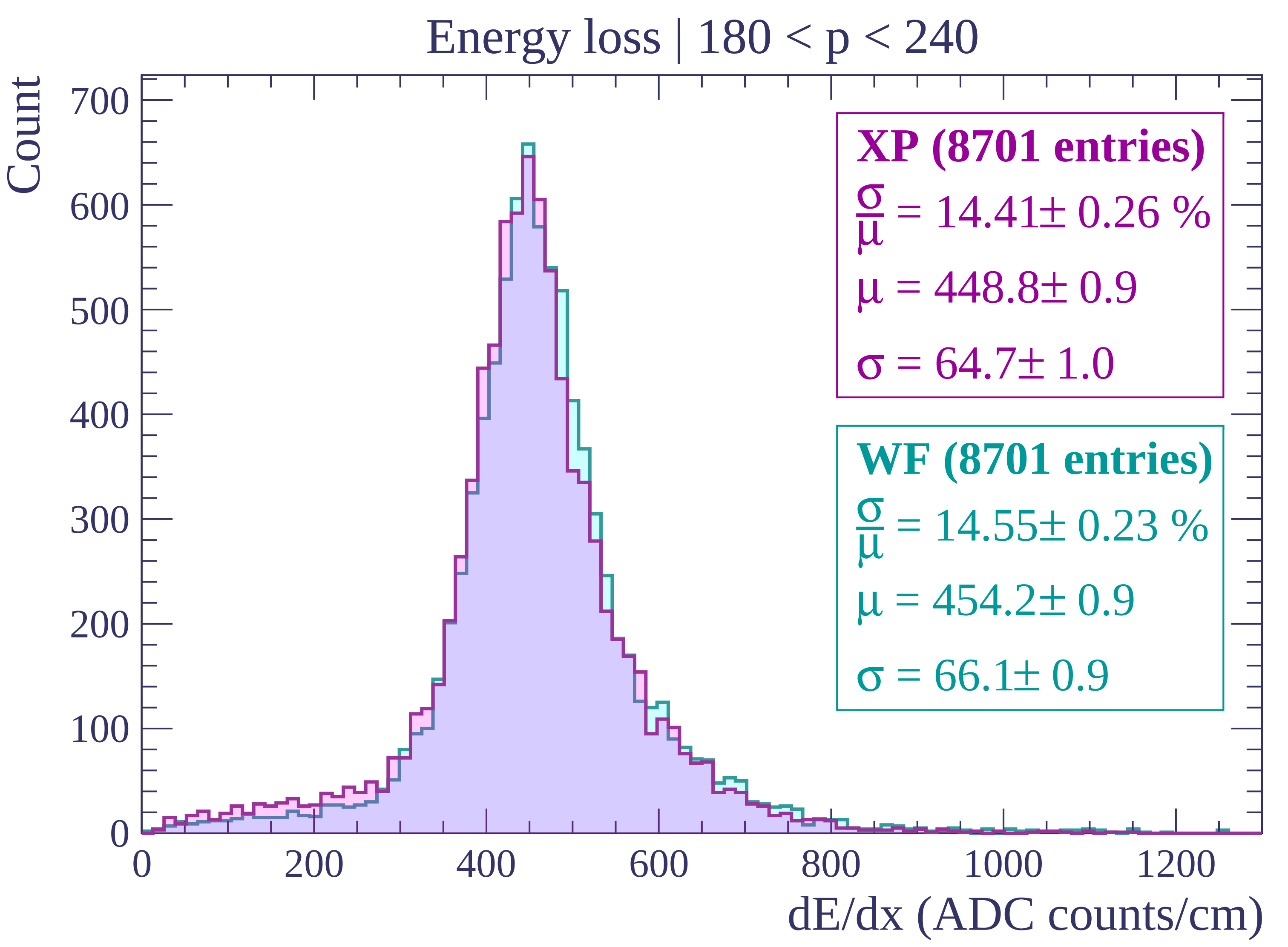
Energy loss | 0 < p < 60
Count
dE/dx (ADC counts/cm)
XP (68192 entries)
σ/μ = 18.23± 0.13 %
μ = 484.3± 0.5
σ = 88.3± 0.5
WF (68192 entries)
σ/μ = 15.85± 0.10 %
μ = 500.3± 0.4
σ = 79.3± 0.4
0
200
400
600
800
1000
1200
500
1000
1500
2000
2500
3000
3500
4000

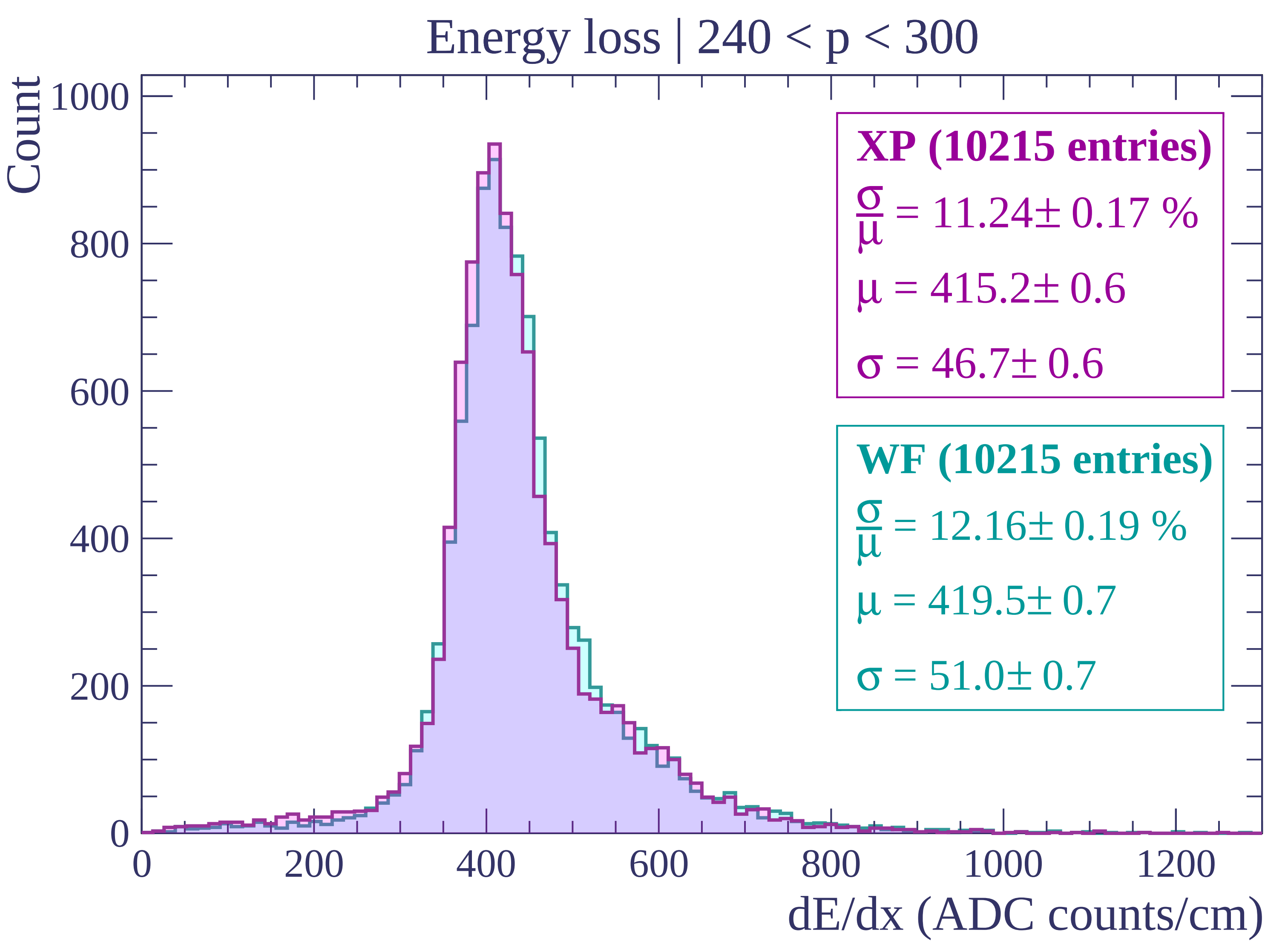


Energy loss | 60 < p < 120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (14478 entries)
$\frac{\sigma}{\mu}$ = 12.20± 0.17 %
μ = 536.4± 0.8
σ = 65.5± 0.8
WF (14478 entries)
$\frac{\sigma}{\mu}$ = 12.07± 0.17 %
μ = 540.3± 0.8
σ = 65.2± 0.8

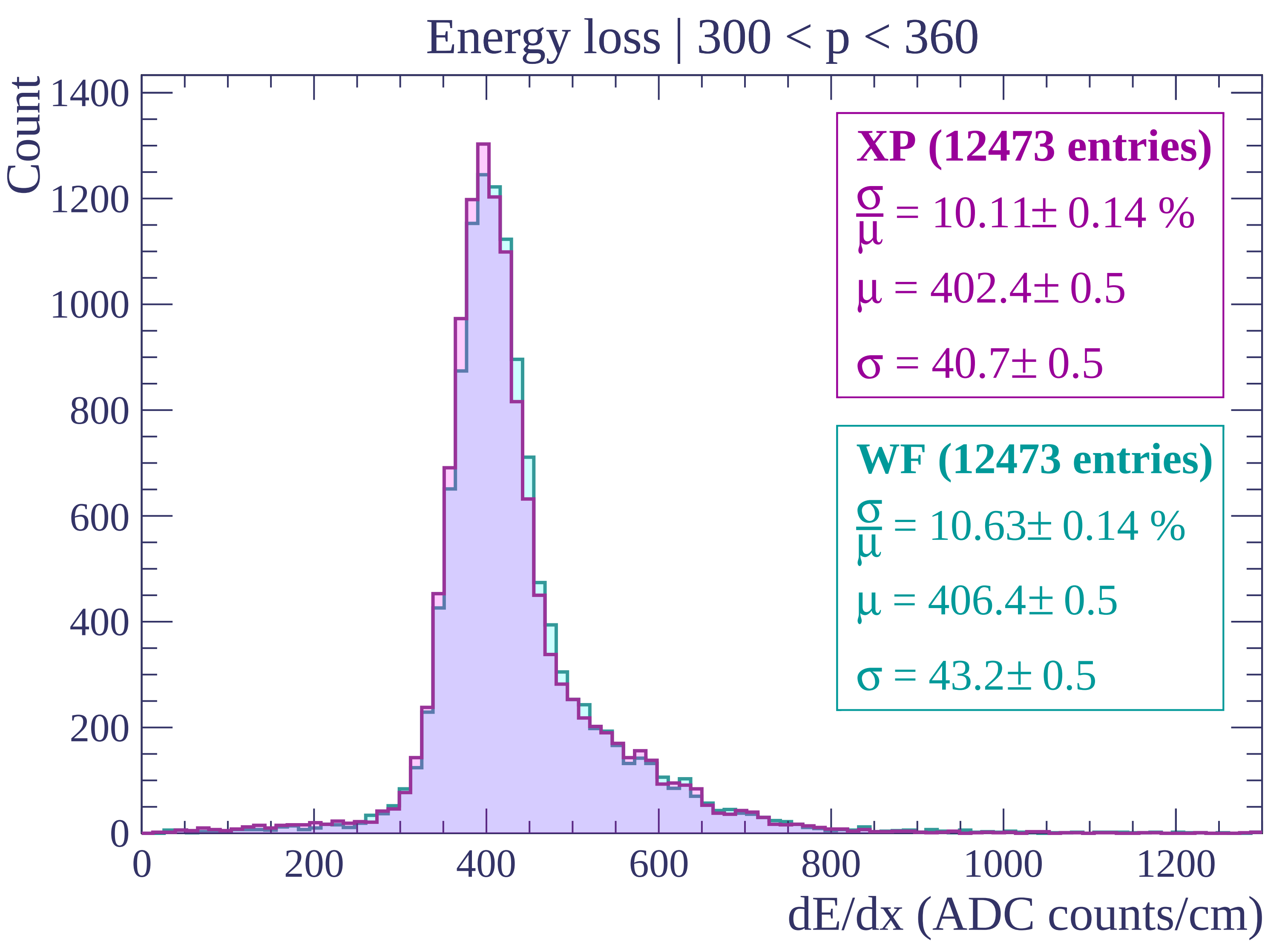


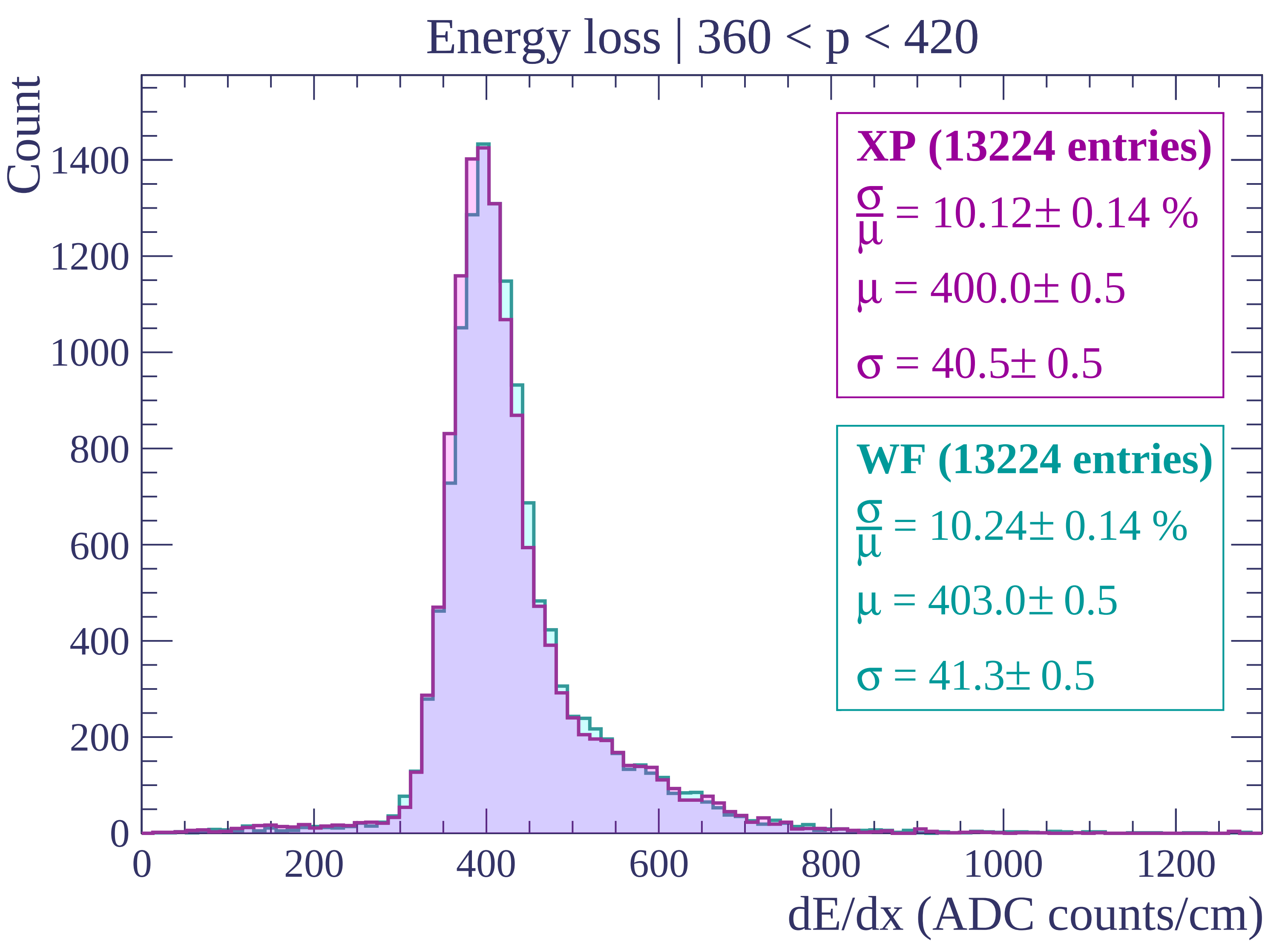
Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (7897 entries)
σ/μ = 23.98± 0.45 %
μ = 502.5± 1.8
σ = 120.5± 1.8
WF (7897 entries)
σ/μ = 22.85± 0.40 %
μ = 513.3± 1.6
σ = 117.3± 1.7


Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
XP (8701 entries)
σ/μ = 14.41± 0.26 %
μ = 448.8± 0.9
σ = 64.7± 1.0
WF (8701 entries)
σ/μ = 14.55± 0.23 %
μ = 454.2± 0.9
σ = 66.1± 0.9
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

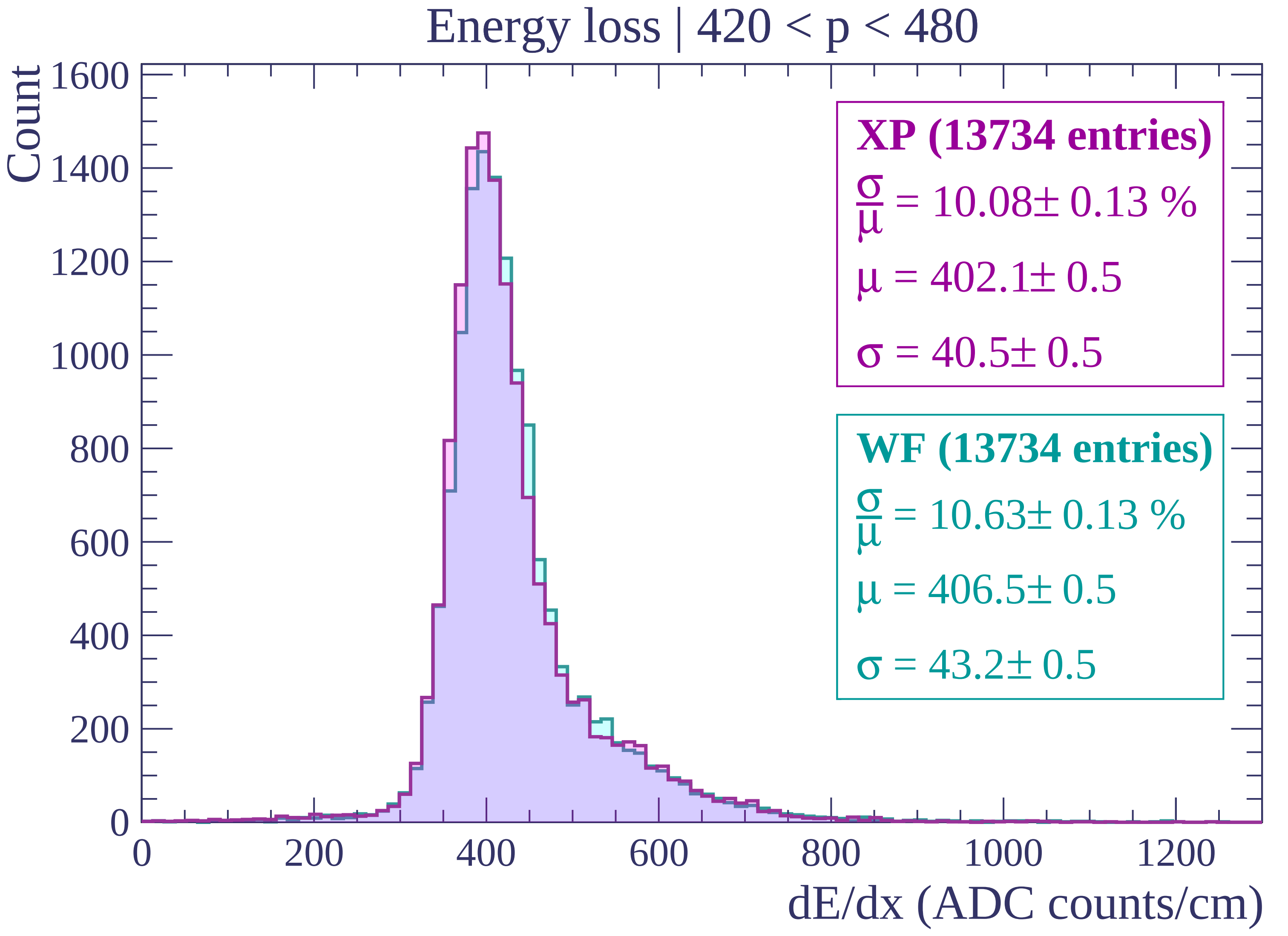


Energy loss | 240 < p < 300
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (10215 entries)
σ/μ = 11.24± 0.17 %
μ = 415.2± 0.6
σ = 46.7± 0.6
WF (10215 entries)
σ/μ = 12.16± 0.19 %
μ = 419.5± 0.7
σ = 51.0± 0.7

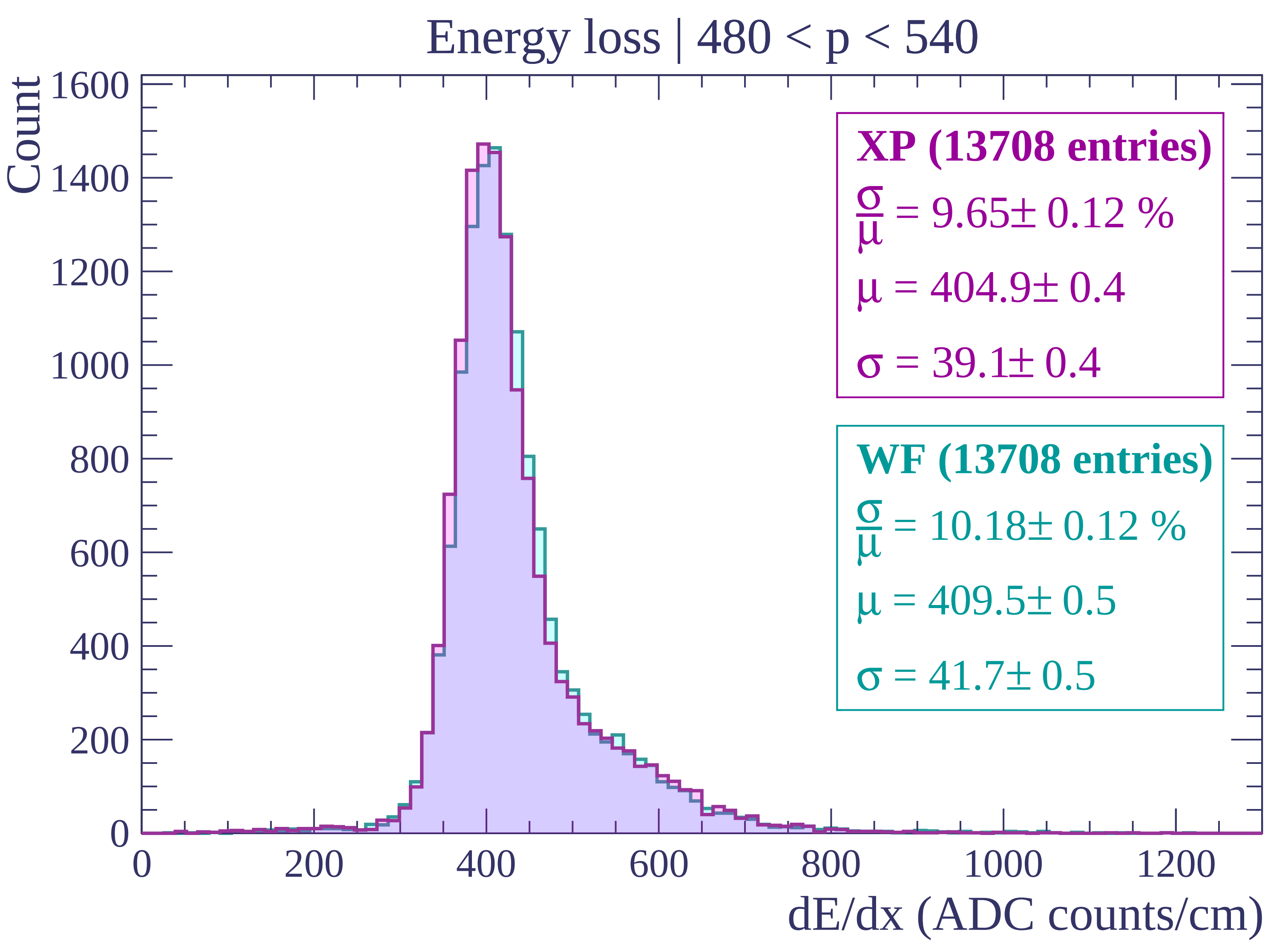

Energy loss | 300 < p < 360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP (12473 entries)
$\frac{\sigma}{\mu}$ = 10.11± 0.14 %
μ = 402.4± 0.5
σ = 40.7± 0.5
WF (12473 entries)
$\frac{\sigma}{\mu}$ = 10.63± 0.14 %
μ = 406.4± 0.5
σ = 43.2± 0.5

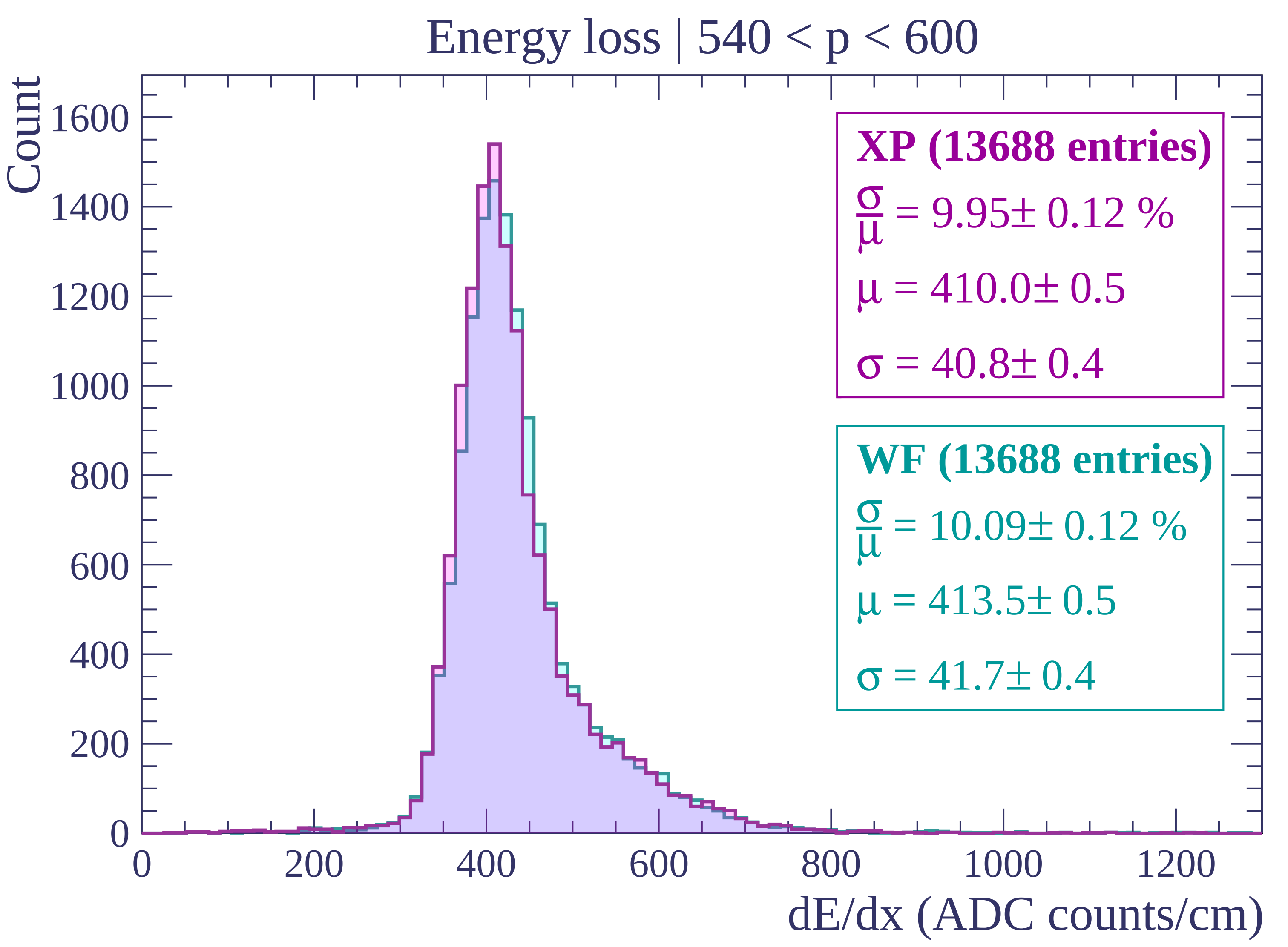


Energy loss | 360 < p < 420
Count
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (13224 entries)
$\frac{\sigma}{\mu}$ = 10.12± 0.14 %
μ = 400.0± 0.5
σ = 40.5± 0.5
WF (13224 entries)
$\frac{\sigma}{\mu}$ = 10.24± 0.14 %
μ = 403.0± 0.5
σ = 41.3± 0.5

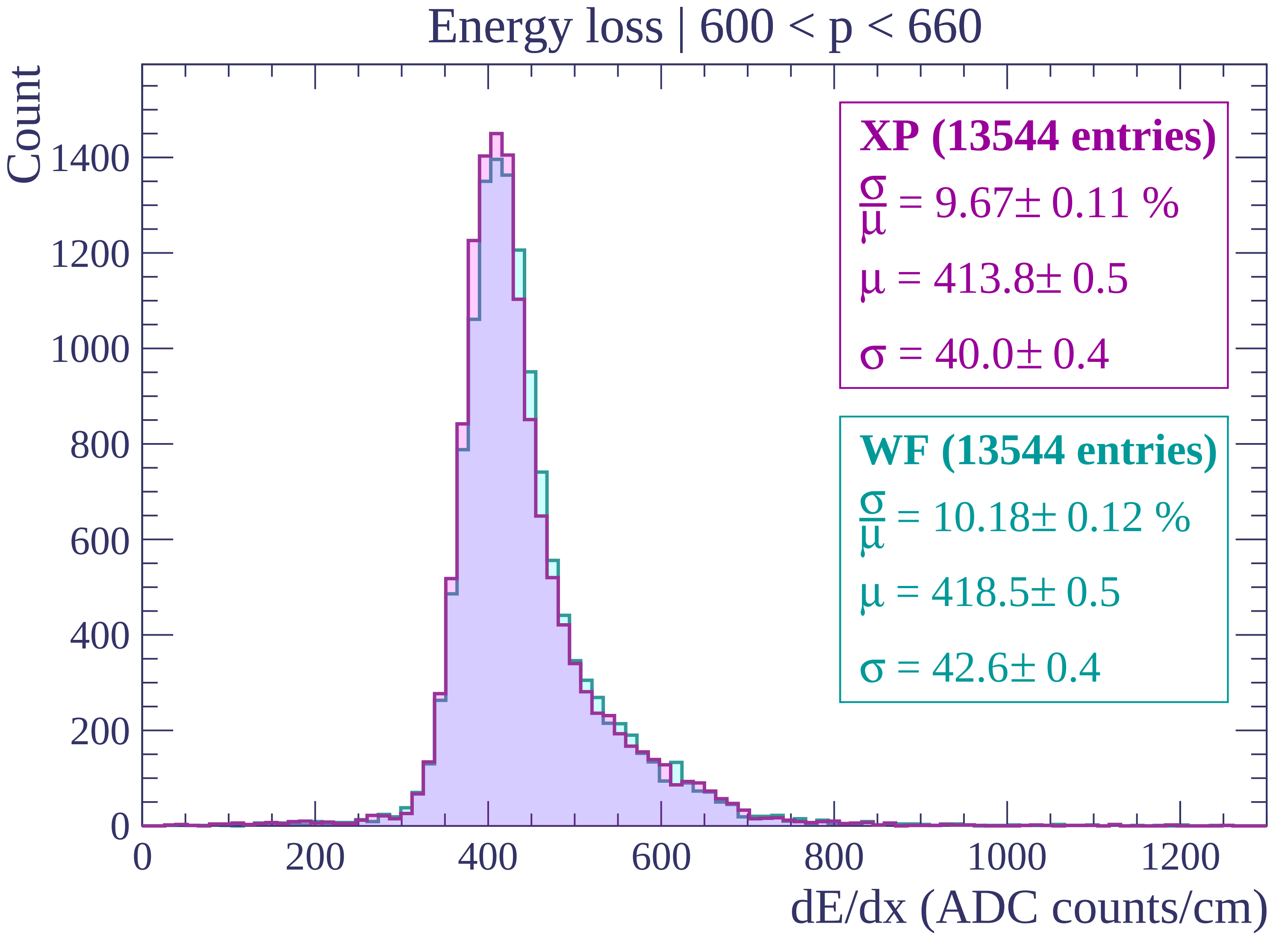


Energy loss | 420 < p < 480
Count
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (13734 entries)
σ/μ = 10.08± 0.13 %
μ = 402.1± 0.5
σ = 40.5± 0.5
WF (13734 entries)
σ/μ = 10.63± 0.13 %
μ = 406.5± 0.5
σ = 43.2± 0.5

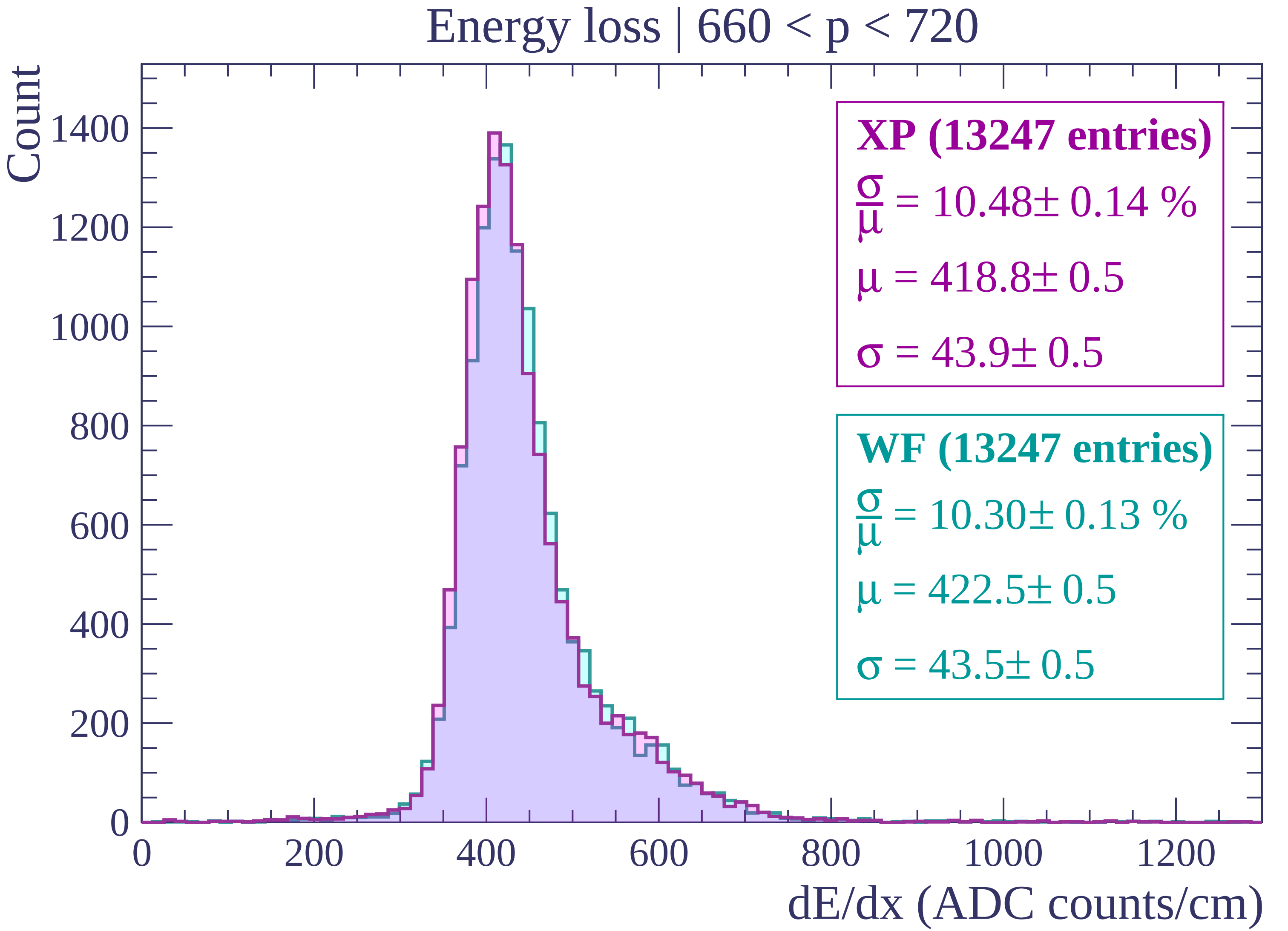

Energy loss | 480 < p < 540
Count
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (13708 entries)
σ/μ = 9.65± 0.12 %
μ = 404.9± 0.4
σ = 39.1± 0.4
WF (13708 entries)
σ/μ = 10.18± 0.12 %
μ = 409.5± 0.5
σ = 41.7± 0.5



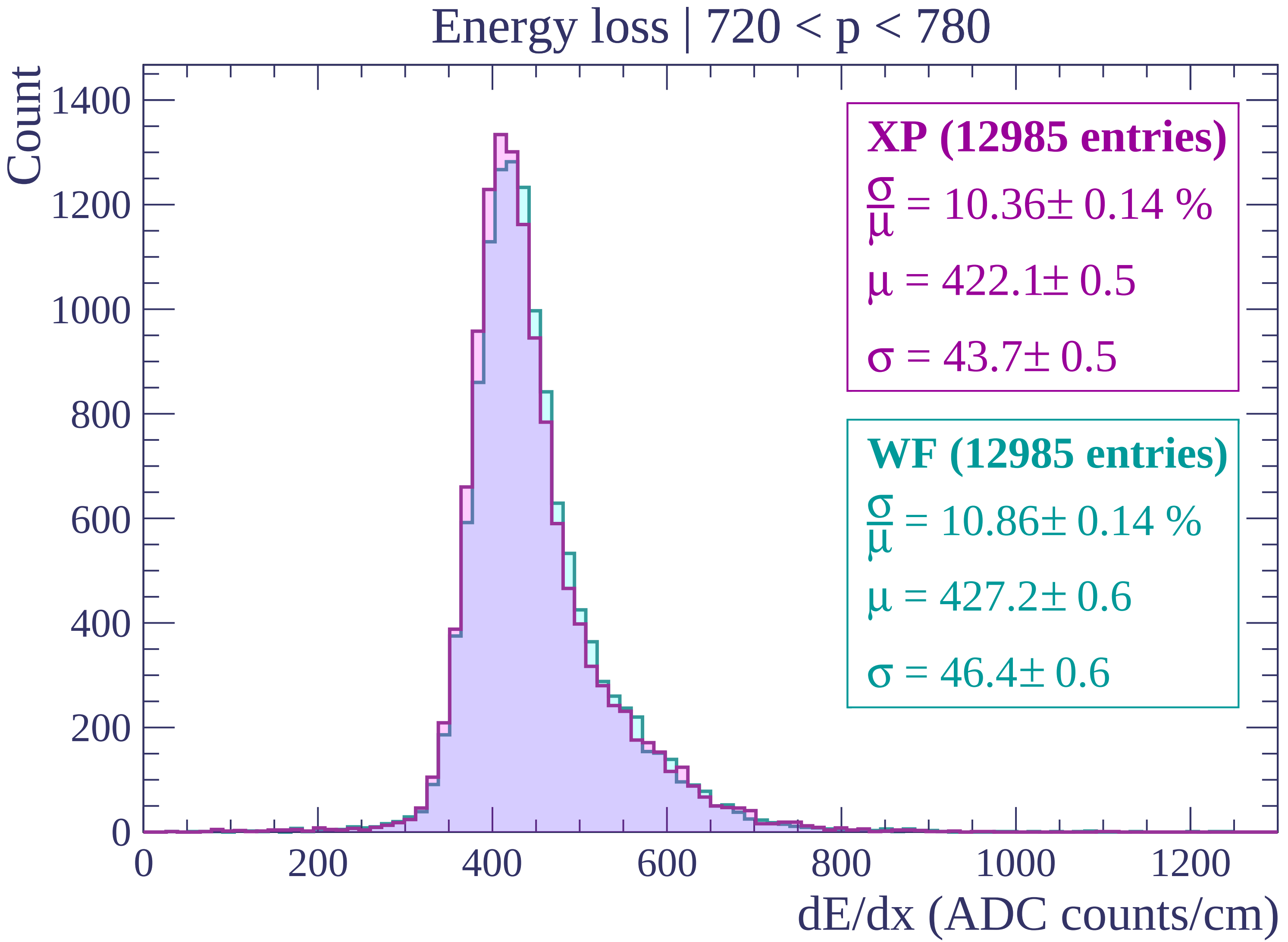
Energy loss | 540 < p < 600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
XP (13688 entries)
σ/μ = 9.95± 0.12 %
μ = 410.0± 0.5
σ = 40.8± 0.4
WF (13688 entries)
σ/μ = 10.09± 0.12 %
μ = 413.5± 0.5
σ = 41.7± 0.4

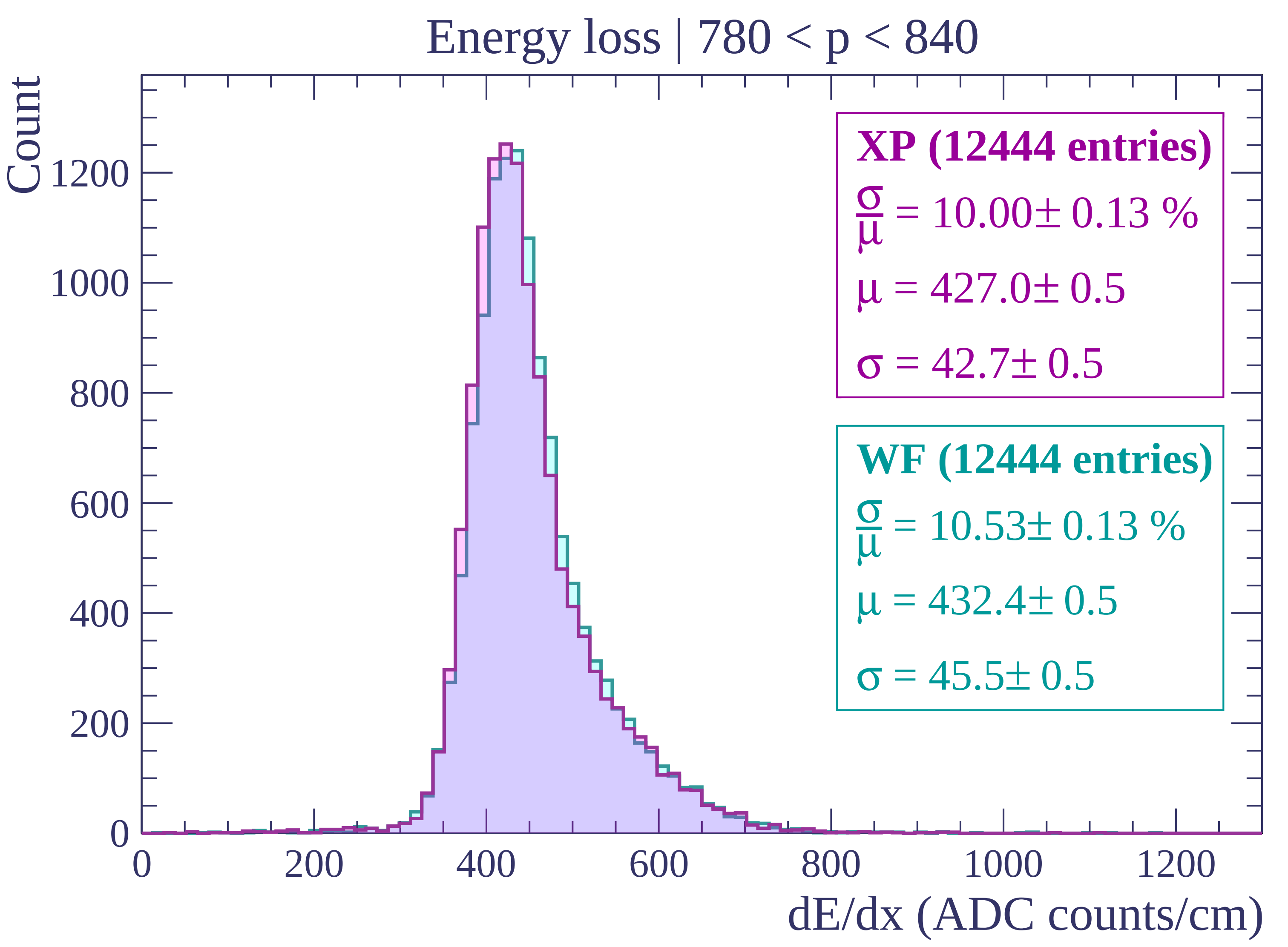


Energy loss | 600 < p < 660
Count
dE/dx (ADC counts/cm)
XP (13544 entries)
$\frac{\sigma}{\mu}$ = 9.67± 0.11 %
μ = 413.8± 0.5
σ = 40.0± 0.4
WF (13544 entries)
$\frac{\sigma}{\mu}$ = 10.18± 0.12 %
μ = 418.5± 0.5
σ = 42.6± 0.4

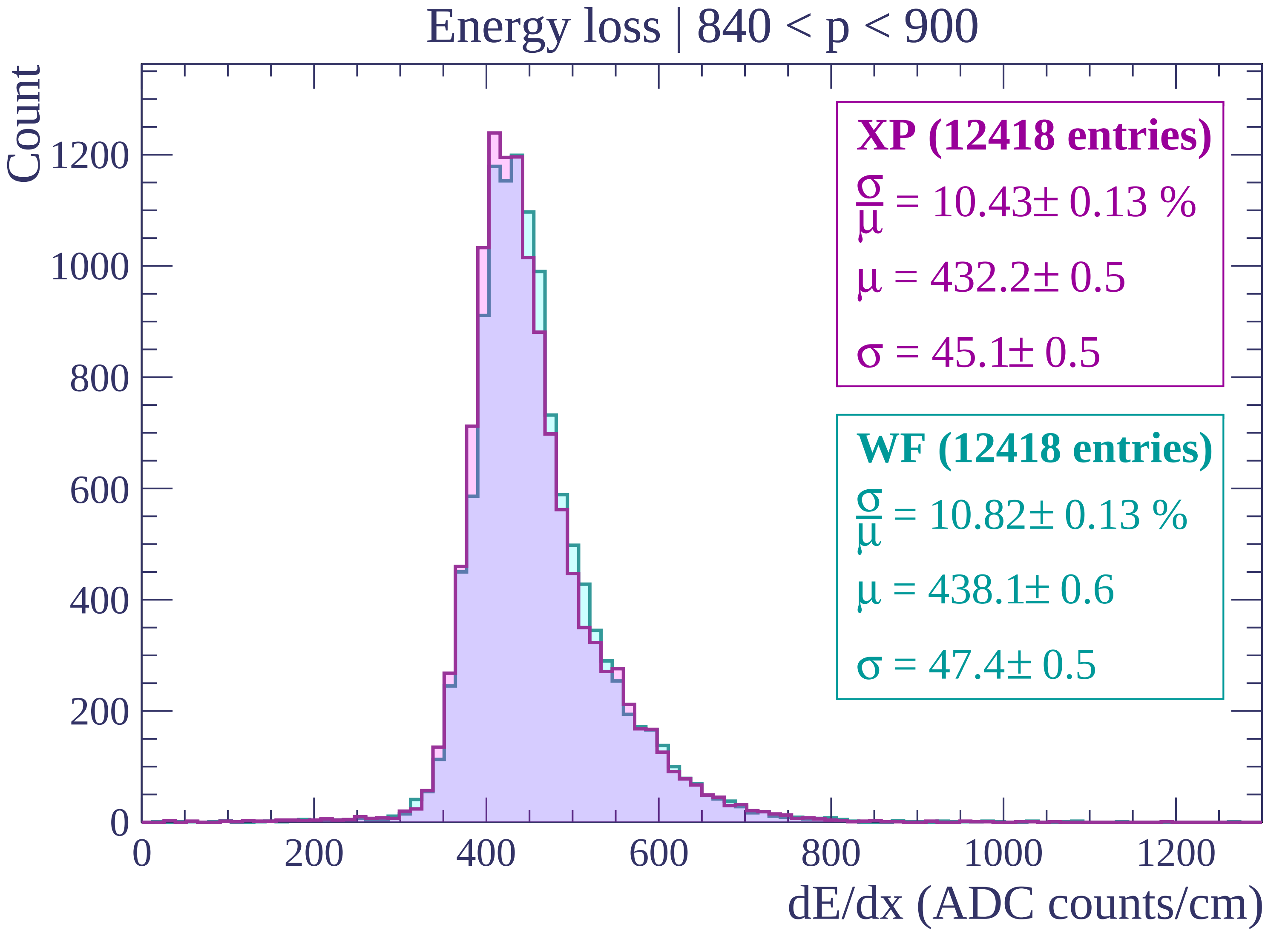


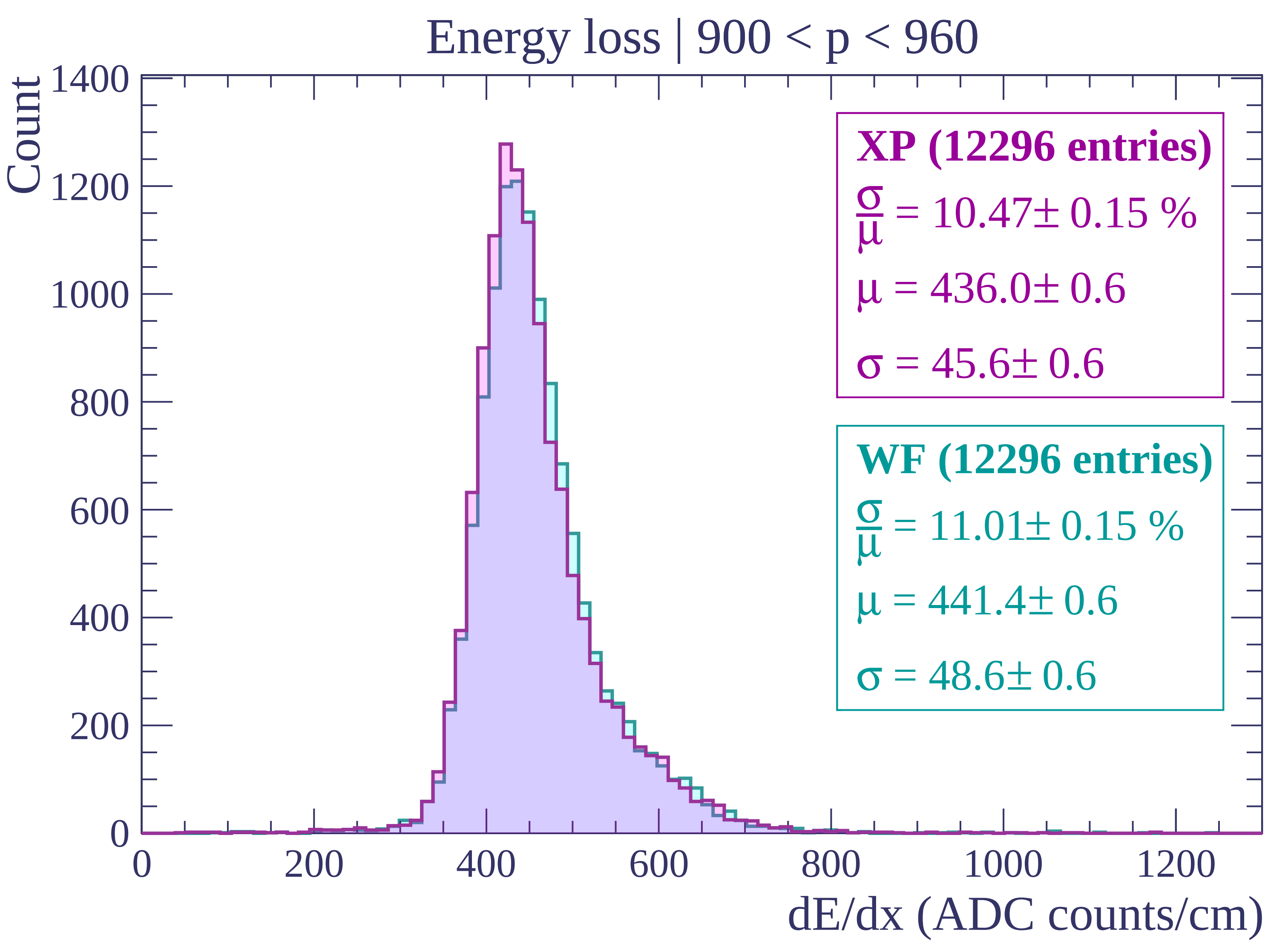
Energy loss | 660 < p < 720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP (13247 entries)
$\frac{\sigma}{\mu}$ = 10.48± 0.14 %
μ = 418.8± 0.5
σ = 43.9± 0.5
WF (13247 entries)
$\frac{\sigma}{\mu}$ = 10.30± 0.13 %
μ = 422.5± 0.5
σ = 43.5± 0.5


Energy loss | 720 < p < 780
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP (12985 entries)
σ/μ = 10.36± 0.14 %
μ = 422.1± 0.5
σ = 43.7± 0.5
WF (12985 entries)
σ/μ = 10.86± 0.14 %
μ = 427.2± 0.6
σ = 46.4± 0.6

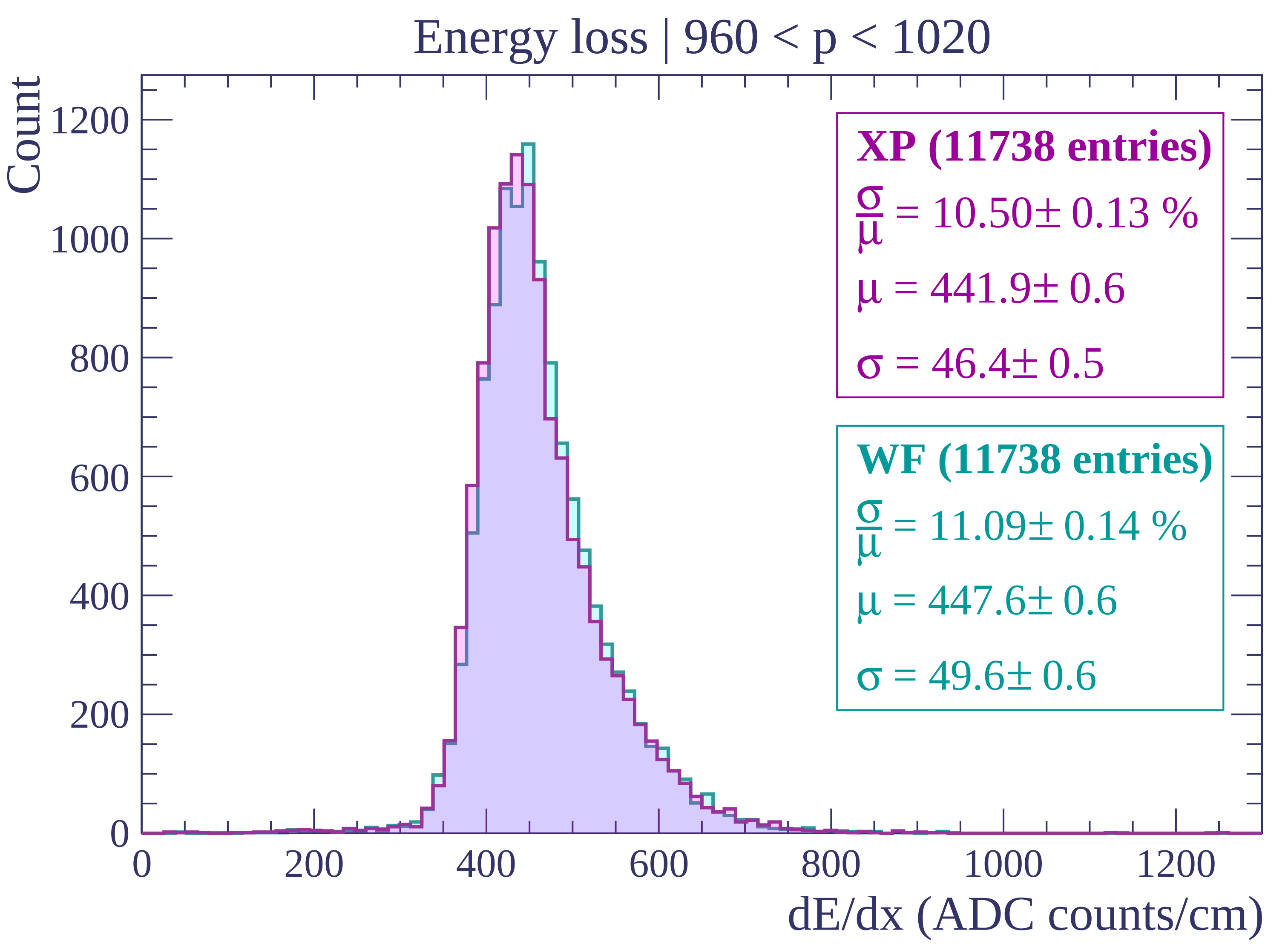


Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (12444 entries)
$\frac{\sigma}{\mu}$ = 10.00± 0.13 %
μ = 427.0± 0.5
σ = 42.7± 0.5
WF (12444 entries)
$\frac{\sigma}{\mu}$ = 10.53± 0.13 %
μ = 432.4± 0.5
σ = 45.5± 0.5

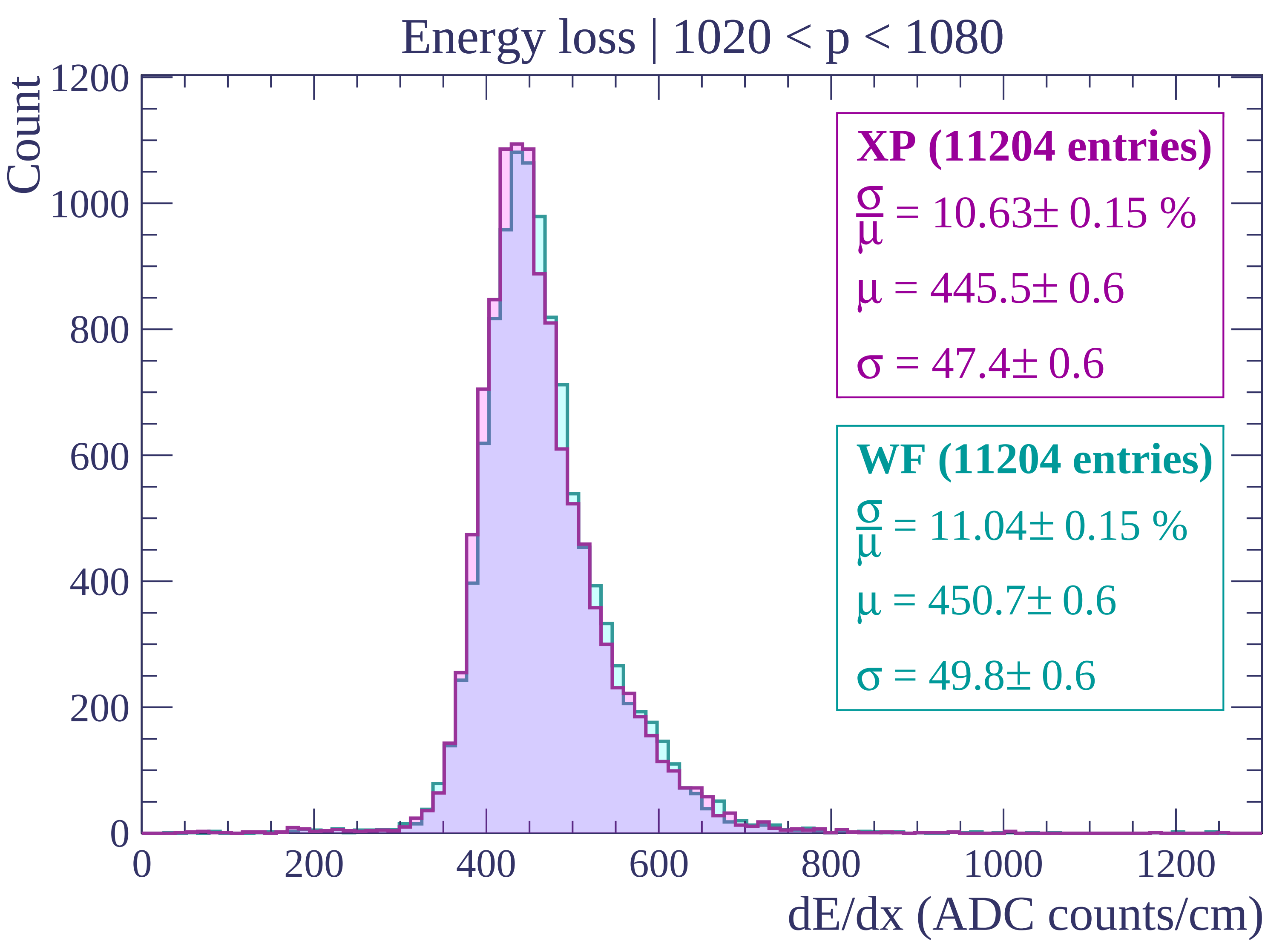

Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
XP (12418 entries)
σ/μ = 10.43± 0.13 %
μ = 432.2± 0.5
σ = 45.1± 0.5
WF (12418 entries)
σ/μ = 10.82± 0.13 %
μ = 438.1± 0.6
σ = 47.4± 0.5
0
200
400
600
800
1000
1200



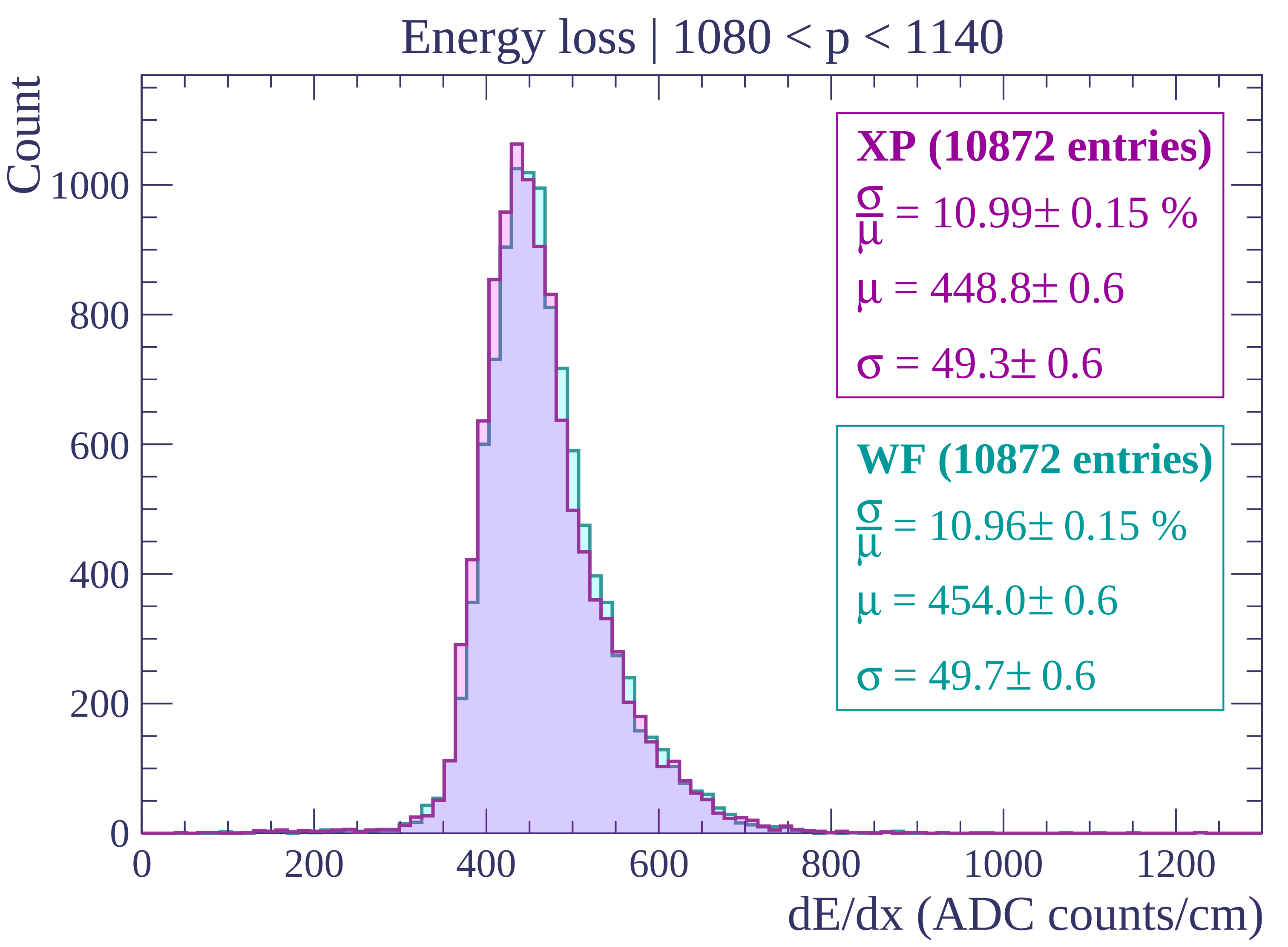
Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
XP (12296 entries)
$\frac{\sigma}{\mu}$ = 10.47± 0.15 %
μ = 436.0± 0.6
σ = 45.6± 0.6
WF (12296 entries)
$\frac{\sigma}{\mu}$ = 11.01± 0.15 %
μ = 441.4± 0.6
σ = 48.6± 0.6
0
200
400
600
800
1000
1200
1400



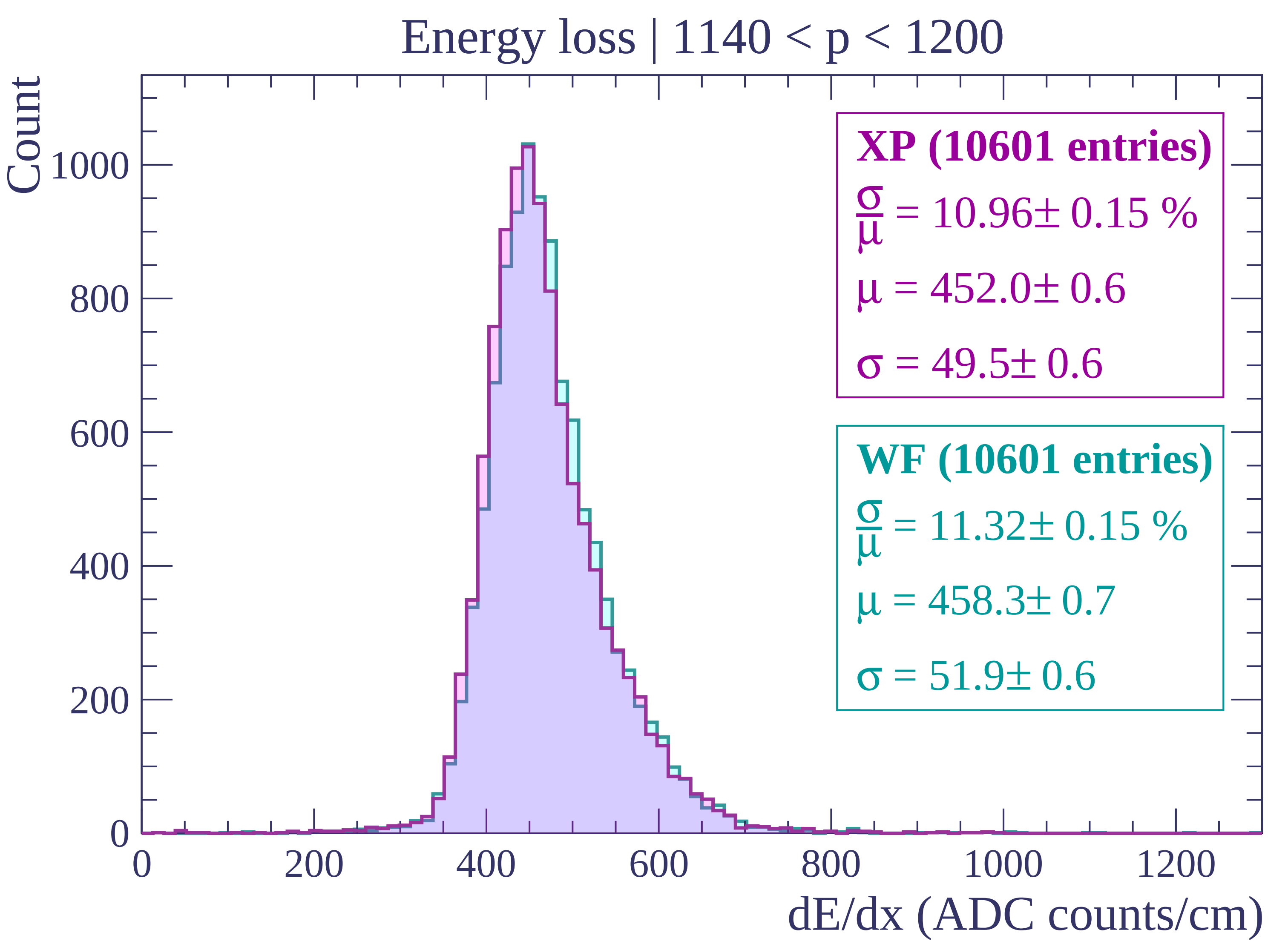
Energy loss | 960 < p < 1020
Count
dE/dx (ADC counts/cm)
XP (11738 entries)
σ/μ = 10.50± 0.13 %
μ = 441.9± 0.6
σ = 46.4± 0.5
WF (11738 entries)
σ/μ = 11.09± 0.14 %
μ = 447.6± 0.6
σ = 49.6± 0.6
0
200
400
600
800
1000
1200

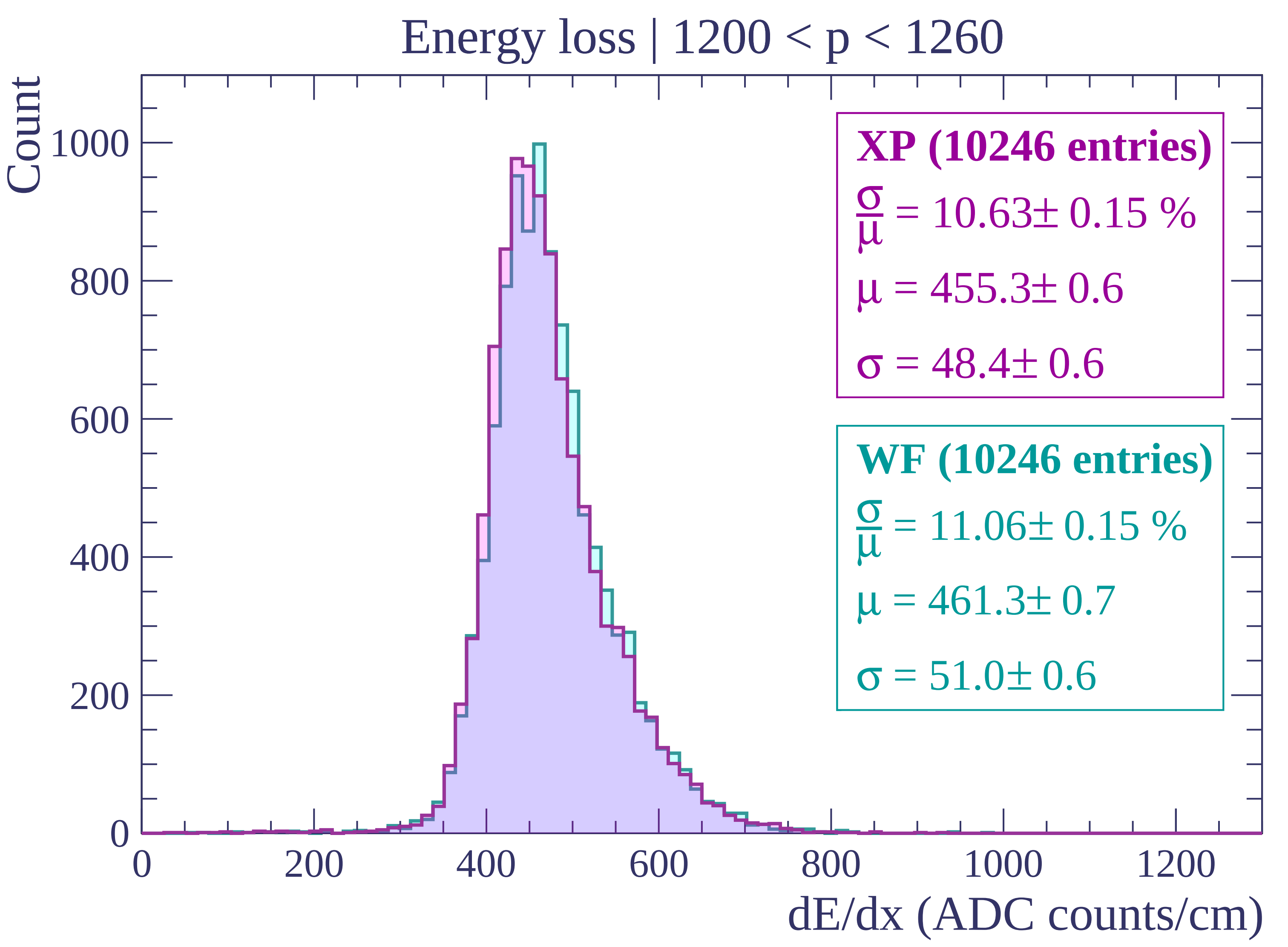


Energy loss | 1020 < p < 1080
Count
dE/dx (ADC counts/cm)
XP (11204 entries)
$\frac{\sigma}{\mu}$ = 10.63± 0.15 %
μ = 445.5± 0.6
σ = 47.4± 0.6
WF (11204 entries)
$\frac{\sigma}{\mu}$ = 11.04± 0.15 %
μ = 450.7± 0.6
σ = 49.8± 0.6

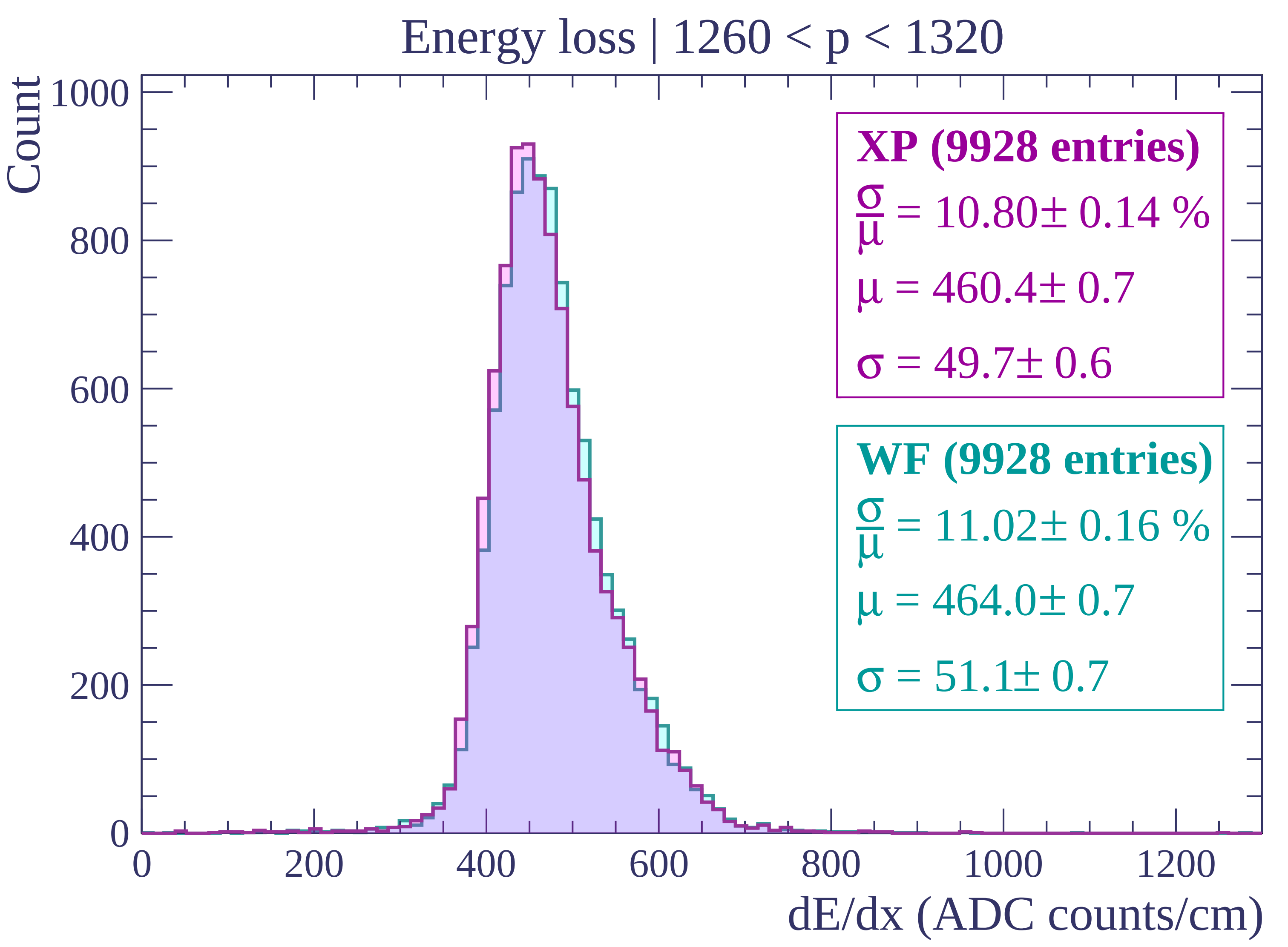


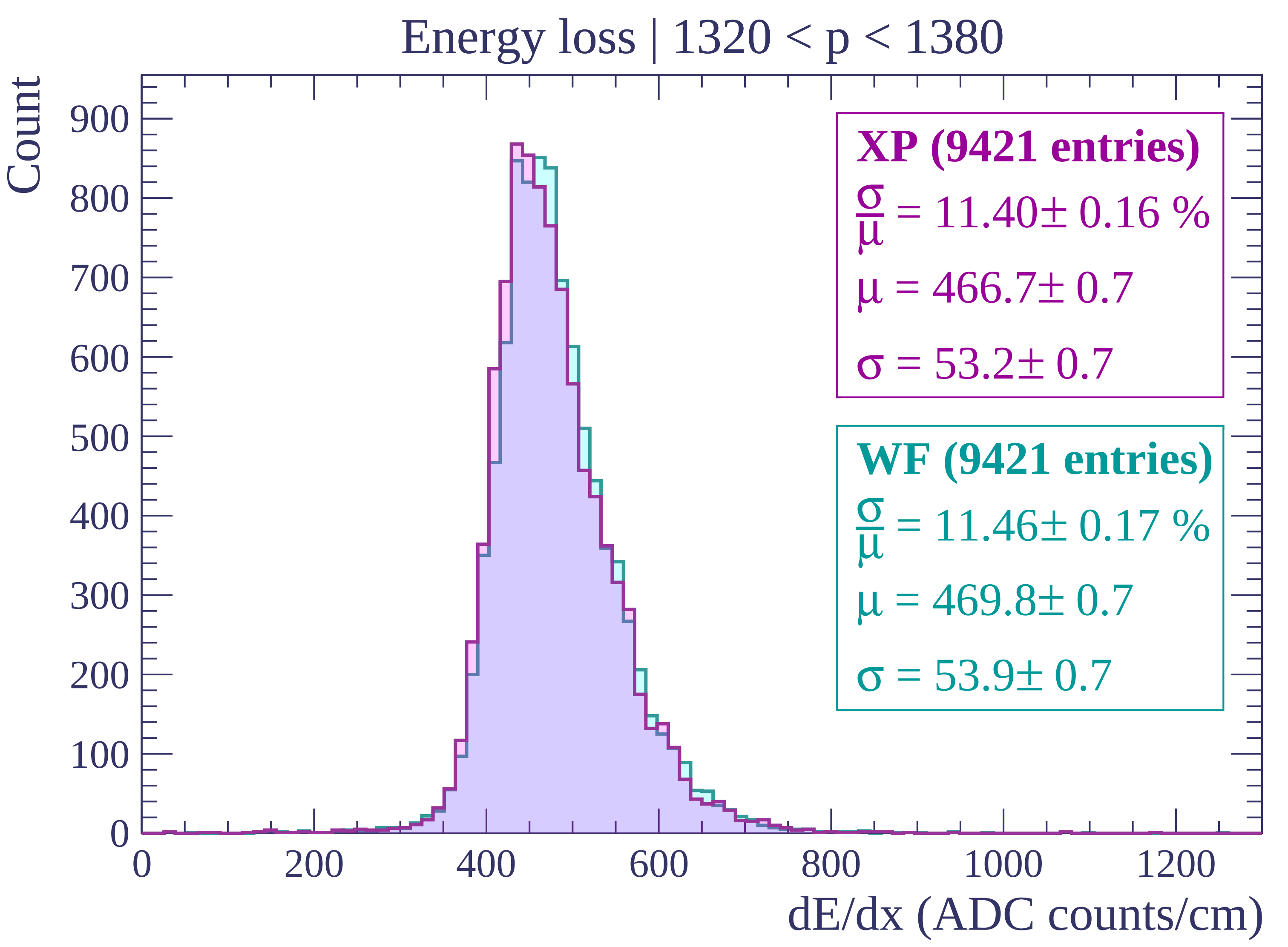
Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
XP (10872 entries)
$\frac{\sigma}{\mu}$ = 10.99± 0.15 %
μ = 448.8± 0.6
σ = 49.3± 0.6
WF (10872 entries)
$\frac{\sigma}{\mu}$ = 10.96± 0.15 %
μ = 454.0± 0.6
σ = 49.7± 0.6


Energy loss | 1140 < p < 1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (10601 entries)
σ/μ = 10.96± 0.15 %
μ = 452.0± 0.6
σ = 49.5± 0.6
WF (10601 entries)
σ/μ = 11.32± 0.15 %
μ = 458.3± 0.7
σ = 51.9± 0.6

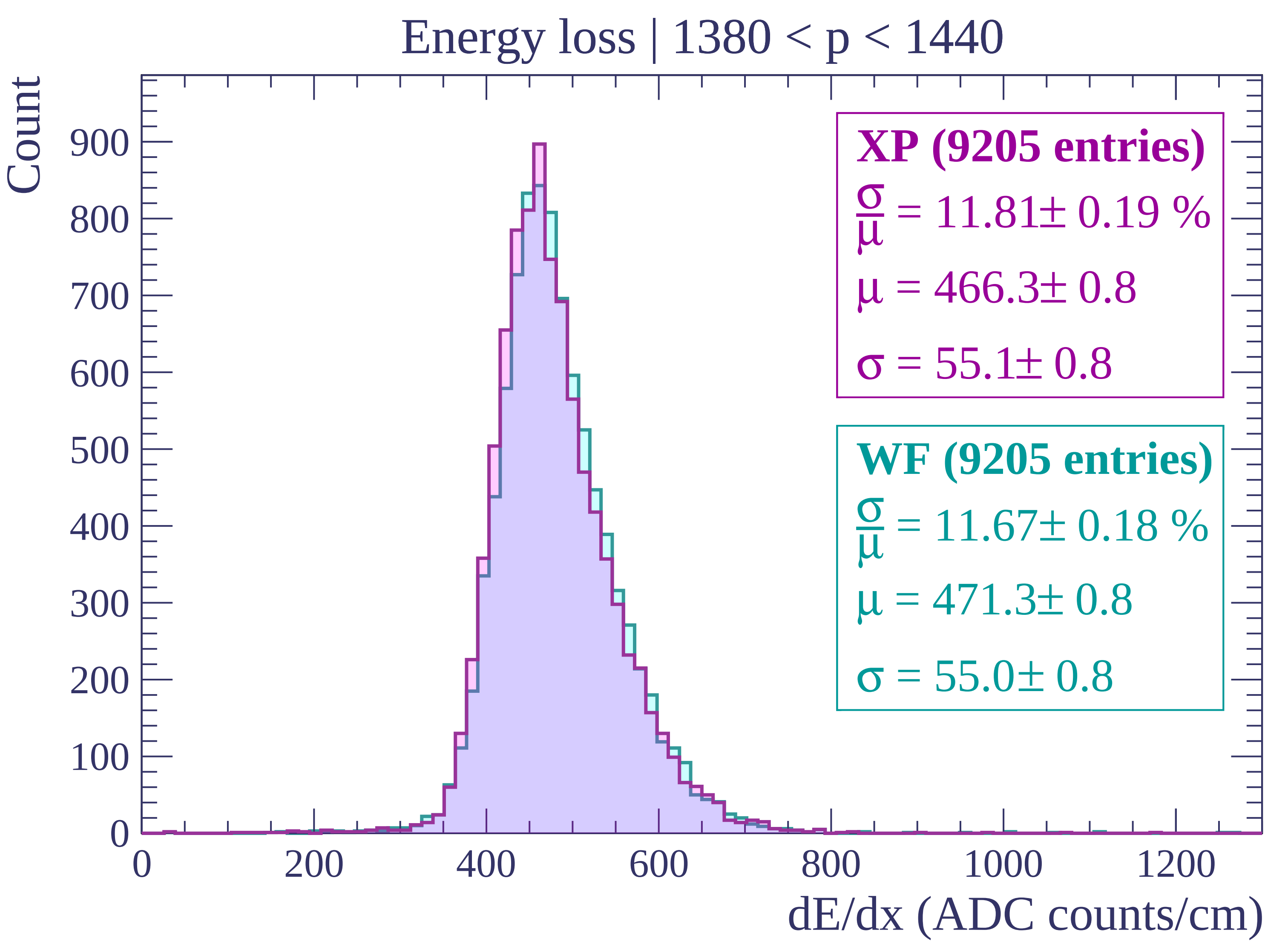


Energy loss | 1200 < p < 1260
Count
dE/dx (ADC counts/cm)
XP (10246 entries)
σ/μ = 10.63± 0.15 %
μ = 455.3± 0.6
σ = 48.4± 0.6
WF (10246 entries)
σ/μ = 11.06± 0.15 %
μ = 461.3± 0.7
σ = 51.0± 0.6

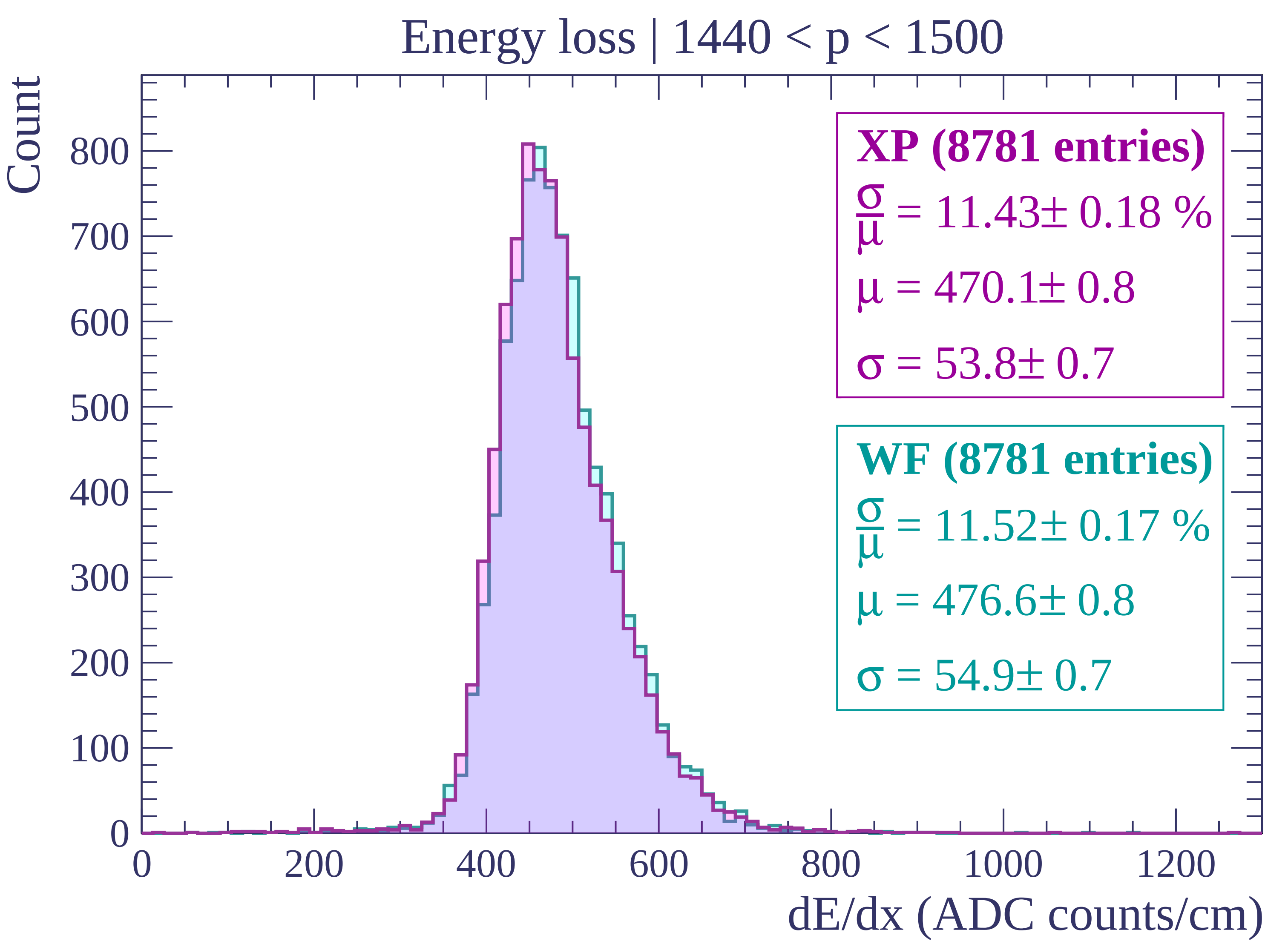

Energy loss | 1260 < p < 1320
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (9928 entries)
σ/μ = 10.80± 0.14 %
μ = 460.4± 0.7
σ = 49.7± 0.6
WF (9928 entries)
σ/μ = 11.02± 0.16 %
μ = 464.0± 0.7
σ = 51.1± 0.7



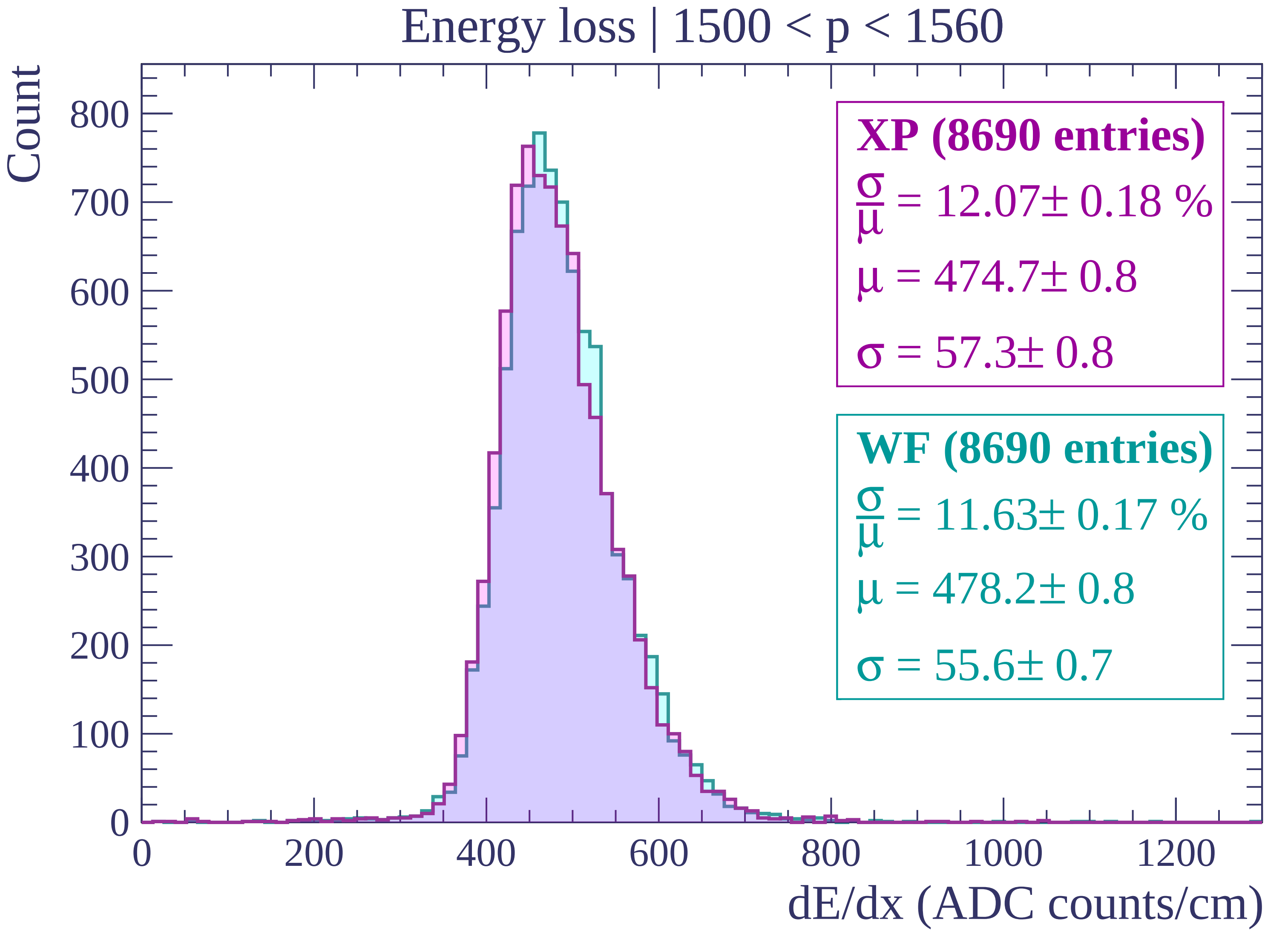
Energy loss | 1320 < p < 1380
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (9421 entries)
σ/μ = 11.40± 0.16 %
μ = 466.7± 0.7
σ = 53.2± 0.7
WF (9421 entries)
σ/μ = 11.46± 0.17 %
μ = 469.8± 0.7
σ = 53.9± 0.7



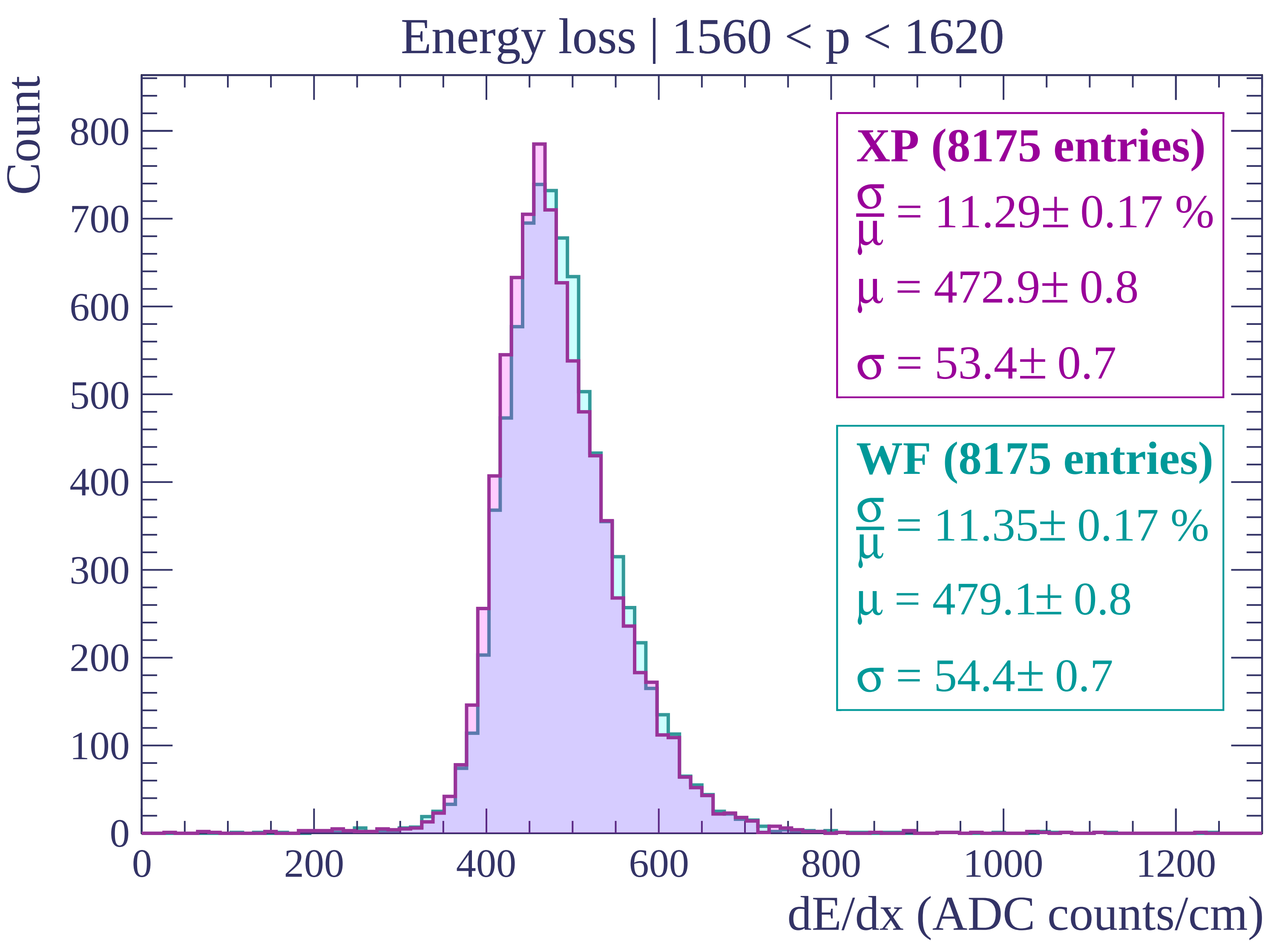
Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (9205 entries)
$\frac{\sigma}{\mu}$ = 11.81± 0.19 %
μ = 466.3± 0.8
σ = 55.1± 0.8
WF (9205 entries)
$\frac{\sigma}{\mu}$ = 11.67± 0.18 %
μ = 471.3± 0.8
σ = 55.0± 0.8



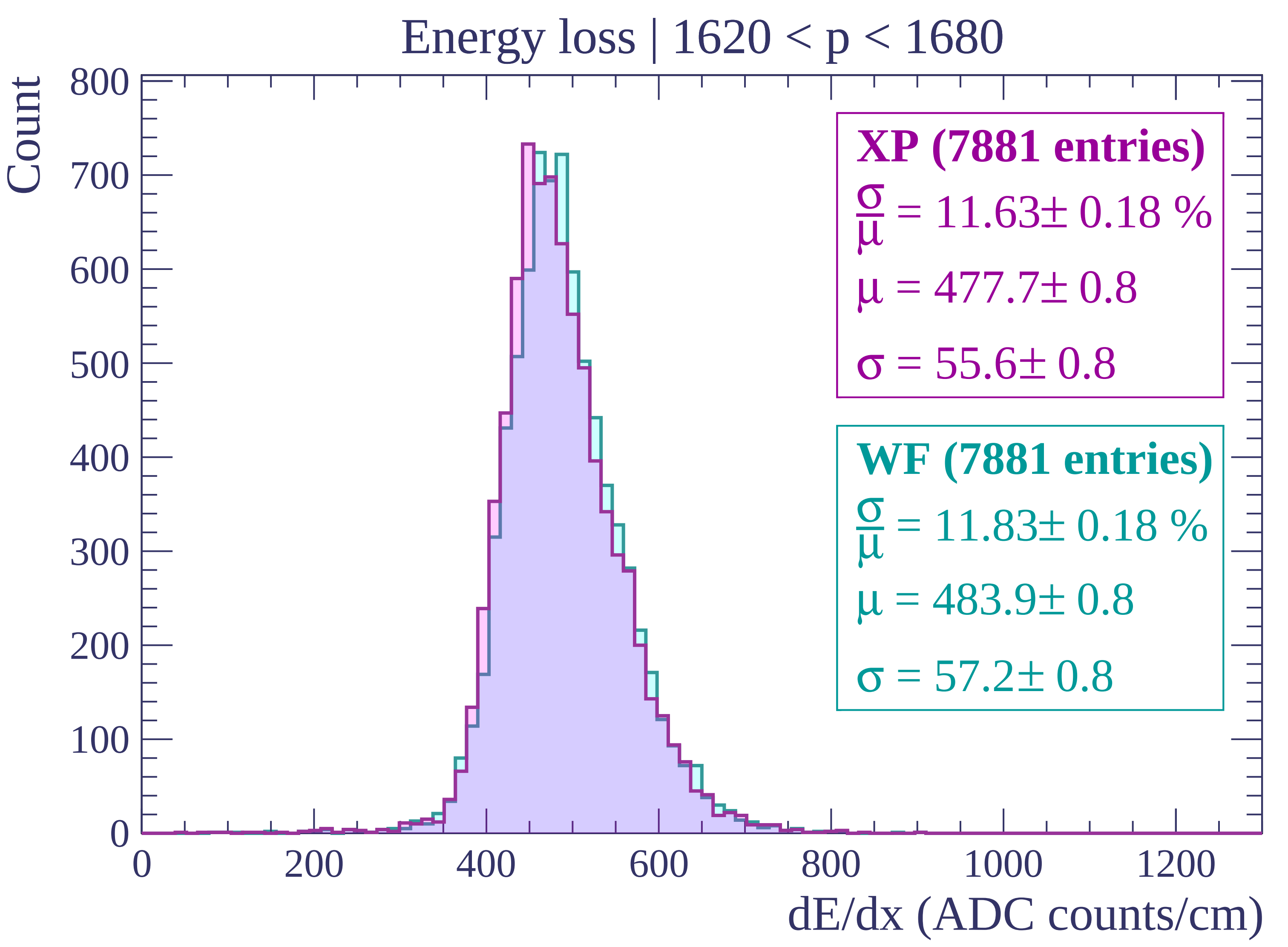
Energy loss | 1440 < p < 1500
Count
dE/dx (ADC counts/cm)
XP (8781 entries)
$\frac{\sigma}{\mu}$ = 11.43± 0.18 %
μ = 470.1± 0.8
σ = 53.8± 0.7
WF (8781 entries)
$\frac{\sigma}{\mu}$ = 11.52± 0.17 %
μ = 476.6± 0.8
σ = 54.9± 0.7



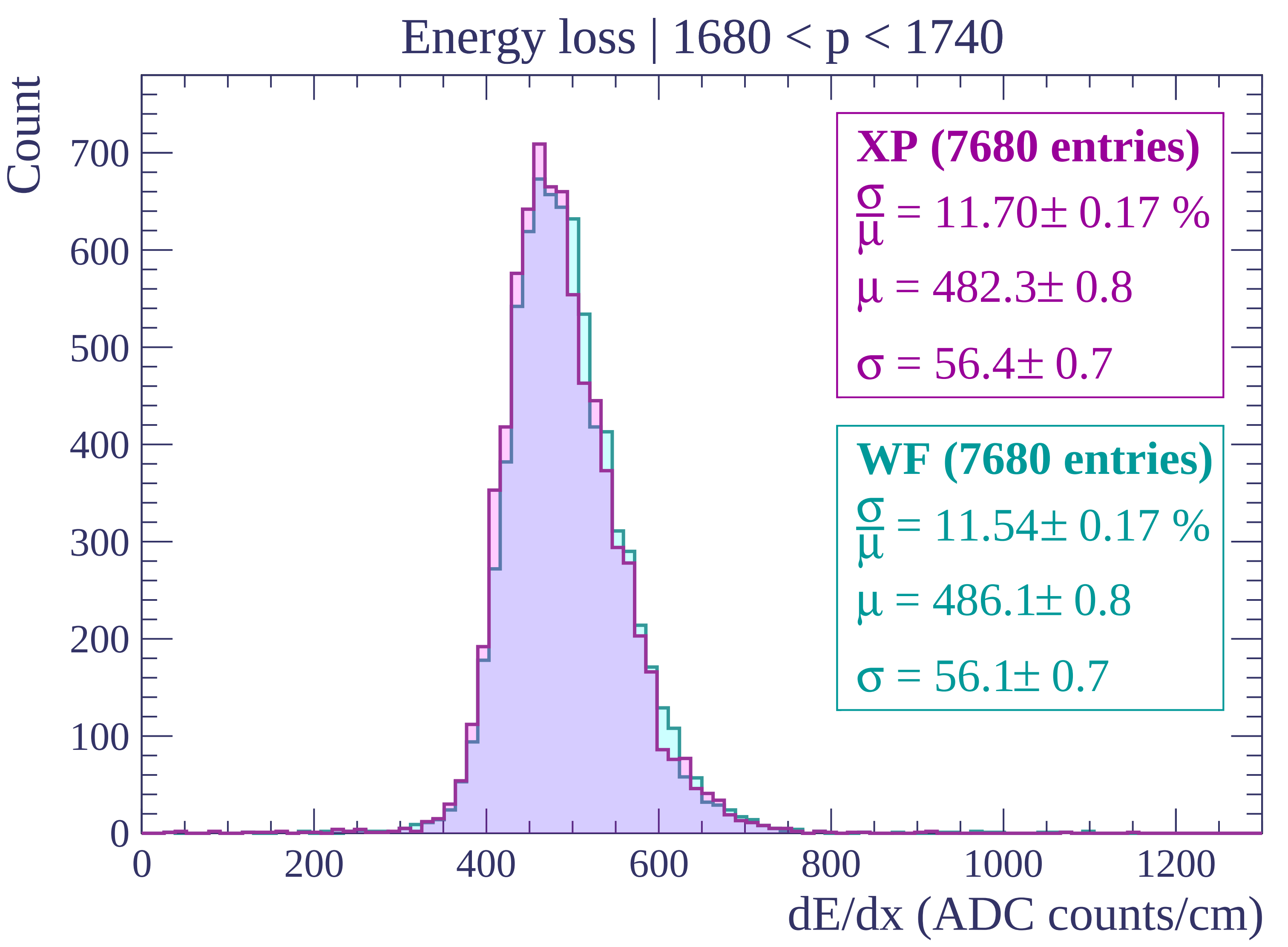
Energy loss | 1500 < p < 1560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (8690 entries)
$\frac{\sigma}{\mu}$ = 12.07± 0.18 %
μ = 474.7± 0.8
σ = 57.3± 0.8
WF (8690 entries)
$\frac{\sigma}{\mu}$ = 11.63± 0.17 %
μ = 478.2± 0.8
σ = 55.6± 0.7

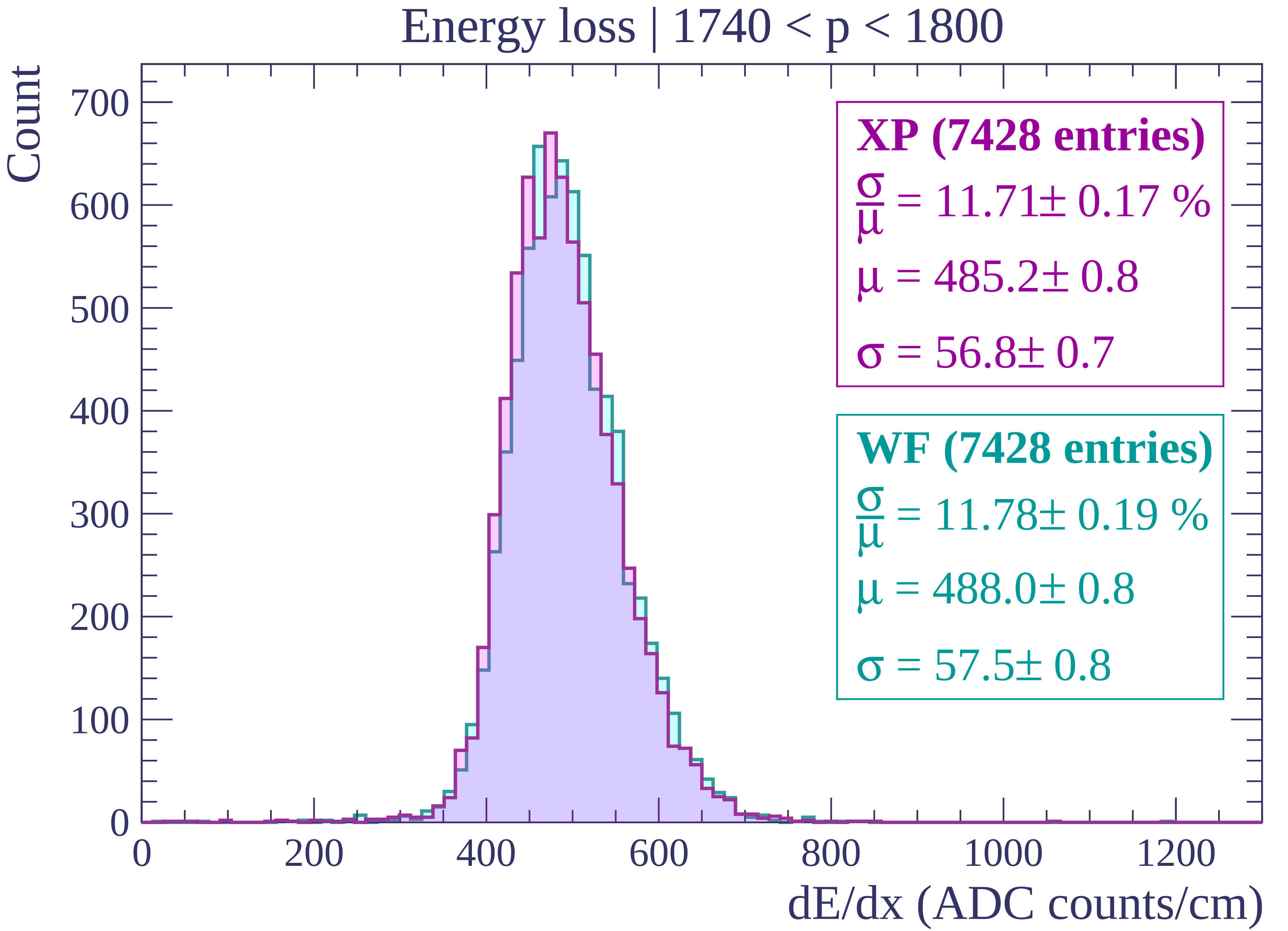


Energy loss | 1560 < p < 1620
Count
dE/dx (ADC counts/cm)
XP (8175 entries)
σ/μ = 11.29± 0.17 %
μ = 472.9± 0.8
σ = 53.4± 0.7
WF (8175 entries)
σ/μ = 11.35± 0.17 %
μ = 479.1± 0.8
σ = 54.4± 0.7
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200

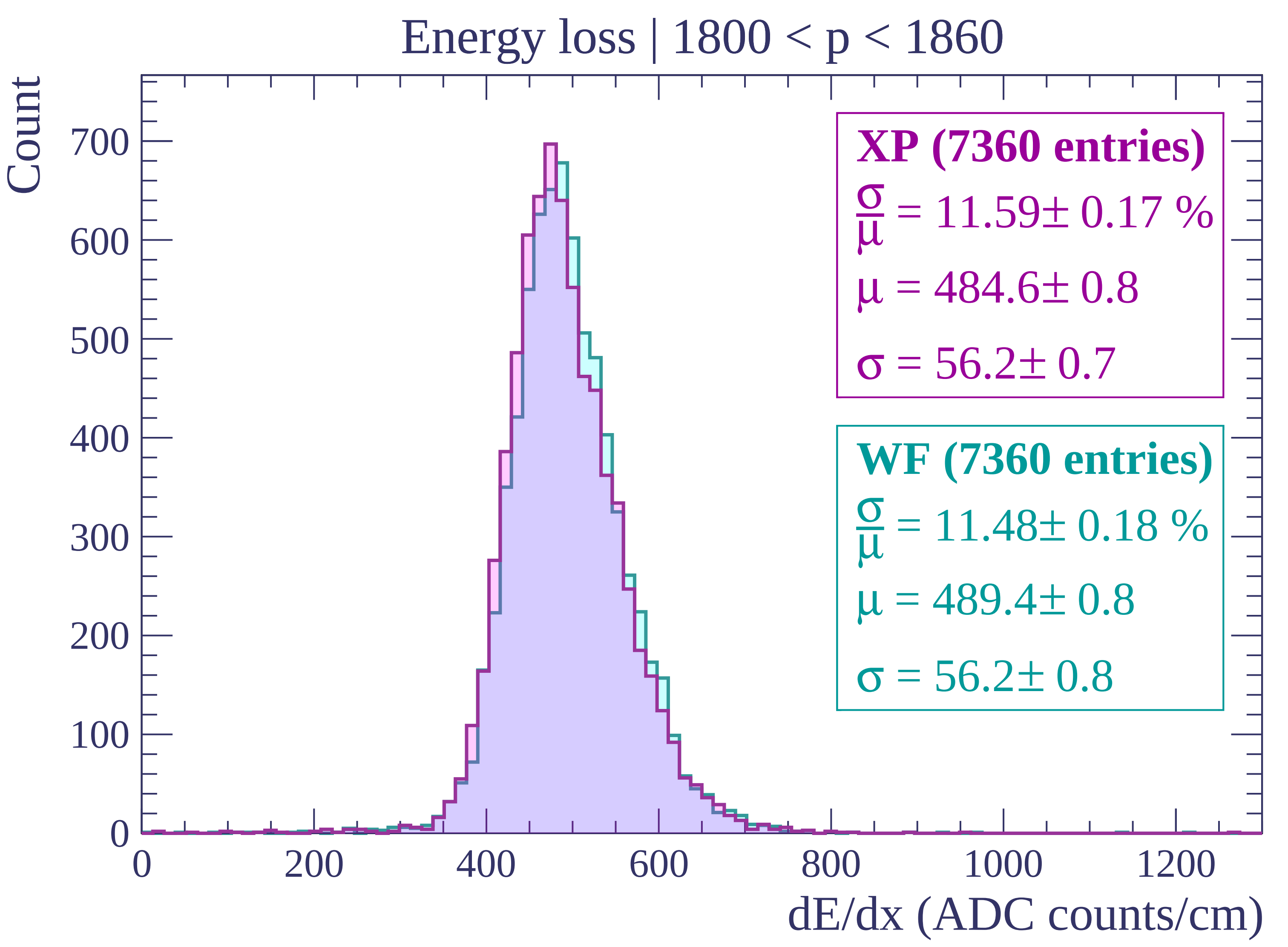


Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
XP (7881 entries)
σ/μ = 11.63± 0.18 %
μ = 477.7± 0.8
σ = 55.6± 0.8
WF (7881 entries)
σ/μ = 11.83± 0.18 %
μ = 483.9± 0.8
σ = 57.2± 0.8
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200

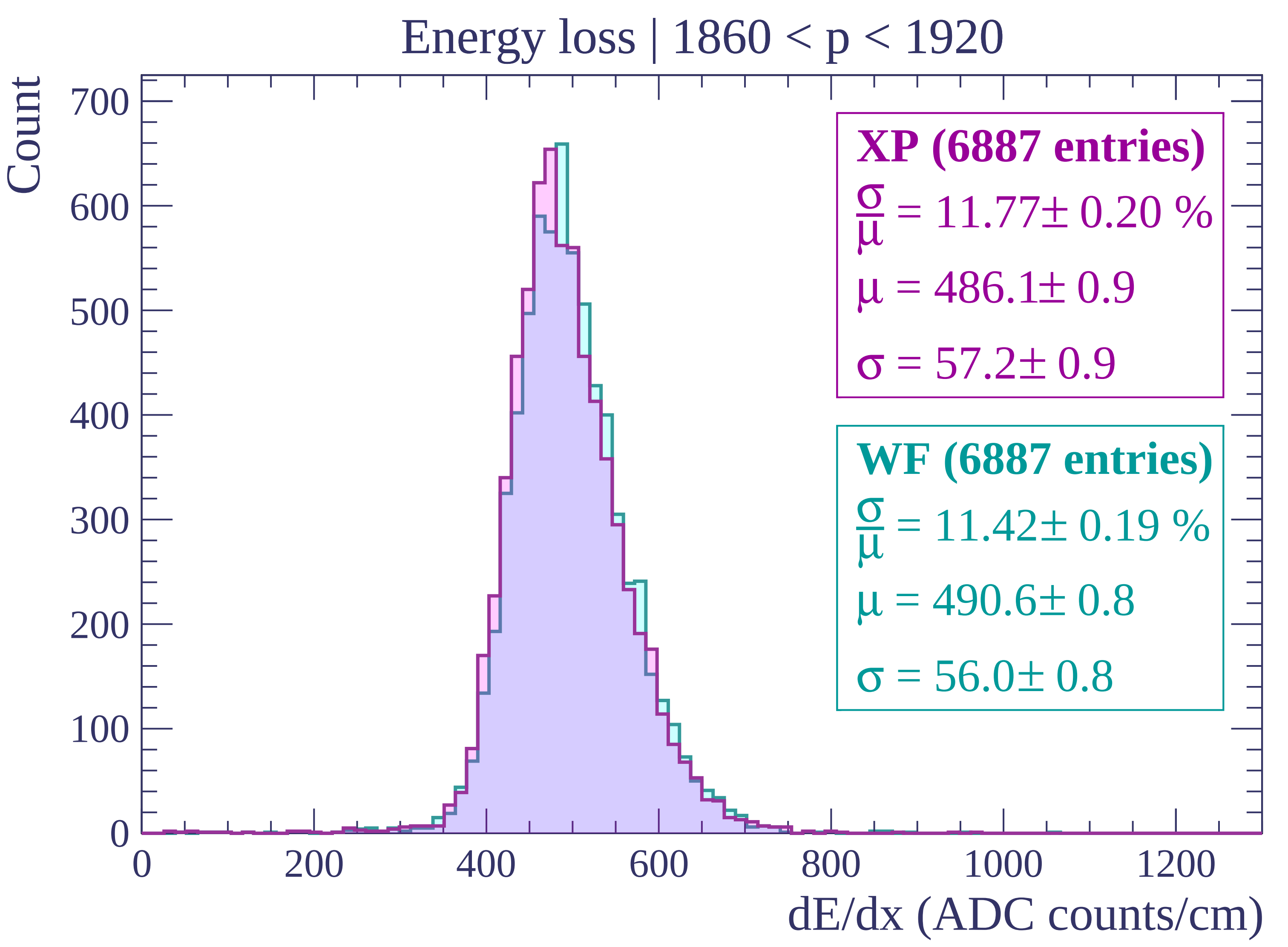

Energy loss | 1680 < p < 1740
Count
dE/dx (ADC counts/cm)
XP (7680 entries)
σ/μ = 11.70± 0.17 %
μ = 482.3± 0.8
σ = 56.4± 0.7
WF (7680 entries)
σ/μ = 11.54± 0.17 %
μ = 486.1± 0.8
σ = 56.1± 0.7

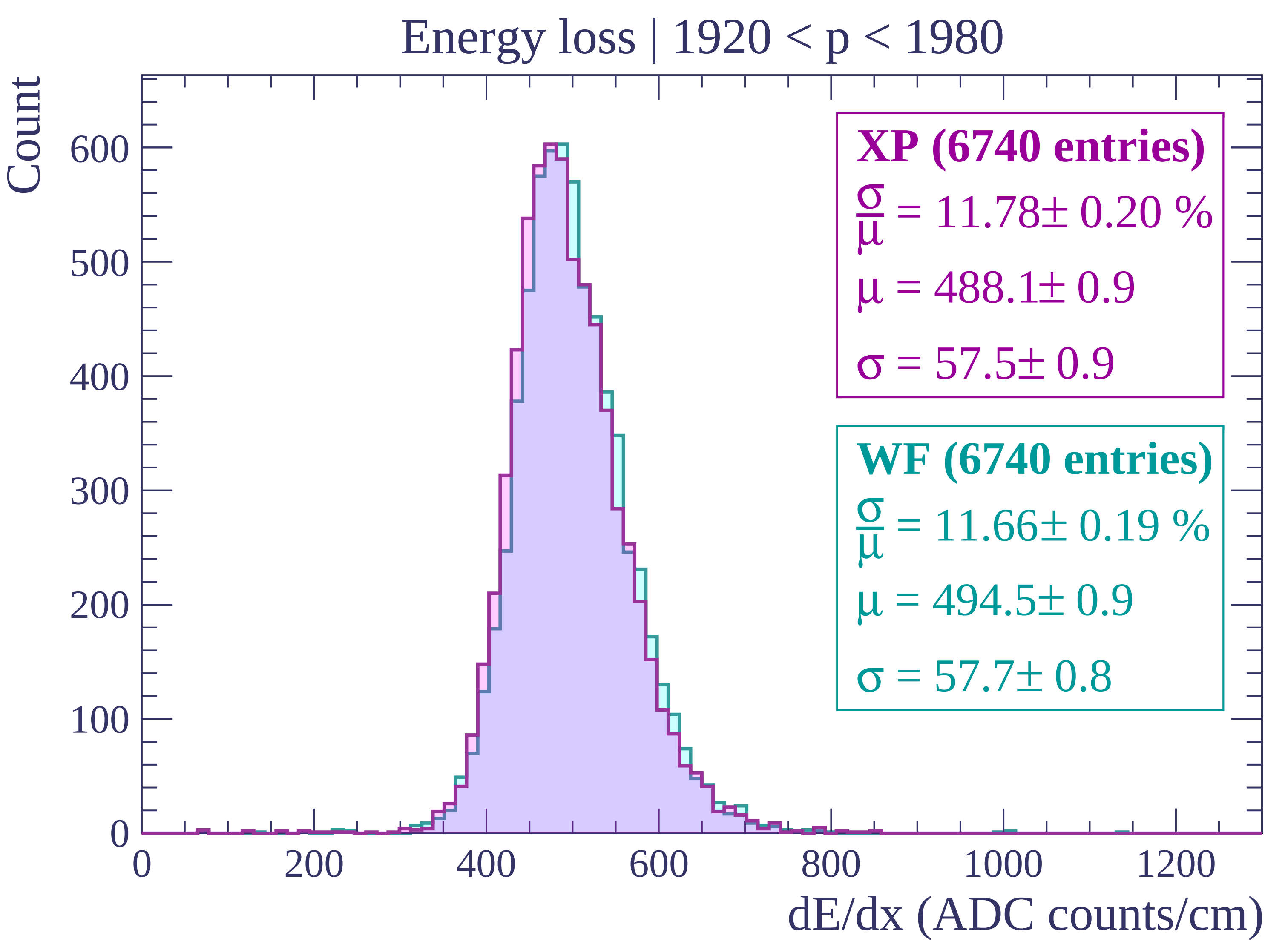


Energy loss | 1740 < p < 1800
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (7428 entries)
σ/μ = 11.71± 0.17 %
μ = 485.2± 0.8
σ = 56.8± 0.7
WF (7428 entries)
σ/μ = 11.78± 0.19 %
μ = 488.0± 0.8
σ = 57.5± 0.8

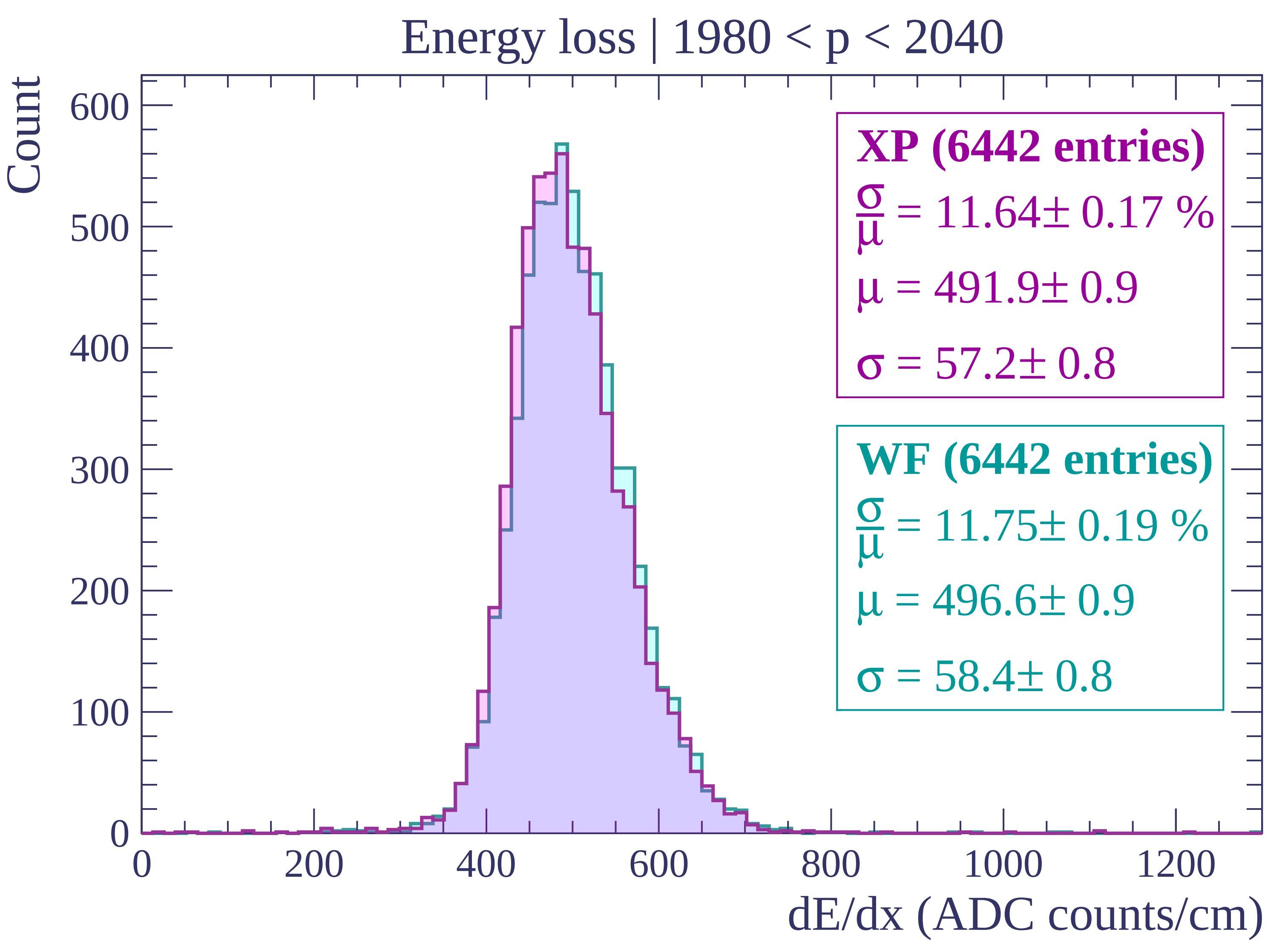


Energy loss | 1800 < p < 1860
Count
700
600
500
400
300
200
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7360 entries)
σ/μ = 11.59± 0.17 %
μ = 484.6± 0.8
σ = 56.2± 0.7
WF (7360 entries)
σ/μ = 11.48± 0.18 %
μ = 489.4± 0.8
σ = 56.2± 0.8

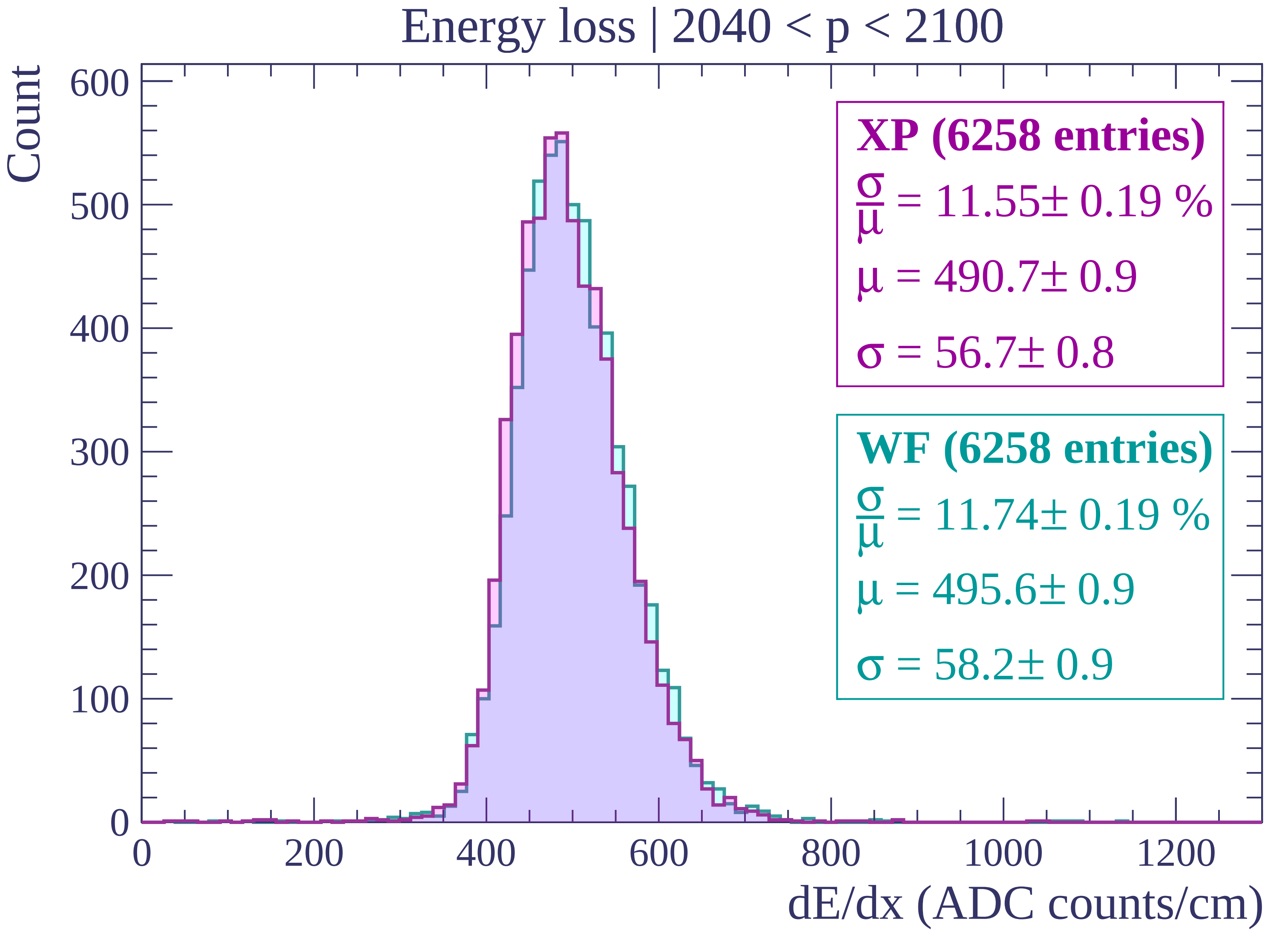

Energy loss | 1860 < p < 1920
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (6887 entries)
σ/μ = 11.77± 0.20 %
μ = 486.1± 0.9
σ = 57.2± 0.9
WF (6887 entries)
σ/μ = 11.42± 0.19 %
μ = 490.6± 0.8
σ = 56.0± 0.8

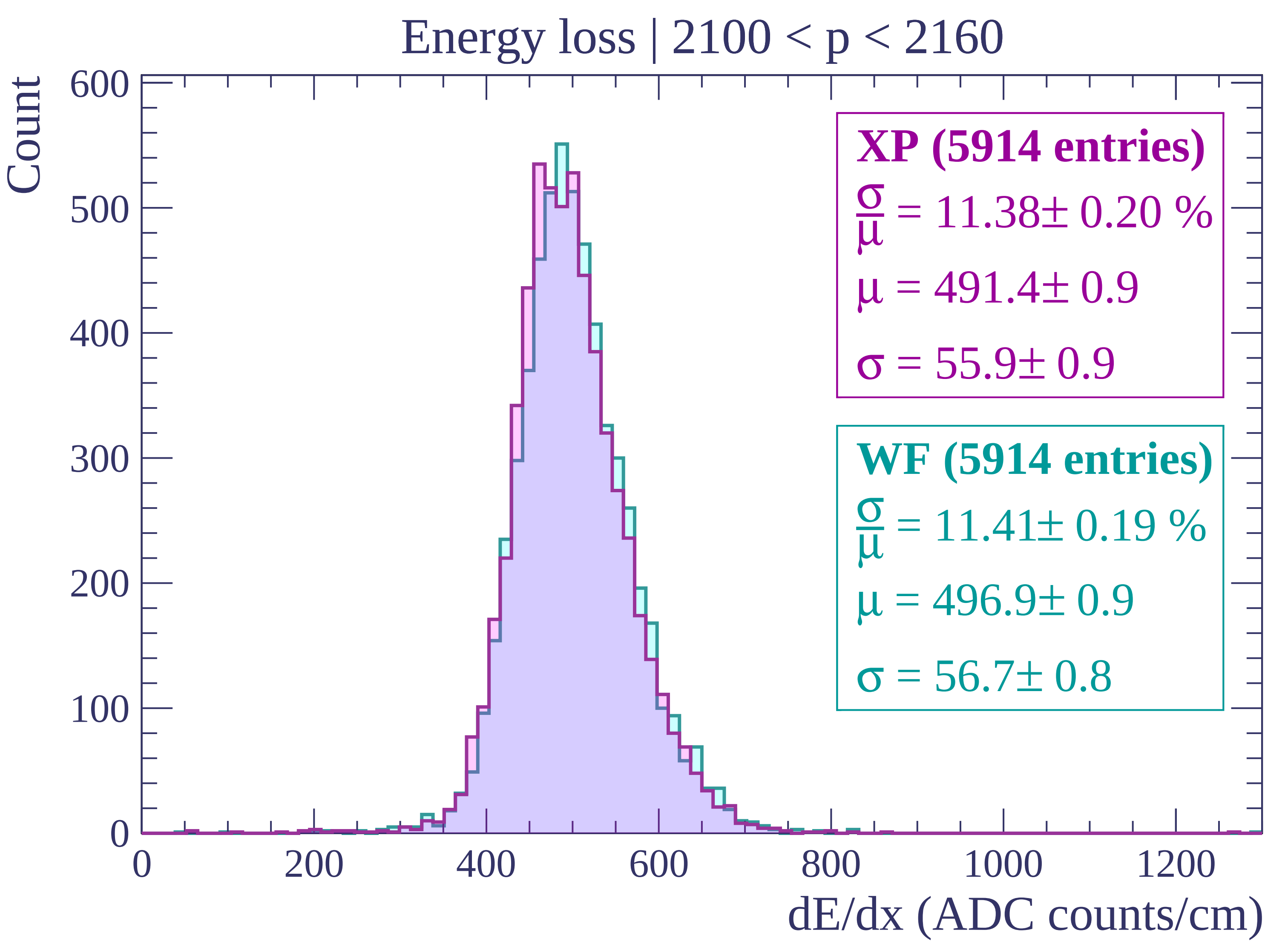


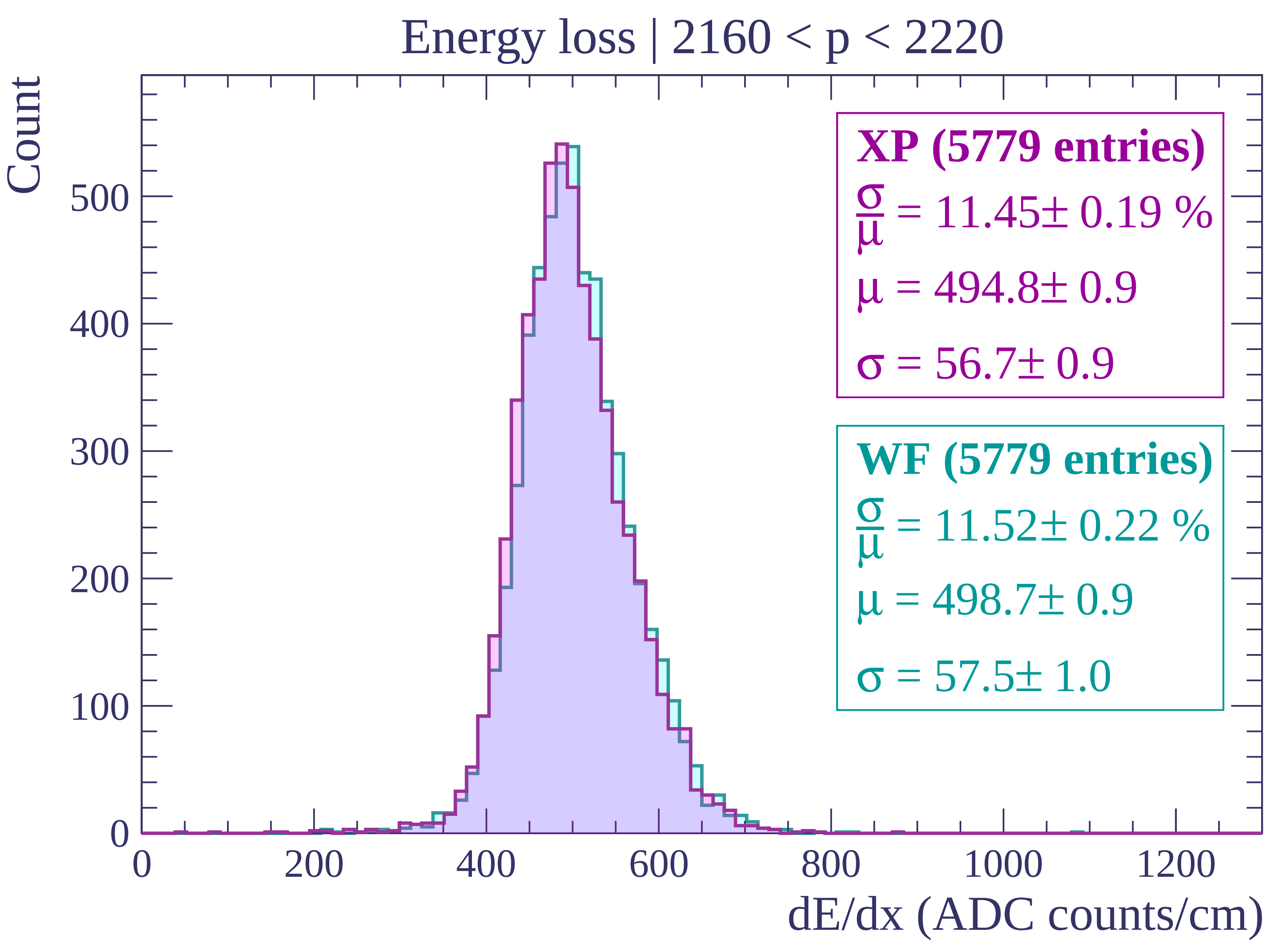
Energy loss | 1920 < p < 1980
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (6740 entries)
$\frac{\sigma}{\mu}$ = 11.78± 0.20 %
μ = 488.1± 0.9
σ = 57.5± 0.9
WF (6740 entries)
$\frac{\sigma}{\mu}$ = 11.66± 0.19 %
μ = 494.5± 0.9
σ = 57.7± 0.8



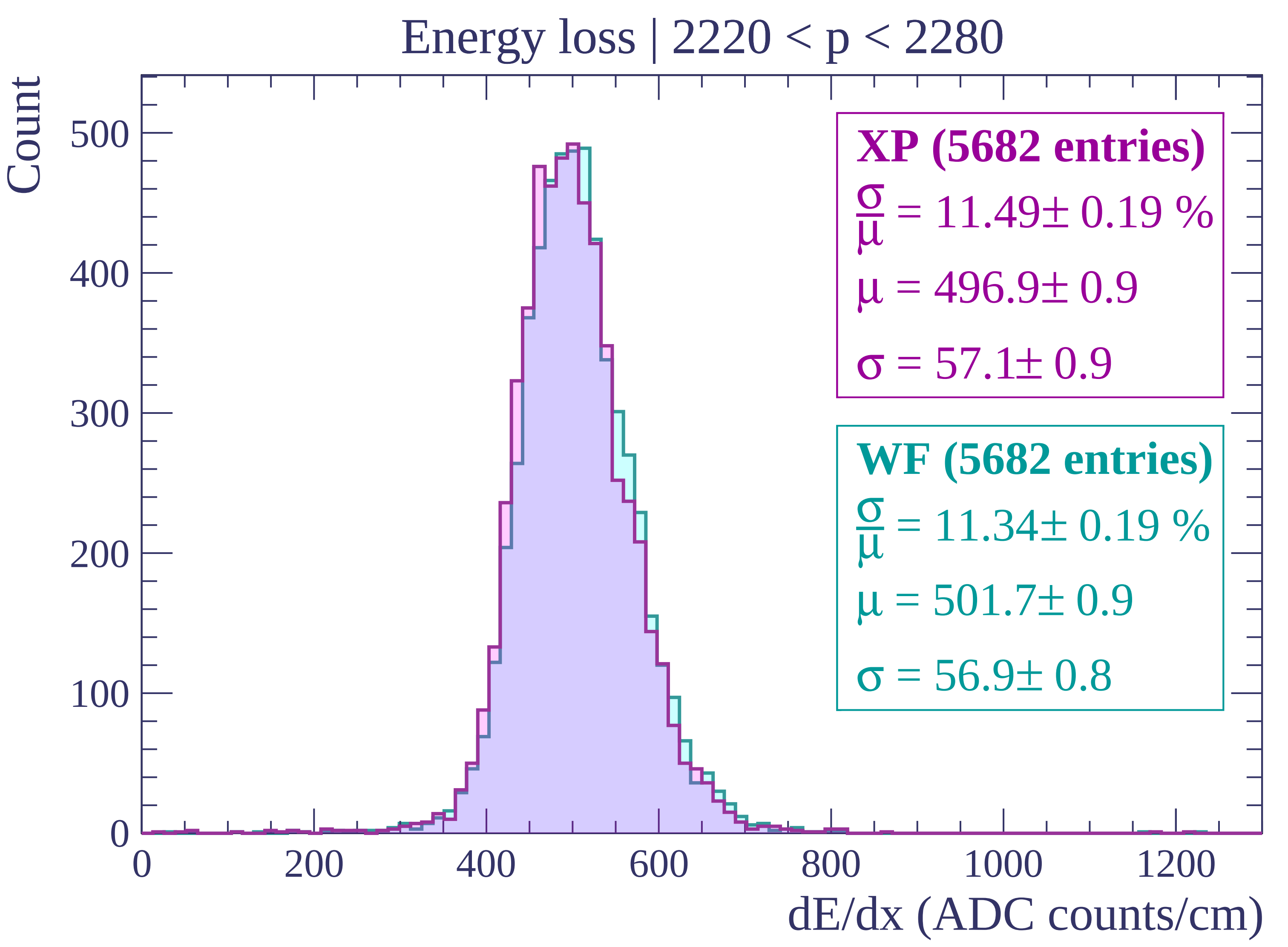
Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (6442 entries)
σ/μ = 11.64± 0.17 %
μ = 491.9± 0.9
σ = 57.2± 0.8
WF (6442 entries)
σ/μ = 11.75± 0.19 %
μ = 496.6± 0.9
σ = 58.4± 0.8


Energy loss | 2040 < p < 2100
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6258 entries)
σ/μ = 11.55± 0.19 %
μ = 490.7± 0.9
σ = 56.7± 0.8
WF (6258 entries)
σ/μ = 11.74± 0.19 %
μ = 495.6± 0.9
σ = 58.2± 0.9

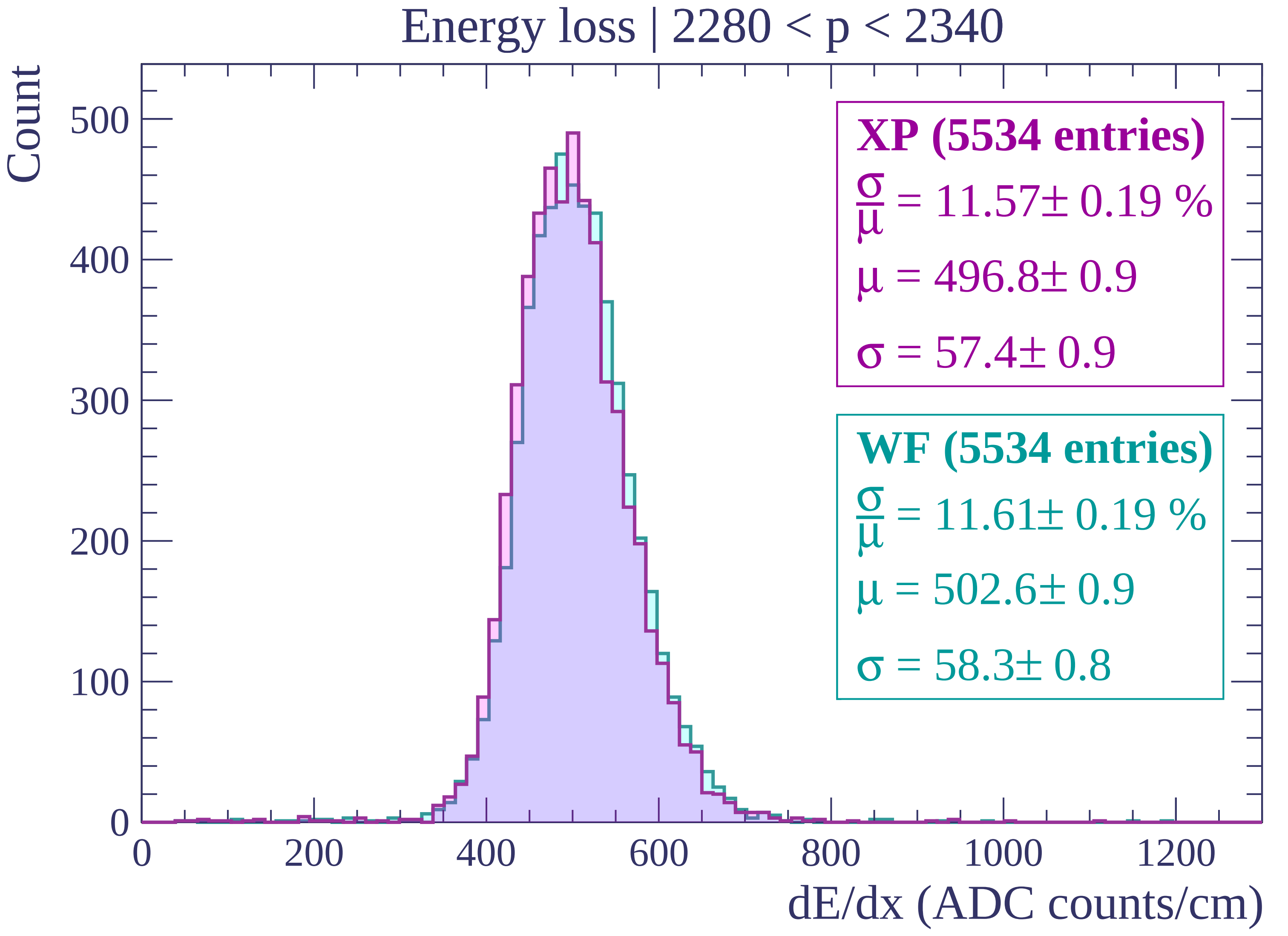

Energy loss | 2100 < p < 2160
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5914 entries)
σ/μ = 11.38± 0.20 %
μ = 491.4± 0.9
σ = 55.9± 0.9
WF (5914 entries)
σ/μ = 11.41± 0.19 %
μ = 496.9± 0.9
σ = 56.7± 0.8

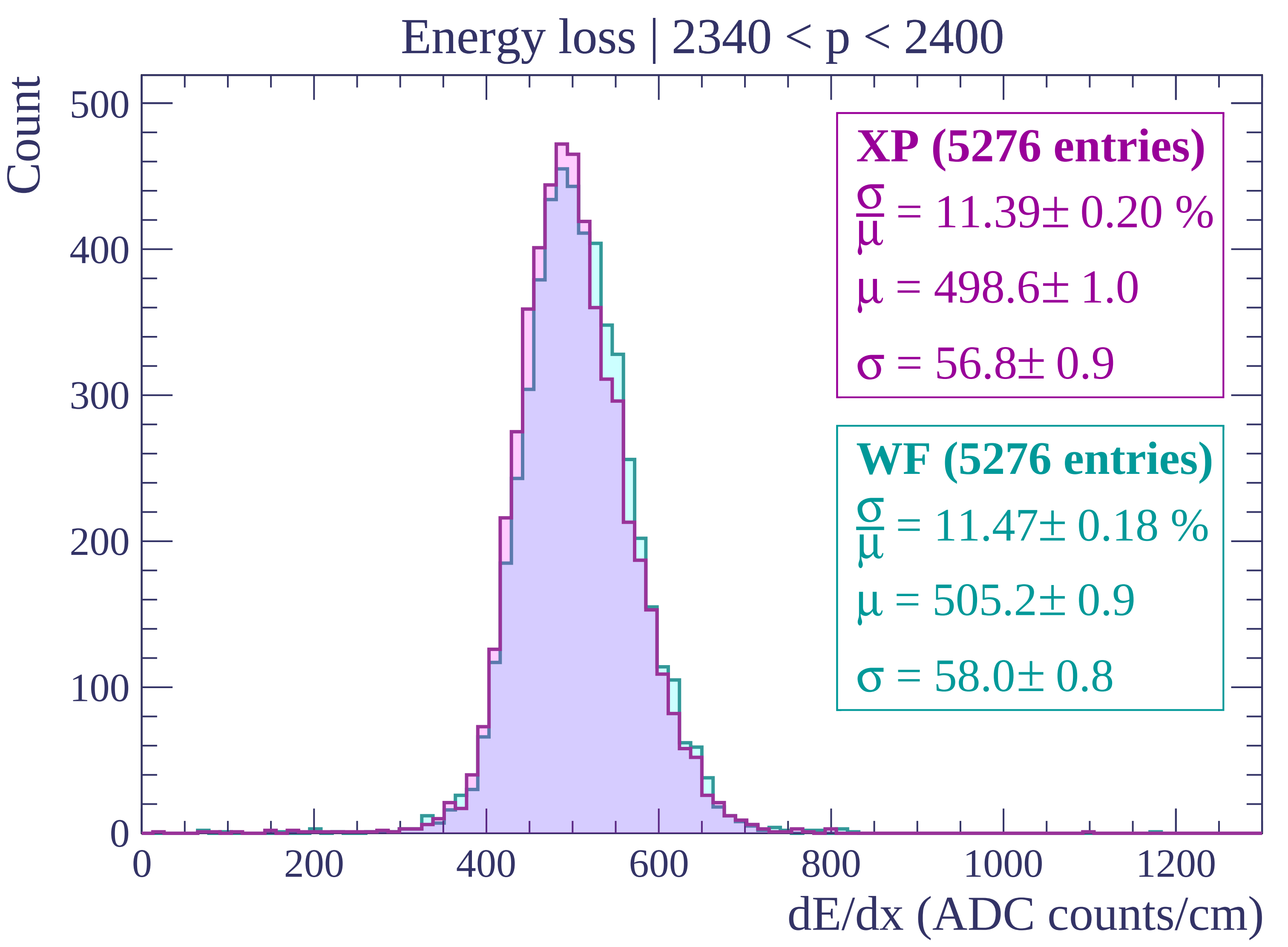


Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5779 entries)
$\frac{\sigma}{\mu}$ = 11.45± 0.19 %
μ = 494.8± 0.9
σ = 56.7± 0.9
WF (5779 entries)
$\frac{\sigma}{\mu}$ = 11.52± 0.22 %
μ = 498.7± 0.9
σ = 57.5± 1.0

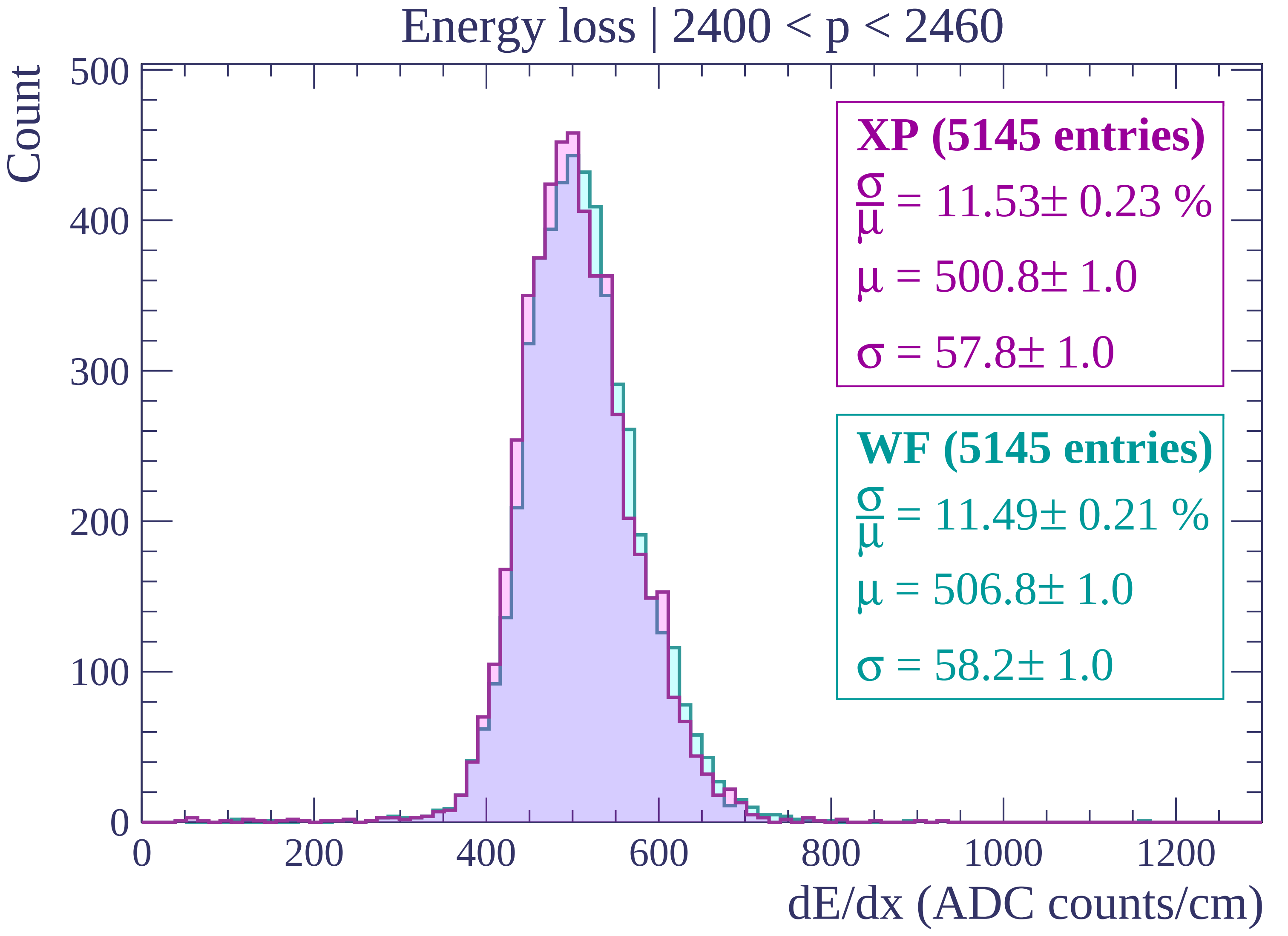


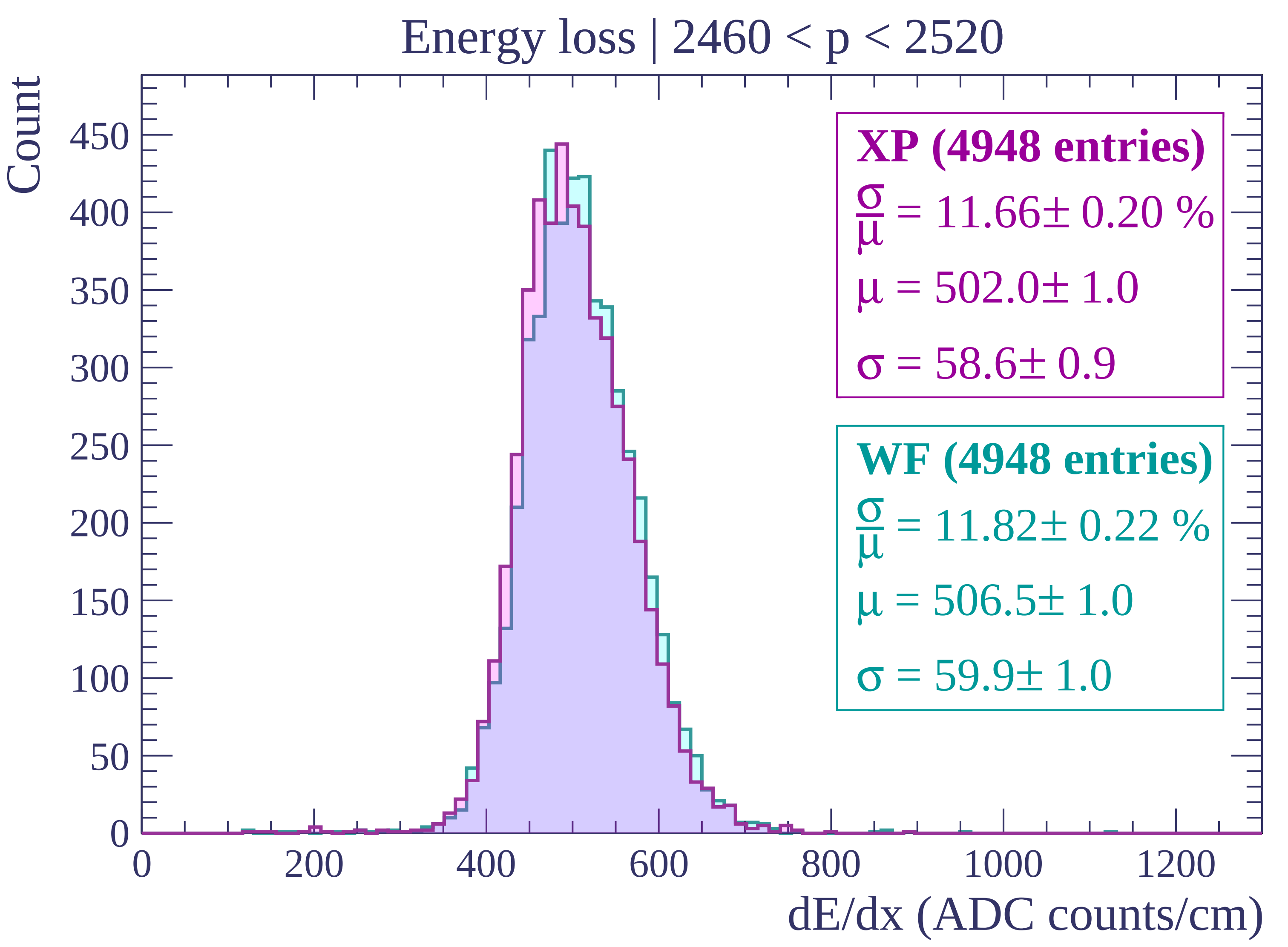
Energy loss | 2220 < p < 2280
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5682 entries)
σ/μ = 11.49± 0.19 %
μ = 496.9± 0.9
σ = 57.1± 0.9
WF (5682 entries)
σ/μ = 11.34± 0.19 %
μ = 501.7± 0.9
σ = 56.9± 0.8


Energy loss | 2280 < p < 2340
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5534 entries)
$\frac{\sigma}{\mu}$ = 11.57± 0.19 %
μ = 496.8± 0.9
σ = 57.4± 0.9
WF (5534 entries)
$\frac{\sigma}{\mu}$ = 11.61± 0.19 %
μ = 502.6± 0.9
σ = 58.3± 0.8

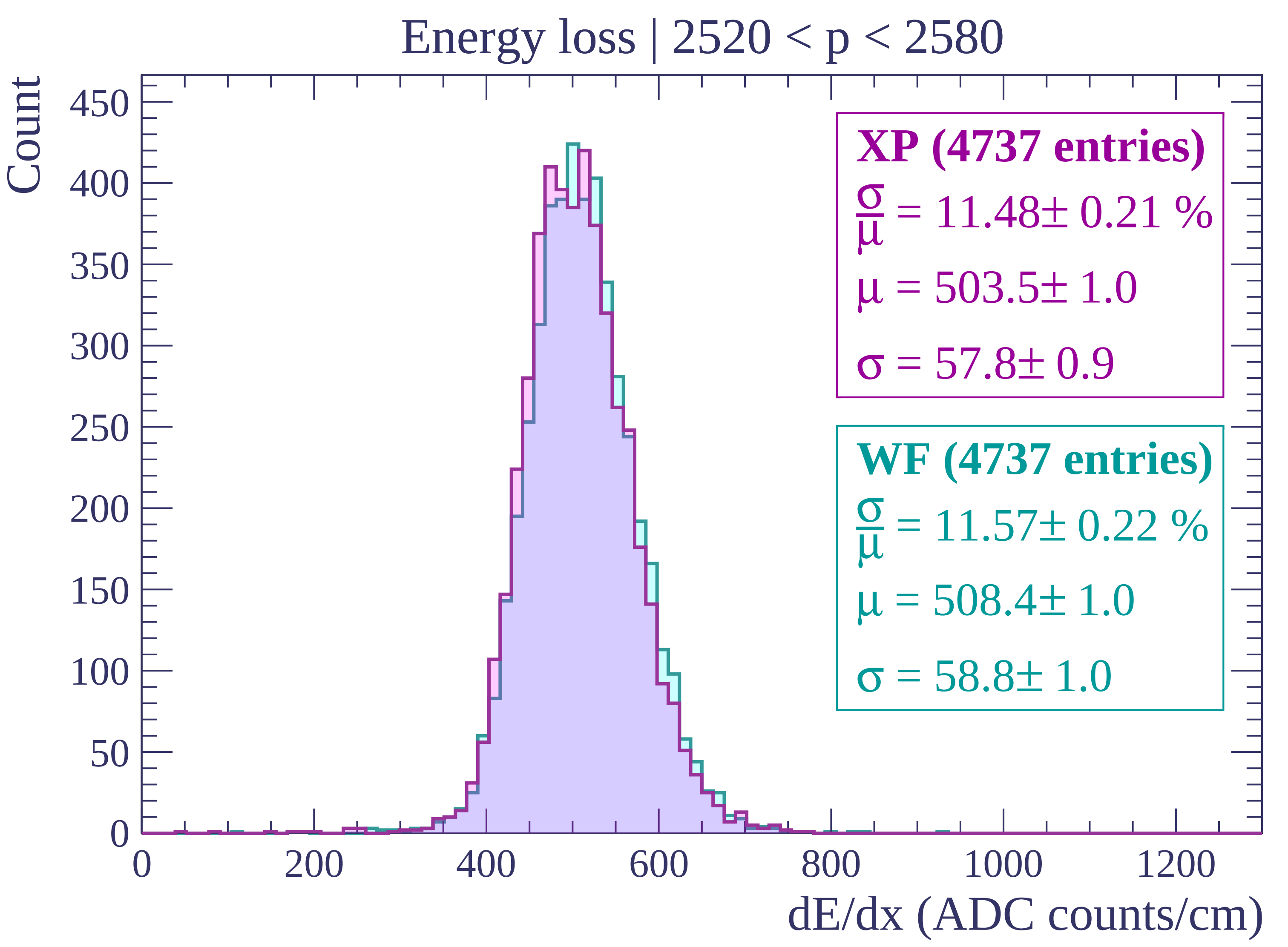


Energy loss | 2340 < p < 2400
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5276 entries)
$\frac{\sigma}{\mu}$ = 11.39± 0.20 %
μ = 498.6± 1.0
σ = 56.8± 0.9
WF (5276 entries)
$\frac{\sigma}{\mu}$ = 11.47± 0.18 %
μ = 505.2± 0.9
σ = 58.0± 0.8

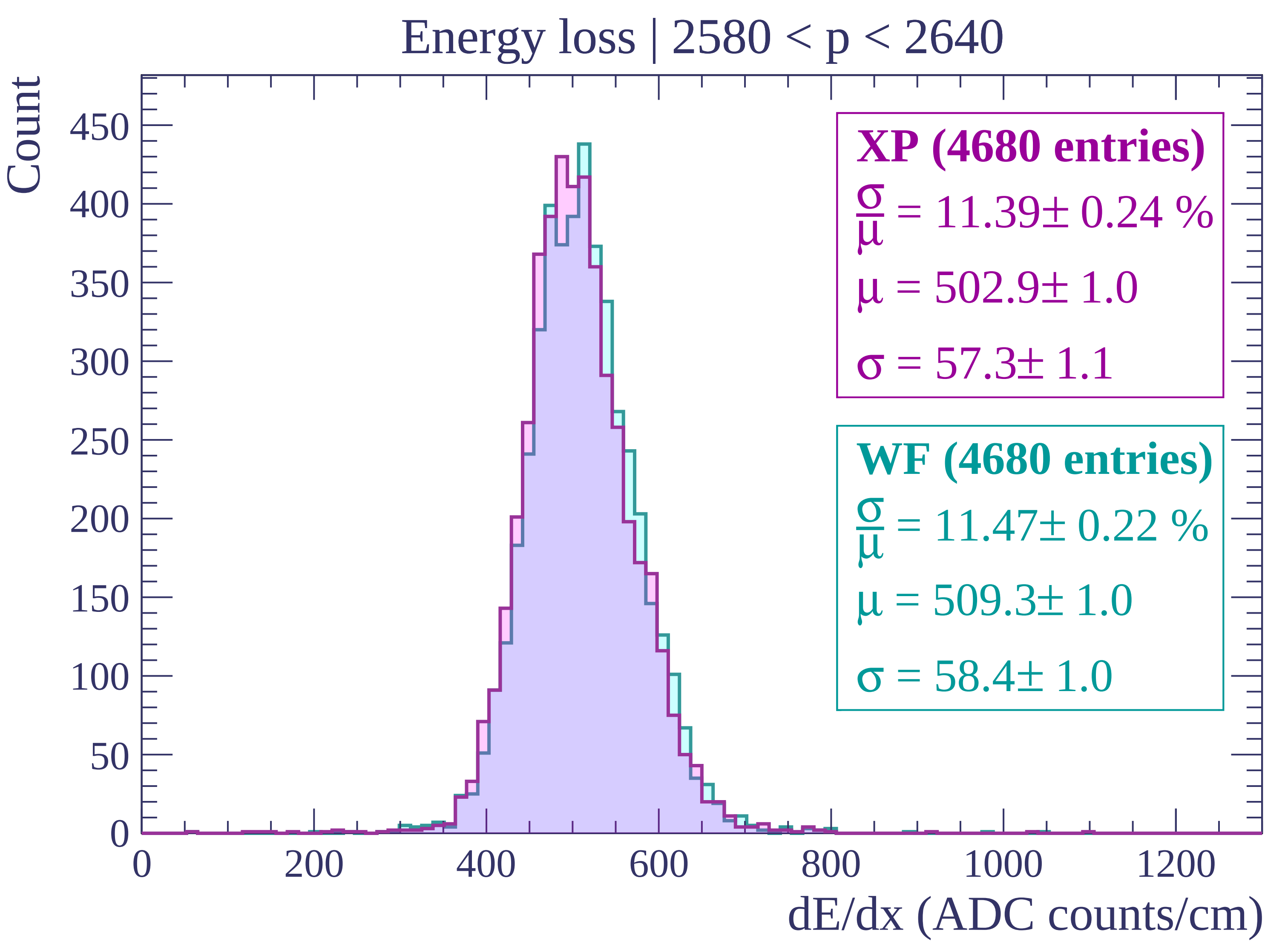

Energy loss | 2400 < p < 2460
Count
dE/dx (ADC counts/cm)
XP (5145 entries)
σ/μ = 11.53± 0.23 %
μ = 500.8± 1.0
σ = 57.8± 1.0
WF (5145 entries)
σ/μ = 11.49± 0.21 %
μ = 506.8± 1.0
σ = 58.2± 1.0



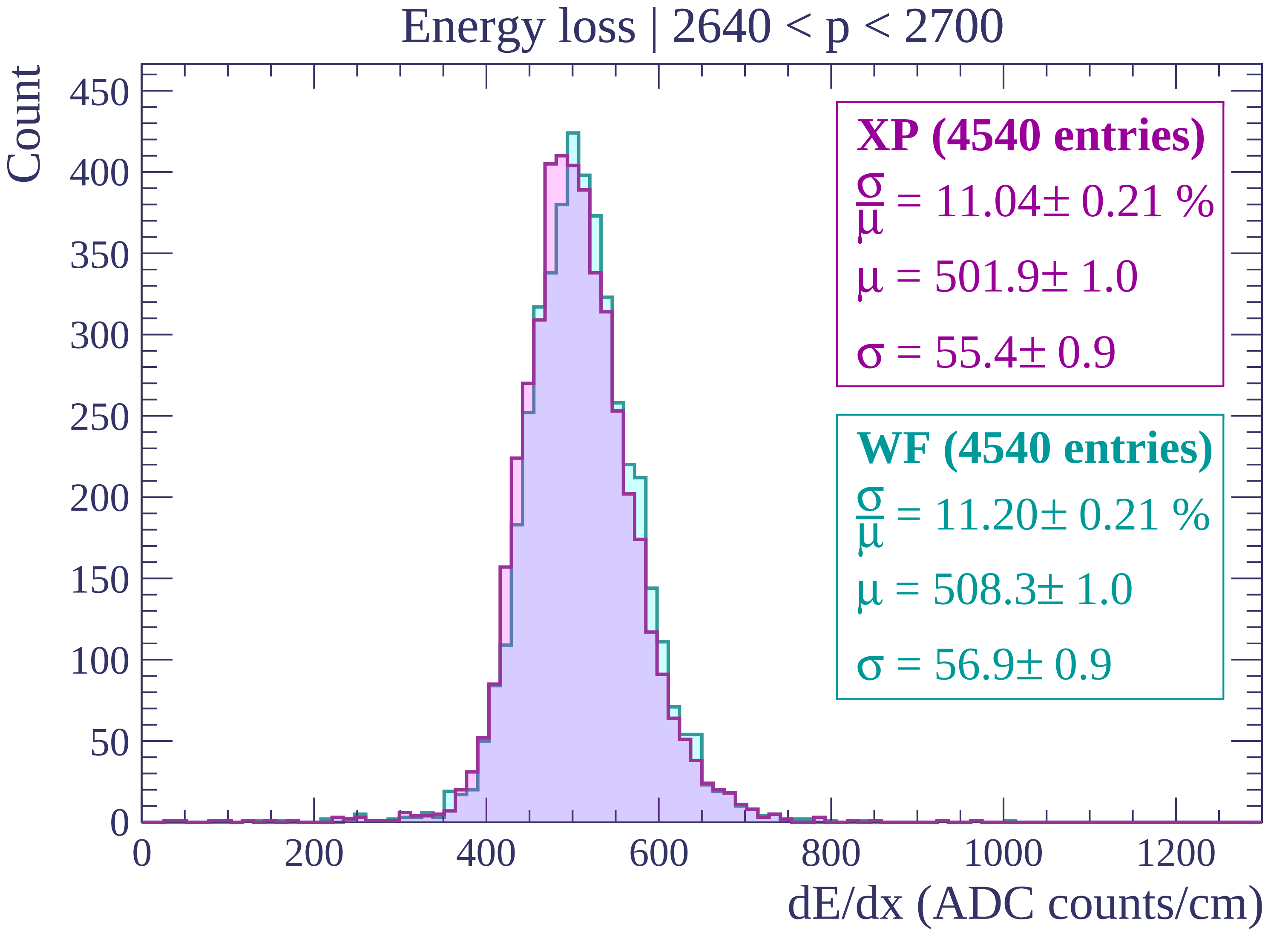
Energy loss | 2460 < p < 2520
Count
dE/dx (ADC counts/cm)
XP (4948 entries)
σ/μ = 11.66± 0.20 %
μ = 502.0± 1.0
σ = 58.6± 0.9
WF (4948 entries)
σ/μ = 11.82± 0.22 %
μ = 506.5± 1.0
σ = 59.9± 1.0
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200

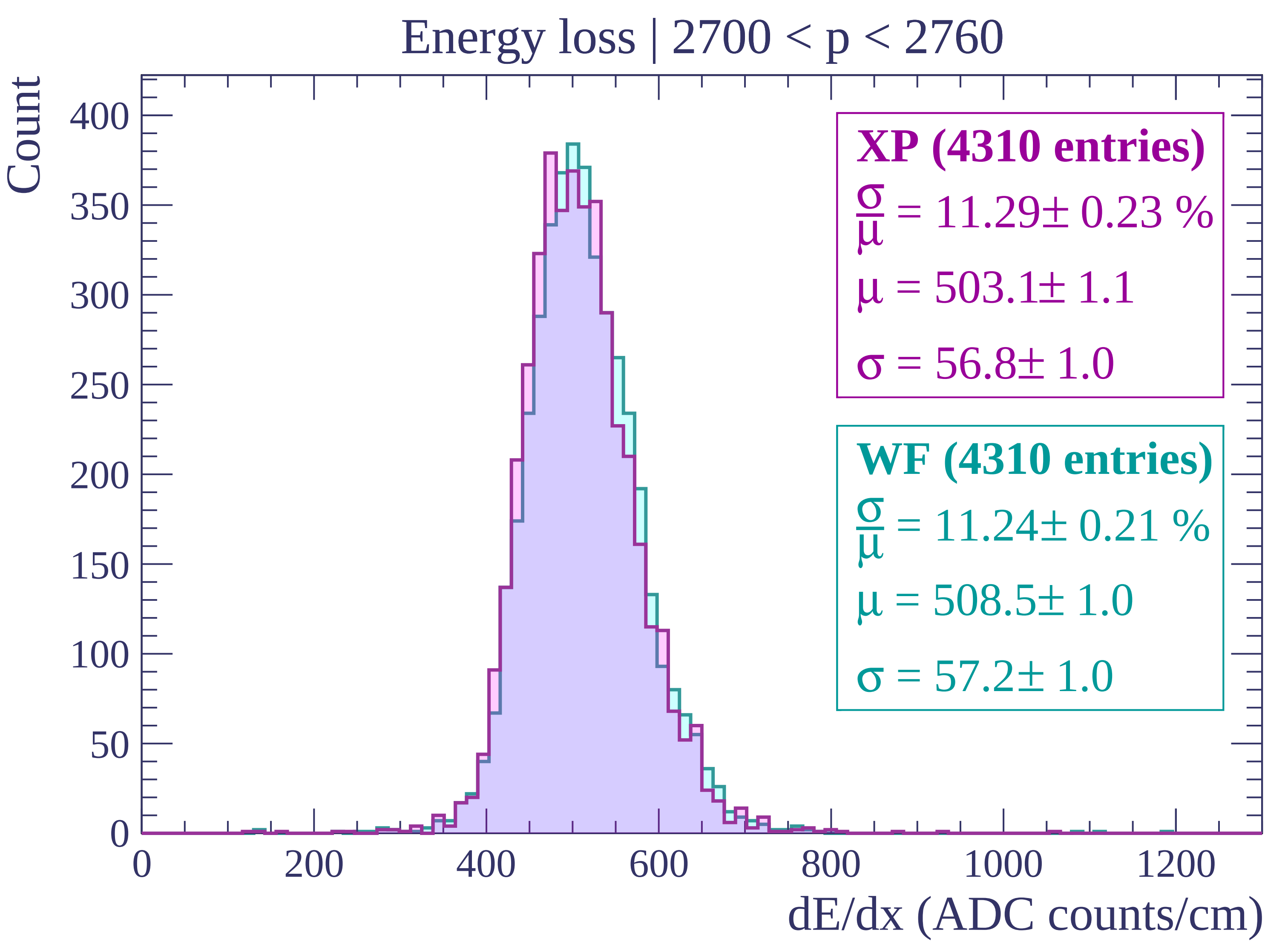


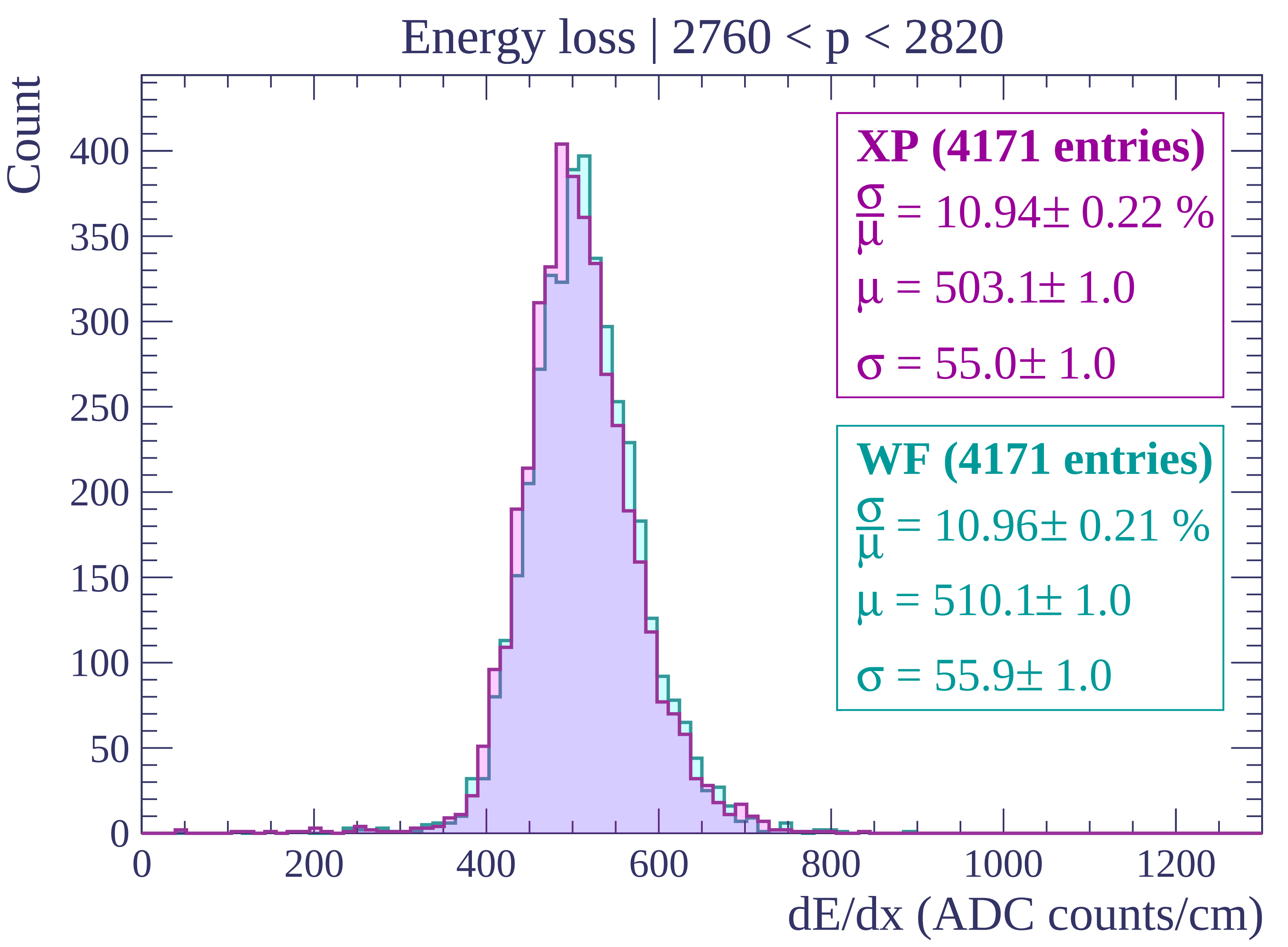
Energy loss | 2520 < p < 2580
Count
XP (4737 entries)
σ/μ = 11.48± 0.21 %
μ = 503.5± 1.0
σ = 57.8± 0.9
WF (4737 entries)
σ/μ = 11.57± 0.22 %
μ = 508.4± 1.0
σ = 58.8± 1.0
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200



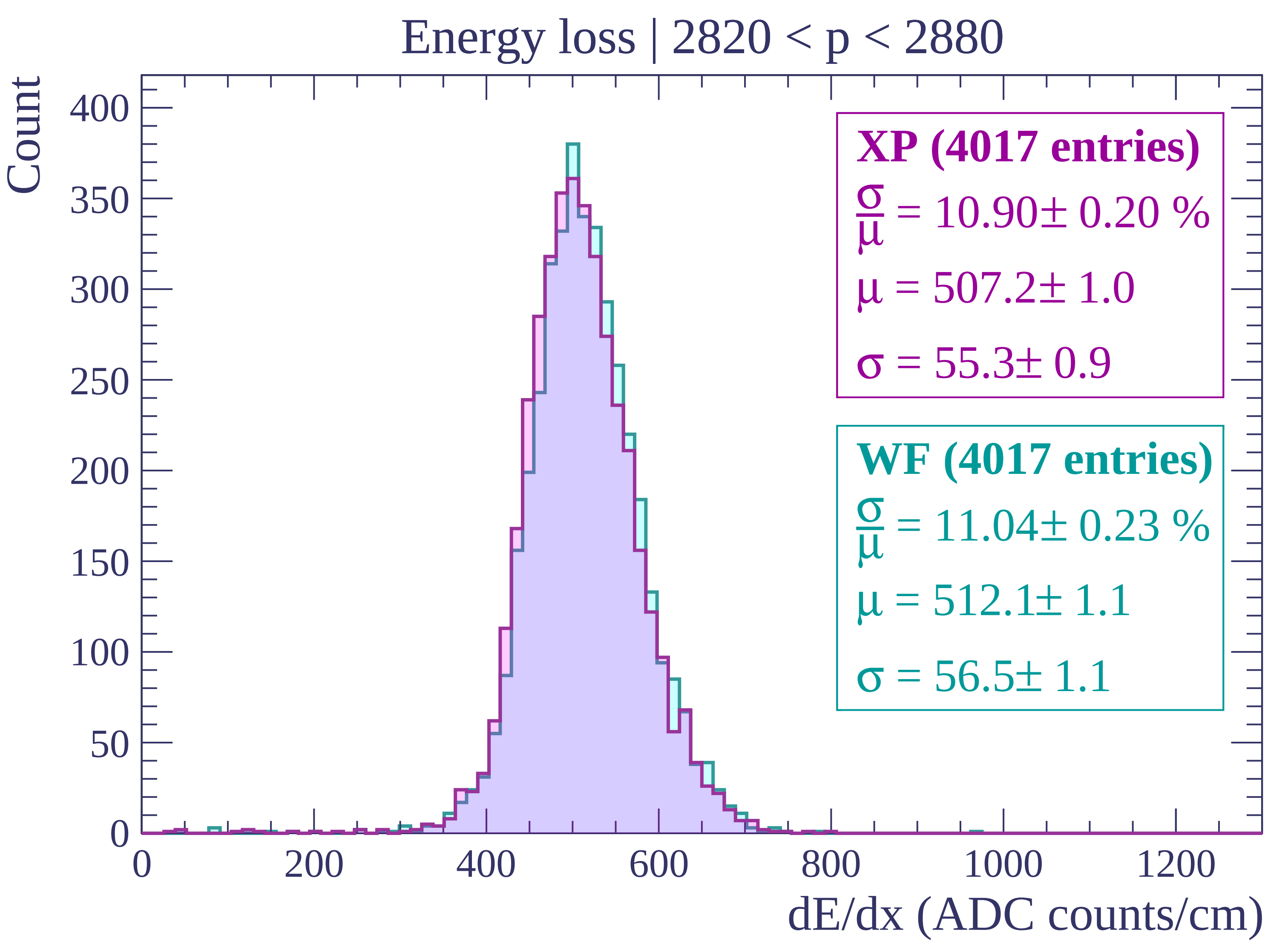
Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4680 entries)
σ/μ = 11.39± 0.24 %
μ = 502.9± 1.0
σ = 57.3± 1.1
WF (4680 entries)
σ/μ = 11.47± 0.22 %
μ = 509.3± 1.0
σ = 58.4± 1.0

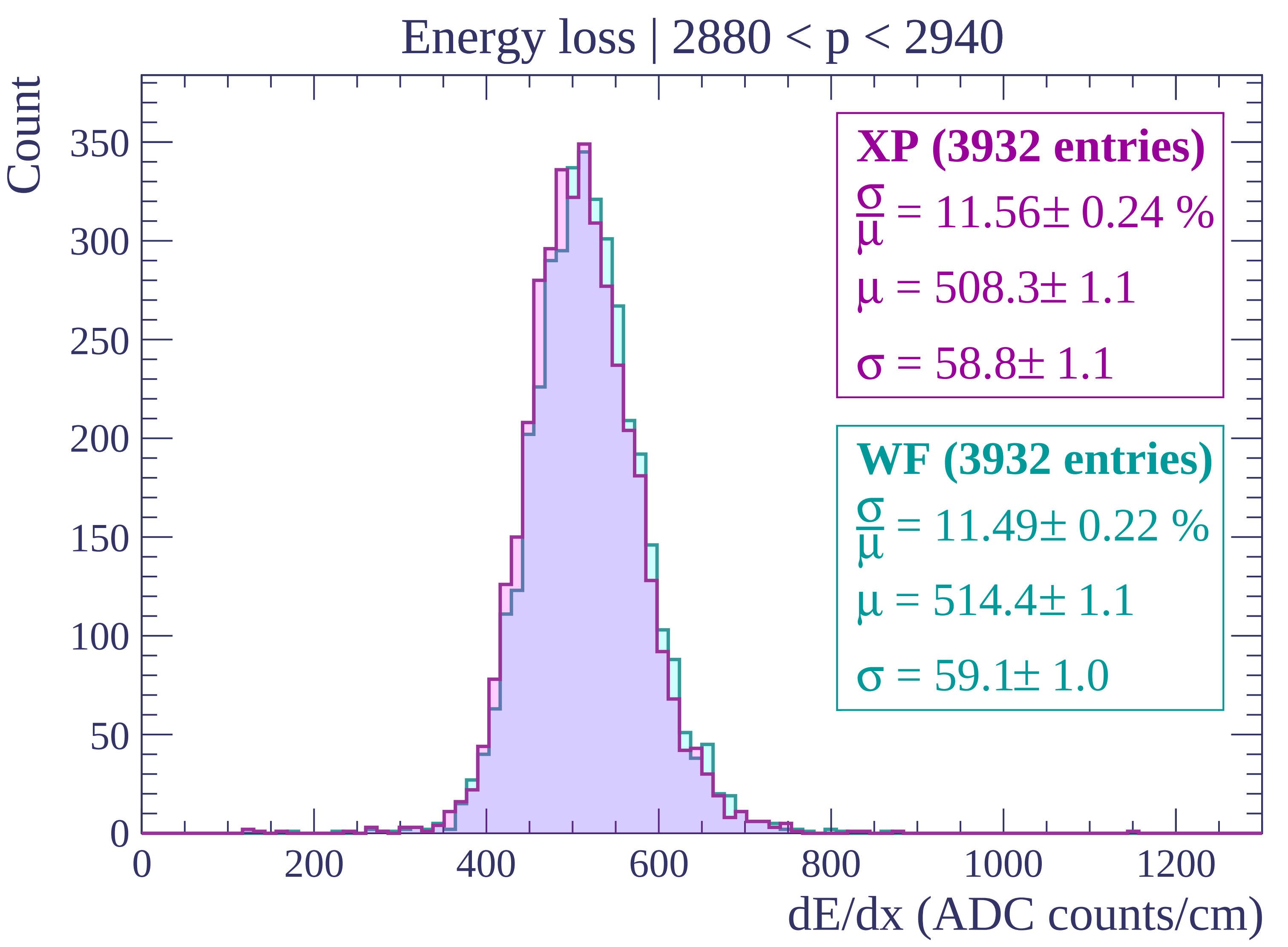

Energy loss | 2640 < p < 2700
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4540 entries)
$\frac{\sigma}{\mu}$ = 11.04± 0.21 %
μ = 501.9± 1.0
σ = 55.4± 0.9
WF (4540 entries)
$\frac{\sigma}{\mu}$ = 11.20± 0.21 %
μ = 508.3± 1.0
σ = 56.9± 0.9

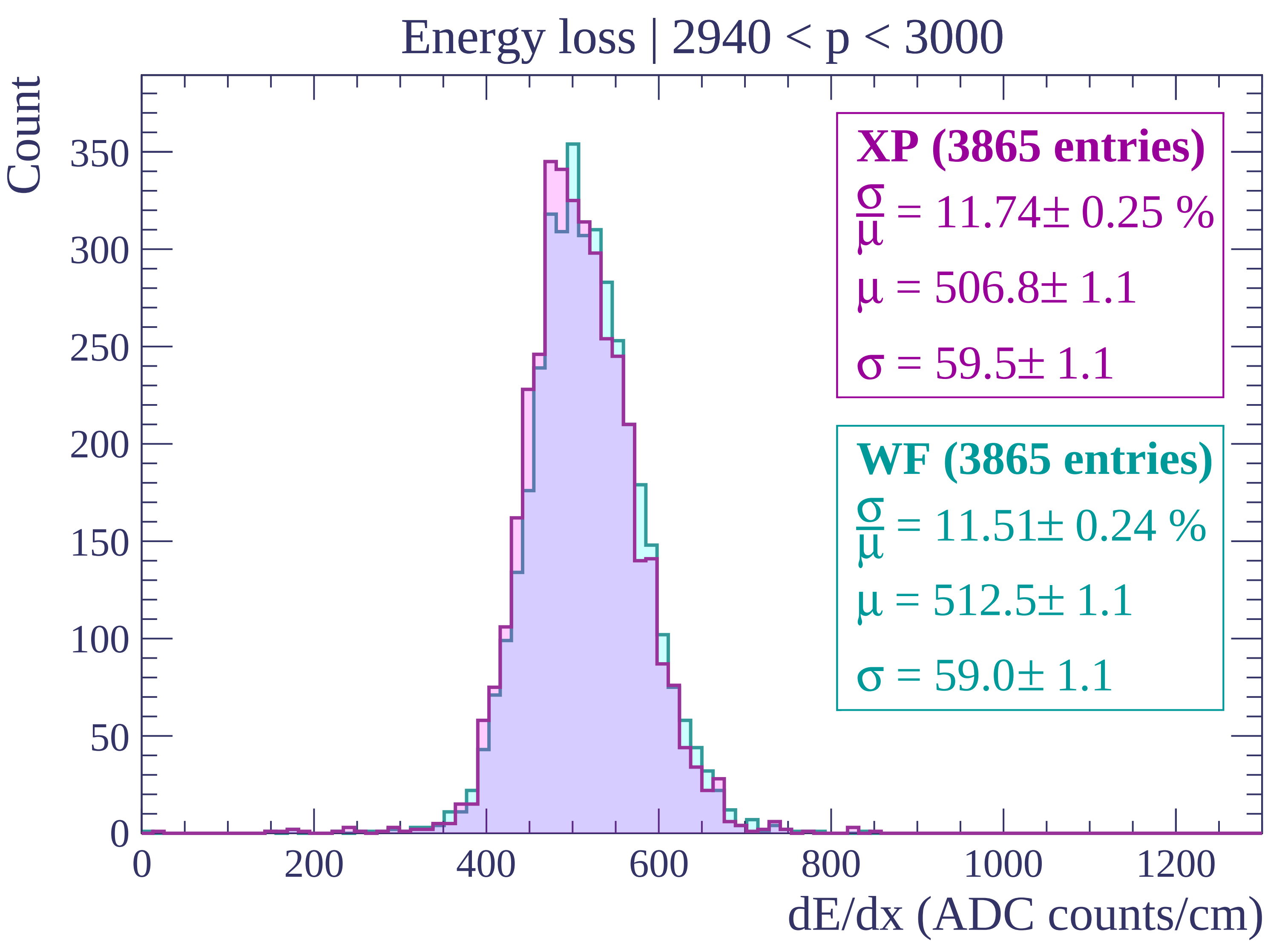

Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
XP (4310 entries)
σ/μ = 11.29± 0.23 %
μ = 503.1± 1.1
σ = 56.8± 1.0
WF (4310 entries)
σ/μ = 11.24± 0.21 %
μ = 508.5± 1.0
σ = 57.2± 1.0

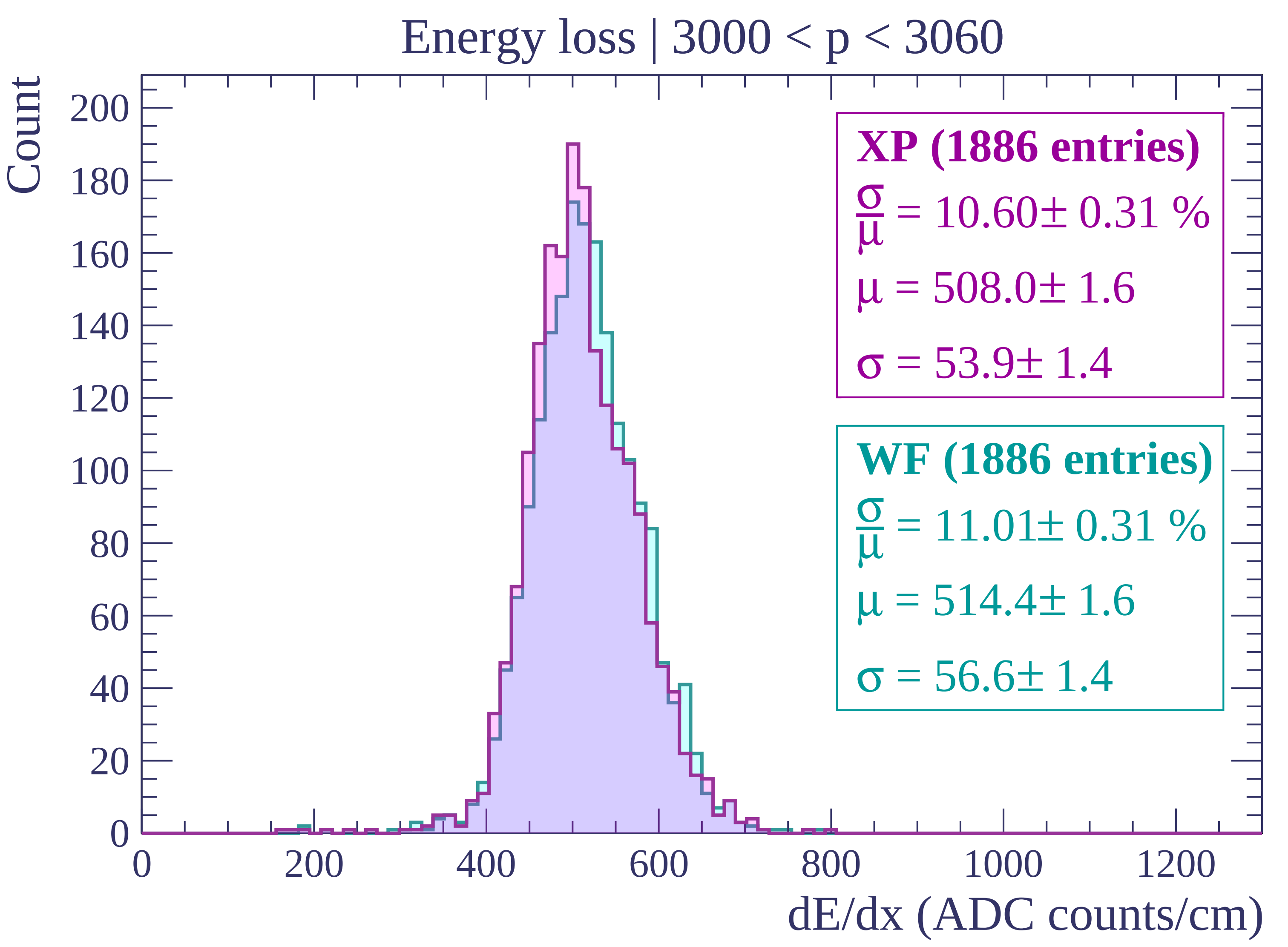


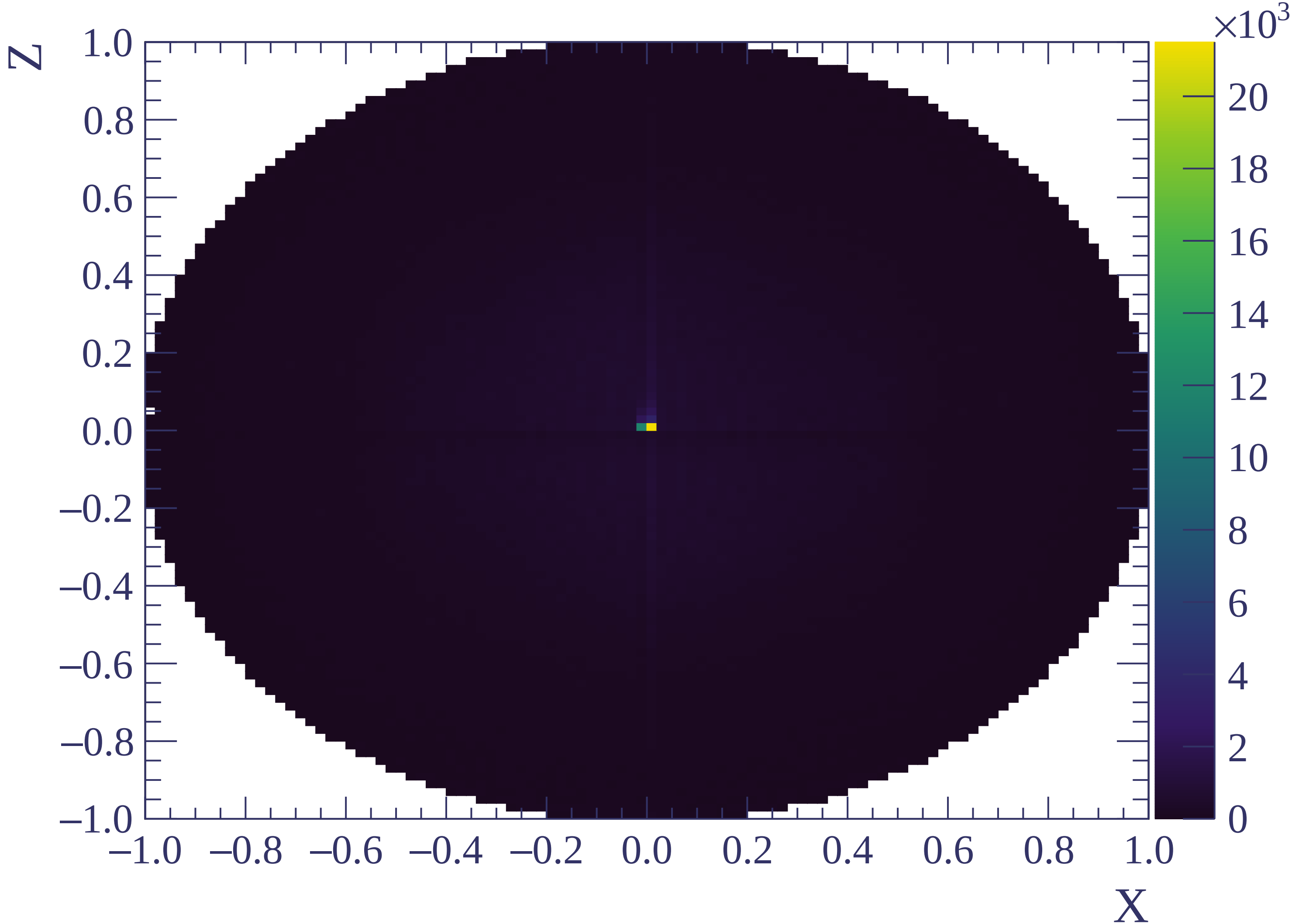
Energy loss | 2760 < p < 2820
Count
dE/dx (ADC counts/cm)
XP (4171 entries)
$\frac{\sigma}{\mu}$ = 10.94 ± 0.22 %
μ = 503.1 ± 1.0
σ = 55.0 ± 1.0
WF (4171 entries)
$\frac{\sigma}{\mu}$ = 10.96 ± 0.21 %
μ = 510.1 ± 1.0
σ = 55.9 ± 1.0

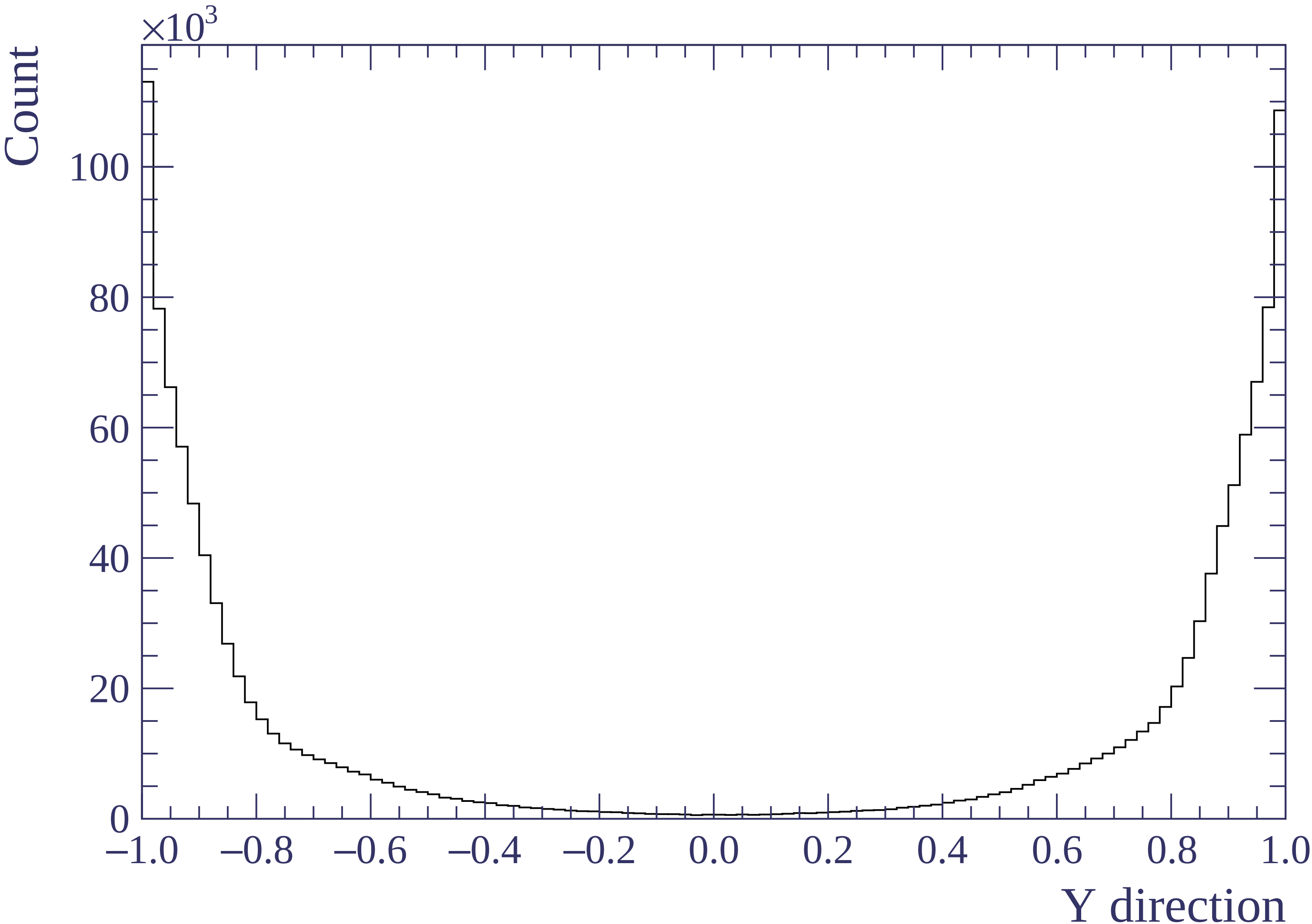

Energy loss | 2820 < p < 2880
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (4017 entries)
σ/μ = 10.90± 0.20 %
μ = 507.2± 1.0
σ = 55.3± 0.9
WF (4017 entries)
σ/μ = 11.04± 0.23 %
μ = 512.1± 1.1
σ = 56.5± 1.1


Energy loss | 2880 < p < 2940
Count
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3932 entries)
σ/μ = 11.56± 0.24 %
μ = 508.3± 1.1
σ = 58.8± 1.1
WF (3932 entries)
σ/μ = 11.49± 0.22 %
μ = 514.4± 1.1
σ = 59.1± 1.0

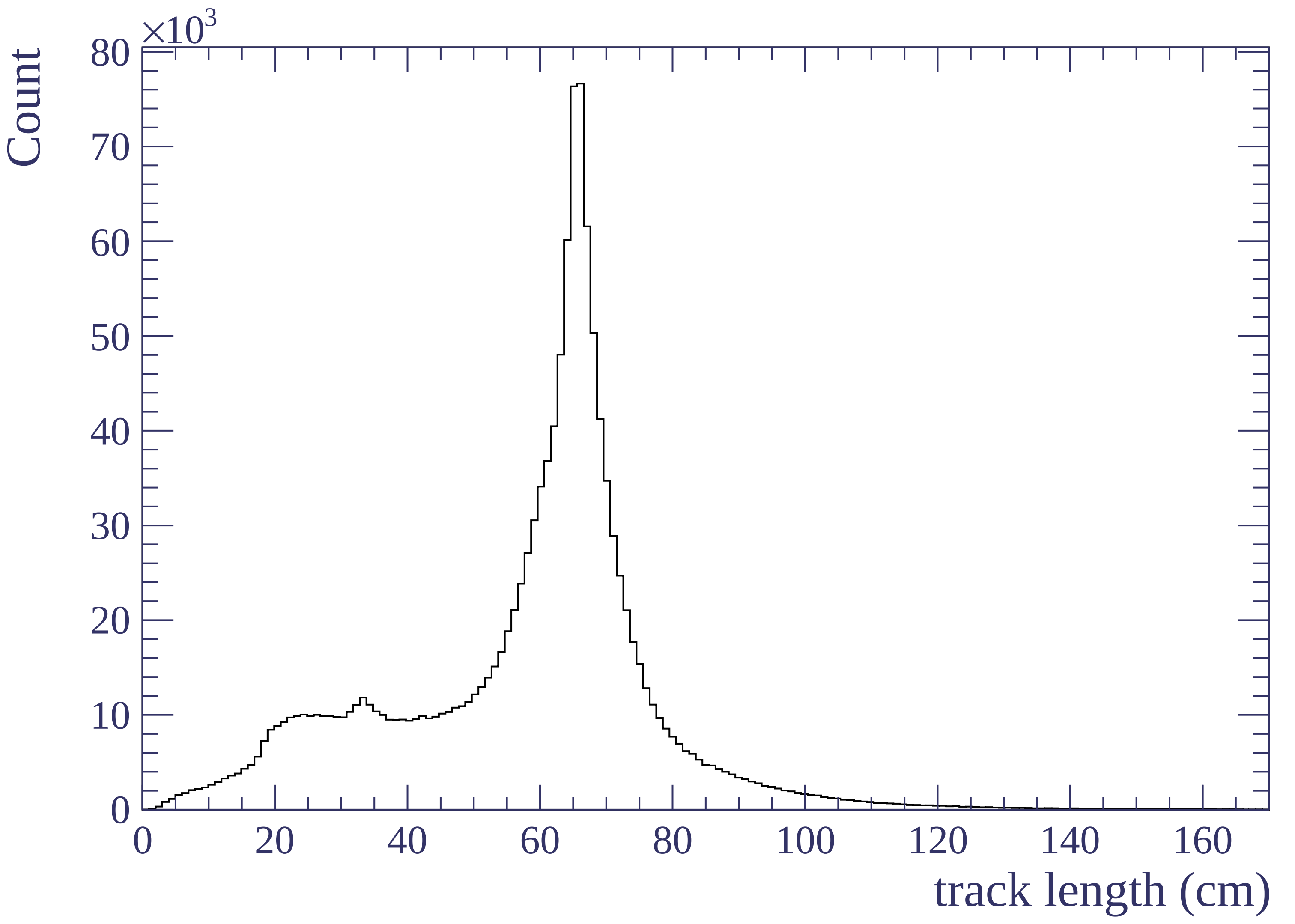

Energy loss | 2940 < p < 3000
Count
dE/dx (ADC counts/cm)
XP (3865 entries)
$\frac{\sigma}{\mu}$ = 11.74± 0.25 %
μ = 506.8± 1.1
σ = 59.5± 1.1
WF (3865 entries)
$\frac{\sigma}{\mu}$ = 11.51± 0.24 %
μ = 512.5± 1.1
σ = 59.0± 1.1
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200


Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
XP (1886 entries)
$\frac{\sigma}{\mu}$ = 10.60± 0.31 %
μ = 508.0± 1.6
σ = 53.9± 1.4
WF (1886 entries)
$\frac{\sigma}{\mu}$ = 11.01± 0.31 %
μ = 514.4± 1.6
σ = 56.6± 1.4
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200

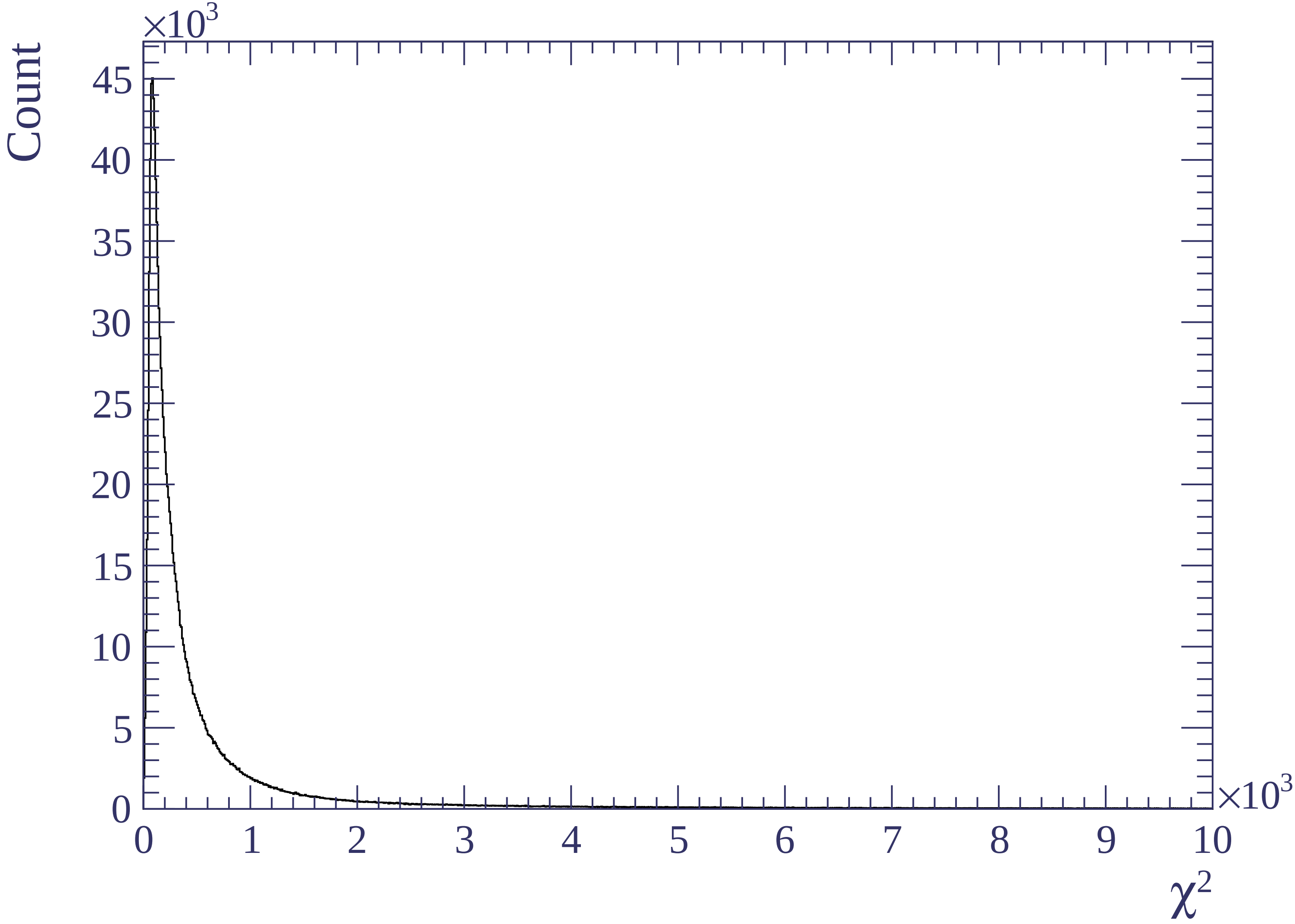

Z
X
×10^3
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
20
18
16
14
12
10
8
6
4
2
0



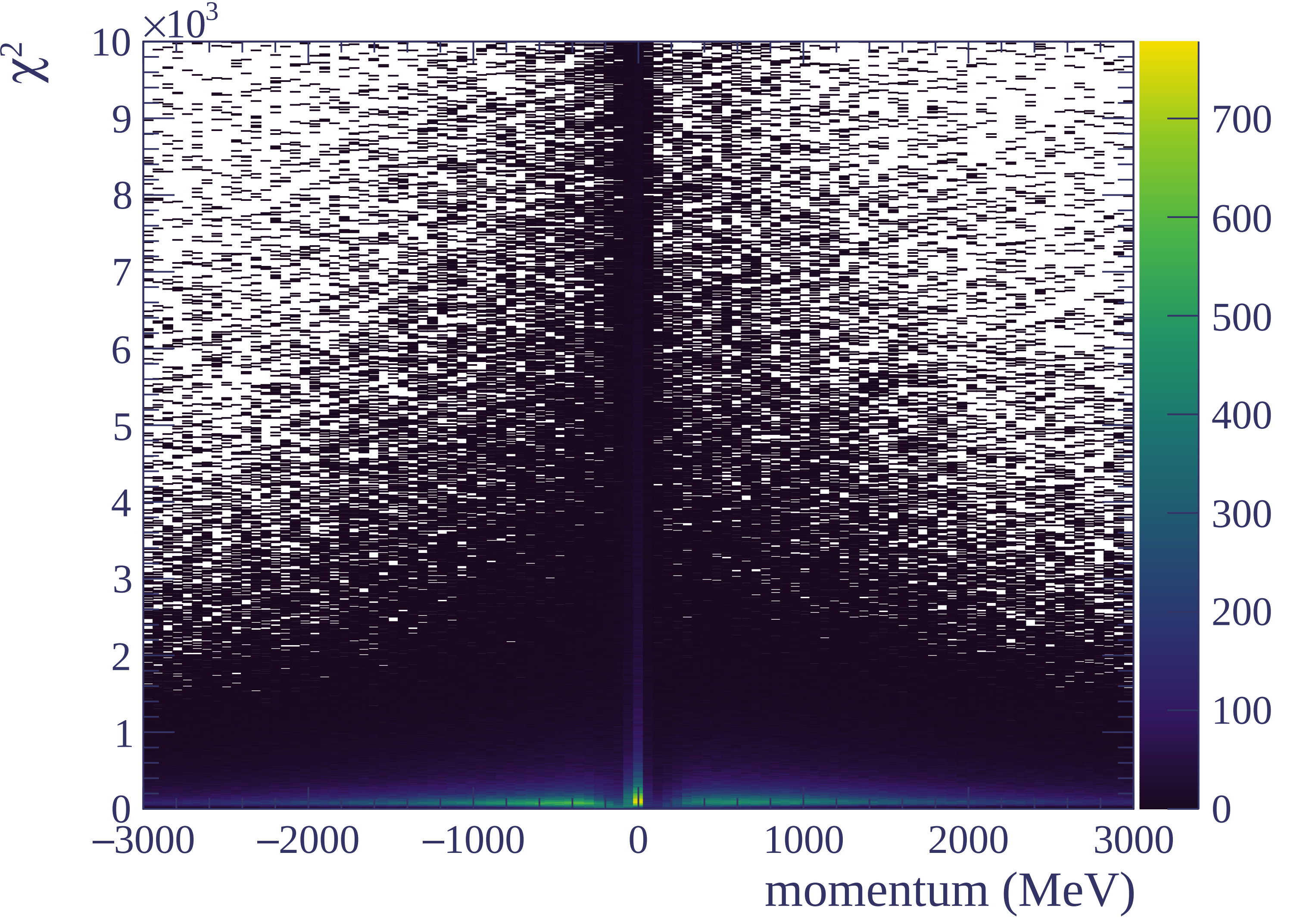
×10^3
Count
100
80
60
40
20
0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction



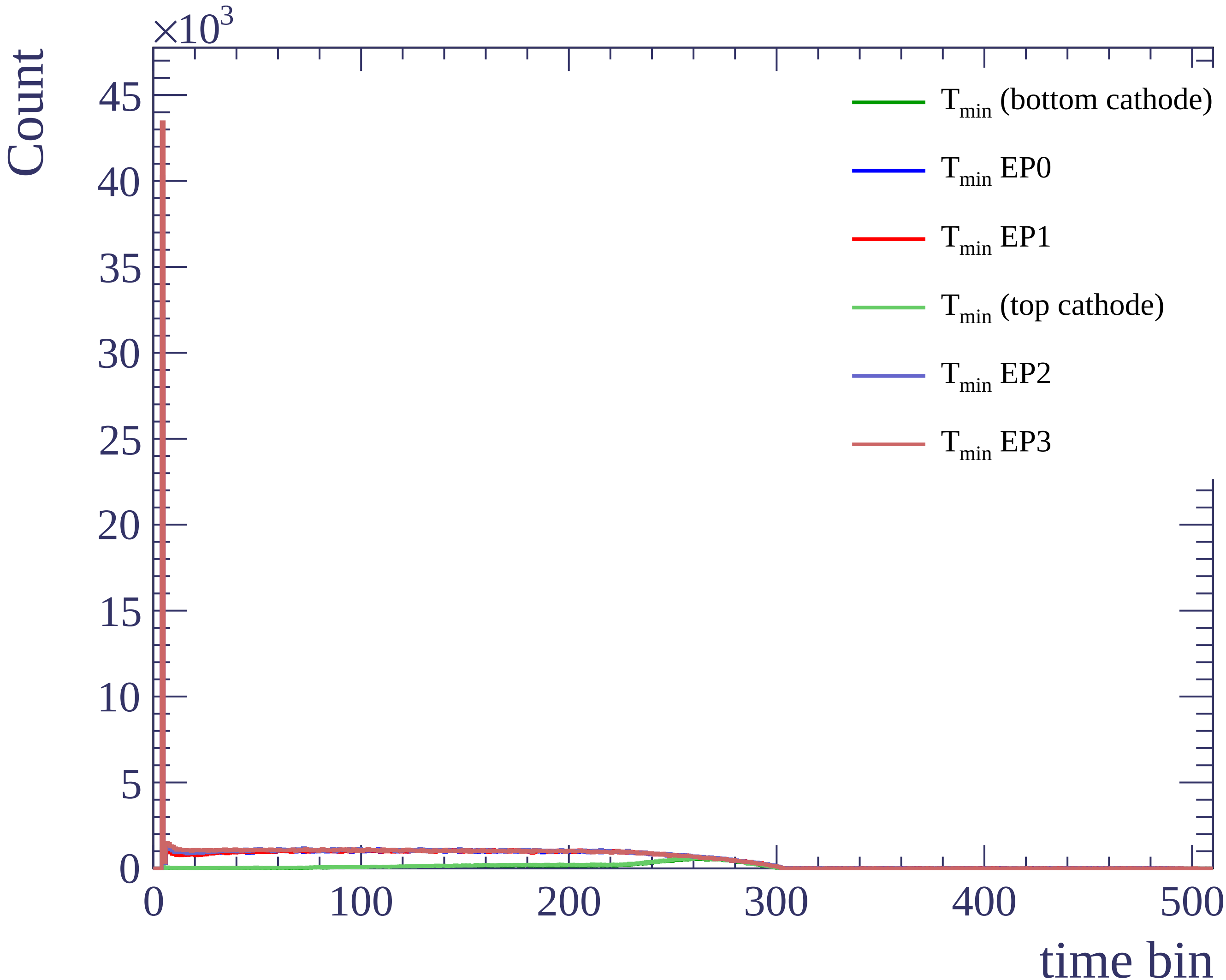
×10^3
Count
0
10
20
30
40
50
60
70
80
0
20
40
60
80
100
120
140
160
track length (cm)



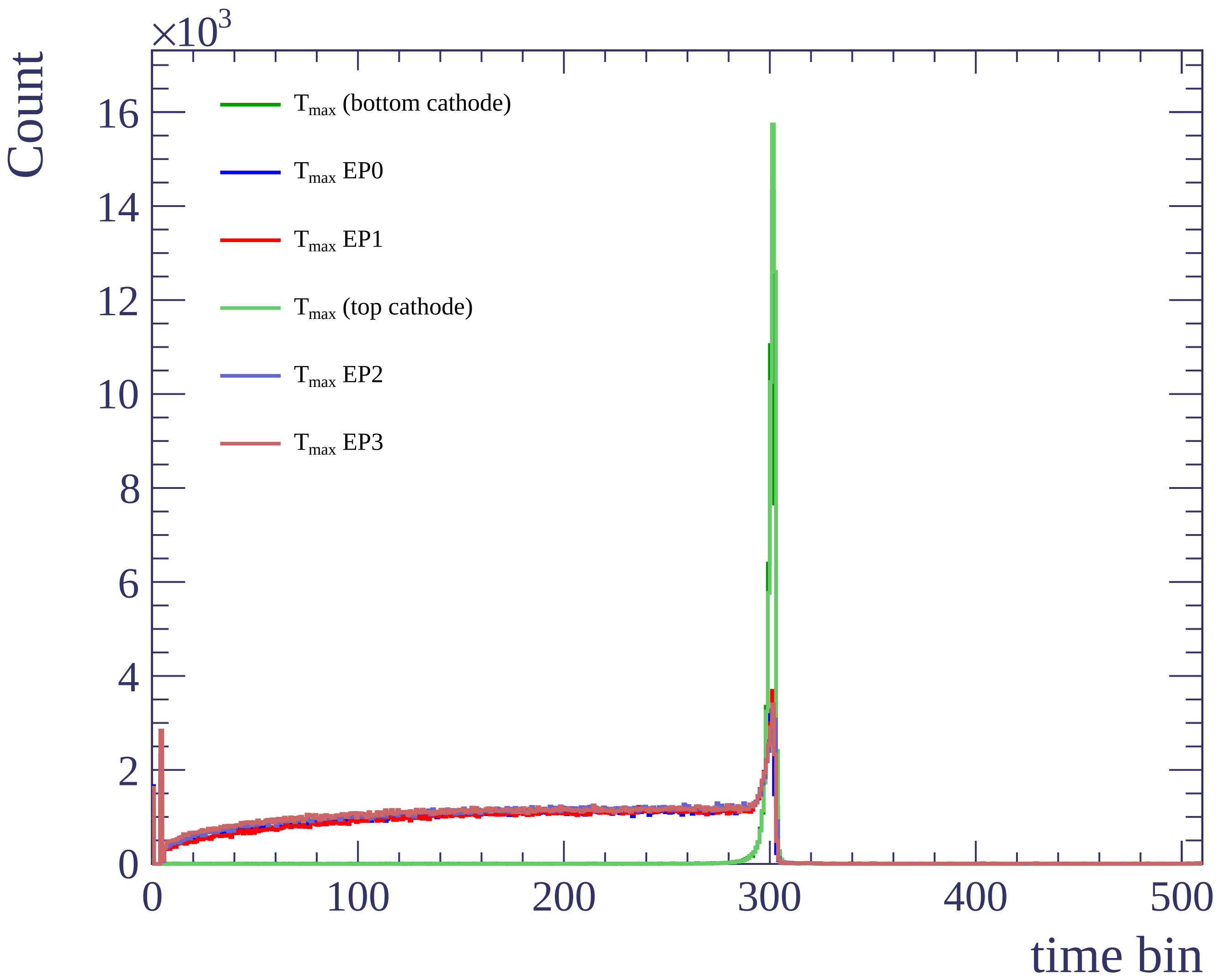
Count
×10³
45
40
35
30
25
20
15
10
5
0
0
1
2
3
4
5
6
7
8
9
10
×10³
χ²



χ^2
$\times 10^3$
10
9
8
7
6
5
4
3
2
1
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
700
600
500
400
300
200
100
0



×10^3
Count
45
40
35
30
25
20
15
10
5
0
0
100
200
300
400
500
time bin
T_min (bottom cathode)
T_min EP0
T_min EP1
T_min (top cathode)
T_min EP2
T_min EP3



$\times10^3$
Count
16
14
12
10
8
6
4
2
0
100
200
300
400
500
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3